Annexure 'E'
Specifications (for water supply related works)

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Chapter 1

General :

1.1 Scope of work

The specifications for this project & various components / materials to be used for the project shall confirm to BIS standards with up to date amendments as given below,

S. No.	IS Code No.	Title
1.	IS 269:1989	33 grade ordinary Portland cement
2.	IS 8112:1989	43 grade ordinary Portland cement
3.	IS 12269:1987	53 grade Ordinary Portland cement
4.	IS 1489:1991	Portland pozzolana cement
	Part I:1991	Fly ash based
	Part II:1991	Calcined clay based
5.	IS 1786:1985	High strength deformed steel bars and wires for concrete reinforcement
6.	IS 875:1987	Code of practice for design loads for building structure
	Part I:1987	Dead loads
	Part II:1987	Imposed loads
	Part III:1987	Wind loads
	Part IV:1987	Snow loads
	Part V:1987	Special loads and load combinations
7.	IS 13920:1993	Ductile detailing of reinforcement concrete structures subjected to seismic forces
8.	IS 1893:2002	Criteria for earthquake resistant design of structures
9.	IS 456:2000	Code of practice for plain and reinforcement concrete(third revision)
10.	IS 457: 1957	Code of practice for general construction of plain and reinforcement concrete for dams and other massive structure.
11.	IS 1343:1980	Code of practice for pre-stressed concrete (first revision)
12.	IS 3370:1965	Code of practice for concrete structure for the storage of liquids.
13.	Part 1:1965	General requirement
	Part 2:1965	Reinforced concrete structure
	Part 3:1967	Pre-stressed concrete structures
	Part 4:1967	Design tables
14.	IS 6518:1972	Code of practice for control of sediment in reservoirs
15.	IS 5330:1984	Criteria for design of anchor block for penstock with joints (first revision)
16.	IS 7357:1974	Code of practice for structural design of tanks.
17.	IS 3913:1966	Suspended sediment load samplers
18.	IS 3917:1966	Scoop type bed material samplers.
19.	IS 4890: 1968	Method for measurement of suspended sediment in open channels.
20.	IS 4926:1976	Ready mix concrete (first revision)
21.	IS 6295:1986	Code of practice for water supply and drainage high altitude and/or sub-zero temperature regions(first revision)
22.	IS 5477	Method for fixing the capacities of reservoir
	Part 1:1969	General Requirement
	Part2:1969	Dead storage

	Part3:1969	Live storage
	Part4:1971	Flood storage
23.	IS 9668:1980	Code of practice for provision and maintenance of water supply for fire fighting
24.	IS 8062	Code of practice for cathodic protection for steel structure
25.	Part1:1976	General principles
26.	Part2: 1976	Underground pipelines
27.	IS 10221:1982	Code of practice for coating and wrapping of underground steel pipes
28.	IS 8329: 2000	Centrifugally cast(spun) ductile iron pressure pipes for water, gas, and sewerage
29.	IS 9523:1980	Ductile iron fittings for pressure pipes for water, gas, and sewerage
30.	IS 11906:1986	Recommendation for cement mortar lining cast iron, mild steel and ductile iron pipes
31.	IS 12288:1987	Code of practice for laying of ductile iron pipes
32.	IS 4984:1994	HDPE pipes for potable water supplies, sewage and industrial effluents(third revision)
33.	IS 7634 Part2:1975	Laying and jointing polyethylene (PE) pipes.
	Part1:1976	General requirement
34.	IS 14333	High Density Polyethylene Pipes for sewerage
35.	IS 2530	Methods of test for polyethylene moulding materials and polyethylene compounds DI K7 Pipes, Joints and Fittings for use for Sewerage applications
36	IS 5382	Rubber sealing rings for gas mains, water mains and sewers.
37	IS 4905	Methods for random sampling
38	IS 8008	Injection moulded HDPE fittings for potable water supplies
	Part1:1976	General requirement
39	IS 7328	High density polyethylene materials for moulding and extrusion

1. BIS specified in integrated Standard Schedule of Rates, **Volume -1** (Water supply, Sewerage and Tubewell works), **Volume -2** (Building works), **Volume - 3** (Road & Bridge Works), **Volume - 4** (Electrical Works) issued by Directorate Urban Administartion and Development effective since **02nd August 2021** with latest amendments shall be applicable.

The specifications for various materials to be used for the project shall confirm to BIS standards. Any other BIS standards as may be required will shall also be applicable. Quality assurance programme of the manufacturer shall have to be enclosed with the detailed design and drawings.

Any other BIS standards as may be required will also be applicable. Quality assurance program of the manufacturer shall have to be enclosed with the detailed design and drawings.

2. The other part of the specifications for various components of the project shall be as per provisions of clauses and sub clauses of chapters of Manual on Water supply and Treatment (Latest edition), CPHEEO Ministry of Urban Development Govt. Of India or any higher/ better standards.

a. Transmission of water, Chapter 6

b. Water treatment, Chapter 7, 8 and 9

c. Distribution system, Chapter 10

d. Pumping Stations and Machinery, Chapter 11

e. Instrumentation and Control in Water treatment plant, Chapter 12

f. Operation and Maintenance, Chapter 13

g. Laboratory Tests and Procedures, Chapter 15

Disclaimer: **Any specifications not covered above shall be as per best Engineering practice or as directed by Engineer In Charge. In the event of any disparity between the written specifications and BIS provisions, the provisions in BIS shall prevail.**

The item wise specifications to be followed by the Contractor are given in the subsequent Chapters. The following are some guidelines for achieving the best possible quality of work under this project. Contractor shall submit the details of all material to be procured meeting BIS standards and specifications for approval from Engineer In charge (prior approval is required)

Chapter 2

Pipeline

1.0 Excavation of trenches, back filling, road restoration

1.1. Excavation for Pipe Line Trenches

The excavation shall be carried out by open cut method, cutting of road shall be carried out by **Road Cutter only**, Horizontal drilling is adopted where permission is not obtained for open excavation such as Railway, NHAI crossing etc. The excavation in hard rock will have to be carried out either by controlled blasting or chiseling, wedging or by mechanical means and the tendered rate is supposed to cover cost of all such means.

1.2. Site Clearance

The pipe line alignment shall be cleared of all bushes, shrubs, roots, grass, weeds and if required trees, coming in the alignment of pipe line in the trench width portion. The rates for excavation shall cover all such site clearance work and no extra payment will be allowed on this account.

1.3. Alignment marking

After the work site is cleared as above, pipe line alignment with required trench width shall be marked on the ground with apex points, curves etc, as shown on the drawings or as directed by the Engineer-in-Charge for the stretch where the work is to be started. The contractor shall provide all labour , survey instruments, and materials such as strings, pegs, nails, bamboos, stones, mortar, concrete etc. required for setting out and establishment of bench marks. The contractor shall be responsible for the maintenance of benchmarks and other marks and stakes as long as they are required for the work in the opinion of the Engineer-in-Charge.

1.4. Working survey

The contractor before start of the excavation work shall carry out working survey of the pipeline alignment. Based on the working survey, the alignments, L-section (depth of laying), grade, and location of specials, valves and chambers shall be finalized and got approved from the competent authority. The gradient and alignment shall be such that minimum horizontal and vertical bends shall be required.

1.5. Use of Machinery:

All excavations shall be carried out by mechanical equipment's / machinery unless, in the opinion of the Engineer-in-Charge, the work involved and time schedule permit manual excavation.

1.6 Trench Width and Depth:

All buried pipelines shall be minimum 1 meter +/- 0.2 mtr below ground level to maintain proper cover & gradient unless other depths are approved by the engineer in charge. The trench width shall be constant throughout the trench depth, & shall provide a minimum clearance of about 0.30 m on either side of the pipe line. The contractor may, for the facility of work or similar other reasons, excavate and also backfill later, if so approved by the Engineer-in-Charges, at his own cost, outside the allowable trench width specified above. Should any excavation be taken below the specified trench bottom, contractor shall fill it up to required level, at his own cost, with the same material available at the trench bottom including watering and compaction.

The excavation shall be as per approved in drawings, having sufficient depth over the pipe cover, bedding below pipe line wherever bedding is required. The trench bottom shall be excavated to proper grade as shown on drawings. The contractor shall provide site rails and leveling instruments required for checking the grade during excavation, bottom bedding and pipe laying Projections in rock excavation shall be removed by chipping.

The contractor shall carryout extra excavation at the pipeline joints to be welded in the trench, as required (minimum 0.6 m deep and 0.9 m lengthwise, all around the pipe), for facilitating proper welding of the bottom joint from outside. The work of trench excavation should be commensurate with laying and jointing of the pipe line. It should not be dug in advance for a length greater than 500 m ahead of work of laying and jointing of pipeline unless otherwise permitted by the Engineer-in-Charge.

1.7 Barricading and Guarding:

To protect persons from injury and to avoid damage to property, adequate barricades, construction signs, red lanterns and guards as required shall be placed and maintained during the progress of work, by the contractor at his cost.

All precautions shall be taken during excavation and laying operation to guard against possible damage to any existing structures, underground cables, pipe lines of water, gas, sewage etc. Any damage done to such properties will have to be repaired / rectified by the contractor at his cost. The Contractor has to ensure the following:

- safety protections as mentioned above have to be incorporated in the work process
- hindrances to the public have to be minimized
- the trench must not be eroded before the pipes are laid
- the trench must not be filled with water when the pipes are laid
- the trench must not be refilled before laying of the pipes

The bed for the laying of the pipes has to be prepared according to the L-Section immediately before laying of the pipes.

1.8. Reuse of surface material

All surface materials, which in the opinion of the Engineer-in-Charge, suitable for reuse in restoring the surface shall be kept separate from the general excavation material, as directed by the Engineer-in-Charge.

1.9. Stacking of excavated material

All excavated materials shall be stacked in such a manner that it does not endanger the work and avoids obstructing foot paths and roads. Hydrants under pressure, surface boxes, fire and other utility controls shall be left unobstructed and accessible until the work is completed. Gutters shall be kept clean or other necessary provisions made for street drainage and natural water courses shall not be obstructed. All the excavated material shall be the property of the Employer and shall be stacked or disposed off as directed by the Engineer-in-Charge.

1.10. Maintenance of traffic

The work of excavation and pipe laying shall be carried in such a manner that it causes the least interruption to traffic and the road / street May be closed in such a manner that it causes the least interruption to the traffic. Where it is necessary for traffic to cross open trenches, suitable bridging arrangement shall be provided. When the street is closed for traffic, suitable signs indicating that street is closed shall be placed and necessary detour signs for proper maintenance of traffic shall be provided.

1.11. Structure protection

Temporary support, adequate protection and maintenance of all underground and surface structures, drains, sewers and other obstructions encountered in the progress of work shall be furnished under the direction of the Engineer-in-Charge. The structures, which have been disturbed, shall be restored upon completion of work.

1.12. Protection of property

Trees, shrubbery fences, poles and all other property shall be protected unless their removal is allowed by the Engineer-in-Charge. When it is necessary to cut roots and tree branches, such cutting shall be done under the supervision and direction of the Engineer-in-Charge.

1.13. Avoidance of existing services

As far as possible, the pipeline shall be laid below existing services, such as water and gas pipes, cables, cable ducts and drains but not below sewers. Excavation of the trenches shall be carried out to the required depth accordingly. If it is unavoidable, the pipeline shall be suitably protected and lesser trench depth in such cases can be allowed. A minimum clearance of 150 mm shall be provided between the pipeline and such other services. When thrust or auger boring is proposed for laying pipeline across roads, railway or other utilities, larger clearance as required shall be provided. Adequate arrangements shall be made to protect and support the other services during excavation and pipe laying operations. The work shall be so carried out as not to obstruct access to the other services for inspection, repair and replacement. When such utilities are met with during excavation, the authority concerned shall be intimated and arrangements made to support the utilities in consultation with them.

1.14. Bailing out of Water

During the excavation if subsoil water is met with, contractor shall provide necessary equipment and labour for dewatering the trenches. If pumping out subsoil water is found necessary, contractor shall provide sufficient number of pumps for the same. The TENDERED rate shall cover all costs for bailing out of water including hire charges of pumps, cost of diesel and labour etc and hence, no extra payment shall be allowed.

1.15. Disposal of loose boulders etc

All loose boulders, semi detached rocks, (along with earthy stuff which might move therewith), not directly in the excavation but close to the area to be excavated, as to be liable, in the opinion of the Engineer-in-Charge, to fall or other wise endanger the workman equipments, or the work etc, shall be stripped off and removed away form the area of the excavation. The method used shall be such as not to shatter or render unstable or unsafe the portion which was originally sound and safe. The Tendered rate is supposed to cover this job and no extra payment will be allowed on this account.

1.16. Moorum / Sand Bedding below Pipeline

In case of hard rock and black cotton soil, before lowering of the pipes in trenches, a layer of selected moorum, available from excavated material under the same contract shall be provided below the pipe line to act as bedding. The bedding shall be compacted properly including required watering and the thickness of well compacted layer shall not be less than 100 mm. The bedding shall be provided for full trench width with proper grade as shown on drawings.

2. Refilling the trenches

2.1. Use of selected excavated material

Filling of excavated material in trenches shall be commenced as soon as the joints of pipes and specials have been tested and passed. The backfilling material shall be properly consolidated by watering and ramming, taking due care that no damage is caused to the pipes and the outer coating.

Selected surplus spoils from excavated material shall be used as backfill. Fill material shall be free from clods, salts, sulphate, organic or other foreign material. All clods of earth shall be broken or removed. Where excavated material is mostly rock, the boulders shall be broken into pieces not larger than 150 mm size, mixed with properly graded fine material consisting of moorum or earth to fill up the voids and the mixture used for filling.

2.2. Filling zones

For the purpose of back-filling, the depth of the trench shall be considered as divided in to the following three zones from the bottom of the trench to its top:

Zone A: From the bottom of the pipe (top of bedding) to the level of the centre line of the pipe	Back-filling by hand with selected approved material available from excavation, placed in layers of 150 mm and compacted by tamping. The back-filling material shall be deposited in the trench for its full width on each side of the pipe, specials and appurtenances simultaneously. Special care shall be taken to avoid damage of the pipe and the coating or moving of the pipe.
Zone B: From the level of the centre line of the pipe to a level 300 mm above the top of the pipe	Back-filling and compaction shall be done by hand or approved mechanical methods in layers of 150 mm; special care shall be taken to avoid damage of the pipe and the coating or moving of the pipe.
Zone C:	Back-filling shall be done by mechanical methods in 15 cm.

2.3. All excavations shall be backfilled to the level of the original ground surfaces unless otherwise shown on the drawings or ordered by the Engineer-in-Charge in Charge, and in accordance with the requirements of the specification. The material used for backfill, the amount thereof, and the manner of depositing and compacting shall be subject to the approval of the Engineer-in-Charge in Charge, but the Contractor will be held responsible for any displacement of pipe or other structures, any damage to their surfaces, or any instability of pipes and structures caused by improper depositing of backfill materials.

The back filled layers shall be wetted and compacted to a density of minimum 90 percent of the maximum dry density at optimum moisture content of the surrounding material. Any deficiency in the quantity of material for backfilling the trenches shall be supplied by the Contractor at his expense.

The Contractor shall at his own expense make good any settlement of the trench backfill occurring after backfilling and until the expiry of the defects liability period.

On completion of pressure and leakage tests exposed joints shall be covered with approved selected backfill placed above the top of the pipe and joints in accordance with the requirements of the above specifications. The Contractor shall not use backfilling for disposal as refuse or unsuitable soil.

2.4. Fillings of the trench excavated in rock

In case of excavation of trenches in rock, the filling up to a level of 30 cm above the top of the pipe shall be done with fine materials, such as soft soil, moorum etc. The filling up of the level of the centre line of the pipe shall be done by hand compaction in layers not exceeding 15 cm, whereas the filing above the centre line of the pipe shall be done by hand compaction or mechanical means in layers not exceeding 15 cm. The filling from a level of 30 cm above the top of the pipe to the top of the trench shall be done by mechanical methods with broken rock filing of size not exceeding 15 cm mixed with fine material as available to fill up the voids.

2.5. Consolidation

The consolidation of the filled material shall be done to attain 95 % proctor density. The density of the filled and compacted material shall be tested regularly and record maintained accordingly.

2.6 Road Restoration

The contractor shall be restoring the road after laying & jointing of pipe & refilling of trench. The restoration work shall include WBM & Bituminous/cc surface of thickness & design mix as directed by EIC

3.0 Supply, laying and jointing of DI Pipes and fittings.

3.1 The laying of pipe shall be as per IS 12288:1987 with up to date amendments.

3.2 The manufacturer and their associates (if any) should have the facility to carry out the internal coating lining and external coating / painting at factory for pipes and specials confirming to IS 11906:1986.

3.3 The DI pipe manufacturer should have valid BIS license with an experience of manufacturing and supplying at least 500 km of various diameters of DI pipe to any State/Central govt. / board/organization of repute in last 3 years conforming to IS 8329-2000 and with up to date amendment.

3.4 The DI pipe manufacturer should have house facility for carry out the following test for size DN 80-DN1000:-

a) C -value determination arrangement

b) Type test for leak tightness as per ISO 2531:2009/BS EN 545/IS 8329:200.

3.5 DI pipe manufacturer should have the ISO 9001:2008 & ISO 2531:2009 certification for manufacture of DI pipe.

3.6 Quality assurance program of the manufacturer shall be enclosed with the Tender.

4.0 Laying and jointing of High Density Polyethylene (HDPE) Pipes and fittings.

4.1. The testing, supplying, laying, jointing and testing at work sites of HDPE pipes shall be as per standards and Codes. If requirements of this Specification conflict with the requirements of the standards / Codes, this Specification shall govern.

Code No.	Title/Specification
IS 4984	High Density Polyethylene Pipes for Water Supply
IS 2530	Methods of test for polyethylene moulding materials and polyethylene compounds DI K7 Pipes, Joints and Fittings for use for Potable Water Supply
IS 5382	Rubber sealing rings for gas mains, water mains and sewers.
IS 4905	Methods for random sampling
IS 7328	High density polyethylene materials for moulding and extrusion
IS 7634	Laying & Jointing of Polyethylene (PE) Pipes

IS 9845	Method of analysis for the determination of specific and/or overall migration of constituents of plastics material and articles intended to come into contact with foodstuffs
IS 10141	Positive list of constituents of polyethylene in contact with food stuffs, pharmaceuticals and drinking water.
IS 10146	Polyethylene for its safe use in contact with foodstuff, Pharmaceuticals and drinking water.

4.2 Handling, Transportation Storage and Lowering of pipes

During handling, transportation, storage and lowering, all sections shall be handled by such means and in such a manner that no distortion or damage is done to the section or to the pipes as a whole.

The following procedures should be followed so as to eliminate potential damage to pipes and fittings and to maintain maximum safety during unloading, lifting and lowering.

- Pipes must not be stored or transported where they are exposed to heat sources likely to exceed 60°C.
- Pipes shall be stored such that they are not in contact with direct sunlight, lubricating or hydraulic oils, petrol, solvents and other aggressive materials.
- Scores or scratches to a depth of greater than 10% or more of wall thickness are not permissible; any pipes having such defects should be strictly rejected.
- PE pipes should not be subjected to rough handling during loading and unloading operations. Rollers shall be used to move, drag the pipes across any surface.
- Only polyester webbing slings should be used to lift heavy PE (>315mm) pipes by crane. Under no circumstances, chains, wire ropes and hooks be used on PE pipes.
- Pipes shall not be dropped to avoid impact or bump. If any time during handling or during installation, any damage, such as gouge, crack or fracture occurs, the pipe shall be repaired if so permitted by the competent authority before installation.
- During coiling care should be taken to maintain the coil diameter at or above the specified minimum to prevent kinks. Coiling shall be done when the pipe attains the ambient temperature from the extruder. In uncoiling or recoiling care should be taken that sharp objects do not scour the pipe.
- When releasing coils, it must be remembered that the coil is under tension and must be released in a controlled manner. The end of the coil should be retained at all times, then the straps released steadily, one at a time. If the coil has bands at different layers of the coil, then they should be released sequentially starting from the outer layers. The amount of the energy locked up in the coil will depend on the size of the pipe, the SDR of the pipe, and the size of the coil.
- Straight lengths should be stored on horizontal racks giving continuous support to prevent the pipe taking on a permanent set
- Bare coils shall be wrapped with hessian cloth for long distance (> 300Kms) transportation. The truck used for transportation of the PE pipes shall be exclusively used of PE pipes only with no other material loaded – especially no metallic, glass and wooden items. The truck shall not have sharp edges that can damage the Pipe.
- Pipes manufactured at factory are to be carried to the site of work directly or stacked suitably and neatly along the alignment/road side/elsewhere near by the work site or as directed by the Engineer-in-Charge.
- Damages during transit, handling, storage will be to the Contractor's account and replacement for such pipes has to be made by the Contractor without any extra cost as directed by the Engineer-in-Charge.

4.3 Lowering, Laying of Pipes

- Each pipe shall be thoroughly checked for any damages before laying and only the pipes which are approved by the Engineer-in-Charge shall be laid.
- While installing the pipes in trenches, the bed of the trench should be level and free from sharp edged stones. In most cases, the bedding is not required, as long as the sharp and protruding stones are removed, by sieving the dug earth, before using the same as backfill material. While laying in rocky areas suitable bed of sand or gravel should be provided. The fill to about 10 to 15 cm above the pipe

should be fine sand or screened excavated material. Where hard rock is met with, bed concrete M15, 15 cm or 20cm thick sand bed as approved by the Engineer-in-Charge May be provided.

- As PE pipes are flexible, long lengths of Electro/Butt-fusion jointed pipes having joints made above ground can be rolled or snaked into narrow trenches. Such trenches can be excavated by narrow buckets.
- During the pipe laying of continuous Electro/Butt-fusion jointed systems, due care and allowance should be made for the movements likely to occur due to the thermal expansion/contraction of the material. This effect is most pronounced at end connections to fixed positions (such as valves etc) and at branch connections. Care should be taken in fixing by finishing the connections at a time the length of the pipe is minimal (lower temperature times of the day.)
- For summer time installations with two fixed connection points, a slightly longer length of PE pipe May be required to compensate for contraction of the pipe in the cooler trench bottom.
- The final tie-in connections should be deferred until the thermal stability of the pipeline is achieved.
- The flexibility of polyethylene pipes allows the pipe to be cold bend. The fusion jointed PE pipe is also flexible as the plain Pipe. Thus the total system enables directional changes within the trench without recourse to the provision of special bends or anchor blocks. However, the pipe should not be cold bend to a radius less than 25 times the OD of the pipe.
- The Installation of flanged fittings such as connections to sluice/air/gate valves and hydrant tees etc., requires the use of stub ends (collars/flange adaptors complete with backing rings and gaskets. Care should be taken when tightening these flanges to provide even and balance torque.
- Provision should be made at all heavy fittings installation points for supports (such as anchoring of the flange in the soil) for the flange joint to avoid the transfer of valve wheel turning torque on to the PE flange joint.
- PE pipe is lighter than water. Hence care should be taken for normal installations where there could be a possibility of flooding of the trench thus the trench shall be kept free of water till the jointing has been properly done
- When flooded, some soils May lose cohesiveness, which May allow the PE pipe to float out of the ground. Several design checks are necessary to see if groundwater flotation May be a concern. Obviously, if the pipeline typically runs full or nearly full of liquid, or if groundwater is always below the pipe, flotation May not be a significant concern.
- However, weights by way of concrete blocks (anchors) are to be provided so that the PE pipe does not float when suddenly the trench is flooded and the soil surrounding the pipe is washed away. Thus site conditions study is necessary to ensure the avoidance of flotation.
- Pipe embedment backfill shall be stone-free excavated material placed and compacted to the minimum 95% of maximum dry density.

4.4 The pipe ends shall be suitable for Electro-fusion/ Butt- Fusion jointing system that shall provide for fluid tightness for the intended service conditions.

4.5 Bedding, Backfilling and Compaction

4.6 Bedding

In case of sandy strata no separate bedding is required. However, the bottom face / trench bed where pipe shall be placed shall be compacted to provide a minimum compaction corresponding to minimum 95% of maximum dry density. The pipe bedding should be placed so as to give complete contact between the bottom of the trench and the pipe. The minimum cover over buried pipe should be 1 m.

4.7 Back Filling

Backfilling should be placed in layers not exceeding 15cm thickness per layer, and should be compacted to a minimum 95% of maximum dry density. The refilling should be done on both sides of pipe together & height difference in earth fill on each side should not be more to cause lateral movement of pipe.

Most coarse grained soil are acceptable. This May comprise of gravel or sand. However silty sand, clayey sand, silty and clayey gravel shall not be used unless proposed to be used in conjunction with gravel or clean sand.

It is very important that the pipe zone backfill material does not wash away or migrate in to the native soil. Likewise, potential migration of the native soil in to the pipe zone backfill must also be prevented.

Heavy earth moving equipment used for backfilling should not be brought until the minimum cover over the pipe is 90 cm in the case of wide tracked bulldozers or 120 cm in the case of wheeled roaders or roller compactors.

4.8 Compaction

Vibratory methods should be used for compaction. Compaction within distances of 15 cm to 45 cm from the pipe should be usually done with hand tempers. The backfill material should be compacted minimum 95% of maximum dry density.

4.9 Thrust Block

RCC thrust blocks, if required, should be suitably designed & provided at bends and at places of reduction in cross section to take care of thrust as required as per the provisions of relevant standards/codes.

4.10 Fittings & Specials

All HDPE fittings/ specials shall be of minimum PN 8 or above Pressure class, fabricated in accordance with IS: 8360 (Part I & III). PE Injection moulded fittings shall be as per IS: 8008 (Part I to IX). All fittings/specials shall be fabricated or moulded at factory only. No fabrication or moulding will be allowed at site, unless specifically permitted by the Engineer-in-Charge. Fittings will be Butt welded / Electrofusion welded on to the pipes or other fittings by use of Electro/Butt fusion process. Recommended makes for PE / PP fittings / specials of reputed manufacturer or should be manufactured by the manufacturer himself to have consistency with pipe material /grade.

4.11 Bends

HDPE bends shall be plain square ended conforming to IS: 8360 Part I & III Specifications. Bends shall be moulded.

4.12 Tees

HDPE Tees shall be plain square ended conforming to IS: 8360 Part I & II Specifications. Tees May be equal tees or reduced take off tees. Tees shall be moulded or fabricated from pipe elements.

4.13 Reducers

HDPE Reducers shall be plain square ended conforming to IS: 8008 Part I & VII Specifications. Reducer must be moulded.

4.14 Flanges/ HDPE Pipe Ends

HDPE Stub ends shall be square ended conforming to IS: 8008 Part I & VI Specifications. Stub ends will be welded on the pipe. Flange will be of slip on flange type as described below.

4.15 Slip-On Flanges

Slip-on flanges shall be metallic flanges covered by epoxy coating or plastic powder coating. Slip-on-flanges shall be conforming to standard mating relevant flange of valves, pipes etc. Nominal pressure rating of flanges will be PN10.

4.16 Welding Procedure

Jointing between HDPE pipes and specials shall be done as per the latest IS: 7634 part II. Method of jointing between the pipes to pipes and pipes to specials shall be with Electro or Butt-fusion welding using automatic or semi automatic, hydraulically operated, superior quality Electro/Butt-fusion machines which will ensure good quality welding of HDPE pipes.

Normally Electro/Butt fusion welding shall include following activities:

- Aligning of pipe on welding M/C
- Surface preparation for welding.
- Heating of pipes/ ends
- Holding pipes for welding
- Cooling etc.

4.17 Hydraulic Testing

Pipes shall be given different hydraulic tests for ensuring quality of manufacture as per clause 11.0 of this chapter.

4.18 Site Fabrication of PE Fittings

Two or more PE specials coming at one place (like PE Tee, Reducer, Flanged end etc.,) shall be jointed at contractor's workshop and transported to the site of works for final installation with proposed PE pipelines. In no case, jointing of three or more welds in one place, at site will be allowed.

4.19 Training

The Contractor shall provide on-site training on PE pipe laying, jointing, testing and maintenance etc., to the personnel authorized by EMPLOYER.

4.20 Manuals

Technical Manual on PE pipes including precautions to be taken during operation of the pipeline shall be prepared and submitted by the contractor immediately on completion of work.

4.21 Flanges

All flanges employed in the project must be compatible whatever material used.

4.22 Marking

All pipes shall be marked at maximum interval of 1 m. The marking shall indicate at least the following information.

1) Manufacturer's name & / or trade mark.
2) The dimensions (nominal outside diameter X nominal wall thickness)
3) The outside diameter tolerance (A or B)
4) The designation of pipes material (PE 100, PE 80 etc)
5) The nominal pressure (PN)
6) The production period (date or code)
7) The number of the International standard.
8) The word "Water" shall also be included.
9) Lot number/Batch Number
10) Project Name and Name of ULB

4.23 Packing & Transport

The pipes should be preferably transported by road from the factory and stored as per the manufacturer specifications to protect damage.

4.24 Summary of quality Tests:

1.	**Quality Mark**	:	Pipe: IS 4984
2.	**Material**	:	As per IS 4984. However only virgin resin is allowed, reworked material is not allowed.
3.	**Grade of Material** material	:	PE 100 as per IS 4984 (Certificate from raw manufacturer is required).
4.	**Pressure Rating**	:	Minimum PN 6 or above as per requirement.
5.	**Colour**	:	as per IS 4984
6.	**Dimensions**	:	
	Diameter	:	The nominal diameter (outside)
	Wall thickness	:	As per IS 4984.
	Length	:	**For diameter upto 110 mm :** Min-6mtr, Max.100 meter
		:	**For diameter more than 110 mm :** Minimum 6 meter.
7.	**Visual Appearance**	:	as per IS 4984.
8.	**Test and sampling**	:	as per IS 4984.
9.	**Special Test**	:	

Notch hydraulic Test for the HDPE pipe made from PE-100 grade raw material as per ASTM 1474 OR ISO 13479 at manufacturers laboratory or independent laboratory and should pass aaathe Hydraulic test as per IS:4984:1995 for a minimum 165 Hours. The test reports shall not be more than three months old.

Pipe shall convey water under variable temperature conditions ranging from 4 degree centigrade to 45 degree centigrade.

10. **Jointing of pipes (pipe end) :**

Electro-fusion Process/ Butt Fusion as per provision made in the section 4 of contract agreement.

11. **Quality Assurance :**

Quality Assurance Plan shall be got approved from the employer before production start.

4.25 Technical Qualifications for procurement of pipes during construction: High Density Polyethylene (HDPE) Pipe:

a. The Pipe manufacturer should have an annual installed production capacity of quantity equal to this TENDER.

b. The manufacturer should hold valid IS license under IS:4984 consecutively for last five years to manufacture ISI marked pipes.

c. The Pipe manufacturer should have manufactured and supplied in India at least HDPE Pipe of minimum 110mm or above dia. more than required in this Tender during past 3 years. Self certified document from HDPE Pipe Manufacturer to be attached along with technical Tender.

4.26 Installation and Commissioning of HDPE pipes

a. Supplying, laying, jointing, testing and commissioning of pipes shall conform to relevant IS codes, as applicable.

b. The alignment of pipelines shown in drawings of the TENDER documents is only indicative and the exact alignment will be as per drawings and/or as directed by the Engineer or his representative.

c. The HDPE Pipes shall be laid in accordance with the latest IS 7634 Part-2

4.27 Field Hydraulic Test

a. The Sectional Hydraulic Test shall be carried out after the pipeline section to be tested has been laid jointed and backfilled to a depth sufficient to prevent floatation

b. Each length of the pipeline to be tested shall be capped or blanked off at each end and securely strutted or restrained to withstand the forces which will be exerted when the test pressure is applied.

c. Proposals for testing where thrusts on structures are involved, even where thrust flanges on the piping are installed, shall be with the prior approval of the Engineer.

d. The proper method of filling the pipeline with water shall be used. The length under test shall be filled making certain that all air is displaced through an air valve or any other appropriate mechanism. The test length shall then remain under constant moderate pressure as per testing method given in the IS 7634.

e. As per IS code water required to built up allowable drop in pressure during test will be treated as a makeup water.

f. The maximum allowable test pressure shall be 1.5 times the system design pressure or pipe rating whichever is higher

g. Notwithstanding the satisfactory completion of the hydraulic test, if there is any discernible leakage of water from any pipe or joint, the Contractor shall, at his own cost, replace the pipe, repair the pipe or re-make the joint and repeat the hydraulic test with cost including the cost of water.

h. Test pressures are to be measured in kg/cm2 at the centre of the blank flange situated at the lowest end of the pipeline under test. Unless otherwise specified the test pressure shall be as stated below.

5. M.S. Pipe :

1.1. Applicable codes & spécifications (M.S. Pipes)

The following specifications, standards and codes are made a part of the specification. all standards, tentative specifications, specifications, codes of practice referred to herein shall be the latest editions including all applicable official amendments and revisions.

In case of discrepancy between this specification and those referred to herein, this specification shall govern.

1.	IS : 2062	Steel for general structural purposes.
2.	IS : 814	Covered Electrodes for manual Metal Arc Welding of carbon and C-Mn steel.
3.	BS EN 499	Welding Consumables covered Electrodes for Manual metal arc welding of non alloy and fine grain steel.classification
4.	AWS : A-5.1	Specification for Mild Steel Covered Arc Welding Electrodes.
5.	IS : 3613	Acceptance Tests for Wire Flux combinations for Submerged – arc welding.
6.	AWS : A-5.17	specification for bare mild steel electrodes and fluxes for submerged arc welding.
7.	IS : 1377 -	technical supply conditions for threaded fasteners
8.	IS : 1367	technical supply conditions for threaded fasteners (parts 1 to 3).
9.	IS : 2074	Ready Mixed Paint, Red Oxide Zinc Chrome and Priming.
10.	IS : 102	Ready Mixed Paint, Brushing, Red Lead, non setting Priming.
11.	IS : 816	Code of practice for use of Metal Arc Welding for General Construction in mild steel. –
12.	IS : 4353	Submerged Arc Welding of Mild Steel & Low Alloy Steels Recommendations.
13.	IS : 817	Code of practice for Training and Testing of Metal Arc Welders.
14.	IS : 1182	Recommended practice for Radiographic examination of Fusion -Welded Butt Joints in steel plants
15.	IS : 2595	Code of Practice for Radiographic Testing.
16.	IS : 3658	Code of Practice for Liquid Penetrant Flaw Detection
17.	IS : 5334	Code of practice for Magnetic Particle Flaw Detection of welds.
18.	ASTM E 94	Guide for Radiographic Testing
19.	ASTM E 709	Guide for Magnetic Particle Examination.
20.	ASTM E 165	Test Method for Liquid Penetrant Examination.
21.	IS : 3600	Methods of Testing Fusion Welded Joints and weld metal in Steel (Parts 1 to 9)
22.	IS : 4853	Recommended Practice for Radiographic Inspection of Fusion Welded Butt Joints in Steel Pipes.
23.	IS : 3589	Seamless or Electrically welded steel pipes for Water Gas And Sewage (168.3 to 2540 Outside Diameter)
24.	IS : 6631	Steel pipes for Hydraulic Purposes
25.	IS : 7343	Code of practice for ultrasonic Testing of Ferrous Welded Pipes And Tubular Products

26.	IS : 2598	Safety Code for Industrial Radiographic Practice
27.	IS : 5822	Code of Practice for Laying of Electrically Welded steel pipes For water supply
28.	IS : 1608	Mechanical testing of Metals.
29.	IS : 9595	Metal Arc welding of Carbon and Carbon-Manganese Steels.
30.	IS : 2825	Code of unfired Pressure Vessels
31.	IS : 5504	Code for Spiral Arc Welded pipes
32.	I.S. 10748	Requirement of weldable hot rolled carbon steel strip in coils.

1.2. Materials

1.2.1 Steel plates / H. R. coil

The steel used for pipes, fittings, specials and stiffeners shall be of mild steel conforming to is: 2062, grade B Fe 410 or H. R. coil confirming to IS: 10748 grade III or equivalent ISO. the dimension of HR coils shall conform IS: 2062.the quality of steel, chemical composition and tensile strength of the steel plates shall be as specified in IS: 3589 for steel of grade Fe 410. positive tolerance as per IS: 3589 shall be allowed for plate thickness, but no negative tolerance shall be allowed.

Steel plates should confirm to IS: 1730, dimensions for steel plates, sheets, strips and flats for general Engineer-in-charge purpose. The test certificate for the plates shall be issued by plate manufacturer.

1.3. Minimum thickness of the MS pipe line:

The thickness of the plate for the manufacturing of the MS rising main shall be governed by the economic sizing of the pipes analysed in line with the guidelines of C PHEEO manual and bis standards.

1.4. Welding Consumables

Welding consumables such as electrodes, filler rods and wires shall conform to IS: 814, IS: 3613, IS: 6419 and IS: 7280.

1.5. Standard length of pipes

The MS pipes shall be manufactured in lengths of 10 to 12 m with bevelled ends, unless otherwise specified or as agreed beetween Engineer in charge and the manifacture for ease in handling. Length of each pipe shall be measured at diametrically opposite four places and average of the four measured lengths shall be considered for measurements of pipe length.

1.6. Welding process

The pipes shall be manufactured by shop welding from steel plates only by qualified welders, butt welded spirally by automatic submerged arc welding process using at least two runs, one of which shall be on the inner side of the pipes. welding shall be so done that there will be through fusion and complete penetration and shall be free from cracks, oxides, and slag inclusion and gas pockets. longitudinally welded pipes shall have circumferential weld at not less than 1.5 m centre to centre. longitudinal weld shall be staggered. minimum distance between spiral welds should be 1.5 m. the

contractor shall certify that the welding of pipes has been done by qualified welders and welding operators.

1.6.1 Welding

For manufacturing of the steel pipes, any of the following types of welding shall be adopted as per IS: 3589.

i. Automatic submerged arc welding.
ii. Electric resistance welding

The welding of pipes in the field and testing should comply with IS: 816. plates shall be held in the correct position. abutting edges shall be properly squared. each deposited layer of the welded metal in the fusion welding process shall be thoroughly cleaned before additional weld metal is supplied to coincide with the centre line of joint and the finished joint shall be free from all defects. All welds shall have complete fusion with the base metal and shall be free from cracks, oxides, slag inclusion and gas pockets.

If welding is stopped for any reason, special care shall be taken when welding is resumed to obtain complete penetration between weld metal, plate and weld metal previously deposited. three percent of all seams of pipes welded in the fabrication shop shall be radiographed (As per IS: 4853) to render visible inspection of any internal defects such as blow holes, slag inclusion or cracks. if any defects are detected the metal at the location shall be chipped out and rewelded. in addition to radiography of the joints, 100 percent testing by ultrasonic equipment (As per IS: 4260) shall also be included in welding tests. any defects found out shall be rectified free of cost. welds found deficient in quality shall be removed by chipping or melting and remade as per specifications. chipping or cutting the weld shall not extend to the base metal.

Sample welds shall be submitted for testing. the weld joints shall be tested in accordance with the procedure laid down in IS: 3600 (Part-I) and as specified in IS: 5822. approval of such tests shall be required prior to welding of the pipe.

Manual welding of special sections and fittings will be permitted when it is impracticable to use an automatic welding machine.

Final welding of closure gaps for buried pipes shall be done after intermediate pipes have been backfilled.

1.6.2 Radiograph of welded joints

i. **Radiographic / ultrasonic test**

Three percent of all seams of pipes, welded in the fabrication shop, shall be radio graphed at the end of pipe (As per IS: 4853) to render visible inspection of any internal defects such as blow holes, slag, inclusion of cracks. if any defects are detected, the metal at the location shall be gouged out and re-welded. in addition to the radiography of the joints, 100 percent testing by ultrasonic equipment (As per IS: 4260) shall also be done for welding tests. any defects found out shall be rectified free of cost. welds found deficient in quality shall be removed by gouging or melting and remade as per specifications. chipping or cutting the weld shall not extend to the base metal. gouging of defective welds should be done carefully by experienced welders so that no damage to the original plate base metal occurs.

Soon after welding, each longitudinal / spiral and circumferential welded length, the minimum length of the weld at random for each pipe shall be radiographed to detect welding defects, as per the requirements of IS 4853 and as directed by engineer. this sampling will be at random to ensure 100% coverage of junctions of longitudinal / spiral

and circumferential joints. if the results of such radiography fail to conform to the requirements, the contractor shall carry out at his expense 100 % radiography test for the pertinent pipe as directed and fully satisfying the engineer-in-charge.

The percentage testing of radiography is as follows:

i) Pipes and specials manufactured at mill/ factory - 10%

ii) Field joints welded at site - 5%

The weld ripples or weld surface irregularities and slag etc., on both inside and outside shall be removed by any suitable mechanical process to a degree such that resulting radiographic contact due to any remaining irregularities cannot mark or be confused with that of objectionable defect. the radiograph shall be made in strict accordance with the latest requirements and as per the latest and most efficient technique either with x-ray or gamma ray equipment. the safety requirements during radiography shall be in accordance with IS: 2598

The radiographs are to be marked in such a way that the corresponding portion of the welded seam and the welder can be readily identified. all radiographs will be reviewed by the engineer to identify the defects and determine those which requires rectification. defects that are not acceptable shall be removed by chipping, grinding or flame gouging to sound metal and the resulting cavities shall be welded. after rectification, the joint is to be radiographed again to prove the quality of the repair. the radiographs will be judged as acceptable or unacceptable by the engineer based on the latest standards prescribed by relevant indian standard specification.

All x-rays shall be taken with equipment and by personnel of the contractor. films shall be developed within 24 hours of exposure and be readily accessible at all times for inspection by the engineer. the contractor shall provide for the use of the engineer suitable x-ray viewing equipment. x-ray films shall be properly maintained by the contractor. a complete set of radiographs and records as described in IS: 2595, for each job shall be retained by the contractor and shall be handed over to the engineer on completion of the contract. all films shall be identified by the number and chart prepared indicating location of any work associated with the pipe erection and such inspection shall be performed by the radiographer at the discretion of the engineer.

ii. **Radiographic inspection of welded joints**

All welded joints to be radiographed shall be examined in accordance with:

(i) IS 2595 - Code of practice for radiographic testing

(ii) IS 4853 - Recommended practice for radiographic examination of fusionwelded butt joints in steel pipes

(iii) IS 2598 - Safety code for industrial radiographic practice.

Radiographers performing radiograph shall be qualified in accordance with snt-tc- 1a. supplements and appendices “recommended practice for non-destructive testing personnel qualification and certification” published by the american society for non-destructive testing as applicable for the technique and methods will be used.

final acceptance of radiographs shall be based on the ability to see the prescribed penetrometer image and the specified hole.

sections of welds that are shown by radiography to have any of the following types of imperfections shall be judged unacceptable and shall be repaired.

1.7. Deviation in length

Finished pipe length shall not deviate from straightness by more than 0.2 percent of the total length.

1.8. Straightness of pipes

Finished pipes shall not deviate from straightness by more than 0.2% of the total length checking for straightness shall be carried out using as taut string or wire from

end to end along the side of the pipe to measure, the greatest deviation. pipe ends shall be at right angle to the axis within 1°.

1.9. Cement concrete and mortar lining of pipes and specials

The cement mortar and cement concrete used for lining of pipes and specials shall conform to relevent IS specifications. The maximum size of aggregate shall be one third the thickness of concrete cover inside the steel pipe or 10mm, whichever is less. the concrete mix shall have a minimum cement content of 450 kg/m3 and a characteristic compressive strength of 25 N/mm2 at 28 days. the cement mortar shall have a minimum cement content of 600 kg/m3 and a characteristic compressive strength of 25 N/mm2 at 28 days.

1.9.1 Preparation of surface before lining or coating.

The surface of pipe to be lined or coated with cement mortar or cement concrete shall be clean. any loose rust, loose millscale, dirt, debris, oil, grease and other detrimental materials present shall be removed by mechanical means.

1.9.2 Cement concrete lining for inside of steel pipes and specials and mortar coatingoutside.

Cement concrete shall be composed of cement, coarse and fine aggregates and water well mixed and of proper consistency to obtain a dense, homogeneous lining that will adhere firmly to the pipe surface. thickness of cement concrete lining inside the pipe shell shall be as specified in tender document and that of the mortar lining outside the pipe shell shall be as per relevant IS specifications. The mixing of_concrete for internal lining shall be mixed as per provisions of IS code. The clear cover to the reinforcement whether steel cylinder or cage shall not be less than 9 mm for lining and 12mm for coating.

1.9.3 Equipment for lining of pipes

Straight sections of pipe shall be lined by use of a spinning machine specifically designed to suit for the purpose of rotating the section and centrifugally applying cement concrete lining tothe interior of steel pipe. mechanical feeding is preferable.

The contractor shall plan for installation of sufficient number of spinning machines/ equipment for inside lining and outside coating of pipes and the proposed factories. if during the progress of works, it becomes necessary for transporting of plates/ pipes from one factory to the other for fabrication/ lining and or coating due to inadequate number of machines or breakdowns or to meet progress targets, the contractor should arrange for such transportation of pipes from one factory to the other at his own cost.

1.9.4 Concreting lining and mortar coating of pipes

This shall be done generally in two stages to give the required thickness of the barrel.

In the first stage concreting shall be done inside the pipe shell and in the second stage placing of the mortar is done outside the pipe shell. placing of concrete inside the steel pipe shell shall always be done at the manufacturer's works and consolidated by spinning, vibrating, spinning combined with vibrations or other appropriate mechanical means whilst the placing of mortar outside the steel pipe shell shall be done later either at the manufacturer's works or at site.

Mortar coating outside the steel pipe shell shall not commence before the expiry of three days after the completion of concreting inside, unless otherwise it is established that the concrete lining inside has attained works cube strength of not less than 110 kg/cm2, earlier than this period. During this entire period the concrete inside the steel pipe shell

shall be under curing. The mortar outside the steel pipe shell shall be either vibrated or applied under pressure.

In the case of pipes with plain ends or slip-in type ends, the portions that should be left exposed without concreting or mortar lining shall be mutually agreed to between the purchaser and the manufacturer.

1.9.5 Lining of specials

Whenever practicable, specials shall be made form cut lengths of matured lined straight pipes.the lining shall be cut back from the end to ends to be bevelled and welded for a sufficient distance to ensure that any of the concrete lining which is intended to remain as part of lining shall not suffer damage by the cutting or welding process. the concrete lining shall be made good by rendering by hand.

Hand rendering of specials shall consist of freshly mixed concrete of a mixture equivalent to that of the lining being repaired and shall be thoroughly compacted and finished to a smooth surface of the correct form.

Cement concrete shall be used both for lining and coating of specials. the thickness of lining and coating shall be as per relevent IS. Specials shall have the nominal dia as that of the connecting pipe after lining. specifications for lining and coating of specials shall be as that of the straight pipes.

The application of cement concrete lining to bends, or specials sections whose shapes preclude other process of lining shall be by mechanical placement, pneumatic placement or hand trowelling finished to produce a smooth dense surface. debris shall be removed as necessary, permitting the application of the lining to a clear surface.

Thickness shall be as required for spun lining of straight sections except that it may be varied by feathering or filleting to affect system lining with adjoining sections of pipe and thickness of lining.

1.9.6 Determination of thickness of lining

Lining thickness shall be determined on the freshly lined pipes, at intervals frequent enough to assure compliance. thickness of lining may be determined by means of a steel pin not larger than 1.5mm in diameter or on a hardened concrete or mortar by means of a non-destructive measurement process. the lining shall be measured at four equidistant points on two cross sections of the barrel at each end of the pipe or fittings. The first set shall be at least 200 mm from the respective ends of the pipe or fitting. The second set shall be made as far into the interior of the pipe or fitting as can be readily reached without injuring the lining

1.9.7. Curing

After completion of concreting or mortar lining, the concrete or mortar shall be kept wet by any suitable means such as immersion in water, covering by wet gunny bags, or by mechanical sprinklers, for a period of not less than 14 days when cement conforming to IS: 269 or IS: 455 or IS: 1489 or IS: 8043 or IS: 6909 is used; not less than 7 days when cement conforming to IS: 8041 or IS: 8112 is used; not less than 3 days when cement conforming to IS: 6452 or IS: 12269 is used; and not less than 21 days when cement conforming to IS: 12330 is used.

Non-pressure steam curing may be permitted provided the requirements of non- pressure steam curing are fulfilled. For non-pressure steam curing, the pipe shall be subjected to the action of thoroughly saturated steam at a temperature of 38°C to 54°C for a period of not less than 24 hours or for such additional time as may be necessary to enable the pipe to meet the strengthrequirements.

1.9.8. Repair of defective or damaged areas of lining

Defective or damaged areas of linings may be patched by cutting out the defective or damaged lining to the metal so that the edges of lining not removed are perpendicular or slightly undercut. A fresh mix of concrete or mortar equivalent to that of the lining being repaired shall be prepared. The cut-out area and the adjoining lining shall be thoroughly wetted, and the concrete or mortar applied to the inside or outside respectively and trowelled smooth with the adjoining lining. After any surface water has evaporated, but while the patch is still moist, it shall be cured as specified.

1.9.9 Protection of work

The lined pipe and fittings shall be protected from extreme heat due to direct sun rays, from impact of rainfall, and from freezing temperatures until the linings have cured sufficiently to withstand these conditions.

1.9.10 Workmanship

All pipes shall be cleanly finished free of cracks, surface flaws, laminations and all other defects. They shall be cylindrical, concentric and straight in axis. The ends shall be accurately cut and shaped for welds. The ends shall be square with the axis of the pipe. The repair of minor defects by welding or otherwise shall be permitted only after obtaining the prior permission from the engineer in writing. No heating shall be permitted.

1.9.11 Handling of pipes and specials.

During manufacturing and during the entire period of the application of concrete or mortar lining protection and the curing thereof, the section shall be carefully supported and handled so as to avoid injury to the fresh lining. If a pipe section must be rolled or otherwise moved, such operation shall be done slowly and with every reasonable precaution against damage. Any portion of the lining, coating that may become damaged shall be cut and replaced. During delivery, all sections shall be handled by such means and in such a manner that no distortion or damage is done to the protection or to the section as a whole

1.9.12 Field jointing of insides and outsides of concrete lined pipes.

Cement mortar shall be used for lining of inside and outside of joints. It shall be of the same specifications as those of outside mortar coating of pipes, cement mortar lining for inside andoutside of field joints shall be done only after testing the pipeline.

1.9.13 Jointing inside

For the inside lining of field joints, wire fabric of 50 x 50mm steel mesh, 13 gauge each way, shall be provided. Inside joints of mortar lined pipe shall be plastered with cement mortar and finished off smooth with the inside surface of the pipe by trowelling or by other equipment means. Before placing the joint mortar material against the surface of the lining, these surfaces shall be carefully cleaned and wetted to ensure good bond between the lining and joint mortar. The field joint shall be cured for a minimum of 14 days. In any case the pipeline shall not be put into serve until the mortar has cured for a minimum of 24 days. The joints shall be finished off smooth with the inside surface of the lining by trowelling.

1.9.14 Jointing outside

For this item, the joints have been classified into two groups:

- Those coming under waterlogged area
- Those coming in other area.

Joints under waterlogged areas: these joints shall be done with cement mortar 1:2 with two layers of m.s. 50mm x 50mm x 13 gauge weld mesh and the thickness of mortar shall be 50mmthick and a suitable water proofing compound shall also be used.
Joints coming in other areas: these shall be done with cement mortar 1:2, with required thickness with one layer of 50mm x 50mm x 13 gauge wire fabric reinforcement.
Outside field joints shall be coated with cement mortar retained by suitable forms so as to bridge the joints. The mortar shall be compacted within the form to produce a dense coating without voids. The joint coating shall be kept moist for a minimum or 14 days.

1.10 Testing of pipeline

Mechanical test (tensile test)

Tensile test shall be carried out as specified in IS: 1894. the tensile strength and percentage elongation of pipes shall not be less than the values specified in the table below. the percentage elongation shall be determined on a gauge length of 5.65 times of square root of original cross- sectional area of the specimen (so).

Steel grade	Tensile strength mpa (min)	Percentage elongation (min)
Fe-330	330	20
Fe-410	410	18
Fe-450	450	12

1.11 Reinforcement

All cage reinforcement used in the pipes and specials shall conform to IS: 432 (part- 1) or IS:432 (part-2). Wire fabric shall be used for external reinforcement. it shall be 50 x 50mm steel wire mesh, 13 gauge each way, and conform to IS: 432 (part-1) or IS: 432 (part-2). the reinforcement cage for the pipes shall extend throughout the pipe barrel and shall consist of spirals or rings and straights. the spirals or rings shall be circular in shape and shall be either wound round the steel pipe shell itself or wound round collapsible frame or drums and then slipped on the steel pipe shell. the fabric shall be wrapped on the shell by tack welding. the outer cage reinforcement for swelled end portions of the pipe shall be wound on them at the time of laying. minimum reinforcement in the lining shall not be less than three percent of the quantity of steel pipe shell. splices shall be made by welding or other suitable means. the fabric shall be wrapped on the shell by tack welding.

1.12 Hydraulic test at works

Each pipe shall be hydraulic tested at manufacturer's works before applying any coating/lining.

1.13 Allowable tolerance

- Allowable tolerance for outside diameter shall be ± 0.75% of the specified diameter.
- Allowable tolerance for ovality shall not be more than 1% of the specified diameter.
- Negative tolerance for plate thickness shall not be allowed.

1.14 Sampling and testing

Sampling and conformity criteria for various tests shall be as given in IS: 4711. the test samples shall be cut from pipes in the final condition of supply. tests for tensile strength, percentage elongation, guided bend test, shall be carried out and the test values shall be in conformity with those specified in IS: 4711.

1.15 Destructive test

The destructive test of pipe joints on the field shall be carried out as per is 3600. it will be one for every rising main and for every two km of manufactured length.

1.16 Third party inspection

The pipes ready in the manufacturer's work yard shall be inspected and tested before dispatch, by the employer and or inspecting agency appointed by the employer. all the tests including hydraulic test shall be carried out on specified number of samples from each lot as specified in IS: 4711, in presence of the inspecting agency. the protective coating as per specification shall be applied to the pipes only after the inspecting agency is satisfied regarding the test results,

which shall be in conformity with the limits specified in IS: 4711. the contractor shall produce the test certificate issued by the inspecting agency.

1.17 Application of red oxide

The pipes cleared by the inspecting agency shall be provided with one shop coat of red oxide as per relevant is, externally, prior to despatch.

1.18 Manufacture, supply, laying and jointing of MS specials

1.18.1 Fabrication of specials and appurtenances

Specials such as bends, tapers, and tees shall confirm to IS: 7322. specials shall be fabricated by cutting plates / hr coils of required thickness to the required shape, obtained by developing the form of specials on ground. abutting profiles shall be obtained using templates, which guide the cutting torches as to obtain a uniform cut. no hand cutting shall be permitted. specifications for using and testing of the plates, electrodes, welding process, etc shall be the same as for the ms pipes. plain ends shall have beveled edges. specials shall be manufactured as per the standard drawings enclosed in the tender documents. stiffeners shall be provided wherever necessary.

1.18.2 Requirement

specials as specified below may be required. slight variation may be possible as per the site conditions.

- Horizontal and vertical bends
- Air valve tees
- Washout valve tees
- Access manhole tees with blank flange
- Eccentric reducers (tapers)
- Steel strap ring and bearing plates at chairs proposed for the above ground pipeline.
- Stiffener rings for pipes and specials as required and as directed by the Engineer-in-charge.
- Blank flanges etc.
-

1.18.3 Bends, tees and tapers

The testing of these specials is not envisaged at the fabrication stage. these shall be subjected to test when the completed pipeline is tested hydraulically. in view of this, the contractor shall fabricate these specials with due care so that there is no failure of any welded joint during testing of the completed pipeline, which will invite dewatering the main, repairing and retesting.

1.18.4 Eccentric tapers (reducers)

The tapers shall be fabricated in one or more pieces as may be necessary according to their lengths and plate sizes available. stiffener rings shall be provided as necessary and as directed by the engineer-in-charge. the tapers shall be eccentric with bottom portion straight and top reducing as shown on drawing.

1.18.5 Stiffener rings

Stiffener rings shall be provided for the pumping main on each side of air valves, washout valves, on line valves and access manholes to strengthen the pipe at provided openings.

1.18.6 Bearing plates and steel strap rings

Bearing plates, equal in width of chairs and thickness as shown on approved drawings shall be provided on the chair / pedestal supports below roller bearings as shown on drawing. steel straps of size and thickness as shown on drawings shall be provided and fixed for pipe anchorage with the chair supports. the straps shall be welded to ms angle on either end and anchored with the chair concrete by providing ms bolt as shown on drawing.

1.18.7 Gas cutting for MS pipes

Cutting of the pipe for providing tees for air valves, washout valves, preparation of manhole openings, etc becomes necessary. the edges of the specials are required to be cut to match with the pipe edge for proper jointing and welding. plate cutting operations at such places are performed with the use of gas cutting.

1.19 Procedure for receiving steel pipes

1.19.1 General

To ensure that the work of erecting pipes is not held up at any stage and place, the contractor shall maintain an adequate stock of standard specials, flange rings, plug plates, manhole covers, etc. and short length of smaller diameter pipelines, etc. at site in his field stores, in consultation with the engineer. wherever possible, the contractor shall arrange one full month's requirement of pipes, specials, etc. stacked along the alignment.

1.19.2 Stacking of pipes, etc. and inspection

The contractor shall keep in each section a responsible representative to take delivery of the pipes, specials and appurtenances, etc. transported from the fabricating stockyard or received from any other work site to the site of laying and to stack along the route on timber skids. padding shall be provided between coated pipes and timber skids to avoid damage to the coating. suitable gaps in the pipes stacked shall be left at intervals to permit access from one side to the other. the pipes, specials, appurtenances so received on site shall be jointly inspected and defects recorded, if any, such as protrusions, grooves, dents, notches, damage to the internal coating etc. shall be

pointed out immediately to the engineer at the site and in the acknowledgement challans. such defects shall be rectified or repaired to the satisfaction of the engineer entirely at the contractor's risk and cost.

1.19.3 Handling of pipes, specials, appurtenances, etc.

It is essential to avoid damage to the pipes, fittings and specials, etc. or their coatings at all stages during handling. the pipes and specials shall be handled in such a manner as not to distort their circularity or cause any damage to their surface treatment. pipes shall not be thrown down from the trucks nor shall they be dragged or rolled along hard surfaces. slings of canvas or equally non-abrasive materials of suitable width of special attachment shaped to fit the pipe ends shall be used to lift and lower coated pipes to prevent damage to the coating. great care shall be taken in handling the pipe right from the first operation of manufacture until they are laid and jointed. the contractor will provide temporary props in order to prevent any sagging of the pipes while they are stacked in their yard and while transporting to the site of delivery, i.e. laying. the props shall be retained until the pipes are laid. if at any time these props are found to be dislodged or disturbed, the contractor shall immediately reinstate them in such a way that the true shape of the pipe shell or specials is maintained to the satisfaction of the engineer. no defective or damaged pipe or special shall be allowed to be used in the work without rectification to the satisfaction of the engineer. any damage to the coating shall be repaired by the contractor at his own cost to the satisfaction ofthe engineer.

1.19.4 Dents

Whenever any dent, i.e. a significant alteration of the curvature of the pipe shell is noticed, the depth of the dent shall be measured between the lowest point of the dent Fand the pipe shell curvature line. all dents exceeding 2 percent of the outer diameter of the pipe shall be removed by cutting out a cylindrical portion of the pipe and replacing the same by an undamaged piece of the pipe. the engineer may permit insert patching if the diameter of the patch is less than 25 percent of the nominal diameter of the pipe. repairs by hammering with or without heating shall not be permitted. any damage to the coating shall also be carefully examined and rectified.

1.19.5 Marking

The component parts of the pipes shall be carefully marked for identification in the field. the marking shall be on the side, which will be the inside of the pipe after bending. the marking operation shall be conducted with full size rulers and templates. only blunt nose punches should be used. the plates used for fabrication of pipes shall be laid out in such a way that when the shells are completed one set of original identification markings for the material will be plainly visible. in case these markings are unavoidably cut out, they shall be accurately transferred by the contractor to a location where these markings will be visible on the completed work. after hydraulic tests on the specials and other items, the number of the shell in the line as it will be erected and the direction of flow shall be stamped in a prominent manner on each piece.

A register shall be maintained in suitable proforma giving the following information for each shell tested:

(a.) Serial no.
(b.) Shell no.
(c.) Date of test
(d.) Thickness and specification of steel

(e.) Weight of shell tested
(f.) Maximum test pressure
(g.) Details of test performance
(h.) Details of radiographic examination of welds
(i.) Name of engineer's representative witnessing tests

A copy of these details shall be furnished to the owner at free of cost.

1.20 **Specification for liquid epoxy coating systems for interior coating of steel water pipelines:**

Providing & applying internal lining / painting / coating to ms pipe with solvent free food grade epoxy having dry film thickness of 406 microns including scrapping the surface of the pipe.

1.21 **General**

Buried steel pipeline shall be coated internally, with a single coat two part solvent free high build liquid epoxy lining as per AWWA C210-07 suitable for potable water application and shall be approved by NSF international standard NSF/ANSI-61 2004 or by water regulations advisory scheme (WRAS), UK or central food research institute, mysore as per standards BS 6920 UK. protective lining shall consist of a coating system consisting of 100% solids epoxy comprising of resin & hardener mixed in a proportion as recommended by the manufacturer. the mixed epoxy shall be applied to abrasive blasted steel surface in the coating plant using plural hot airless spray machine with extended boom to apply the lining to a smooth finish without any drip or sag. the epoxy shall also be capable of application by brush at field joint locations and repair of damages to the spray applied epoxy.

The contractor shall perform all work in accordance with these specifications and the latest pipeline coating practices, and shall complete the work in all respects to the full satisfaction of the Owner / Owner's Representative. The entire coating operation starting from cleaning and surface preparation till coating shall be performed under the supervision of skilled personnel who are well conversant with the work. pipes which have been cleaned and primed, or cleaned, primed and coated, without having been inspected and approved shall be rejected.

This specification is not intended to be all inclusive and the use of guidelines set forth here does not relieve the contractor of his responsibility for the quality and performance of the applied coating system, and to supply coating material capable of performing its intended service.

All steel special sections, connections fittings and field joints to be used for underground steel pipeline shall be internally coated with same liquid epoxy.

1.22 Reference standards

The following standards (latest revision) referenced below are a part of this specification.

In case of conflict between this specification and the referenced standards, this specification shall apply

(a) ANSI/AWWA C210 - Standard for liquid epoxy coating systems for the interior & exterior of steel water pipelines.
(b) NSF/ ANSI-61 - Standard for drinking water system components-health effects.

(c) ANSI /ASTM D149 - Standard test method for dielectric breakdown voltage and dielectric strength of solid electrical insulating materials at commercial power frequencies.

(d) ANSI /ASTM D3359 - Standard test method for measuring adhesion by tape test.

(e) ANSI/ASTM D3363 - Standard test method for film hardness by pencil test.

(f) ANSI/ASTM D4417 - Standard test method for field measurement of surface profile of blast cleaned steel.

(g) ANSI/ASTM D4541 - Standard test method for pull-off strength of coatings using portable adhesion testers.

(h) ANSI/ASTM D2240 - Standard test method for rubber property – durometer hardness

(i) ANSI/ASTM D5000 - Standard test method for abrasion resistance of organic coatings by the taber abraser

(j) SSPC-PA 2 - Measurement of dry paint thickness with magnetic gauges

(k) SSPC-SP L - Solvent cleaning

(l) SSPC-SP 6/NACE NO. 3 - Commercial blast cleaning

(m) IS 16676:2017 Specification for solvent less liquid epoxy system for application on the interior and exterior surface of steel water pipeline.

1.23 Internal lining system

The internal epoxy lining shall consist of chemically cured resin (epoxy) & hardener (curing agent), mixed in the ratio as recommended by the manufacturer and applied to a dry film thickness (DFT) of minimum 406µ.

Table-1 physical properties of the internal lining shall be as follows.

long term contact with potable water	no effects for human consumption as certified by nsf international	NSF / ANSI -61
thickness (applied min)	406µ	SSPC-PA 2
thickness deviation (min)	0 / + 200µ	SSPC-PA 2
dielectric strength (min)	450 V/mi l (15 V/µm)	ASTM D149
hardness shore d	85 (min)	ASTM D2240
adhesion to steel (min)	800 psi	ASTM D4541
tabor abrasion	1560 cycles / mil	ASTM D4060
compressive strength	70 mpa	ASTM C109
Property	**Requirement**	**Test method /Reference**

1.23.1 COATING APPLICATION FOR BENDS, JOINTS ETC.

General

For internals of bends, short section of pipes, tees, fittings etc., internal epoxy will be manually spray applied either in the coating plant or in the field using suitable two component airless plural spray machine.

Internal field joint coating shall be brush applied.

The internal epoxy lining shall be applied only on blasted steel surface.

1.23.2 Pipe preparation

Metal surface condition

Internal surface of the pipe shall be free from mud, mill scale, mill lacquer, wax, oil, grease, or any other foreign material. before blast cleaning, surfaces shall be inspected and pre-cleaned according to sspc-sp 1 to remove oil, grease, and loosely adhering deposits. visible oil and grease spots shall be removed using a solvent. only solvents that do not leave a residue shall be used. preheating to remove oil, grease, and mill scale may be used provided that all pipe is preheated in a uniform manner to avoid distortion.

After drying and removing all loose adhering foreign materials, the pipe surface shall be cleaned by blasting with grit or steel shots to achieve a surface preparation at least equal to that specified in SSPC: SP6/NACE3. The blast anchor pattern or profile depth shall be 2 mil to 3 mills (50µmto 75µm) measured in accordance with ASTM D-4417.

For consistent surface finish, a stabilized working mix shall be maintained in abrasive recycling blasting machines by frequent small additions of new grit, shot infrequent large editions shall be avoided. the abrasive working mix, abrasive recycling blasting machines shall be maintained clean of contaminants by continuous effective operation of blasting machine scalping and air-wash separators.

The cleaned interior pipe surface shall be in inspected for adequate surface preparation. surface imperfections, such as slivers, scabs, burrs, weld spatter, and

gouges, shall be removed by hand filing or grinding if necessary to prevent holidays.

Blast-cleaned pipe surfaces shall be protected from conditions of high humidity, rainfall, or surface moisture. no pipe shall be allowed to flash rust before coating. to ensure a dry pipe surface at the time of liquid epoxy application, the minimum steel substrate temperature shall be 10°c and at least 3°c above the dew point.

1.23.3 Application temperature

The temperature of the mixed coating material and of the pipe at the time of application shall be not lower than 10°C. preheating of the coating material by using in line heaters to heat the coating material may be used to facilitate the application. heating shall conform to the recommendations of the epoxy coating manufacturer.

1.23.4 Application of epoxy

The epoxy shall be applied directly to the abrasive roughened steel surface pipe using a hot plural component airless spray equipment, with all necessary ancillary equipment like spraying boom etc, in accordance with the epoxy coating manufacturer's recommendations. Pipe will be rotated at a suitable speed and the boom of the spray machine carrying the spray gun will travel inside the pipe at a pre-determined speed to ensure that thickness of minimum 406 microns is achieved in one single spray application.

1.23.5 Curing

After application of epoxy, curing of the epoxy shall be natural air convection. typical dry time of the epoxy at ambient temperature of 25-400°C shall be between one hour to three hours. shore d hardness of the epoxy after 10 hours shall be > 65.

Internal coating cutback on either side of the pipe shall be 75 mm +/- 25 mm.

1.23.6 Field joints

After the main pipes have been welded in the trench, internal epoxy coating shall be applied on the inside of the pipeline at the girth welds using brush application. internal epoxy for field joint coating shall be supplied by the same manufacturer and shall be NSF approved. the epoxy shall be supplied in small cans of appropriate size required per joint with applicator pad. the internal surface of the field joint shall be inspected and weld spatter or slag shall be removed by grinding.

The internal surface shall then be manually blasted to remove rust / contamination etc. The resin & hardener will be mixed and the coating applied to a minimum thickness of 406µ. The internet field joint coating shall overlap the fectory applied main line internal coating by 50mm on either side.

1.24 Material acceptance, inspection and testing

1.24.1 Material acceptance

Proposed internal epoxy lining materials shall be approved by the Owner / Owner's representative. test certificate from independent third party test laboratory shall be submitted conforming compliance to the physical properties as listed in table i. no deviation to the physical properties shall be acceptable.

When the material is supplied, acceptance of the material shall be based on submission of certificate of conformance of the internal epoxy to this specification along with manufacturing acceptance test certificates for various lots as per manufacturer's quality assurance and quality control requirements.

1.24.2 Coating application inspection

the entire internal lining operation by the contractor will be supervised by qualified experts from the manufacturer. the cv of the manufacturer's expert shall be approved be approved by the Owner / Owner's representative. All coating work will be done by the Owner / Owner's represnative. All coating work will be done in the presence of er / Owner's represnative.

1.24.3 Thickness

Thickness of the coating system shall be checked in accordance with SSPC-PA 2. thickness shall be in accordance with the values given in the table i and shall chacked at a frequency specified by the owner/ Owner's representative.

1.24.4 Pull off adhesion

The pull off adhesion of the internal epoxy lining to steel shall be checked in accordance with ASTM d-4541. the average value below the limits stated in the table i shall constitute a failure of the system to meet the adhesion requirement. the pull off adhesion test shall be conducted at a frequency specified by the owner/ Owner's representative.

1.24.5 Field procedures

At all times during construction of the pipeline, the contractor shall use caution to prevent damage to the internal lining on the inside of the pipe. no metal tools or heavy objects shall be permitted to unnecessarily contact the finished coating. workmen shall not be permitted to walk on the internal coating except when necessary. in these cases, they shall wear shoes with rubber or composition soles and heels or other suitable footwear that will not damage the coating. any damage to the pipe or the protective coating from any cause during the installation of the pipeline shall be repaired.

1.24.6 Coating repair in field

All defects visually or electrically discovered either at the coating plant or in the field shall be repaired by applying the same liquid epoxy coating using brush. the minimum over-lap at the damaged area shall be 100 mm all around. the repaired area shall be tested with a holiday detector as per specifications after the repair is completed. The thickness of the coating at the repair area shall be minimum 406µ.

1.25 Epoxy painting for internal surface of pipes and specials

The epoxy coating shall be solvent-less, liquid epoxy coating of at least 406 microns thickness, for the interior surface of steel pipe lines & conforming to IS 3589: 2001. epoxy paint to be used for this work shall be suitable to local environmental conditions / ambient conditions.

1.25.1 General

All steel pipes and fittings outside the pumping station shall be internally lined with a spray applied solvent free epoxy approved for contact with potable water under the united kingdom water regulations advisory scheme authorized for use under

regulation 31 (4)(a) of the water supply (water quality) regulations 2000 for contact be as per manufacturer's specifications.
with potable water. the applicable specifications for the coating system shall be suitable to potable water at elevated ambient / air temperature tested as per BS 6920. AWWA C210 is referred standard for coating application as per sec. 1.24. Surface preparation application condition, post treatment of coated surface to paint NSF standard 61 NACE RP 01 75. paint should be suitable for long time storing or potable water in the pipeline.

1.25.2 Materials & workmanship

The coating system shall be factory lined and the lining shall be suitable for application in an environment with black bulb temperature up to 85 deg c.
All steel pipes shall be lined at the factory. pipes welded on site shall be lined on site to the same standards as for pipes.
Pipe ends –coating shall be held back in from the ends of pipe sections to be jointed by field welding.
Detailed proposals of the lining method, materials and apparatus to be used for both factory and site application shall be submitted to and approved by the engineer before work starts. storage and application shall be accordance with recommendations of the coating manufacturer, but as a minimum:

a) a visual examination of the surface to be coated shall be carried out and any slivers or similar deposits removed.
b) prior to blasting all oil and grease shall be removed from the surface to be coated.
c) preparation of steel surfaces for both factory and site application shall be to a minimum of sa 2 ½ in accordance with bs 7079/ iso 8501-1:2007 specifications /or as per client specifications and roughness should 50-75 micron.
d) the surface to be coated shall be dry, clean and free from foreign material and coating shall take place before any surface rusting and at least within 4 hours of blasting.
e) surface preparation and coating shall not be carried out when the relative humidity exceeds 85% or when the surface to be coated is less than 3 deg c above the dew point.
f) the coating shall have a minimum mean dry film thickness of 406 microns.
g) coatings shall be used within the pot life specified by the manufacturer.
h) the materials to be used for the work of each batch, test certificate shall be reviewed.
i) paint shall be applied by brush / air less or conventional spray method.

Pipe linings shall be inspected on site and damaged, defective or otherwise unsatisfactory linings may be condemned. all defective areas shall be made good to the satisfaction of the engineer.

1.26 Inspection and testing:

It will be detailed in the contractor quality plan and will be under taken as minimum to the following:

a) Visual inspection of blasted profile on every pipe to be coated.

b) Visual inspection of finished coating on every pipe.

c) Measurement of coating thickness at four points on each pipe.

d) The completed pipe shall be subjected to holiday detection tests which shall be carried out to the satisfaction of the engineer. the applied voltage shall be hose appropriate to the coating under test as recommended bythe coating material manufacturer.

e) The adhesion test as per clause no. 5.1.6 of AWWA-C-210.07 shall be carried out on one pipe per lot of every 10 pipes. two parallel knife cuts about 100 mm long and 20 mm apart shall be made through the coating. if necessary, the test knife may be heated to make the cut. the painted surface between two cuts shall be lifted off the pipe with a stiff blade. if the paint film does not pill-off, more than the width of the cut, the bond shall be deemed to be satisfactory. if the width of the peel exceeds the width of the cut, two additional tests shall be made on the same pipe at two different locations. the painted surface shall be accepted if both the tests are satisfactory. if the results of either of the tests are un-satisfactory,the painting work of pipeline shall be rejected. adhesion test shall be performed on 1 pipe out of 10 pipes. if the adhesion test fails for one pipe, then this test shall be performed on all the remaining 9 pipes of that particular lot.

f) Prior to acceptance and application of the material, the epoxy coating material shall be submitted by the contractor & shall be tested in an

independent third party / vendor's test laboratory designated by the purchaser, in presence of client / consultant and the results shall be in accordance with the specified standards. the frequency of the test shall be minimum 1 sample in every 10000 liters or per batch of the coating material. tests like: specific gravity, hardness (shore - D), adhesion to steel, tabor abrasion, pull-off adhesion tests, will be carried out.

g) Rejection: if the sample of coating material does not comply with the standard, the coating material represented by such a sample shall be rejected.

h) The pull-off adhesion of the internal epoxy lining to steel shall be checked in accordance with the ASTM D–4541.

i) All holidays visually or electrically discovered either at the coating plant or in the field shall be repaired by applying the same liquid epoxy coating using the brush. the minimum over-lap at the damaged area shall be 100 mm all around. the repaired area shall be tested with a holiday detector as per specification after the repair is completed. the thickness of the coating at the repair area shall be minimum 406 micron.

1.27 Preservation, marking and shipping preservation

1.27.1 P reservation

The bare ends of each pipe shall be painted outside with a removable varnish as temporary corrosion protection during transportation.

bevel protectors of a type to be approved by the engineer shall protect the bare ends of each pipe.

1.27.2 Marking

In addition to the marking required by API 5L, the specification MO1 "Steel Pipes for main lines and other applicable project specifications, the pipe coating

the Contractor's unique coating number shall be marked to the internal surface or pipe with synthetic resin paint.

further marking details like colour coding etc. shall be agreed upon with the engineer.

the marking shall have at least a distance or 150 mm to the pipe end.

1.27.3 Shipping

Shipping and loading preparation shall be in accordance with api specification 5l or otherwise stated in the contract documents.

the pipe coating contractor shall submit detailed loading, stacking- and shipping procedures for approval by the engineer.

1.28 Documentation

1.28.1 Pre-production documentation

the pipe coating contractor shall submit the following documents to the engineer for approval prior to commencing production:

a) The manufacturer's trade name and data sheets for all proposed coating materials. this includes cleaning and abrasive blasting consumables.

b) Procedure for identifying or maintaining the identification of each coated item.

c) Handling procedure.

d) Stacking procedure.

e) Materials control and traceability procedure for the batches of coating materials.

f) Materials control and traceability procedure (pipe and coating materials).

g) Procedure for steel surface preparation including materials, cleaning, inspection, verification of cleanliness and surface profile.

h) Coating application procedures.

i) The results of the batch tests for batches to be used for pre-qualification tests.

j) Details of testing methods including instrument types and copies orcurrent calibration certificates.

k) Details of inspection methods for bare and coated pipe.

l) Full test results from the coating procedure qualification test (PQT).

m) Repair procedure and results or tests on demonstration of repairs.

n) Project specific quality plan. work shall not commence until these Procedures have been reviewed and approved by the engineer.

The selection of proposed coating materials shall be subject to Engineer's approval

1.29 Production records

A daily log containing the following data shall be maintained and be available for inspection by the engineer during and/or after production. data shall be recorded against the pipe unique identification number.

a) Bare pipe inspection data

b) Ambient temperature (every 4 hours)

c) Humidity (every 4 hours)

d) Coating progress (no. Of items coated. Including item serial numbers)

e) Blast pipe surface amplitude

e) Tests for cleanliness of blast surface

f) Test for cleanliness or blast medium

g) Film thickness measurements

h) Average, maximum and minimum coating thickness during eachshift

i) Details of any coating repairs

j) the unique identification number of all items that are stripped for recoating – 'RP'

k) Pipe coating test results

This log shall be available to the engineer throughout all coating operations**.**

1.30 Release documentation

The pipe supplier and/or pipe coating contractor shall submit to the engineer the following documentation in hard copy and softcopy (format to be agreed upon with the engineer) with each batch of pipes released:

a) Mill certificates for line pipe
b) Unique pipe identification numbers
c) Unique coating identification number (if different)
d) Pipe length
e) Length of the coated portion of each pipe and total coated lengths of allpipes.
f) Reductions in length due to use in tests, damage or repairs, recorded againstpipe unique identification number
g) Date or coating
h) Batch numbers of coating materials used

This shall be followed within two weeks by the following:

a) Manufacturer's certificates for each batch of coating materials
b) Certification / calibration certificates for all testing and coating equipment
c) Inspection and test records, results, and other documentation of allmaterials and coating tests

All reports shall be signed by the pipe coating contractor to signify compliance with the requirements of this specification.

The finished epoxy lining surface shall be able to deliver a guaranteed 'c_r' value of

(1) One in modified Hazen- William's formula for estimating pipe friction loss. The arrangement for 'C' value testing is to be arranged by the contractor.

1.31 700 MICRON POLYURETHANE EXTERNAL COATING

1.31.1 Scope

Technical specification governs requirements for material, equipment, application, inspection, repair and handling aspects associated with the external coating.

1.32.2 External coating:

External surface of the pipe and specials to be painted with one coat of two part chemically cured polyurethane primer of min 30 micron dry film thickness followed by one of two coats of two part solvent free polyurethane provide a high build coating of dry film thickness of 700 micron including primer coat. polyurethane coating should have good resistance to ultraviolet degradation along with corrosion resistance. pu coating should provide hard yet flexible, with very low water absorption with outstanding adhesion as well as impact and abrasion resistance.

Solventless pu system is to be used eliminating solvent health hazards.

External coating should be carried out at site by paint manufacturer as during transportation coating will get damaged.

1.32.3 General requirement:

I) The work shall conform to the following documents / specifications (latest revisionor as specified)

a) SSPC SP 10 / NACE 2 / SA2$^1/_2$ Near white blast cleaning .
b) SSPC PA 2 Measurement of dry film coating thickness with magnetic gauges
c) ASTM D 4541 Method for pull off strength using portable adhesion tester.

d) NACE RP 0274 2004–High voltage electrical inspection of pipeline coatings.
e) NACE SP 0188 – 2006 Discontinuity (holiday) testing of new protective coatings on conductive substrates.
f) NACE RP 0169 2002–control of external corrosion on underground orsubmerged metallic piping systems.
g) AWWAC 210 - 2007 liquid epoxy coating systems for the interior and exteriorof steel water pipelines.
h) IS 3589: 2001 annexure b steel pipes for water and sewage specification.
i) AWWA C222 2000–polyurethane coating for the interior and exterior of steelwater pipe and fittings.
j) Is 13213 : 2000polyurethane full gloss enamel (two pack)
k) Is 10221 : 2008coating and wrapping of underground mild steel pipelines
l) Bis Doc : chd 20 (1938) p solvent less liquid epoxy system for interior &exterior of steel water pipeline.
m) Is 16719:2018 specification for polyurethane coating for interior and exterior ofsteel pipe and fittings.

II) The steel surface preparation prior to actual commencement of coating shallconform to SSPC SP 10 / NACE 2 / Sa $2^1/_2$ (near white metal) with sand blasting.

III) The EPC contactor shall submit a detailed written description in the form of a manual covering coating equipment, procedures, materials inspection test, and repair etc. to the owner for approval.

IV) The contractor shall also provide copies of test reports from nabl approved laboratory (like national test house, kolkata) evidencing that materials (paints / primer) shall conform to the technical specification.

V) The contractor shall also provide certificates from coating manufacturer quoting batch numbers, date of manufacture and shelf life etc. the contractor shall provide manufacturing quality plan (MQP) and field field quality plan (FQP) before supply of material and starting of work.

VI) The coating shall be applied either by coating / paint manufacturer or by their authorized applicator. if done by applicator, then the same has to be approved by the authorized applicator should have been trained by and certified by manufacturer and shall possesses the necessary specialized equipment, trained crew and experience in spraying epoxy coating.

VII) Applied coating shall be tested for dry film thickness, holiday (electrical inspection for continuity) and adhesion as per relevant standard such as SSPC PA 2, NACERP 0274 and ASTM 4541.

VIII) The material may be heated if necessary and applied by airless spray / pluralcomponent spray system with minimum wastage.

1.32. 4 Coating procedure and application

1.32.4.1 Surface preparation:

Pipe shall be blast cleaned by sand. the cleanliness achieved prior to application shall be in accordance with the requirement of SSPC SP 10 / NACE 2 / SA21/2OF IS0 8501 (near white metal)

a) The blast pattern or profile depth shall be 40 to 100 micron and shall be measured by dial micrometer.
b) Before sand blasting is started or during blasting or coating, temperature of the pipe surface should be more than 3^0C above dew point temperature. blast cleaned surface should be primed within 4 hours and shall be protected from rainfall or surface moisture and shall not be allowed to flash rust. if the rust occurs, the surface again to be prepared by sand blasting or wire brushing.

1.32.4.2 Application of PU coating

a) PU coating shall be applied when the pipe surface temperature at least 30 c above due point temperature (when r.h. is more than 85 %)
b) Material preparation and application shall be done as per manufacturer recommendation and shall

be applied by plural component machine.

1.32.4.3 Test requirements:

1.32.4.3.1 Measurement of dry film thickness

Measurement of dry film thickness of coating: coating thickness shall be in the range of ± 20% and as per SSPC PA 2.

(i) Apparatus/ instrument: the instrument used for dry film thicknessmay be type i pull of gauges or type 2 electronic gauges.

(ii) **procedures**

a) No. of measurements. for 100 square feet (9.29 square meters) five (5) spots per test area (each spot is 3.8 cm.) in diameter. three gauge readings per spot (average becomes the spot measurement).

b) If the structure is less than 300 square feet, each 100 squarefeet should be measured.

c) If the structure is between 300 and 1000 sq. ft. select 3 random 100 square feet test areas and measure.

d) For structure exceeding 1000 square feet, select 3 random 100 square feet testing areas for the first 1000 sq. ft. and select i random 100 square feet testing area for each additional 1000 square feet.

e) Coating thickness tolerance: individual reading taken to get a representative measurement for the spot are unrestricted (usually low or high readings are discarded). spot measurements (the average of 3 gauge readings) must be within 80 % of the minimum thickness and 120 % of the maximum thickness. area measurement must be within specified range.

1.32.4.3.2 Electrical inspection (holiday) test:

a) All the coated/ lined pipes shall be tested with an approved high voltage holiday detector preferably equipped with an audio visual signalling device to indicate any faults, holes, breaks or conductive particles in the protective coating.

b) The applied output voltage of holiday detector shall have a sparkdischarge of thickness equal to at least twice the thickness of the coating to assure adequate inspection voltage and compensate for any variation in coating thickness. The electrode shall be passed over the coated surface at approximately half the spark discharge distance from the coated surface only one time at the rate of approximately 10 to 20m/min. the edge effect shall be ignored. excessive voltage shall be avoided as it tends to induce holiday in the coated surface thereby giving erroneous readings.

c) While selecting test voltages, consideration should be given to the tolerance on coating thickness and voltage should be selected on the basis of maximum coating thickness likely to be encountered during testing of a particular pipe.

d) The testing voltage shall be calculated by using following formula (as per NACE 0274:2004)

e) Testing voltage V=7900_T± 10 percent where t is The average coating thickness in mm.

f) Any audio visual sound or spark leads to indicate pinhole, break or conductive particle.

1.32.4.3.3 Adhesive pull off test:

After holiday the coated surface is subjected to adhesion pull off test as per ASTMD 4541

a) Apparatus/ instrument: adhesion tester consists of three basic components: a hand wheel, a black column containing a dragging indicator pin and scale in the middle and a base containing three legs and a pulling jaw at the bottom and also dollies.

b) Prepare the test surface: once test area is selected, test area shall be free of grease, oil, dirt, water. the area should be flat surfaces and large enough to accommodate the specified number of replicate test.

c) Prepare dolly (test pull stub): the dolly is a round, two sided aluminium fixture. both sides of the dolly looks same, however,one side sloped on top surface while flat on bottom surface. as the surface of the dolly is polished aluminium, roughen the same using a coarse sand paper.

d) Select an adhesive: use araldite, a 100 % solid epoxy adhesive. this adhesive requires at least 24 hours at room temperature to cure.

e) Attach the dolly to the surface.

- Using a wooden stick, apply an even layer of adhesive to the entire contact surface area of the dolly.
- Carefully remove the excessive adhesive by using a cotton swab. Allow the adhesive to fully cure before performing the adhesion test.
- Attach the dolly to the coated surface and gently push downward to displace any excessive adhesive.
- Push the dolly inward against the surface, then apply tape across the head of the dolly.

f) Determine whether scoring is required.

g) Perform the adhesion test.

- Attach the adhesion tester to the dolly by rotating the hand wheel counter clockwise to lower the jaw of the device.
- Slide the jaw completely under the head of the dolly. position the three legs of the instruments so that they are sitting flat on the coated surface.
- Slide the dragging indicator pin on the black column to zero by pushing it downward.
- Firmly hold the base of the instrument in one hand and rotate the hand wheel clock wise to raise the jaw of the device that is attached to the head of the dolly. the dragging indicator pin will move upward on the black column as the force is increased and will hold the reading. apply the tension using a moderate speed. continue to increase the tension on the head of the dolly until (a) the minimum psi/mpa/kg.cm2 required by project specification is exceeded and the test is discontinued (b) the maximum psi/mpa/kg/cm2 of adhesion tester has been achieved and dolly is still attached (c) the force applied by the adhesion tester causes the dolly to dislodge.
- Read the scale and record the adhesion value.

h) The black column near the middle of the adhesion tester contains two scales and a dragging indicator pin in between. right scale is PSI and left scale is MPa. the dragging indicator pin will hold the reading until push back to zero. when achieve the end point of test, read the scale across the bottom of the dragging pin.

1.32.4.4 Coating repair

Defecting coating: coating shall be repaired in accordance with the following subsections.

Surface preparation: accessible areas of pipe requiring coating repairs shall be cleaned to remove debris and damaged coating using surface grinders or other means. the adjacent coating shall be feathered by sanding, grinding or other method. accumulated debris shall be removed by blowing with contaminant free air or wiping with clean rages.

Areas not accessible for coating repair such as interior surface of small diameter pipe shall be reprocessed and recoated**.**

Coating application: the coating system shall be applied to the prepared areas in accordance with procedure.

Repair inspection: repaired portion shall be electrically inspected using a holiday detector.

1.32.4.5 Welded field joints

Preparation : The weld joints shall be cleaned so as to be free from mud, oil, grease, welding flux, weld spatter and other foreign contaminants. the cleaned metal surfaces of the weld joint shall then be abraded using rotary abrading pads or emery paper. the adjacent liquid Epoxy / **pu** coating shall be feathered by abrading the coating surface for a distance of 25 mm.

Electrical inspection: After curing the coating system applied to the welding joints shall be holiday tested. any holidays is indicated by the detector shall be marked with chalk to identify the area of repair

1.32.5 Lowering, Laying & Jointing Of Pipe

1 General

Unless specified otherwise, the pipeline shall be buried with minimum cover of 1.2 m at top, as shown on drawings. no material shall be erected unless it has been previously passed by the engineer.

Erection of fabricated shells shall be carried out by the contractor who shall equip himself, at his cost, with all necessary tools, machinery, labour etc. required for the purpose.

2 Welding

Except for routine welding of joints, no other work shall be done in the absence of contractor's engineer, either during the daytime or at night.

chipping shall not be kept in arrears for more than 15 joints.

Saddle pieces shall be fixed in position after checking bolts holes, by means of templates. these works shall be done together with the pipe laying work, if pipeline is to be laid above ground in unavoidable circumstances.

3 Temperature

The components of the pipeline such as base plates, top plates and pedestals have been so designed that the centers of the plates and pedestals shall coincide at the mean temperature (30°).

For this reason, all works such as fixing flanges, base plate etc. in true alignment, and incorrect position and tack welding pipes shall be done at the mean temperature. for ascertaining the temperature, the contractor shall provide mercury cups and fix them to the pipe shell from outside and shall also provide thermometers of the required type and range. no extra payment shall be made for this.

4 Saddle supports

Unless otherwise specified pipeline shall be underground. however at unavoidable reaches it shall be on R.C.C. saddles spaced at about 7 m centre to centre. the material and construction of R.C.C./STEEL structures such as saddles, anchor blocks, crossings etc. associated with the work of pipe line shall conform with the relevant

I.S. codes, good engineering practice and as directed by the engineer. the pipes to be laid on saddle supports shall be erected at mean temperature. saddle supports

Shall either be sliding type or fixed type. for both the types of supports a 10mm thick double plate shall be welded to the part circumference of the pipeline that will make contact with the saddle and another similar plate shall also be embedded in the concrete saddle with necessary arrangement to facilitate welding it to the double plate welded to the pipe, in case of fixed support. in case of sliding support, the

pipe shall be allowed to slide freely over the plate embedded in the saddle. alternatively to achieve fixity, the pipe shall be anchored by providing suitable anchor block. the rate for laying the pipe on saddle support shall include for laying, aligning, tack welding, provision of rigging screws with screw eyes etc., complete.

In addition to above, the pipe shall be held in position on saddles with two numbers 50mm x 8mm thick holding down traps fixed to the saddles with holding down bolts and nuts which shall be paid separately under relevant item of bill of quantities.

5 Erection of shells

The erection shall be true to position, lines and grade as shown in the drawings or as modified by the engineer. the contractor shall provide at his cost necessary saddles, pads, spider etc., all necessary instruments and other materials and labour required for proper erection of shells in position and for the engineer in checking the correctness of the erection.

Alignment of sections at edges to be butt welded shall be such that the maximum offset is not greater than the values given below:

Thickness 'T' (mm)	**Offset in Longitudinal joints (mm)**	**Offset in Girth Joints (mm)**
upto 12	0.25 t	0.25 t
12 to 20	3 mm	0.25 t
20 to 40	3 mm	5 mm
40 to 50	3 mm	1/8 t
over 50	**lesser of 0.0625 t or 10 mm**	lesser of 0.125 t or 20 mm

The best of welders as selected from their work in the contractor's workshop shall be selected for in-situ welding of the shells. the relevant provision under welding such as qualification standard for welding procedures, tests on welder's work and removal of defects etc., shall also apply to in-situ welding.

6. General sequence of operations

Before commencing the work of pipe laying, the contractor shall study the L- section of the pipeline for the section concerned. he shall also study the details of laying i.e. underground or aboveground. the underground pipeline shall be laid on sand cushioning/bedding as shown on the drawing. the difference in depth due to uneven excavations shall be made up by sand cushioning.

Pipe laying shall generally start from the fixity points on either side, the expansion joints if required for pipeline aboveground being provided last. fixing points are at all anchor blocks. where such blocks are not required for long lengths, fixity shall be achieved by fixing the pipeline to the special type of R.C.C. or steel saddles as directed. the distance between successive fixity points shall not exceed 300 m.

Anchor blocks shall be constructed before commencing the pipe laying work in any section. the construction of the blocks shall be carried out in 3 stages: in the first stage the lower part upto 150 mm below the invert of the pipeline including concrete chairs to support it shall be constructed; in the second stage the pipeline on this part of the block shall be laid; and lastly, the remaining block around and over the pipeline shall be constructed.

The fixity saddles and ordinary saddles if the pipeline is aboveground shall be cast- at least 3 weeks before the pipeline is laid on them. after all saddles between successive fixity points have been cast, a line plan showing the actual position thereof shall be prepared, after taking levels and measuring

distances. in case of any errors in casting the pedestals, corrections shall be applied. the pipe laying work shall then start from the fixity points and shall proceed towards the expansion joints. the method of jointing the pipes and erecting them on previously cast R.C.C. saddles shall be determined by the contractor depending upon the type of plant equipment and personnel available with them.

The pipe strakes shall be assembled in position on the saddles either by the cranes, portable gantries, shear legs or any other equipment approved by the engineer. normally, not more than two pipes shall be aligned, tacked and kept in position on temporary supports. the contractor shall not proceed with further work, until the circumferential joints of these pipes are fully welded. during assembly, the pipeline shall be supported on wooden sleepers and wedges, with the free end of the pipeline held in position by slings to avoid deflection due to temperature variations during the day. in general, the assembly of pipe strakes and one run of welding shall be done during the day time while full welding including the external gouging and sealing runs shall be done after 5 p.m. or so. the contractor shall maintain the continuity of the work by adding two more pipes on the second day in a similar manner, after full welding of the previous joints is completed during the night. while this new work is being done, the contractor shall proceed with the work of providing permanent supports for the pipeline assembled and welded previously.

7. Fixing expansion joint

The work of laying pipeline in aboveground, laying starts from the fixity points and proceeds towards the expansion joints. it shall be continued until the gap between the pipe ends is less than the lengths of the expansion joint plus pipe strake length. at this stage, the exact gap between the pipe ends shall be measured at mean temperature of that locality. let it be `X'. similarly, the exact length of the pipe strake and the expansion joint bought at site shall be measured at the same temperature let these be `Y' and `Z' respectively. normally, the length of the expansion joint (`Z') is standard.

(a) case when `Y' plus `Z' is more than `X' or equal to `X' (i.e. fixing of expansion joint without strip)

At mean temperature the exact gap between pipes shall be measured. free ends of pipes shall be brought in a correct line and level; lateral movement, if any, shall be corrected. then the gap between the free ends shall be made equal to the exact length of the expansion joint by cutting one of the pipe ends. choice of the end to be cut must be made from the point of view of bringing the expansion joint to a central position.

The expansion joints are normally supplied without packing. the normal length of the expansion joint shall be reduced by about 100 mm by cutting the inside locks and inserting the inner strake by means of turnbuckles. at mean temperature this expansion joint shall be inserted inside the gap (care being taken to keep the *volume* tapered portion on the down-stream side), and both ends shall be tack welded to the pipe ends, after pulling the expansion joint. (tacks of these two joints shall be of longer length, approximately 100 mm long).

Welding of these two joints of the expansion joints shall be started only after it is ascertained by taking observations that the expansion joint is functioning properly.

The procedure to be followed for taking observations is given in clause 8.5 (c).

(b) case when `Y' plus `z' is less than `x' (i.e. fixing of expansion joints withstrap)

The expansion joint shall be laid in locked position. before laying the pipes adjacent to the expansion joint, the exact gap between the pipes shall be calculated by taking measurements of the first pipe

(upstream of the expansion joint), and the second pipe (downstream of the expansion joint) at mean temperature.

If the gap is less than 100 mm, the second pipe shall be cut to make the desired gap of at least 100 mm. if the gap is more than 200 mm, suitable distance piece of not less than 700 mm shall be inserted after cutting necessary length of the first pipe. the second pipe shall then be laid in position. then a strap of length equal to three times The gap length shall be welded to the pipe, overlapping the second pipe by the gap length. the other end of the strip shall be kept free.

At mean temperature the other end of the strap shall be tacked to the first pipe, after checking of the line and level. simultaneously, all the locks of the expansion joint shall be removed and chipped off properly.

Welding of the joints between the strap and the first pipe shall be started only after observations are over and it is ascertained that the expansion joint is functioning properly.

(c) Observations

Before fixing the expansion joint, two mercury cups - one on the left and the other on the right side - shall be fixed on the pipe near the upstream side of the expansion joint.

Immediately after the expansion joint in case (a) above or the strap in case (b) above is tack welded, observations for total expansion or contraction shall be started and continued for 48 hours round the clock. similarly, the central and end fixity pedestals shall be kept constantly under observation.

The expansion and contraction shall be measured by making a temporary marking on the inner strake (on the upstream side) and measuring the distance between this mark and the edge of the gland of the expansion joint.

The observations shall be recorded in the following proforma;

reading no.	**time**	**shell temp on upstream side**	**shell temp on downstream side**	**atmospheric temperature**	**dist between edge of glandand marking**
1	2	3	4	5	6

In case the pipeline is laid in trenches as shown on the drawing, after welding and field testing, the trench shall be filled with selected material upto 300 mm above pipes. this backfill shall be provided in layers not more than 150 mm, with a density more than 95% of the standard proctor density. samples shall be tested as directed by the engineer. remaining depth of trenches shall be filled with ground backfill material.

1.32.6 SPECIALS

1. General

Specials, such as tees, y-pieces, bends (single or composite), tapers, etc. shall necessarily be in steel and shall be manufactured as per standards and tested and laid in the same manner as the pipes. small branches, single piece bends, etc. may be fabricated at site, care being taken to ensure that the fabricated fittings have at least the same strength as the pipeline to which they are to be jointed.

2. Bends

(a) Bends bends shall be fabricated taking into account the vertical and horizontal angles for each case.

(b) the bends shall have welded joints and the upstream and downstream ends of each bend shall have a straight piece of variable lengths as required.

(c) bends shall be designed with deflection angle of maximum 10 deg. between segments.

(d) when the point of intersection of a horizontal angle coincides with that of a vertical angle, or when these points can be made to coincide, a single combined or compound bend shall be used, designed to accommodate both the angles. the combined bend should have a pipe angle equal to the developed angle, arrived at from appropriate formula.

(e) all joints in bends shall be thermally stress relieved as specified.

(f) details of thrust collars anchor bolts, holding down straps, saddle plates should be furnished together with full specifications in contractor's fabrication drawing.

3. Manholes

(a) Manholes of 600 mm diameter shall be provided at locations generally 2.5 km apart, with additional manholes at both the sides of butterfly valves. Manholes in the pipeline shall be placed in suitable position in the top quadrant.

(b) The contractor shall fabricate different parts of manhole in conformity with relevant is specification, well established practices and as directed by the engineer.

(c) The manhole shall be provided with MS blank flange on top. The required MS nut, bolts & rubber packing shall be fixed of standard quality for fixing of flange.

4. Closing or make up sections

Closing or make up sections shall be furnished at appropriate locations on the line to permit field adjustments in pipeline length to compensate for shrinkage in field welded joints, differences between actual and theoretical lengths anddiscrepancies in measurements.

5. Heads

Test heads may be ellipsoidal, standard dished as per ASME code or hemispherical heads. they shall be welded in the shop and removed after the test. allowance should be made in the length of the pipe section receiving the test head for the welding and removal of the head and preparation of the plate edges for the final weld after testing.
No separate payment will be made for such test heads**.**

6. **Walkways, stairs, ladders, hand rails etc**.

Walkways, stairs, rungs, ladders, hand rails, etc. shall be provided as shown in the drawings and/or as directed by the engineer. they shall conform to well established design and construction for each accessory concerned.

7. Flanges

Flanges shall be provided at the end of pipes or special where, butterfly, air valve, scour valve, sluice valves, blank flanges, tapers, etc. have to be introduced. the flanges received from the manufacturers will have necessary boltholes drilled. the contractor shall assemble the flanges in the exact position by marginal cutting, if necessary, so as to get the desired position of the valves, etc. either vertical or horizontal and shall then fully weld the flanges from both sides in such a way that no part of the welding protrudes beyond the face of the flanges. in case the welding protrudes beyond the flanges and

if the engineer orders that such protrusions shall be removed, the contractor shall file or chip them off. if required and when ordered by the engineer, the contractor shall provide and weld gusset stiffeners, as directed on site. the drilling pattern shall be matching with the drilling pattern of flanges of valves.

8. Blank flanges

Blank flanges shall be provided at all ends left unattended for the temporary closure of work and also for commissioning a section of the pipeline or for testing the pipeline laid. for temporary closures, non-pressure blank flanges consisting of mild steel plates, tack welded at the pipe ends may be used. for pipes subjected to pressures, the blank flanges or domes suitably designed as per engineer's requirements shall be provided.

9. Stiffener rings

The contractor shall provide stiffener rings wherever required by design. the contractor shall weld the same to the pipes with one circumferential run on each side. all fillet welds shall have a throat thickness of not less than 0.7 times the width of welding.

1.32.7 FIELD HYDRAULIC TEST

After each section of the pipeline has been completed, it shall be tested for water tightness before being covered. the contractor shall at his own cost fill up water in pipeline & give necessary hydraulic test section by section & the pipeline shall stand the pressure which shall exceed the working pressure by (a) 50% of the highest pressure in the section (b) 30 m whichever us less without showing any leakage or sweating anywhere in the pipe joints, specials, valves, etc. if any defect is found, the contractor shall make good the same at his own cost.

Any leaking joint shall be made good & above test pressure is to be lowered gradually after satisfactory test is over.

The hydraulic test shall be given again if considered necessary by the superintending engineer or his representative to show that no further leakages or sweating is there. the contractor shall have to make necessary arrangements for water testing as well as plugging the opened pipe, etc. as directed without claiming any extra cost.

It the contractor does not able to manage section wise hydraulic test, hydraulic test shall be observed as below:

After completion of entire pipeline work and commissioning of pump sets (i.e. the project is ready for commission), the entire pipeline shall be hydro tested.

The contractor shall have to take the permission for drawal of water for field- testing of pipe. the cost of hydraulic testing of the installation by providing necessary testing equipments, creating and maintaining pressure and the necessary bulk heads and their fixtures and their subsequent removal and restoring the installation to working stream shall be included in the rate for laying and testing of the pipe. water charges and electricity charges for field hydraulic testing shall be borne by the contractor.

The pressure test shall be conducted in dry weather to observe whether any leakage takes place or not. for field hydraulic test water will pass through entire pipe line with a working pressure of maximum working pumps for a period of seven days continuously after attaining the required pressure and water reaches at the outlet of transmission main. a leakage equivalent to 0.267 litre per mm of pipe dia. per kilometre of pipeline per day for testing at working pressure applied is permissible. any leakage beyond that specified herein above shall be rectified to the satisfaction of the engineer.

Water shall not spout, ooze or sweat either through joints-welded or bolted or the body of the pipe. if any leakage noticed shall be repaired by the contractor, which shall include coating and repairing of the damaged portion. repairs and replacements and further testing including the cost of the plates and other raw materials shall be carried out by the contractor at his own cost. if any leakages are observed during the defects liability period due to defective workmanship or material supplied by the contractor, he shall repair the same to the entire satisfaction of the employer, at his own cost.

After completion of field hydraulic test as specified above and if no leakage is found in the entire pipeline / joints and all welded portion, then this may be considered as field hydraulic test and after commissioning of the pipeline system, amount retained at the rate of 3% can be released. (maximum number of working pumps will work continuously to achieve required pressure specified in the tender).

Till the contractor gives satisfactory field hydraulic test & commissioning of the pipeline an amount at the rate of 3% of testing commissioning as per price schedule shall be withheld.

1.32.8 PROGRESS IN LAYING

The contractor shall submit along with the tender his detailed bar chart for manufacturing and laying of the pipeline. while preparing his bar chart, the contractor shall plan his activities such that the laying of pipes shall closely follow the manufacturing schedule and no pipes shall remain stacked in factory or at site for a period more than one month.

It is mandatory that he shall submit an approach note on how he will carry out this work within the contractual period and on the compatible resources in terms of construction equipment and other facilities that he shall utilize to complete the tendered work.

1. Lowering and Jointing

The pipe shall be lowered into the trenches by removing only one or two struts at a time. it shall be seen that no part of the shoring is disturbed or damaged and, if necessary, additional temporary struts may be fixed during the lowering operations. it shall also be necessary to see that the gunite coating of pipe is not damaged in any way during the lowering and assembling. after the pipe is lowered into the trench, it shall be laid in correct line and level by using the leveling instruments, sight rails, theodolite, etc. care shall be taken to see that the longitudinal joints of two consecutive pipes at each circumferential joints are staggered by 90°. while assembling the pipes, the ends shall have to be brought close enough to leave a uniform gap not exceeding 4mm. if necessary, a marginal cut may be taken to ensure a close fit of the pipe faces. for this purpose, only experienced cutters who can make uniform and straight cuts, shall be permitted to cut the faces of the pipes. no extra payment shall be made for such marginal cutting. there shall be no lateral displacement between the pipe faces to be joined. if necessary, spiders from inside and tightening rings from outside shall be used to bring the two ends in perfect contact and alignment. it may also be necessary to use jacks for this purpose. in no case shall hammering or longitudinal slitting be permitted. when the pipe is properly assembled and checked for correct line and level, it shall be firmly supported on wooden beams and wedges and tack welded. some portion of the trench may be refilled at this stage so as to prevent the pipeline from losing its alignment. the tack welded circumferential joints shall then be welded fully. only experienced welders, who shall be tested from time to time shall be permitted to carry out the welding work.

On completion of the pipe jointing and external protection, the trench and the welding pits shall be cleaned of guniting rebound. the welding shall be filled and compacted in 150mm layers with the bedding material.

Backfilling shall be carried out as detailed here under.

Providing steel props inside the pipeline (dia. 1200 mm and above).

It is necessary that the roundness of the pipes is maintained circular. to achieve the same, steel adjustable screw type props of screw or similar approved make consisting of minimum six (three on each side) legs shall be fixed inside the pipe. the deflection of the pipe should be limited to 1% of the average diameter. in no case shall the limit be exceeded, even under the full load, in case of pipes laid underground. the design and drawings of the props that the contractor intends to use should be got approved by the engineer before starting the work. while laying the pipes underground, the contractor shall provide this propping arrangement from inside to maintain circularity. these props shall be fixed vertically and at intervals of not more than 1.8 metres or as directed by the engineer. in case the engineer finds it necessary, they will have to be fixed in any position. the props should be kept in position at least for three days after the encasing of the pipe in that section is completed or till refilling is done to the full height of fill over the pipe in case the pipes are not encased. the props shall be removed only after obtaining permission from the engineer. the height of earth fill over the pipe top shall normally be such as to avoid floatation under submerged conditions and to have a minimum earth cushion of about 1.00 metres over the pipe whichever is greater. it is also necessary that, in case of buried pipe, adequate side supports from the backfilled materials is developed to keep the diametrical deflection withinthe specified limits.

Backfilling of the excavated trenches, particularly below the pipe and along the sides shall, therefore, have to be done with proper care and compaction as desired. no extra payment will be made for the above work.

1.1 Precautions against floatation

When the pipeline laid underground or above ground in a long narrow cutting gets submerged in water collected in the trench of cutting it is subjected to an uplift pressure due to buoyancy and is likely to float if completely or partly empty. in the design of pipelines, provision is to be made to safeguard against floatation providing sufficient overburden or by providing sufficient dead weight by means of blocks, etc. factor of safety for calculations for check against floating shall be takenas 1.2.

In the case of works extending over one or more monsoon seasons, however, special care and precautions are necessary during the progress of work on this account. the contractor shall close down pipe laying operations well in time for the monsoon. the work of providing blocks, refilling the earth to the required level, compacting the same, etc. shall always be done as soon as the pipeline in the cutting has been laid.

The contractor shall see that the water shall not be allowed to accumulate in open trenches. where work is in an incomplete stage, precautionary work, such as blank-flanging in the open ends of the pipeline and filling the pipeline with water etc. shall be taken up as directed by the engineer.

Such works shall be to the contractor's account and no separate payment shall be made for the same. the contractor's rate for pipe laying shall be deemed to include such precautionary measures against floatation.

Protection of the pipeline against floatation during the contract period shall be the responsibility of the contractor. should any section of the pipeline float due to his negligence, etc. the entire cost of laying it again to the correct line and level shall be to his account.

Electric and telephone line, cable and pole if required to be shifted, necessary charges for shifting to be paid to respective authorities shall initially be paid by the contractor to the concern department and the same will be reimbursed by govt. to contractor on producing receipt of challan of concern department.

L - section of route is tentative. contractor has to prepare a fresh longitudinal section before starting the work, showing all service lines coming across or parallel.

Alignment of pipeline may required to be changed on same route due to any obstructs or service lines. no extra payment made to contractor.

Manhole chamber and all other structure required to be constructed on road are to be design for 'AA' class loading and same shall be got approved From Engineer-in-charge.

All necessary actions required for diversion of the traffic on state high way, or any other routes are to arranged by the contractor at his own cost as per the requirement of the traffic rules.

During the execution of work any damage occurred to the private / Govt. property is to be reimbursed by the contractor to the owner.

Asphalt cutting for pipeline excavation included in the scope of work and the debris of asphalt cutting shall have to be carted and dumped at the place shown by the engineer in charge so that Govt. can sort it out and reuse the same in other work. the contractor has to bring the road to its original position without any extra cost.

1.32.9 ADDITIONAL REQUIREMENT FOR PIPE TO BE LAID BY PUSHING METHOD

For insertion of casing pipe boring shall be carried out by auger type boring device with cutting head to drill a horizontal bore. the hole drilled shall be of suitable size to accommodate casing pipe. the casing pipe shall be inserted along with boring to keep the formation supported to prevent any settlement. the casing pipe shall be installed with even bearing throughout its length. the work will be done under the supervision of canal authority as per restrictions & precautions as may be stipulated by the canal authority.

The casing pipe will be minimum 1.2 m below the canal bed level & 0.9 m below the natural G.L. the length of Casing pipe shall be upto the boundary of canal bank. casing pipe shall be provided with minimum 50 mm dia gi pipe of 2 m height above G.L. with screen at top. rates are inclusive of bottom & side rollers of suitable size to be provided inside casing pipe at 4 m interval to facilitate insertion of carrier pipe of required dia. the ends of casing pipe shall be suitably sealed.

The casing pipe is to be provided with epoxy treatment as below on outer face of the pipe before lowering.

Zinc Rich Epoxy Primer

Description	Two pack zinc rich consisting of
Base	Fine zinc dust ground in epoxy resin solution, supplied in paste Form
Catalyst	Abduct type- the non volatile portion of the material (mixed) should Consist of 92 % zinc dust & 8-10% epoxy resin & curing agent
Shade	Grey
Characteristics	The paint shall provide a complete rust inhibitive barrier coating of high mechanical & abrasion resistance. The film shall be compatible For fusion & spot weld.
Pot-life	4-6 hours

Mixing ratio	The proportion of mixing of base & hardener should be as specified by the manufacturer by weight & volume. The mixed primer shall Conform to the specifications detailed.
Viscosity of ready Mixed paint	15-22 in fort cup no. 4 at 30°c
Drying time	i. Dust free 10-15 minutes ii. Chamber curing 24-28 hours
Thickness	4 mils (406 microns)

6. PRESTRESSED CONCRETE (PSC) PIPES :

This Specification covers the requirements for design, manufacturing, testing, supplying, laying, jointing and testing at works and site of Prestressed Concrete (PSC) pipes used for water supply mains.

Applicable Codes

The manufacturing, testing, supplying, jointing and testing at work sites of PSC pipes shall comply with all currently applicable statutes, regulations, standards and Codes. In particular, the following standards, unless otherwise specified herein, shall be referred. In all cases, the latest revision of the Codes shall be referred to. If requirements of this Specification conflict with the requirements of the standards / Codes, Specification of standard / codes with latest amendments shall govern.

IS:784	Specifications for Prestressed Concrete Pipes (Including Fittings)
IS: 783	Code of Practice for Laying of Concrete Pipes
IS: 1343	Code of Practice for Prestressed Concrete
IS: 7322	Specifications for Specials for Steel Cylinder Reinforced Concrete Pipes
IS: 3597	Methods of Test for Concrete Pipes
IS: 5382	Specifications for Rubber Sealing Rings for Gas Mains, Water Mains and Sewers
IS: 226	Specifications for Structural Steel (Standard Quality)

Others I.S. Codes not specifically mentioned here but pertaining to the use of PSC pipes form part of these Specifications.

Design

The Contractor shall design PSC pipes as per the Specifications for PSC pipes, to withstand the likely extreme conditions of stresses that may arise during all stages of manufacturing, handling and service. The designs shall be submitted to the Engineer for approval. Only on written approval of the designs by the Engineer, the Contractor has to start manufacturing the pipes. Any alterations / additions suggested by the Engineer are to be carried out by the Contractor without any extra cost.

All pipes shall be designed to withstand the combined effects of internal water pressure and the external loads. The design of prestressed concrete pipes shall be in accordance with the basic assumptions and general requirements stipulated in IS: 1343.

The design of PSC pipes shall take into consideration all such stages which may cause stress in any section of the pipe. For investigation of these design stages, the likely extreme conditions of stresses shall be considered in the order of their occurrence, during the process of manufacture, handling, erection and under service, giving due regards to the worst accompanying conditions.

The wall of PSC pipes shall be sufficiently thick to ensure that during and immediately upon completion of the prestressing operation, the stress in the concrete shall not exceed the limit specified in relevant clauses of IS: 1343 and IS: 784. The maximum permissible stress due to all forces, after deducting full losses, shall confirm to relevant clauses of IS: 1343 and IS: 784, under combined conditions of full prestress with or without hydrostatic pressure and stresses due to erection or handling. All combination of direct forces in any plane section of the pipe shall remain always compressive. The values of this compressive stresses shall not be less than 10 kg/cm2.

No weightage on the difference in the concrete core thickness as well as the reinforcement schedule provided from one tender to another shall he considered for evaluation. This is entirely for the test pressure indicated in the schedule.

For any other details in the design of the pipe, Specifications in IS: 1343 and IS: 784 are binding.

Material

A. Cement

The cement used in the manufacture of PSC pipes shall conform to one of the following:

1. Ordinary Portland cement conforming to IS: 269
2. Rapid Hardening Portland Cement conforming to IS: 8041
3. Slag cement (with not more than 50 % slag) conforming to IS: 455

B. Aggregate

The coarse and fine aggregates shall conform to I.S.383. The maximum size of aggregate shall not exceed one third the thickness of the pipe or 20 mm whichever is smaller.

C. Water

The water used in the preparation of concrete mix shall be clean and free from harmful or injurious compounds such as acids, alkali, oil, organic material or other substances and shall conform to the requirements of mixing water as per IS: 456.

D. Admixtures

No admixtures shall be used in the preparation of concrete mix.

E. Concrete

Suitable mix of concrete shall be proportioned such that the pipes and fittings made from it shall conform to all the requirements of the Specifications for PSC pipes and the Specifications laid down for design mix concrete as per IS: 1343. The quality control of the concrete shall be exercised in accordance with IS: 456.

F. Steel reinforcement

The reinforcement steel used for manufacturing PSC pipes shall conform to the Specifications of IS: 784.

G. Rubber gaskets

Rubber gaskets used for joints shall comply with IS: 5382. The pipe manufacturer shall test each gasket by stretching it to 33.3 % in excess of its original length and examining it visually for defects, particularly at any joint. Gaskets shall be clearly labelled in bundles. The label shall indicate the type of gasket, the type of joint and the diameter of the pipe with which they are to be used, and that they have been tested.

1. Dimensions and Tolerances

The internal diameter of the pipes shall be as per Specifications in IS: 784.

Pipe sections shall be manufactured in lengths of 5 m unless otherwise specified or approved by Engineer.

The tolerances over length, diameter and thickness of pipes shall be as per IS: 784.

A. Manufacture Moulds

The Moulds and method of manufacture shall be such that the form and dimensions of the finished pipe conform to the requirements of Specifications of dimensions and tolerances and workmanship**.** The surface and edges of the pipes are clean and true to the dimensions**.**

B. Concrete Core

Unless the design calls for higher concrete strength, the minimum compressive strength of concrete shall not be less than 45 N/mm2 at 28 days**.**

Clear cover

The clear cover of concrete over all steel reinforcement including the ends of the longitudinal prestressing wires shall not be less than 12 mm.

Prestressing

1. **Pretensioning and release of Longitudinal wires**

- The concrete core shall be longitudinally prestressed throughout its length, including the socket, by means of high tensile steel wires, which shall be provided with permanent anchorages embedded within the joint portion at each end.
- The centre line spacing between the longitudinal wires measured along the arc shall not exceed twice the core thickness or 150 mm, whichever is greater. Where reinforcement is used, the pretension stress shall be maintained by suitable supports, during the placing and curing of the pipe core until the concrete in the core has attained strength equal to 1.8 times the longitudinal induced compression in the concrete.

- The longitudinal wires shall be stressed to the design tension, taking into account the yield of slip of the temporary anchorages on the pipe moulds, and the tension shall be maintained by positive means during the curing of the concrete.

- The tensioned wires shall not be released until the concrete in the core has attained a compressive strength of concrete in compression at 28 days as mentioned in Clause 15.6.13.2.

2. **Circumferential Prestressing**

- Circumferential prestressing shall not take place until the concrete in the core has reached a sufficient compressive strength to resist without damage the force acting upon it, nor until the concrete in the core has reached a minimum cube strength of 32 N/mm2.

- The initial compressive stress induced in the concrete core shall not exceed 55 % of the compressive strength of the concrete in the pipe at the time of transfer.

- Methods and equipment for applying the wire shall be such that wire shall be wound around the core in a helical form at the predetermined design spacing and capable of controlling the tension. Splicing or jointing (not welding), developing the full strength of the wire, may be permitted.

- The clear spacing between successive turns of the circumferential prestressing wire shall not be less than the greatest of the following:
  (i) Diameter of the prestressing wire,
  (ii) One and half times the maximum nominal size of the aggregate used for the outer coat concrete, and
  (iii) 5 mm.

Cement mortar cover coating

1. After circumferential prestressing wire is wound on the pipes, it shall be given a coating of cement mortar, to protect it from corrosion. The cover coating shall preferably be applied within 16 hours after the prestressing wire has been wound. Minimum cement content in coating mortar shall be 600 Kg/m3. The thickness of this coating shall not be less than 25 mm, over the circumferential reinforcement.

2. Cement mortar coating shall be done by rotary brushes or belts or by other approved means. The sand and cement shall be thoroughly mixed before being fed into the cover coating machine. Pneumatic process in which mixing of all ingredients is carried out at the nozzles or gun, shall not be permitted.

Handling and Transportation of pipes

Pipes shall be handled and transported to the site carefully as per the general Specifications for Laying of Pipes and Fittings given in this chapter, as per IS: 783.
Pipes manufactured at factory are to be carried to the site of work directly or stacked suitably and neatly along the alignment / road side / elsewhere near by the work site, as directed by the Engineer.
All pipes shall be loaded in trucks by mechanical crane / tripod and unloaded carefully using crane / tripod. No unloading using crow bars or on tyres will be allowed in any case. Rubber belt may be used instead of crow bars or chains.

Extreme care shall be taken while handling the pipes. Damages during transit will be to the Contractor's account and replacement for such pipes has to be made by the Contractor without any extra cost.

The rates should be inclusive of payment of entry tax, octroi, work contract tax and all other statutory taxes prevailing at the time of tendering. All such taxes are to be paid by the Contractor at no extra cost to the Engineer. This does not include any increase of slabs of levy due to turn over.

Lowering, Laying and Jointing of Pipes

Pipes shall be lowered, laid and jointed carefully as per the general Specifications for Laying of Pipes and Fittings given in this chapter, as per IS: 783.

Each pipe shall be thoroughly checked for any damages before laying. and only the pipes which are approved by the Engineer shall be laid.

As directed by the Engineer, moorum / sand bedding has to be done at the places shown. At other places, consolidation of bottom depth to the required grade will be done, as far as possible.

The trench shall be kept free of water till the jointing material is properly set.

Walking on completed line shall not be permitted until trenches are back filled.

Bedding for PSC Pipes

- In case of sandy soil / soft soil / hard soil strata no separate bedding is required. However the bottom face / trench bed where pipe shall be placed shall be compacted to provide a minimum compaction corresponding to 95% of maximum dry density.
- The trench bottom should be flat and free from sharp projection and stones/ rocks bigger than 30 mm in diameter or of other potentially damaging debris.
- As directed by the Engineer, approved sand / moorum bedding has to be done in case of rock strata. At other places, consolidation of bottom depth to the required grade will be done, as far as possible.
- The pipe should be uniformly and continuously supported throughout its whole length with approved uniform bedding material. The pipe bedding should be placed so as to give complete contact between the bottom of the trench and the pipe.

Jointing of Pipes

Pipes shall be provided with Flexible Joints unless otherwise specified. Joining shall be done in accordance with the Specifications of IS: 783, the general Specifications of Laying of Pipes and Fittings given in this chapter. The sections of the pipe should be joined together in such a manner that there shall be as little unevenness as possible along the inside of the pipe. The basic requirements for jointing are:

1. Cleanliness of all parts, particularly joint surface,
2. Correct location of components,
3. Centralisation of spigot within socket,
4. Provision of correct gap between one end of the spigot and the back of the socket (for rubber ring joint) to ensure flexibility at each end, and
5. any lubricant used shall be approved as to composition and method of application.

The following Specifications for jointing of PSC pipes are to be considered in addition to the above:

1. The joints shall be Confined Joints as per the approved Design and Drawing, with a continuous rubber ring made of a special composition rubber of such size and cross section as to fill completely the recess provided for it, over the spigot portion of the pipes. The gasket shall be the sole element to make the joint water tight and shall have smooth surface free from pits, blisters, porosity and other imperfections. The rubber compound shall contain not less than 5% by volume of first grade natural crude or synthetic rubber. The remainder of the compound shall consist of pulverised fillers free from rubber substitutes, reclaimed rubber and other deleterious substances.

2. Rubber ring used for jointing shall comply with IS: 5382. The manufacturer shall test each ring by stretching it to 33.3 % in excess of its original length.

3. Every rubber ring shall be visually checked particularly at joints, for any defects.

4. No jointing shall be made under water. The jointing surfaces shall be dried and cleaned before the joint and the prepared joint shall be kept clean.

5. All jointing work shall be done in an approved manner by skilled workmen so that complete pipeline shall have a continuous, smooth and uniform interior surface. Extended joint materials shall be removed from the inside of the pipes.

Specials and Fittings :

Specials and fittings for PSC pipes shall be made out of steel plate fabricated to the required dimensions and given a coating of cement mortar inside and outside and shall be as per the Specifications below:

1. These specials shall be suitable for fixing prestressed concrete pipeline and shall conform to the requirements of IS: 784 and IS: 7322.

2. The special such as bends, tapers, tees, and branches shall be fabricated by cutting steel plates of 10 mm thickness and shall be manufactured with spigot and socket type joints with rubber gasket, suitable for jointing with a prestressed pipe, as per Specifications.

3. The steel for fabricated steel plate fittings is cut, shaped and welded so that the finished fitting has the required shape and interior dimensions. Adjacent segments shall be jointed by means of lap or butt welding.

4. The steel used for manufacturing of specials shall conform to IS: 226.

5. The specials shall be lined inside and outside with required thickness cement mortar, reinforced with 50 x 50 mm weld mesh of specified guage. The cement mortar inside shall be 1:1.5 (1 part cement: 1.5 part sand) and cement mortar outside shall be 1:2 (1 part cement:2 part sand).

6. All specials shall be hydrostatically tested before using as per Clause 8 of IS: 7322. In case of specials manufactured at site, they shall be tested by penetration oil or other approved means.

7. All the specials shall be tested for hydrostatic pressure as specified for PSC pipes and to the pressure specified for pipes in the reaches where the specials are fitted.

8. All fittings shall be inspected by the Engineer at the place of manufacture, either at site or at factory. Engineer must have free access to the place of manufacture for the purpose of examining, testing and marking of the fittings.

9. The Contractor is fully responsible for any defects in manufacturing not conforming to IS Specifications. All defective specials shall be replaced by the Contractor free of cost.

10. Extreme care shall be taken while carting the specials to site, so that inside or outside coating is not damaged. If damages are observed the Contractor shall have to make it good at his cost.

11. For all other Specifications, the IS. 7322 or relevant amendments are binding.

12. Payment shall be done on per meter basis measured along the central line of the special/fittings. The quantity is as mentioned in Bills of quantities.

13. The valves like sluice valves, non return valves, scour valves, air valves shall be arranged by the Contractor as per the quoted rates and as per Bill of Quantities.

14. The quantity of specials shown in Bill of Quantities is tentative and the Contractor is required to supply and fix the specials as per actual requirement on site, as applicable.

Hydraulic Testing

Pipes shall be given different tests for ensuring quality of manufacture as per CPHEEO manual and IS code.

Measurement

The net length of pipes as laid or fixed shall be measured in running meters correct to a cm.

7. MDPE Pipes :

This specification covers the requirements for successfully designing, manufacturing, supplying, laying, jointing and testing at works and site of Medium Density Polyethylene Pipes used for water supply upto 50mm dia pipes.

Applicable Codes

The manufacturing, testing, supplying, laying, jointing and testing at work sites of MDPE pipes shall comply with all currently applicable statutes, regulations, standards and Codes. In particular, the following standards, unless otherwise specified herein, shall be referred. In all cases the latest revision of the Codes shall be referred to. If requirements of this Specification conflict with the requirements of the standards / Codes, Specification of standard / codes with latest amendments shall govern.

Code No.	Title/Specification
ISO 4427	Medium Density Polyethylene Pipes for Water Supply
IS 2530	Methods of test for polyethylene moulding materials and polyethylene compounds
IS 5382	Rubber sealing rings for gas mains, water mains and sewers.
IS 4905	Methods for random sampling
IS 7634	Laying & Jointing of Polyethylene (PE) Pipes
IS 9845	Method of analysis for the determination of specific and/or overall migration of constituents of plastics material and articles intended to come into contact with foodstuffs
IS 10141	Positive list of constituents of polyethylene in contact with food stuffs, pharmaceuticals and drinking water.
IS 10146	Polyethylene for its safe use in contact with foodstuff, Pharmaceuticals and drinking water.

Others Codes not specifically mentioned here but pertaining to the use of MDPE pipes form part of these Specifications**.**

Designation

Pipes shall be designated as per ISO 4427, according to the grade of material and pressure class.

Colour

The colour of the pipe shall be **blue**.

Materials

The material used for the manufacturer of pipes should not constitute toxicity hazard, should not support microbial growth, should not give rise to unpleasant taste or odour, cloudiness or discoloration of water. Pipe manufacturers shall obtain a certificate to this effect from the manufacturers of raw material by any internationally reputed organization as per the satisfaction of the engineer in charge. The resin shall be supplied by any of the reputed manufacturer:

Raw Material

Raw material used to manufacture the MDPE pipes shall be 100% virgin blue compounded PE 80 polyethylene grade in accordance with ISO 4427-1996 or Natural resin with blue master batch as per IS 4984:1995 and should be approved by an organization of international repute for use in potable water. Every meter length of pipe should contain the Grade of PE resin, Pressure Rating, Name of project and Client or as directed by the Engineer.

Characteristics of the PE compound

Characteristics	Units	Requirements	Test methods
Density	Kg/m^3	930 – 940	ISO 1183
		(Basic Polymer)	
MFR	gm/10 minute	0.5 – 1.0	ISO 1133
Thermal Stability	minute	>20	ISO TR 10837
Volatile content	mg/kg	$\leq$ 350	Annex A of ISO 4437
Pigment Dispersion	grade	$\leq$ 3	ISO 13949

Performance Requirements:

Characteristics	Requirements	Test Parameter	Test methods
Hydrostatic Strength			
	Failure time >100 hrs	20^0 / PE 80 σ = 9.0 MPa	ISO 1167
	Failure time > 165h	80^0 C/PE80 σ = 4.6 MPa	- do -
	Failure time > 1000h	80^0 C/PE80 σ = 4.6 MPa	- do -
	Elongation at break	> 350%	ISO6259 / 3
	Tensile strength at yield	>15 MPa	ISO6259 / 3
Resistance to weathering	After weathering Thermal stability % Elongation & HS (165H / 80^0 C) shall be met	E $\geq$ 3.5 GJ/m^2	Annex-A of ISO 4427 ISO/TR/10837 ISO6259/3 ISO 1167
Conventional Density	$\geq$ 930 Kg/m^3	23^0 C	ISO 1183
Thermal Stability	20 minute	200^0 C	ISO/TR 10837
Melt Flow Rate (MFR)	1. Change in MFR by processing < 20% 2. $\pm$ 30% of the value nominated by raw material the manufacturer	190^0 C	ISO 1133
Heat Reversion	Longitudinally reversion shall be not greater than 3%	110 $\pm$ 2^0 C	ISO 2505 - 1

Reworked Material

No addition of **Reworked/ Recycled Material** from any source including the manufacturer's own rework material resulting from the manufacture of pipes is allowed and the vendor is required to use only 100% virgin resin compound.

Coiling

The pipes supplied in coils shall be coiled on drums of minimum diameter of 25 times the nominal diameter of the pipe ensuring that kinking of pipe is prevented.

Workmanship / Appearance

Pipes shall be free from all defect including indentations, delaminating, bubbles, pinholes, cracks, pits, blisters, foreign inclusions that due to their nature degree or extent detrimentally affect the strength and serviceability of the pipe. The pipe shall be as uniform as commercially practicable in colour opacity, density and other physical properties as per relevant ISO 4427 or equivalent International Code. The inside surface of each pipe shall be free of scouring, cavities, bulges, dents, ridges and other defects that result in a variation of inside diameter from that obtained on adjacent unaffected portions of the surface. The pipe ends shall be cut clearly and square to the axis of the pipe.

Handling, Transportation Storage and Lowering of pipes

During handling, transportation, storage and installation, all sections shall be handled by such means and in such a manner that no distortion or damage is done to the section or to the pipes as a whole.
The following procedures should be followed so as to eliminate potential damage to pipes and fittings and to maintain maximum safety during transportation, storage and installation.

- Pipes must not be stored or transported where they are exposed to heat sources likely to
- exceed 60° C.
- Pipes shall be stored such that they are not in contact with direct sunlight, lubricating or hydraulic oils, petrol, solvents and other aggressive materials.
- Scores or scratches to a depth of greater than 10% or more of wall thickness are not permissible; any pipes having such defects should be strictly rejected.
- PE pipes should not be subjected to rough handling during loading and unloading operations.

- During coiling care should be taken to maintain the coil diameter at or above the specified minimum to prevent kinks. Coiling shall be done when the pipe attains the ambient temperature from the extruder. In uncoiling or recoiling care should be taken that sharp objects do not scour the pipe.

- When releasing coils, it must be remembered that the coil is under tension and must be released in a controlled manner. The end of the coil should be retained at all times, then the straps released steadily, one at a time. If the coil has bands at different layers of the coil, then they should be released sequentially starting from the outer layers. The amount of the energy locked up in the coil will depend on the size of the pipe, the SDR of the pipe, and the size of the coil.

- Bare coils shall be wrapped with hesian cloth for long distance (> 300Kms) transportation. The truck used for transportation of the PE pipes shall be exclusively used of PE pipes only with no other material loaded – especially no metallic, glass and wooden items. The truck shall not have sharp edges that can damage the pipe.

- Damages during transit, handling, storage will be to the Contractor's account and replacement for such pipes has to be made by the Contractor without any extra cost as directed by the Engineer.

Jointing of Pipes

The pipe shall have a jointing system that shall provide for fluid tightness for the intended service conditions. Appropriate jointing for MDPE pipe as per ISO 4427 & ISO 14236 shall be selected considering the existing working condition, pressure and flow of liquids.

Compression Fittings

The Compression Fittings should be as per following guidelines:

a. The body of the compression fittings should be made from Poly Propylene.
b. Compression fittings should be manufactured in accordance with ISO14236 for use with HDPE/MDPE/GI pipes.

c. All types of compression fittings shall be certified by third party for drinking water usage.

d. The fitting body shall be opaque. The composition of the plastics parts exposed to sun/ultra-violet radiation shall include UV stabilizer.

e. The minimum bore of the fittings shall be such that there shall be no obstruction for the passage of water from the pipe to pipe fitting.

f. The fittings shall conform to the requirements for leak tightness for 1 hour when the pipe and fittings are assembled and hydraulically tested to the working pressure of the pipe.

The material used for manufacture of Compression Fittings and the material of the PP fitting components, which are in contact with the water, shall not contain toxic additives. The PP compression fittings shall be certified for drinking water usage. A certification to that effect shall be submitted by the vendor of the fittings.

PP Saddles

a. The body of the tapping saddle should be made from Poly propylene.
b. The rubber ring to arrest leakage, provided around the tapping location of the saddle shall be made with Neoprene.
c. The fasteners used to fix the saddle to the PE water mains shall be maid with stainless steel / carbon steel.
d. The tapping saddles be suitable to tap PE water mains of diameter 63 to 200 mm and the pressure rating of the saddle shall be PN16.\

Electrofusion Saddles

a. The Electrofusion saddles should confirm to pr EN12201-3
b. It should be manufactured from virgin compounded blue / black resin or natural resin with blue or black master batch
c. The body of the tapping saddle should be made from Poly Propylene.
d. The tapping saddles shall be suitable to tap PE water mains of diameter 63 to 200 mm and the pressure rating of the saddle

Tests to Establish Portability of Work

Pipe specimen shall be subjected to tests specified below in order to establish the suitability of these pipes for use in carrying potable water:

i. Smell of the extract
ii. Clarity of the colour of the extract
iii. Acidity and alkality
iv. Global migration UV absorbing material Heavy metals
v. Unreacted monomers (styrens) and Biological tests

Hydraulic Testing

Pipes shall be given different tests for ensuring quality of manufacture as per CPHEEO manual and IS code

8 Galvanised Iron Pipes

This Specification covers the requirements for manufacturing, supplying, laying, jointing and testing at works and site of Galvanised Iron pipes used for water supply.

Applicable Codes

The laying GI pipes and fittings / specials shall comply with all currently applicable statutes, regulations, standards and Codes. In particular, the following standards, unless otherwise specified herein, shall be referred. In all cases, the latest revision of the standards / Codes shall be referred to. If requirements of this Specification conflict with the requirements of the standards / Codes, Specification of standard / codes with latest amendments shall govern.

IS: 1239 (Part-I)	Specification for Medium Grade GI Pipes
IS: 1239 (Part-	MS Tubular or Wrought Steel Fittings for GI Pipes

II)	
IS: 4736	Specification for Galvanizing
IS: 554.	Specifications for Pipe Threads of Screwed Tubes and Sockets
IS: 778	Gate Valve & Check Valve
IS: 2692	Ferule for Water Supply

Others I.S. Codes not specifically mentioned here but pertaining to the use of GI pipes form part of these Specifications.

Manufacture

The pipes shall be Galvanised mild steel hot finished seamless (HFS) or welded ERW, HRIWor HFW screwed and socketed conforming to IS: 1239 (Part-I) for medium grade. The zinc coating shall be uniform adherent, reasonably smooth and free from imperfections.
All screwed pipes and sockets shall have pipe threads conforming to the requirements of IS: 554. Screwed tubes shall have taper threads while the sockets shall have parallel threads.
Where the pipes have to be cut or threaded, the ends shall be carefully filed out so that no obstruction to bore is offered. The ends of the pipe shall then be carefully threaded conforming to the requirements of IS: 554 with pipe dies and tapes in such a manner that it will not result in slackness of joints when two pieces of pipes are screwed together. The taps and dies shall be used only for straightening screw threads which have become bent or damaged and shall not be used for turning of the threads as to make them slack, as the later procedure may not result in a water tight joint. The screw threads of the pipes and fittings shall be protected from damage until they are fitted.

Dimensions

The dimensions and weights of medium grade GI pipes and sockets and tolerances shall be as prescribed below:

Nominal bore (mm)	Dimension of pipes			Weight of pipe	
	Outside diameter (mm)		Thickness (mm)	Plain end Kg/m	Screwed end socket Kg/m
	Max.	Min.			
6	10.6	9.8	2.0	0.427	0.430
8	14.0	13.2	2.35	0.667	0.681
10	17.5	16.7	2.35	0.886	0.892
15	21.8	21.0	2.65	1.27	1.28
20	27.3	26.5	2.65	1.64	1.65
25	34.2	33.3	3.24	2.51	2.53
32	42.9	42.0	3.25	3.23	3.26
40	48.8	47.9	3.25	3.72	3.76
50	60.8	59.7	3.65	5.24	5.31
65	76.6	75.3	3.65	6.69	6.81
80	89.9	88.0	4.05	8.68	8.85
100	115.0	113.1	4.50	12.40	12.70
125	140.8	138.5	4.85	16.50	17.00
150	166.5	163.9	4.85	19.60	20.20

Tolerance

Tolerance in Thickness

Butt welded medium tubes	+ not limited	- 10.0 %
Seamless tubes	+ not limited	- 12.5 %

Tolerance in Weight

Single tube (irrespective of quantity)	+ 10 %	- 8 %
For quantities of less than 150 m of one size	+ 10 %	- 8 %
For quantities of 150 m and over of one size	+ 4 %	- 4 %

Specials and fittings

The fittings for GI pipes shall be of mild steel tubular or wrought steel fittings conforming to IS: 1239 (Part-II). The fittings shall be designated by the respective nominal bores of the pipes for which they are intended.

Jointing

The pipes shall be cleaned and cleared of all foreign matter before being laid. While jointing the pipes, the inside of the socket and the screwed end of the pipes shall be oiled and rubbed over with white lead and a few turns of spun yarn wrapped round the screwed end of the pipe. The end shall then be screwed in the socket, tee etc., with the pipe wrench. Care shall be taken that all pipes and fittings are properly jointed so as to make the joints completely water tight and pipes are kept at all times free from dust and dirt during fixing. Burr from the joint shall be removed after screwing. After laying, the open ends of the pipe shall be temporarily plugged to prevent access of water, soil or any other foreign matter.

Laying of GI pipes and fittings

External work

1. GI pipes if used for external work, shall be laid according to specifications. If they are laid in trenches, the widths and depths, for different diameters of the pipes shall be as follows:

Dia. of Pipe (mm)	Width of Trench (cm)	Depth of Trench (cm)
15 to 50	30	60
65 to 100	45	75

2. At joints the trench width shall be widened where necessary. The work of excavation and refilling shall be done true to line and gradient in accordance with the Specifications on laying of pipes given in this chapter.

3. Pipes shall be laid on a cushion of sand minimum 7.5 cm deep and filled upto 15 cm above the pipes. The remaining portion of the trench shall then be backfilled as described in Chapter 5.

Internal work

1. For internal work, the Galvanised iron pipes and fittings shall run on the surface of the walls or ceiling (not in chase) unless otherwise specified. The fixing shall be done by means of standard pattern holder bat clamps, keeping the pipes about 1.5 cm clear of the wall. When it is found necessary to conceal the pipes, chasing may be adopted or pipes fixed on the ducts or recess etc., provided there is sufficient space to work on the pipes with the usual tools. The pipes shall not ordinarily be buried in walls or solid floors. Where unavoidable, pipes may be buried for short distances provided adequate protection is given against damage and where so required joints are not buried. Where directed by the Engineer, a MS tube sleeve shall be fixed at the place where the pipe is passing through a wall or floor, for reception of the pipe and to allow freedom for expansion and contraction and other movements. In case the pipe is embedded in walls or floors it should be painted with anticorrosive bitumastic paints of approved quality. The pipe shall not come in contact with lime mortar or lime concrete as the pipe is affected by lime. Under the floors the pipes shall be laid in layer of sand filling as done under concrete floors.

2. All pipes and fittings shall be fixed truly vertical and horizontal unless unavoidable. The pipes shall be fixed to walls with standard pattern holder bat clamps of required shape and size so as to fit tightly on the pipes when tightened with screwed bolts. These clamps shall be embedded in the brick work of the walls, and shall be spaced at regular intervals in straight lengths as shown in the table ahead:

Dia of Pipe (mm)	Horizontal length (m)	Vertical length (m)
15	2	2.5
20	2.5	3
25	2.5	3
32	2.5	3
40	3	3.5
50	3	3.5
65	3.5	5
80	3.5	5

3. The clamps shall be fixed at shorter lengths near the fittings as directed by the Engineer.

4. For G.I. pipes 15 mm diameter, the holes in the walls and floors shall be made by drilling with chisel or jumper and not by dismantling the brick work or concrete. However, for bigger dimension pipes the holes shall be carefully made of the smallest size as directed by the Engineer. After fixing the pipes the holes shall be made good with cement mortar 1:3 (1 cement: 3 coarse sand) and properly finished to match the adjacent surface.

5. Unions will be provided to facilitate connections, additions and alternations as well as for maintenance and for change of pipes. The locations where unions are to be provided will be decided with prior within approval of the Engineer.

Testing

The pipes and fittings after they are being laid and jointed shall be tested to a hydraulic test pressure of 60 kg/cm2 (600 m). The pipes shall be slowly and carefully charged with water allowing all air to escape and avoiding all shocks or water hammer, which may develop otherwise. The draw off taps and stop cocks shall then be closed and specified hydraulic pressure shall be applied gradually. A calibrated and accurate pressure gauge shall be used for testing the pressure. The test pump having been stopped, the test pressure should be maintained for at least half an hour. The pipes and fittings shall be tested in sections as the work of laying proceeds, having the joints exposed for inspection during the testing. Pipes or fittings which are found defective shall be replaced and joints found leaking shall be redone, without any extra payment.

The water for testing shall be provided by the Contractor. The quality of water should be approved by the Engineer.

9 . <u>Field Hydraulic testing of the pipelines</u>

a. **Sectional tests**

After laying and jointing the pipeline shall be tested for tightness of barrels and joints, and stability of thrust blocks in sections approved by the Engineer-in-Charge in Charge. The length of the sections depends on the topographical conditions. Preferably the pipeline stretches to be tested shall be between two chambers (air valve, scour valve, bifurcation, other chamber).At the beginning, the Contractor shall test stretches not exceeding 1 km. After successful organizing and execution of tests the length May be extended to more than 1 km after approval of the Engineer-in-Charge in Charge. The hydraulic testing shall have to be commenced immediately after laying and jointing of 1 km reach is completed.

The water required for testing shall be arranged by the contractor himself. The Contractor shall fill the pipe and compensate the leakage during testing. The Contractor shall provide and maintain all requisite facilities, instruments, etc. for the field testing of the pipelines. The testing of the pipelines generally consists in three phases: preparation, pre-test/saturation and test, immediately following the pre-test.

10. Flushing and disinfecting of pipelines

After testing and commissioning the contractor shall flush the pipes with a velocity not less than 1 m/s or as approved by the Engineer-in-Charge in Charge. Disinfection of drinking water pipelines should be done by Contractor.

11. Above ground pipeline

DI K9 pipe of appropriate size, conforming to IS: 8329 or MS pipes of appropriate size and thickness, conforming to IS:3589 will be used wherever the pipeline is laid above ground. MS pipes will be in-lined and out-coated with 15 mm thick Cement concrete mortar or Epoxy coating of 400 Micron DFT.

12.0 INSTALLATION OF VALVES:

General

The installation of valves shall be made according to the instructions of the manufacturer and the Engineer.

Installation of valves

Butterfly/Sluice valves shall be installed between flanges according to the instructions of the manufacturer.

Valves shall be placed on a support of concrete so that no shear stress is in the flanges. In case of axial thrust due to closure of a valve against pressure the valve shall be anchored in the support in a suitable manner to transfer the thrust into the floor slab of the chamber.

Air valves shall be installed on top of air valve tees.

SLUICE VALVES

DESIGN REQUIREMENT

A. Sluice valves shall generally conform to IS 14846/BS EN 1171/DIN 3352. Additionally, they should also meet specific requirement as stated.

B. Spindle, thrust collar and operating arrangement including hand wheel should be designed in such a way that one adult male is able to operate the valve against full differential pressure by exerting no more than 8 kgf effort (pull and push) on the hand wheel.

FEATURES OF CONSTRUCTION

a. Valves shall have inside screw, non rising spindle.

b. Valves shall be with appropriate bushing arrangement for replacement of packing without leakage (350 mm ø and above), up to 350 mm ø valves shall be gland.

c. Valves 450 mm ø & above shall be provided with an antifriction device / ball thrust bearing arrangement to minimize friction between spindle collar and casting. These should be housed away from wet chamber and should have facility for periodic greasing.

d. Valves of size 450 mm ø and above shall be provided with enclosed, grease packed spur gear box.

e. Valves 450 mm ø and above shall be provided with a drain and air plug.

f. All valve doors when fully closed would ensure door faces are riding on body seat ring by at least 50% of the width of seat ring and there is sufficient room for wear travel. Applicable for valves 350 mm and above, up to 300 mm valves shall be resilient seated.

g. All face and seat rings will be force/press fitted and additionally riveted (300 ø & above) to the recess in the CI casting. No screws are allowed.

h. Spindle, thrust collar and operating arrangement including hand wheel should be designed in such a way that one adult male is able to operate the valve against full differential pressure by exerting no more than 80 N effort (pull and push) on the hand wheel. Only single start, square threads with a pitch not exceeding 12 mm in the spindle be used.

i. Manufacturer to give details of gear box proposed – no. of spur pair, ratio, efficiency etc.

j. Manufacturer to justify with calculation that the valve proposed is operable within the effort parameters specified and no. of turns to ensure the time required to operate the valve from full open to full close is within reasonable limits. This is a vital requirement.

k. Nominal size of the valve shall be cast on the body of the valve.

DATA :

1.	Size	:	150 mm to 350 mm
2.	Rating (Kg/sq.cm)	:	PN 1.0

3.	Drilling	:	IS 1538 Table 4 & 6 / relevant ISO with latest amendments/ BS EN 1092-2
4.	Material of construction	:	
	Body	:	DI IS 1865 Gr. 500/7 ; 400/15 or CI IS 210 Gr. FG 200 for PN 1.0 (all sizes)
	Wedge	:	DI IS 1865 Gr. 500/7 ; 400/15 (fully rubber lived EPDM, up to 300 mm) or CI IS 210 Gr. FG 200 for PN 1.0
	Spindle	:	St. St. AISI 410 / 316 / relevant ISO with latest amendments
	Seat & face rings	:	Bronze IS 318 LTB II 6 / relevant ISO with latest amendments (for 350 mm above)
	Drain & air plug	:	Bronze IS 318 LTB II 6 / relevant ISO with latest amendments
	Ball thrust bearing	:	SKF or equivalent
	Bushing arrangement	:	Halprene on bronze
	Rivets	:	Soft annealed brass
	Gland packing	:	Teflon coated / graphite asbestos / hemp
	Fasteners	:	Carbon Steel

SHOP TESTING :

HYDROTEST	:	
Seat leakage	:	10 Kg/cm² (5 min) – for PN 1.0
Back seat leakage	:	5 Kg/cm² (2 min) – for PN 1.0
Body	:	15 Kg/cm² (5 min) – for PN 1.0

BUTTERFLY VALVES

a. Butterfly valves shall be of double eccentric and resilient sealed type generally as per BS EN 593, BS 5155 and IS 13095. Valves shall be installed in valve chambers. Valves shall be provided with stainless steel extension spindle so that valves can be operated from ground level and without entering the chamber.

b. Material of construction of butterfly valves shall comply with following requirements:

Item	**PN 1.0 Valves**
Body	Ductile IRON DIN 1693 – GGG40/spheriodal graphite icon IS 1865 Gr. 400/12
Disk	Ductile IRON DIN 1693 – GGG40/spheriodal graphite icon IS 1865 Gr 400/12
Shaft	Stainless steel BS 970 Grade 431 S 29
Body Seat	Nickel weld overlay micro finished
Disc Seal	EPDM
Seal retaining ring	Ductile iron DIN 1693 – GGG40
Shaft bearing	Bronze with EPDM ‘O’ ring seal
Internal Fasteners	Stainless steel SS 316
Nuts, Bolts & washers for pipe flanges	High Tensile steel hot dip galvanized for valve in chamber. Stainless steel SS 316 for buried valves
Coating	Internal and external with power of liquid epoxy coating with minimum dry film thickness of 250 microns

c. Butterfly valves shall be suitable for mounting in any position. The valve shall be free from induced vibration.

d. Butterfly valve shall be suitable for bi directional pressure testing with head tight shut off even after long period of operation of 5 years. The valves shall be of double flanged long type.

e. The valve seal shall be of replaceable design. When the valve is fully closed, the seal shall seat firmly. The seat surfaces shall be machined smooth to provide a long life for the seal. All fasteners shall be set flush so as to offer the least resistance possible to the flow through the valve.

f. The shaft shall be stainless steel with bronze or equivalent seal with self lubricating bearings. Disc pin shall be stainless steel. Ring shall be Ten directional seal adjusting suitable for pressure and vacuum service. Removal and replacement of steel shall be possible without removing the operating mechanism, valve shaft and without removing the valve from the pipeline. Valve shaft shall be of one piece unit extending completely through the valve disc hubs.

g. All valve spindle and head wheels shall be positioned to give access for operational personnel. Valves shall be provided with enclosed gear arrangement for ease of operation. The gear box shall be worm and worm wheel design type totally enclosed grease filled and weather proof. The operation with gearing shall be such that they can be opened and closed by one man against an unbalanced head of 1.15 times the specified ratings. Valves and gearing shall be such as to permit manual operation in a reasonable time and not exceed a required rim pull of 200 N. The valve disc shall be 90 deg turn.

h. The disc shall be designed to withstand the maximum pressure differential across the valve in either direction of flow. The disc shall be contoured to ensure the lowest possible resistance to flow and shall be suitable for throttling operation.

i. It should be possible to open the valve with upstream pipe fully filled and downstream pipe fully empty. The shaft shall be designed to withstand the maximum torque that will be imposed by the operator. It shall be secured to the disc by tapered stainless steel cotter pins.

j. Valve shall be provided with mechanism position indicator to show the position of the disc mounted on the driven shaft end.

k. Rigid adjustable stop mechanism shall be provided within the gear box or elsewhere on the valve to prevent movement of the disc beyond the fully opened or closed position (i.e. set points).

l. Valve shall be capable of closing against the maximum flow that can occur in practice. The break way torque against maximum differential head conditions shall be within the manufacturer]s limits.

m. All hand wheels shall be arranged to turn in a clock wise direction to close the valve, the direction of rotation for opening and closing being indicated on the hand wheels.

n. All hand wheels shall be provided with an internal locking device to prevent operation device by unauthorized person.

AIR VALVES

A SCOPE AND GENERAL DESIGN FEATURE

This section covers the requirements of double orifice type air valves with tamper proof cover to be used for evacuation of accumulation of air in water mains under pressure, for the exhaust of air when such mains are being charged with water and for inlet of air when they are emptied of water.

The working pressure of the air valves shall be PN 1.0 or PN 1.6 as per design requirements.

The size of Air valves shall be governed by the size of pipeline as IS 14845.

B FUNCTION

Automatic air valves generally conforming to IS 14845 / relevant ISO with latest amendments are to be used for evacuation of accumulated air in water mains under pressure, for the exhaust of air when such mains are being charged with water and for ventilating the mains when they are being emptied of water.

C DESIGN FEATURES

a) Air valves shall be double orifice type and tamper proof unless otherwise directed by Engineer. A buoyant rigid float shall seal the large orifice and the chamber housing shall be designed to avoid premature closing of the valve by the air whilst being discharged. Small orifice shall discharge small air volume during operation under full internal pressures. All air valves shall be provided with isolating sluice valves and flanged end connection.

b) The valve shall be capable of exhausting air from pipe work automatically when being filled, the air being released at a sufficiently high rate to prevent the restriction of the inflow rate. Similarly the valve shall be capable of ventilating pipe work automatically when being emptied or under water hammer condition, the air inflow rate being sufficiently high to prevent the development of a vacuum in the pipelines. The valve shall also automatically release air accumulating in pipe work during normal working conditions.

c) The valves shall be designed to prevent premature closure prior to all air having been discharges from the line. The orifice shall be positively sealed in the closed position with the float only raised by the liquid and not by a mixture of air and liquid spray. The seating shall be so designed to prevent the float sticking after a long period in the closed position.

d) Air valves shall thus be designed to automatically operate so that they will;

- Positively open under internal pressure less than atmospheric pressure to admit air in bulk during pipeline draining operation;
- Exhaust air in bulk and positively close as water, under low head, fills the body of the valve during filling operation;
- Not blow shut under high velocity air discharge; and
- Exhaust accumulated air under pressure while the pipe is flowing full of water.

D CONSTRUCTION FEATURES

Material of construction of air valves shall comply with following requirements:

a) All air valves shall be constructed so that internal working parts which May become necessary for repairs shall be readily accessible, removable, and replaceable without used special tools and removing the valve from the line.

b) Valves with air intake or exhaust facilities shall have an integral protecting cover top shall be supplied to prevent dirt and debris from entering the outlet of the valve.

c) The contractor shall verify with the supplier of the valves that the valves have the capacity to sustain the pipeline test pressure prior to testing. In the event that the valves do not sustain the pressure they shall be removed and the stub pipes from the main pipeline blanked off before pressure testing the pipeline.

E DATA

a)	Valve size	:	50 to 400 mm dia.
b)	Suitable for max. differential pressure (kg/cm²)	:	10
c)	Material of construction Body and cover iron 1865 Gr 400/12 or grade GGG40	:	CI IS 210 Gr FG 200 **or** SG

Float	:	Rubber coated timber **or** Polycarbonate up to 50 NB/ SS 304 above 50 NB
Internal Linkages	:	SS 304
Seat Ring	:	Dexine (Nitrile Rubber) or bronze seat
Isolating Sluice Valve	:	Generally conforming to IS 14846/relevant ISO with latest amendments
Spindle for Sluice Valve	:	St. St. AISI 410
Bolts & Nuts	:	M.S.

TESTING AND PERFORMANCE

a) When tested as per clauses 11.6.d.1, the air passage and the function of ball floats in a valve shall be satisfactory, and the valve shall work smoothly.

b) Hydrostatic test of valve body, when tested in accordance with 11.6.d.4 there shall be no leakage through pressure sustaining components and joints.There shall be no permanent deformation of any part.

c) Valve seat and cock, when tested in accordance with 11.6.d.2 and 11.6.d.3 shall not show any leakage.

d) Function and Performance Test

d.1 The valve shall be fitted on a test bench. The pressure of the water in pipe shall be developed to working pressure, and the main valve shall be gradually opened to check the air release and float function. Compressed air shall then be slowly put into the valve through underside of the valve, and check the function of floats.

d.2 High Pressure Orifice Seat Test

Subsequent to high pressure orifice performance test, hydraulic pressure shall be reduced up to half of the working pressure to check leakage of orifice seat for a duration of three minutes.

d.3 Low Pressure Orifice Seat Test

Subsequent to high pressure orifice performance test, hydraulic pressure shall be reduced up to half of the working pressure to check leakage of orifice seat for duration of three minutes.

d.4 Body Test

The valve body (without cover and ball floats) shall be covered by a blank flange, keeping isolating valve open. Hydrostatic pressure of 1.5 times the pressure class of the valve shall be applied for duration of 5 minutes to check the water tightness of the body.

CAST IRON OPEN CHANNEL GATES :

A. GENERAL :

The construction of cast iron sluice gates shall be strictly in accordance with the IS-13349-1992 Indian standard specifications. The Sluice gates shall be capable of performing the isolation duties in water treatment plant & pumping stations for isolation of flow in & out of a closed conduit as well as in those applications where water head is more than the height of shutter / opening. They shall be so constructed that there is no undue wear or deterioration during its operative life and so designed that the maintenance is kept to a minimum. The preferred manufacturers shall be as stated in data sheet. The specification given hereunder shall supersede other specification in case given elsewhere in tender document.

B. DESIGN & CONRUCTIONAL DETAILS :

The sluice gates shall be manufactured generally as per IS-13349-1992 Indian standard. The constructional features and details of components of the required gates are to be as under:

B.1 GATE FRAME :

a) The gate frame will be made from cast iron and shall be sufficiently rigid to withstand the designated water head. The gate frame shall either be flat back type or flange back type to suit the designed head and site condition.

b) Back flange of the gate aperture frame to be precisely machined flat and drilled to engage with the Cast iron wall thimble mounted on the wall. A rubber gasket will be provided between the wall thimble and the gate for ease in future dismounting of the gate for repairs / replacement and seal any leakage between the flange of frame and wall thimble.

c) The gate frame of these sluice gates shall either be self contained type or non self contained type depending upon site requirement. In case of non self contained gates the frames shall have short length extension guides and shall be without yoke at their top. The length of extension guides in such cases shall be sufficient to engage atleast half the overall vertical height of door when the gate is full open and shall be in accordance with the relevant provisions of IS-13349. In case of self contained gates the frames shall have full length extension guides and shall be provided with a yoke at their top. The length of extension guides in such cases shall be sufficient to engage the overall vertical height of door when the gate is full open position.

B.2 WALL THIMBLE :

a) The Wall thimble will be made from cast iron for placement in the concrete wall. Its front flange will be machined, drilled and tapped to match with the frame flange.

b) The cross section of the thimble shall be F shaped and the depth of thimble shall be maximum 300 mm long or lesser in case wall thickness is less than 300mm. Gates subjected to high unseating heads shall have thimble cross section shaped E.

c) To permit entrapped air to escape as the thimble is being encased in concrete, cast holes of 40mm diameter shall be provided at the bottom of wall thimble in each entrapment zone.

B.3 GATE SLIDE / SHUTTER / DOOR :

a) The gate slide / shutter / door will be made from cast iron and shall be sufficiently ribbed to withstand the designated water head.

b) The gate slide / shutter will be provided with integral pocket to house the thrust nut used to connect the stem with the slide.

B.4 SEATING/SEALING FACES :

a) **Materials** : These should be of Stainless steel or Bronze or as specified.

b) **Fitment :** The facings shall be attached to Plain machined surfaces / finished rectangular grooves of gate frame and door, depending upon the applicable water head, and be secured in place using taper screws. The taper screws adopted for facings shall be of same material as that of the seat facings.

c) The front faces of integral extension guides which can come in contact with the sealing faces of door while opening, shall also be fitted with sealing faces of the same material as that of the sealing faces on door. This is required to offer non corroding smooth sliding surfaces to the sealing faces of door/shutter during its vertical travel for opening and enhance the effective life of gate.

d) **Finish** : The mating seating/sealing faces on the gate frame and door shall be precisely finished for proper contact. They should be so finished that the clearance or gap, if any, between the mating sealing faces, in gate closed position, does not exceed 0.1mm.

B.5 WEDGING DEVICES :

a) The Sluice gates shall be provided with individually adjustable wedging devices to ensure forced contact between frame and shutter seat facings, when the gate is in closed position.

b) The gates meant for seating head shall be provided only with side wedging devices. Gates meant for higher unseating head of sizes larger than 600 mm, shall be provided with side, top and bottom wedging devices or with side and top wedging devices and flush bottom closing arrangement as required.

c) The wedging devices comprise of wedge brackets fitted on gate aperture frame and door. The wedge bracket on frame shall remain in fixed position and those on door shall be adjustable or vice versa. A sort of slot and tennon arrangement shall be provided on base of wedge brackets to prevent any tendency to shift. Provision shall be made to clamp the adjustable brackets firmly in adjusted position.

d) The wedging devices shall be made of cast iron. If the wedges/wedge blocks of wedging devices are of Cast Iron, then these are to be lined with contacting faces of the same material as that of sealing faces attached to the gate frame and door.

B.6 CONVENTIONAL OR FLUSH BOTTOM CLOSING :

The sluice gates shall be provided with conventional or flush bottom closure arrangement as required.

The sluice gates provided with conventional bottom closing arrangement involve corrosion resistant metallic contacting sealing faces at the bottom sill of gate. In such cases, the invert of the gate is required to be kept above the floor of the channel / chamber by at least 200mm to 300mm depending upon the size and type of gate. The contractor should verify whether this clearance is available at the site of installation for fitting a conventional bottom closure gate.

In case of conventional closing gate, if the invert of the gate is kept at the same level as that of the channel/chamber floor, then there remains a slot or a groove at the invert of the gate. Debris, dirt etc. which may settle in this slot and may not allow the gate to close properly and this may give rise to heavy leakages while in operation. With a view to avoid this, in situations where the invert of the gate is to remain at the same level as that of the channel/floor, a Flush Bottom closing gate instead of Conventional Bottom Closing gate should be provided.

Flush Bottom Closing shall involve a flexible rubber seal at the bottom of the gate, mounted either on the shutter or on the frame, ensuring that the sealing face remains flush with the floor. The cast iron bar fitted at the bottom of the frame is required to be embedded in the channel / chamber floor and for this a cut out / recess of ample dimensions is required to be provided beneath the waterway opening along the gate invert, while constructing the floor. The dimensions of this cut out shall be provided depending upon the feasibility to do so as per actual site conditions.

This cut out/recess is to be later on filled up with removable asphalt or loose concrete mixed with sand/saw dust or vermiculate after putting the gate in position so that it is possible to break open this second stage grout for removal of the gate in future.

The rubber seal employed shall be made of EPDM or Neoprene rubber and the rubber seal retainer bar (with SS fasteners) for fitting the rubber seal and the retainer bar are of cast iron.

B.7 GATE OPERATING HEADSTOCK/LIFT MECHANISM :

a) The operating headstocks shall be designed in such a manner as to permit the gate operation by a single person under the specified maximum operating head with an effort of less than 18kgs on the crank / handwheel.

b) The headstock may be ungeared or geared type, as might be necessary to make it convenient for one person to open or close the gate as fast as practicable.

c) Geared headstock shall be supplied with easily removable crank handle or handwheel with a radius not exceeding 375mm.

d) All the gears of geared headstock shall be kept completely encased in cast iron housing to protect them from damage, dirt, dust, water etc. and other atmospheric effects and thus ensure their smooth operation. Grease nipples shall be provided at proper places for lubricating with grease.

e) Headstock meant for mounting on operating platform shall be supplied with a pedestal/floor stand to provide a convenient operating height of approximately 900 mm. The pedestal of the headstock shall be provided with a covered window opening to enable cleaning and greasing of stem threads.

B.8 LIFTING SPINDLE/STEM :

The sluice gates shall be supplied with Rising type lifting spindles/stems. The stem shall be provided with acme / square threading, length of threaded portion being about 400 mm more than the height of waterway opening. This much extra length is required to allow for a minor variation of approximately 100mm on either side of the specified height of operating platform.

The design of stem will be done as per the provision in IS-13349.

B.9 STEM BLOCK / CONNECTING BLOCK / THRUST NUT :

The rising type stem shall be connected to the door through a stem block/thrust nut housed in a ribbed pocket cast integral with the door. The bottom end of stem shall thread into the stem block and is locked in place by a set screw to prevent the stem from unscrewing.

B.10 SAFETY STOP NUT :

The stem shall be provided with a safety stop nut to prevent the chances of over closing of gate which may otherwise damage either the stem or the lifting platform. The stop nut shall be furnished with a set screw for setting it in a fixed position after the gate is installed. Upon installation the safety stop nut should be set in such a way that its bottom remains about 1 to 2 mm away from the top of headstock, in gate closed position.
In case of stainless steel stem, the stop nut shall also be of stainless steel material of the same grade.

B.11 STEM/SPINDLE COUPLINGS :

For ease in transportation and handling, maximum length of one piece stem shall be restricted within 5 meter length. Where the stem are required to be furnished in more than one piece, threaded stem couplings shall be furnished to interconnect different sections of the stem. The couplings shall have provision for pinning after inserting in the threaded end of the stem.

In case of stainless steel stem, the couplings shall also be of stainless steel material of the same grade.

B.12 STEM GUIDES & STEM GUIDE BRACKETS :

Longer stems shall be provided with sufficient number of stem guides to prevent buckling of stem. If required, the stem guide bracket to be provided shall be Adjustable Channel Type - wherein a separate stem guide is bolted on to the wall bracket. The stem guide shall be adjustable in the slots on wall bracket in a direction perpendicular to the face of wall. Wall bracket should also offers minor adjustment in the direction parallel to the wall.
The stem guides shall have machine bored split journals to facilitate erection. The journal shall be lined with brass/gunmetal bush.

B.13 PIPE HOOD FOR STEM :

A Pipehood shall be provided on the top of headstock in case of rising spindle/stem gates to cover the spindle threads for protection against damage, dirt, dust, water etc. It shall be made of transparent fracture resistant polycarbonate material. The pipe hood shall have vent holes to prevent condensation.

B.14 GATE OPENING INDICATING ARRANGEMENT :

In case of manual operation, Gate opening indicating arrangement shall be provided to indicate the position of the shutter.

B.15 MATERIAL OF CONSTRUCTION :

The material of construction for various components shall be as under.

Gate frame, Shutter, Thimble,	: Plain Cast iron IS 210 FG 200
Wedge Blocks	: Plain Cast iron IS 210 FG 200
Seating faces, Wedge Lining	: Stainless Steel ASTM A 240 type 304*
Rubber Seals (If applicable)	: EPDM Rubber to ASTM D 2000
Rubber seal retainer bar (If applicable)	: Plain Cast iron IS 210 FG 200
Assembly bolts, nuts and fastener	: Stainless Steel ASTM A 276 type 304
Stem/Spindle & Stem Coupling	: Stainless Steel ASTM A 276 type 304*
Yoke (If applicable), Stem Guide Bracket	: Mild Steel to IS : 2062 grade A/Br, Epoxy Painted
Stem Block / Thrust nut	: Stainless Steel ASTM A 276 type 304
Pillar, Stem Guides	: Plain Cast iron IS 210 FG 200
Lift nut	: Leaded Tin Bronze IS 318 Type LTB-2 /As per Gear Box Make/ Actuator Make
Pipehood /Stem Cover	: Polycarbonate / MS galvanized pipe

* Positive Material Identification (PMI) test to be carried out for these components at manufacturer works during the inspection.

B.16 PAINTING: Following painting procedure shall be adopted for the gates:
Surface Preparation: Blast clean to near white metal finish.
Priming: 1 coat of epoxy primer before and after shop testing.
Finish Painting for gate assembly: Black coal tar epoxy paint. Minimum DFT 250 microns inclusive of priming.
Painting for headstock pillar/bench/thrust plate/Yoke: Epoxy red oxide primer and epoxy Grey paint. Minimum DFT 150 microns inclusive of priming.

B.17 SHOP TESTING: Following shop tests at manufacturers place will be conducted.

a)	Movement Test	Movement test should be conducted in horizontal/vertical assembled condition using stems & headstock. The gate should be operated once from full close to full open and back to full close condition with a max. force of 135 Newton-meter on the crank or hand wheel.

b)	Shop leakage test	Shop leakage test by applying unseating hydraulic pressure will be conducted at manufacturer's shop with gate mounted vertically on a test bench. A hydrostatic pressure equal to maximum seating/unseating head shall be applied to gate at center line of gate opening from the back, i.e. Unseating face of the gate in closed position, through pump. A suitable scaled calibrated pressure gauge put on the unseating face of the gate shall indicate reading equal to unseating pressure head. Water leakage through the gate under above condition shall be collected in a collection pan and measured. The leakage so measured should not exceed the limit of 2.5, 3.5 and 4.5 lpm per meter sealing perimeter for class-1, class II and class III sluice gates as stated in the IS:13349-1992. No alternate testing arrangement will be permitted in place of above method. Gates can be applied with a coat of primer to prevent rusting due to water exposure during testing.
c)	Hydrostatic test	After the leakage test Hydrostatic body test will be conducted at manufacturer's shop. A hydrostatic pressure equal to 1.5 times the maximum operating head should be applied on the gate for 5 minutes continuously. No permanent deformation in casting should be observed.
d)	Torque test at operating Head	Torque test at operating head would be conducted at applicable head at manufacturer's shop for gates up to 2000x2000mm size
e)	Dimensional Check	Important Dimensions shall be checked with reference to approved GA drawing.
f)	Seat clearance check	With the gate in closed condition 0.1 mm thick feeler gauge should not pass through between seat facings.
g)	Material Test Certificates	Material tests certificates for all important components of gates such as Thimble, Frame, Shutter, Seat facings, Spindle, & Rubber seals etc. to be furnished at the time of inspection.
\h)	PMI Test	Positive Material Identification (PMI) test to be conducted for Sealing / Seating Faces & Stem / Spindle during the inspection.

NON RETURN VALVE :

5.1 GENERAL

The non-return valves shall be single door type, free acting quick opening, giving rapid non-clam closure & with low head loss characteristics when in open position. The valves shall be provided with by passes and isolating valves conforming to relevant Indian Standard.

Specification of **the** valves shall generally conform to IS: 5321 (Part-II)

5.2 MATERIALS OF CONSTRUCTION

Body, Cover, Door and Hinges	**:** Cast steel construction
Hinge pins, Door pins & Door **Suspension pins**	: 12% Chromium steel conforming to IS:1570
Bearing Bushes	: Leaded Gun Metal Gr. 2, Conforming to IS: 318
Body rings and door faces	: Leaded Gun Metal Gr.2, conforming to IS: 318

5.3 TESTING

The valves shall be subjected to closed ends tests as per relevant IS standard. Test certificate in triplicate shall be furnished. If necessary, test shall be witnessed by the Engineer's representative.

WORKING PRESSURE :20 KG/SQ.CM.

TEST PRESSURE BODY :40 KG/SQ.CM.
SEATING :20 KG/SQ.CM.

KNIFE GATE VALVES :

1.1 GENERAL SPECIFICATIONS :
The construction of Knife gate valves shall be strictly in accordance with the relevant IS or specifications mentioned hereunder. The Knife gate valves shall be capable of performing the isolation duties in water treatment plant & pumping stations. These shall be suitable for use at suction and delivery side of pumps as well as in branch lines in sludge handling application of treatment plants and pumping stations and shall be so constructed that there is no undue wear or deterioration during its operative life and so designed that the maintenance is kept to a minimum. The contractor shall provide these valves manufactured by an ISO: 9001-2008 certified company manufacturing the underspecified product for at least 5 years. The preferred manufacturer shall be as stated in data sheet. The specification given hereunder shall supersede other specification in case given elsewhere in tender document.

1.2 DESIGN & CONRUCTIONAL DETAILS :
The Knife gate valves shall be manufactured generally as per the latest AWWA C520-14 standard. The valve should be bonneted up to 600 mm size and bonnetless for higher sizes. Valves should be lug type construction up to 600 mm size and full flanged construction for higher sizes. The valves should be provided with flange drilling suitable to mount between flanges as per ANSI B 16.5 150# with raised face or DIN PN10 or IS 1538-1993 Table VI.

Other constructional features and details of components of the required valves are to be as under:
1.2.1 The valve body should be of Cast ductile iron construction as specified. Valves up to 600 mm size should be designed to withstand 10 bar pressure and valves above 600 mm size should be designed to withstand 4 bar pressure generally applicable in sewerage pumping stations and treatment plants. If the line / system pressure is more than 4 bars then valve design / material of construction should be suitably modified to meet this requirement when specifically pre-informed by the client.

1.2.2 The valve should be provided with gate made of stainless steel of grade as specified and the gate should have beveled knife edge at the bottom to cut through and easily enter in the solids settled in the bottom and ensure positive shut-off / closure in sewage environment.

1.2.3. The Valve should be designed for sealing in uni-directional flow application.

1.2.4 The valve should be provided with replaceable type flexible sealing arrangement to offer drop tight shut off. The seals should be made of EPDM rubber and should be held in place by an easily removable type seal retainer ring. The sealing system should be field replaceable at site.

1.2.5 The valve housing should have integral as cast tapered lugs provided for pushing the gate towards the flexible rubber seal only at the verge of closure with a view to avoid seal wear and achieve drop tight shut off. The surface of the gate coming in contact with the seal should be polished & buffed.

1.2.6 The bonneted valves up to 600 mm size shall have glandless design to avoid repeated tightening / replacement of gland packing. The stuffing “O” ring seal should be pressurized by thrust washer from both the sides and should be pressurized / tighten by thrust retainer nut.

1.2.7 The bonnet-less valve above 600 mm size shall be provided with sufficient ply of gland packings in the in-built stuffing box to seal the rear opening. The packings should be of graphited synthetic yarn to reduce the friction and offer higher life. Provision shall be made to enable tighten the gland packing by means of a pusher arrangement to minimize the leakage through the back of the valve. Replacement of gland packings should be possible in installed condition of the valve without there being line pressure.

1.2.8 THE SPINDLE SHOULD BE DOUBLE START / SINGLE START AND NON-RISING TYPE FOR COMPACT & SAFE OPERATION. GATE OPENING INDICATING ARRANGEMENT SHOULD BE PROVIDED TO FIND OUT THE EXTENT OF GATE OPENING /CLOSING.

1.2.9 THE OPERATION OF THE VALVES SHALL BE MANUAL / MOTORIZED AS SPECIFIED ELSEWHERE IN THE TENDER SPECIFICATION.

1.2. 10 IN CASE THE VALVE IS TO BE OPERATED FROM A FAR THEN IT SHALL BE PROVIDED WITH JOINING COUPLINGS, EXTENSION SPINDLE, SPINDLE GUIDES AND MOUNTING PEDESTAL WITH SUITABLE ANCHOR BOLTS / ANCHOR FASTENERS ON THE PLATFORM.

1.3 MATERIALS OF CONSTRUCTION:

The material of construction for various components of valves shall be as under.

Body	: Ductile iron to BS 2789 Grade 420/12
Seal Retainer ring	: Stainless steel to ASTM A351 Gr CF: 8
Inlet Seal / Rubber Seals / O rings	: EPDM Rubber
Knife gate	: Stainless Steel ASTM A 240 type 304
Spindle	: Stainless Steel ASTM A 276 type 304
Assembly bolts, nuts and fastener	: Stainless Steel ASTM A 276 type 304
Spindle nut	: Brass/DI
Bracket / Adapter plate	: Mild Steel to IS : 2062 grade A, Epoxy Painted
Packing	: Synthetic yarn with PTFE

1.4 PAINTING : Following painting procedure shall be adopted for the valves :

Surface Preparation: Blast clean to near white metal finish.
Finish Painting: Suitable coats of epoxy paint to achieve minimum DFT 100 microns inclusive of priming.
Primer painting / pre-painting before leakage testing shall be allowed.

1.5 DATA SHEET:
Following data sheet duly filled, signed and stamped by valve manufacturer to be submitted by contractor along with technical bid:

13 VALVE CHAMBERS

Valve chambers shall be constructed according to the typical drawings suitable for the respective valve and special arrangement if any shall be approved by Engineer. They shall be constructed in brick masonry as shown in the drawing. The chambers shall be constructed after the laying of the pipes and the assembly of specials and valves. The size of the chambers shall be according to the following criteria/ as per direction of Engineer.

- Minimum distance of flanges from walls : 45 cm
- Minimum distance of sockets from walls : 45 cm
- Minimum distance between highest point of equipment and roof slab : 30 cm
- Maximum distance between highest point of equipment and roof slab : 50 cm

Pipes passing through walls should be coated by two layer of soft material (Hessian felt) to allow for differential settling and longitudinal expansion if directed by Engineer. Only metallic pipes May be cast into the walls for anchoring purposes.

The work shall include excavation, consolidation, leveling, lean concrete as per drawing in foundations, finishing, refilling. It shall include all labour and material required for the complete chamber.

14 DISMANTLING JOINTS

Double flanged Dismantling joints shall be of Cast Iron in such a manner that valves (300 mm and above dia) can be dismantled without stress to the joints. These shall be for working pressures of 10 kg/cm2 (1 Mpa) and shall be completely leak proof with proper gasket arrangement. Flange dimensions shall conform to IS 1538 (part I to XXII). Flanged specials shall be supplied with required nuts, bolts and rubber gaskets. The nuts and bolts shall be of best quality carbon steel, machined on the shank and electro-galvanized. Rubber gasket shall be as per IS 5382. Dimensions and drilling of flat gasket will be as per IS 1538: 1993, suitable for making flanged joint. The dismantling pieces shall provide minimum clearance of + 25 mm (total distance 50 mm. The dismantling joint shall be internally and externally coated with hot applied (dip) bituminous paint.

15 THRUST BLOCKS

The thrust blocks shall be of plain/reinforced cement concrete on site as per design and drawings to be given by the Contractor and approved by the Engineer in Charge. The thrust blocks shall be cast directly against the undisturbed soil. Restrained jointed DI pipes and Fittings (Boltless Double Chamber type) can be used in place of thrust blocks with approval from Engineer In-charge.

16. WATER METERS

AMR Electromagnetic domestic water meters of approved type shall be permitted to be used subject to such conditions prescribed by the Implementing Agency or the local bye-laws**.** Meters of domestic type shall conform to the requirements of IS**:** 779**.**The meters shall be installed in accordance with as per the guidelines of the meter manufacturer. The meter shall be fitted beyond the stopcock with unions to facilitate the necessary periodic changing of the meter. If fitted in an exposed position outside the building, the meter shall be housed in water meter boxes conforming to IS: 2104.
The AMR compatibility of meter shall be universal, such that in future if provision for AMR is made then it should be easily fitted with the meter and compatible with the software**.**

A. **Detailed Specification of Electromagnetic Smart AMR Water Meters (DN15 to DN40)**
Providing, Installing and Giving satisfactory field testing of domestic Battery operated, MID approved with minimum Dynamic ratio (Q3/Q1) of 400:1, AMR Electromagnetic Water Meters from 15 mm to 40 mm having no moving part in contact with water, with fully body made up of composite material, confirming to ISO 4064 along with manufacturing test certificate and Guarantee certificate with Battery life 10 years with RF Technology for communication.

ITEM	SPECIFICATIONS
a) Measuring Principle	A battery operated inline non- Intrusive Electro magnetic water meter with no moving parts.
b) Power Supply	Power-Supply-Battery operation for complete meter including AMR with battery life of more than 10 years to ensure recording at all times
c) Meter Lifetime	More than 10 years
d) Protection Class	Must comply to IP68 Standard for indoor and outdoor operation, including fully submerged installations
e) Approvals and certifications	The meter should be type approved and verified according to international water meter Standard OIML R 49 and or ISO 4064. The MID Certificate from International Notified body should be submitted along with the bid. Manufacturing plant - Quality Management System Certificate - ISO 9001 & ISO 14001. Performance Certificate and Endurance Certificate from FCRI should be submitted along with the bid.
f) Accuracy	+/-2% typical operating range and temperatures. The Electromagnetic water meter should maintain its accuracy over its lifetime. Accuracy class 2 standards of OIML R49/ ISO 4064.
g) Calibration	3-Point calibration with calibration certificate available for each unit.
h) Dynamic Ratio (Q3/Q1)	Minimum of 400:1. Measuring range shall be as per ISO

	4064/OIML R 49.
i) Material	The water meter body shall be made of Composite material or Engineering plastic.
j) Pressure Rating	Pressure rating shall be ≥ PN16
k) Environmental Temperature	0 °C to 50 °C
l) Data Protection and tamper proof	The meter should be tamper proof with suitable data protection of calibration and revenue parameters
m) Self-diagnostics for error detection.	The smart meter should have advanced diagnostics with active alarm(s) indicated on display
n) Access to information	Display with ≥8 digits for main information. Index, menu and status symbols for dedicated information
o) Measuring Units	The measuring units should be m^3 for volume and m^3/h or l/h for flow rate.
p) Facility for Remote Communication interface	The Electromagnetic water meter should be configured with battery operated remote reading capability using point-to-point RF.
q) Data Storage	The Electromagnetic water meter should record Hourly data for minimum 90 days and same should be able to read over AMR using RF communication. Fixed Day readings of Last 24 months.
r) Indicators / Alarms	Tampering, Reverse Flow, Leakage etc.

B. AMR SYSTEM

1. The water meters shall have the anti – magnetic properties / immunity, as specified in ISO-4064:2005, when tested with 4000 gauss magnet. The AMR system shall remain unaffected with application of 4000 gauss magnet, as specified in ISO-4064:2005.
2. The remote readings of AMR water meter need two-way communications without affecting battery life and reading performance throughout O&M period.
3. The remote readings of AMR water meter should be obtainable by 'Walk by/Drive by methods. The Walk by mode shall be used only in case the meter is unreadable by Drive by method.
4. The data Communication from the meters shall comply with the European standard on wireless M-bus/Radian protocol/Manufacturer specific protocol supporting interoperability. The frequency used shall be de-Licensed band frequency in India as per Govt of India regulations. "Equipment type approval from WPC wing of DoT should be provided during bid submission."
5. The AMR water meter / trans-receivers shall be wireless and have IP 68 protection category i.e. no ingress of water after submerging AMR water meter for 48 hours under 3 meters of water column.
6. The remote Meter reading device shall have instant reading facility. The remote readings and physical meter readings of water meters shall match at all the times.
7. All AMR readings shall show the date and time of the reading recorded.
8. The AMR system shall have facility to record the reverse flow in water meters readings and it shall show the quantum of reverse flow on the Meter Reading device i.e. AMR reading device as well as computer system**.**
9. The AMR system shall have the facility to record the abnormalities like application of external magnetic effect, very high consumptions, water leakages, tampering, etc. along with necessary alarms in meter reading device and in software.
10. The battery of AMR water meter shall be inbuilt and its life shall not be less than 10 years from successful installation of said AMR water meter along with its AMR system, the battery life shall be calculated by considering the temperature conditions of installation site and remote reading.
11. During remote reading, meter reading device will show low battery alarm/alert of AMR water meter.
12. The AMR water meter shall not get affected for its AMR functioning due to High Tension or High Voltage line concentration.
13. All water meters shall be fitted with RF based wireless remote Trans-receiver for AMR reading. Meter manufacturing company will assure that the frequency is FREE TO USE and necessary documentation with Department of Telecom is available and should be provided during bid submission.

14. The AMR water meters shall have the facility to transmit reading in submerged condition & the remote readings should be obtained with water meter in submerged condition & lid of the chamber closed.
15. All the time electronic index of the water meter shall match with the physical reading, available on water meter.
16. The AMR system if required in future, should be able to be upgraded to a Fixed Network system (AMI), which will provide hourly interval data on daily basis to control centre of water utility.
17. The AMR system should date and time synchronize every meter.
18. The AMR system should be compatible to inter-operability with other brand AMR meters.
19. The AMR meters and the AMR module should be of the same brand and should use open source protocol along with manufacture specific protocol.
20. AMR system should operate in free frequency band available in India as per notification of Department of Telecommunication Govt. of India. The necessary documents should be submitted with the proposal.
21. The Water Meter Manufacture should give undertaking on non-judicial stamp paper of Rs. 100 stating that the frequency used for AMR system is in free frequency band available in India.
22. The meter manufacturer should provide signed warranty contract for AMR Water meters for the entire period of contract.

C. AMR SOFTWARE

1. The software shall give output, at least in the CSV (Comma Separated Value)/Txt/Xls format.
2. The Route Management software must be capable of running on a standard PC compatible with minimum Pentium processor; in addition, the software must run under latest version of windows operating system.
3. The Route Management software should be cloud based and should have web portal access so that user can view customer data through browser.
4. The software shall allow the PC operator to review and edit any account in Route Management/ AMR software database. Also, the PC operator shall be able to generate routes/groups as per zones or areas and activity reports.
5. The software shall alert the meter reader for unread accounts in that route.
6. The software should have the facility to export CSV/Txt/Xls files in the pre-defined format by utility to billing system.
7. The software shall enable the user to specify the data to be exported from the database for transferring to billing system.
8. The software shall select the routes to be read, and assignment of routes to a reading device and dynamic updating of routes and sub-routes to be enabled.
9. The software shall upload routes from the reading device through GSM/GPRS/USB/Cable. And Meter Reading Device shall send meter reading data to the server through GSM/GPRS/USB/Cable.
10. The software shall post the reading from the reading device onto appropriate accounts within the database.
11. The software shall have provision of taking a backup copy of the route data or exporting the route data.
12. Software shall be able to set meter status as meter not okay, reading not reliable, meter maintenance required etc. Meter reader must be able to mention the irregularity in meter reading device related to meter/meter reading/condition of meter which will reflect in reading software.
13. The software should be able to display reading data on screen.
14. The software should have capability to add additional customer information and create customizable data fields.
15. The software should manage GPS data of AMR Meters.
16. The software should manage customizable list of message codes for Meter Reader.
17. The meter manufacturer should provide O&M for software of AMR Water meters for the entire period of contract.

D. Meter Reading Device

1) Minimum System Requirement for Android Device
    a. 5” Screen
    b. 2.2 GHz Processor
    c. 4 GB of RAM
    d. 16 GB of Flash Disk

e. GPS Sensor
f. Wi-Fi Connection
g. Bluetooth 4.0 LTE
h. Android 5.0 or Higher (Highly Recommended)

2) Meter reading device shall have the sufficient memory for storage of data / meter readings along with sufficient power back up
3) Meter reading device shall have the onsite search facility, to locate the exact physical location of water meter area and to obtain the corresponding details of it
4) The Meter reading device should have adjustable back light, sun light readable, colour display and touch screen.
5) The Meter Reading Device must come with an integrated intelligent fast charge capability that allows full charge within 3 Hours.
6) The Meter Reading Device must have a 3G/GPRS connectivity for transmitting data directly from field to cloud or web server for route monitoring.

Chapter - 3

Elevated Service Reservoir (ESR)/ Overhead Tank (OHT) ,Sump and Intakewell :

1. Preparation of structural design, drawing, construction, testing & commissioning and trial-run of RCC Elevated Service Reservoir/Over Head Tank (ESR/OHT) of various capacities and staging/height Complying with provisions of relevant Indian standards and Engineer-In-Chief, MPPHED, guidelines issued vide circular No.8674 Date 05.10.2018. Using latest Soil [investigation Report of proposed site, Seismic zone, Wind speed Zone. Including:
2. General guideline for construction of OHT (except design part) from E.N.C. PHED, Bhopal Circular no. 236, and Dated 21.05.1997 shall be followed along with IS 456:2000 and IS 3370-2009 Part I to IV.
3. List of Indian Standards for Design of ESR/ OHT:
   Note: The structural design of ESR shall be in accordance with provisions of relevant Indian standards
   (i) I.S.3370 part I to IV 1965 or Its latest revision.
   (ii) IS 456-2000 Reaffirmed 2016.
   (iii) IS 11682- 1985 Reaffirmed 2003.
   (iv) IS 1893-20l6 part I to V.
   (v) IS 13920-1993, or Its latest revision.
   (vi) IS 875 part I to III, l987 or Its latest revision.
   (vii) IS I1089- 1984 Reaffirmed 2002
4. Container shape of any suitable type (or as specified), preferably Circular.
5. Water depth in container shall be adopted as per the requirement of capacity of Tank specified in the tender or as per required Design. Capacity shall be calculated as storage between the designed full supply Level excluding free board of the reservoir and Lowest supply Level (that is, the level of the lip of the outlet pipe).
6. Staging consisting of column brace trestle / combination column- brace trestle as appropriate (or as specified)
7. In case of column -brace trestle type staging having more than 6 columns internal horizontal bracing is obligatory. One bracing shall be at foundation level in case of individual footings.
8. Min. size thickness of various components shall be provided as per design criteria /specifications / IS Code (or as per std. practice). Capacity of the ESR shall be considered excluding free board.
9. Minimum dimensions specified for various components in tender data /specifications should be provided.
10. Maximum spacing between horizontal bracings shall be 5 m (story height).
11. For ESR-having capacity above 1000KL the spiral / doglegged staircase shall be provided outside the staging with effective tie beams in more than one direction.
12. All types of labor & material charges of lowering, laying, erecting / hoisting & jointing of pipe assembly of Inlet, Outlet, overflow, washout and bye pass arrangement as per hydraulic design are including.
13. Providing and fixing of any accessories (specified), CI Manhole frame and covers, water level indicator, lightening conductor, GI Pipe railing around walk way, at roof level, at gallery and both sides of staircase. Adequate cowl type ventilators or lantern type ventilator with stainless steel Jali. Aluminum ladder to provide access inside the tank.
14. The rate shall include providing and fixing pipes, specials, and valves required for inlet, outlet, washout, over flow and Bye- pass arrangement as stipulated in tender documents. The scope of work includes constructing supporting RCC Pedestal for Duck foot Bend, erecting, laying, fixing and joining pipes and special up to Min. 3 m length from face of staging (outer most columns). RCC chambers for valves. Drain out for scour Pipe/ Overflow Pipe min. up to boundary of campus.
15. The rate shall include cost of dewatering during execution making all arrangement with any dewatering technique.
16. Scope of work includes constructing RCC spiral staircase with adequate tie beams, staircase footing, Ventilating shaft and ventilators as well as door after first flight of Staircase to prevent intruders SS grating to be provided to outlet pipe (inside container) for safety.

17. An outside Door/ GATE of minimum 1.8 m height to be provided at 1st landing of staircase in order to avoid entry of anti-social elements.
18. Water level indictor shall be provided and fixed float type /electronic (as specified).
19. This will also include cement plaster in CM l:2 with approved water proofing compound all over inside container (i.e. walls, base, top slab/dome bottom etc. all). Including providing and applying three coats of all weather exterior paint (as specified) to the whole structure.
20. It also includes satisfactory water tightness test as per relevant I.S. Code and painting name of scheme & capacity on the tank as per direction of Engineer in charge. Approval of design/drawing from Government Institution will be the responsibility of contractor, if provided by contractor.
21. The structure shall be designed properly for up lift due to Ground water table specified in data or GWT encountered during execution. No extra payment shall be paid for the same.
22. Agency shall engage qualified (at least graduate) consulting Engineer for designing the structure and he/she shall visit the site for guidance of work at all levels (i.e. below foundation, up to GL, above GL for all lifts up to container).
23. **Sand/ Fine Aggregate**

    Fine Aggregate shall be consisting of naturally occurring (Crushed or uncrushed) stones, gravel and sand or combination thereof. They shall be hard, strong, dense, durable, clear and free from veins and adherent coating and free from injurious amount of disintegrated pieces, alkali, vegetable matter and other deleterious substances.
    The sand shall be as per IS: 383. The preferable sand shall be Narmada river sand. The sand for plaster shall be confirming IS: 1542.
    The coarse sand should be free from soil. This can be checked easily by half filling a transparent glass with the sand sample and the other half by clean water. Stir the sand vigorously. Silt in the sand can then be easily seen in the top water portion. Use always tested sand.

24. **Metal/ Coarse Aggregate-**
    Coarse Aggregate shall be consisting of naturally occurring (Crushed or uncrushed) stones, gravel and sand or combination thereof. They shall be hard, strong, dense, durable, clear and free from veins and adherent coating and free from injurious amount of disintegrated pieces, alkali, vegetable matter and other deleterious substances. Metals/ Coarse Aggregate shall be confirming to IS: 383.
    It is advisable to use metal derived from igneous rock preferably of basaltic of granite origin.

25. **Steel for Reinforcement**
    All the iron and steel required for the work will be procured by the contractor at his own cost. The contractor shall be required to produce the test certificate of the manufacturers to the department before use of steel for the work. No untested steel will be allowed to be used under any circumstances. The Department however reserves the right to get the received/ supplied steel tested at the cost of the contractor. Steel for reinforcement shall be confirming to latest BIS specification IS: 1786 for Deformed Steel Bars and wires. All the steel above 6mm dia shall be deformed bars with strength grade of Fe-500. If the steel being used is of 6mm diameter bars at any place, then it shall be plain steel conforming to IS: 432. The contractor shall be required to produce the test certificate from manufacturer to the Engineer-in-Charge, before use of steel for the work. No untested steel shall be allowed to be used under any circumstance.
    HYSD Fe 4l5/500 grade reinforcing bars confirming to I.S. 1786-2008 shall be considered in design. CRS/TMT bars shall be provided. In saline atmosphere corrosion resistance stainless steel/HCR rebar shall be provided. Any other steel can be used with approval of C.E./in the situation of non-availability in market without extra cost.

26. **Cement**
    The cement to be used in the work shall be 43 grade Ordinary Portland Cement (OPC) conforming to IS: 8112 or IS: 12269 or PPC conforming to IS: 1489 approved by the Engineer-in-Charge, for all important and water retaining structures. It shall be tested for following test at contractor's cost. For under water concreting rapid hardening cement shall be used.

27. **Concrete in water retaining structures:**
    The Min. concrete grade for RCC shall be M-30 for water retaining structure and M-25 for remaining structure of concrete ingredients shall be as per Mix design using weigh batching.

All concrete shall be mixed in concrete mixer and compacted by mechanical vibrators. Slump tests shall be carried out during concreting and sample test cubes prepared as per direction of Engineer-in-Charge tested by the contractor at his own cost. The results of the tests shall conform with the required standards and if the test results are not found satisfactory or otherwise also if the Engineer-in-Charge considers that a structural test is necessary, the same shall be carried out as instructed by the **Engineer-in-Charge** at contractor's expenses and if the results of this be unsatisfactory the contractor will be bound to dismantle and reconstruct the particular portion of work which has given unsatisfactory test results. The contractor shall make arrangement for testing of construction material and concrete at site itself. In general concrete shall be designed as per IS: 456 (latest edition) and concrete for water retaining structures shall be designed as per IS: 3370-Part-II (latest edition). All the components, which are in contact of water and its roof slabs or domes, shall be of minimum Grade M-30. All other components not in contact with water shall be of minimum M-25 Grade. The cement content per cubic meter of concrete shall not be less than the provisions of IS: 456 latest addition. The concrete shall be prepared as per mix design. All ingredients of concrete shall be weighed and mixed as per the mix design. All concrete shall be mixed by concrete mixer and compacted with concrete vibrator only.

The contractor shall setup a laboratory for testing of metal/ aggregate, concrete, cement etc. on the site. Slump test shall be carried out by contractor at its own cost. During concreting, sample test cube shall be prepared as per the frequency prescribed in IS: 456 and shall be tested by contractor at his own cost. To assess the strength of cube immediately, accelerated curing testing may also be conducted as required by the Engineer-in-Charge. If the result of the test shall not be in conformity with the required standard and if the Engineer-in-Charge considers that the structural test is necessary, the same shall be carried out by the contractor at its own cost. If the result of this comes unsatisfactory again then the contractor shall be bound to dismantle and reconstruct the particular portion of work.

It is extremely essential that the contractors undertaking the work should have a concrete mixer with them. No hand mixing should be allowed.

In no case the concrete should be laid without vibration. It is describable to keep two concrete vibrators at the construction site so that in case of a break down the other vibrator can be used. It is desirable that the divisions have with them at least two concrete vibrators, which is an essential T & P for laying concrete.

It is extremely important to make arrangements for supply of sufficient water at the construction site for curing of the concrete. Continuous and efficient curing for minimum required period is extremely important for development of good compressing strength in any concrete structure.

28. Testing of Construction Materials

i. Testing of Sand: -

Following test shall be conducted before the starting of construction work from any Engineering College/ NABL approved lab;

S.No.	Type of Test	Reference & latest amendments up to date
1.	Particle Size and shape	As per IS-2386- Part- I- 1963
2.	Fineness modulus	As per IS-2386- Part- I- 1963
3.	Bulking tests (Dry Bulk density)	As per IS-2386- Part- III- 1963
4.	Silt content	As per IS-2386- Part- I- 1963
5.	Specific Gravity	As per IS-2386- Part- III- 1963
7.	Soundness Test	As per IS-2386- Part- V- 1963
8.	Deleterious Materials	As per IS-2386- Part- II- 1963

ii. Testing of Metal/ Coarse aggregate (20 mm Graded): -

Following test shall be conducted before the starting of construction work from any Engineering College/ NABL approved lab;

S. No.	Type of Test	Reference & latest amendments up to date
1.	Aggregate impact value	As per IS-2386- Part- IV- 1963
2.	Particle size distribution (20mm graded)	As per IS-2386- Part- I- 1963
3.	Flakiness and Elongation Index Test	As per IS-2386- Part- I- 1963
4.	Specific Gravity & Water Absorption	As per IS-2386- Part- III- 1963
5.	Bulk Density	As per IS-2386- Part- III- 1963

6.	Los-Angeles abrasion value	As per IS-2386- Part- IV- 1963
7.	Soundness Test	As per IS-2386- Part- V- 1963

iii. Testing of Cement: -

Following test shall be conducted before the starting of construction work from any Engineering College/ NABL approved lab;

S. No.	Type of test	Reference & latest amendments up to date
a)	Test for initial and final setting time	As per IS: 4031- Part 5- 1988
b)	Test for determination of Compressive strength of cement as	As per IS: 4031- Part 6- 1988
c)	Specific Gravity	As per IS: 4031- Part 11- 1988
d)	Soundness	As per IS: 4031- Part 3- 1988

iv. Water:-

Following test shall be conducted before the starting of construction work from any Engineering College/ NABL approved lab;

S.NO.	Type of Test	
1.	ALKALINITY	As per IS-3025- Part- 23- 1986
2.	ACIDITY	As per IS-3025- Part- 22- 1986
3.	ORGANIC SOLIDS	As per IS-3025- Part- 17- 1984
4.	INORGANIC SOLID	As per IS-3025- Part- 17- 1984
5.	SULPHATE as CI	As per IS-3025- Part- 28- 1988
6.	CHLORIDE as CI	As per IS-3025- Part- 32- 1988
7.	SUSPENDED SOLID	As per IS-3025- Part- 17- 1984
8.	TOTAL HARDNESS	As per IS-3025- Part- 21- 1983

v. Testing of Steel Reinforcement: -

Following test shall be conducted before the starting of construction work from any Engineering College/ NABL approved lab;

S. No.	Type of Test	Reference & latest amendments up to date
1.	Tensile Strength	As per IS-1786- 2008
2.	Elongation Test	As per IS-1786- 2008
3.	Bend & Re-bend Test	As per IS-1786- 2008
4.	Unit Weight	As per IS-1786- 2008

vi. Testing of Concrete Mix:-

FOR M-15 grade of concrete mix, nominal mix may be used with ratio of Cement – Sand – Metal @ 1:2:4 FOR M-25 & M-30 grade of Concrete, Mix to be designed for desired strength & workability as per IS 456:2000. Contractor may propose proportion of Trail mix limiting to minimum cement content & W/C ratio as per the E-IN-C PHED circular vide no.236, dated 21.05.1997. Such proposed mix may be verified by the engineer in charge at site before construction starts or A suitable concrete mix may be designed from any Govt. Engineering collage or polytechnic and same mix shall be verified for desired strength & workability by the Engineer in charge.

Concrete cube test for compressive strength & slump test shall be done for each concreting work at the frequency as per IS 456:2000.

29. SBC & FOUNDATION & STRUCTURE UP TO GL –

1. The Safe bearing capacity (SBC) /allowable pressure on soil shall be refined from latest SBC test report or tender datasheet. During execution. If poor soil strata or ground water table is encountered, the SBC shall have to bere-ascertained and the design should be revised accordingly.

2. Appropriate foundation system. This includes excavation in all types of soil strata (including hard rock), casting 100 mm thick P.C.C. leveling course in M-10, Refilling the pit with proper soil and disposing of the surplus stuff at all required lead and lift.

3. Before the starting of work the bearing capacity of soil must be tested/ verified and accordingly standard drawing shall be adopted. SBC shall be tested at the foundation level by Plate load test method as per IS 1888-1982 from any Engineering collage or NABL approved lab. For black cotton soil or filled up soil, in addition to plate load test, the SPT and any other laboratory test may be adopted as per IS 6304-1981 to find out the actual SBC for the purpose of design.

4. Staging height shall be considered from the bottom of GL brace. Bottom of GL brace shall be fixed at highest level of top of existing surrounding Road. Normally site should be selected at higher elevated places. So that it should be free from any chance of water accumulation and rain water or over flow etc can be easily drain out from the OHT boundary. However due to unavoidable reasons, if the location of OHT selected in depression then the depression should be filled up to GL brace by non-compressive soils like moorum / stone dust/ sand etc with proper compaction up to OMC about 98% but in all such cases the length of column below GL brace from top of foundation shall not be more than 4.0 m.

5. The minimum depth of foundation for all type soils (other than black cotton soil) shall be placed (-2.0) meter from the lowest Ground level, For Hard rock type soils the foundation can be placed at minimum 1.0 m below the top of hard rock & for the soils mixed partly with or fully black cotton soils, the Raft foundation shall be placed at minimum (-3.0) m from the lowest Ground level. In depressed area of OHT , filling is done up to such height that top of plinth protection should match with top of GL brace and bottom of GL brace fixed at highest level of surrounding Road. The maximum depression of selected site should not be below (- 4.0) m from the highest level of surrounding road.

6. Provide raft foundation for soil having S.B.C. below 10T/m2 (for SBC ranging between 7.50 M/T2 to 9.99 T/M2) & it shall be designed for min. SBC @ 7.5 T/m2. Sites having SBC below 7.50 T/ M2 should be avoided for constructing OHT. If it is unavoidable then in such case for below SBC, separate design of raft foundation with complete OHT to be prepared after proper soil investigation.

31. Pipe line Arrangement:

i Inlet, Outlet, Over flow , Scour Pipes, specials & valves i/c chambers etc. shall be as per the contract document. All the Pipes shall be supported on Duckfoot Bend.

S. No.	Particulars	Type of Pipe	Internal Nominal Diameter in mm.
1.	Inlet pipelines	C.I. D/F, IS 7181/1536 Class B or D.I. D/F Class K-9/ K12 IS 8329-2000 latest version	As per Diameter of Incoming pipe.
2.	Outlet pipelines	-do-	Next higher Diameter than inlet or size of main pipe of distribution network, whichever is higher
3.	Overflow pipe	-do-	Next higher Diameter than inlet
4.	Scour pipe	-do-	Min. 100 mm or above suitable to capacity

On the bottom floor of the water container cast iron or stainless-steel grate should be provided over the supply outlet and scour outlet. This is essential to prevent any accident for the labor attending to periodical cleaning of the tanks., Cast iron grates 20x20mm or stainless steel square 20x20mm can be used with square frames on top of the outlet.

The over flow outlets should not be connected to the distribution system. Connection of over flow pipe to the distribution system can result in over filling of the elevated service reservoirs in case supply valves of the distribution system are not open. The over flow outlet should always be kept open for draining any excess storage in the tank.

R.C.C. chamber with RCC cover surrounded outside by M. S. steel angle frame shall be designed for suitable size as tabulated below.

Dia. of Valve	Chamber Size		
	Length(mm)	Width(mm)	Depth(mm)
500	1500	1500	1700
450	1500	1500	1700
400	1250	1250	1400
300	1250	1250	1400
200	1000	1000	1000
150	1000	1000	1000
100	1000	1000	1000

32. Lightening Arrestor:

Lightning arrestor shall be copper bar of 25mm dia and 2m long connected by aluminum/ GI strip of 25mm width and 3mm thickness and the strip shall be connected to a 600x600x6mm thick copper/ GI plate embedded below ground with 40mm dia GI pipe in earthing etc. complete. Lightening arrestor shall comply IS: 2309 & Earthing shall comply to IS: 3043.

33. Water Level Indicator:

Each tank should be provided with water level indicator. Water level indicator shall comprise of PVC float which should be 10cm more in diameter than outlet pipe. The plate shall have turned edged to accommodate and to make easy movement of counter weight made of iron pointer fixed with guide pulley provided with white enamel paint write up with radium blue or black color letters. It shall be fixed on container wall.

34. Stainless Steel Ladder:

The stainless steel ladder shall be of min. SS 310 grade and provided from top of roof to the inside bottom of container. It shall comprise of not less than 600 mm long double round bar of 12mm dia @ 250mm c/c. The ladder shall be rigidly fixed.

35. M.S. Staircase:

M.S. Staircase from balcony to top of roof shall be provided. It shall be of 600 wide angle frame of 50X50X6, angle frame for steps of 40X40X6 with 50X6mm flat and with railing both sides.

36. Railing:

The railing, wherever provided shall consist of 1.2m high 100mm dia RCC M-30 posts/ 50mm dia medium class G.I pipe posts at 1.5m c/c welded with reinforcement and embedded in cement concrete with 3 rows of 32mm diameter medium class G.I pipes (@3.13kg/ Rmt) railing duly painted in two coats, so as to give smooth and even finish, with synthetic enamel paint over prime coat. The specification of GI pipe shall be as per IS 1239 part-1.

37. Manhole covers:

Manholes of minimum size 60cm x 75cm shall be provided. The covers shall comprise of 40x40x5 angle iron frame and 10-gauge thick M.S. sheet, crossed by suitable flat inside, with locking arrangement and a frame of 50x50x5 angle iron shall be grouted in concrete.

38. Ventilators:

1000 mm dia. 1000 high ventilation on top of roof. Side covered with mosquito proof stainless steel Wire/flies mesh and top covered with RCC cover.

39. Painting & Finishing:

Two coats of Weather Shield Apex painting on the entire civil works shall be done outside above ground level, while 2 coats enamel exterior oil painting shall be done on doors, windows, ventilator and all steel fabrication work i.e., railing, gantry girder, manhole cover etc. over one coat of primer. The number of coats may be more than the above specified coats to give smooth and even surface. All MS items of works in contact with water shall have epoxy coating.

40. MS Clamp:

Clamping of inlet, outlet, overflow & scour pipe, with 75x8 MS flat hoops supports duly supported from RCC column.

41. Waterproofing treatment:

Cement based polymer modified waterproofing slurry Coating of thickness as recommended by approved IS or ISO/9001 recognized manufacturer or food grade epoxy may also be used on all concrete surface exposed to water/water vapors i.e. inside surface of tank wall, bottom and top dome after testing the tank for leakage/seepage.

42. Protection work around Tank:

Protection work all around the tank shall be provided. It shall be circular in shape and minimum 2.0 m in width around the outer edge of wall. It shall have 1:60 slope from center and a drain be constructed all around the tank. The protection work shall be with 50mm thick M-15 grade plain

cement concrete over 100mm thick M-10 grade plain cement concrete. The drain shall have minimum dimension of 250x300 constructed in brick masonry in cement mortar 1:4 with cement plaster in 1:4 ratio over 100mm thick M-10 grade bed concrete.

43. Water Tightness Test

Testing of Tank shall be carried out as per IS 3370 part-I (with up to date amendment).

After the completion of structure it shall be tested for water tightness. Initially the tank shall be filled gradually to ensure uniform settlement all over the area. The full supply level should reach in a period of not less than 72 hours. At the time of testing verticality of tank structure should be checked by theodolite over water level at any horizontal line marked before filling. After seven days period for observation after filling with water the external face of sump should not show any sign of leakage and remain apparently dry.

44. Following order or priority regarding specification shall be followed by the contractor,

i. Circular of E.N.C. P.H.E.D., Bhopal vide circular no.236, dated 21.05.1997.
ii. Relevant IS code with latest version. (IS 456, IS 3370 etc.)
iii. Specification in Contact document
iv. Any other specification not covered under the above the said standard shall be fixed by the engineer in charge.

SPECIFICATION FOR GSR / SUMP

1. The structural design of GSR shall be in accordance with provisions relevant I.S standards. List of Indian Standards for Design of GSR / Sump well:

   (I) I.S. 3370 part I & II 2009 or its latest revision

   (II) I.S. 3370 part III & IV 1965 or its latest revision

   (III) IS 456-2000 Reaffirmed 2016

   (III) IS 1893-20 | 6 part I to V

   (IV) I.S. 875, Part - 1 to 3, 1987 or Its latest revision.

2. Water depth in container shall be adopted as per data of tender. Capacity shall be calculated excluding free board of the reservoir. If water depth is not specified, the suitable water depth/acceptable to Engineer in Charge of the work shall be provided.

3. Shape of container (in plan) specified in Tender shall be adopted; in absence of any data circular shape shall be adopted.

4. Size shall be fixed as per availability of space (land area) at site / acceptable engineer in charge.

5. Effect of overlapping of pressure bulbs on soil due to nearby structure and proposed sump well should be considered.

6. Care shall be taken so that no damage occurs to nearby existing structure. Compensation shall be paid for the same by the contractor.

7. The minimum concrete grade for RCC shall be M-30.

8. HYSD Fe 4l 5/500 grade reinforcing bars confirming to I.S. 1893-1984 shall be considered in

9. design. CRS/TMT bars shall be provided. In saline atmosphere corrosion resistance stainless steel/HCR rebar shall be provided. Any other steel can be used with approval of C.E./in the situation of non availability in market without extra cost.

10. Minimum size (or thickness) of various components shall be provided as per tender criteria/specifications in absence as per I.S./ Std. practice of MPPHED Minimum dimensions specified for various components in tender data/specifications shall be provided without fail.

11. The safe bearing capacity (SBC) shall be referred from SBC test report. In absence of report it shall be referred from data sheet. If poor soil is found/water table is met with during excavation SBC shall be scientifically as curtained and design shall be revised. No extra shall be paid for increase in quantity.

12. Pipes and special shall only be used as stipulated in tender documents.

13. The rate shall include cost of dewatering during excavation making all arrangement when water table meets within depth.
14. The structure shall be designed properly to resist uplift due to ground water table specified in data or actual ground water table meets with during excavation.
15. 20 mm GI pipes railing in three rows with 100 mm Dia vertical RCC Pole shall be provided over GSR periphery when GSR height is >= 2 meter above ground level.
16. RCC stair case / RCC Steps should be provided from GL to GSR tops slab (It will be applicable for GSR).
17. Aluminum ladder, main hole cover with lock to be provided.
18. Appearance of structure should be aesthetically good looking acceptable to authority.
19. Any change in size, shape, depth below GL, height above GL, water depth, F.B., size of member etc can be permitted in exceptional case due to site condition or hydraulic design requirement by C.E.
20. No extra shall be paid for change.
21. Any change in data, dimensions, shape, and water depth, reduction in size if permitted by competent authority and if it reduces quantity then payment shall be reduced prorata.
22. When capacity of GSR/Sump capacity is > 10 lakh liters two or suitable compartments acceptable to executive engineer shall be designed and provided.
23. Agency shall engage qualified (at least graduate) consulting engineer for designing the structure and he/she shall visit the site for guidance of work.
24. Rates for this item are for all seismic zone as prevailing in states of MP.
25. This chapter is introduced only for preparation of estimate for their item. The tenders for this item will be called on turn-key basis only.
26. The General specifications for the construction of Ground Service Reservoir shall be similar to the specification for OHT mentioned in this chapter.

Specifications for Intakewell and Pump House

Detailed investigation

The Contractor shall make detailed Survey, topographical, Soil and hydrological investigations to decide most suitable location of Intake well in proposed site. In this regard the contractor will have to visit the site to get him acquainted with the prevailing site conditions and will have to contact the concerned office of Water Resource Department of Government of MP to collect all relevant information for planning and designing the intake well. The floor level of pump house will be 1.5m above HFL.

Under this contract, an Intake well with Pump House is to be constructed in proposed location , at the deepest point in order to take optimum utility from the source. An intake well cum pump house is to be constructed at the location to draw raw water from topmost layers in different seasons and also from the lowest level of water.

Detail Designs and Drawings

A general layout drawing, showing the arrangement of installations of suitable pumping sets, piping arrangement at discharge floor, location of starters, control panels, main switch board, cable laying etc. should be supplied by the contractor. Entire civil works related to these items viz. foundations for pump, motor, cable trenches, ducts and opening in steining wall of Intake, opening in slabs for pumps, gantry cables and pipes etc. in all respects are to be provided by the contractor and shall be a part of this contract. The responsibility for design, construction, structural stability and water tightness shall rest solely with the contractor. In designing with various loads,

applicable seismic load has to be considered. Scour depth of Intake well shall be calculated as per provisions of SP-13 (Special publication) and IRC 78.

Detailed design shall include all the required calculations for all the components including thefollowing:

i. Foundation depth and its design for trash rack structure, Intake well cum pumphouse Staining thickness and its design for various hydraulic conditions.

ii. Pipe floor and motor floor design for anticipated loading conditions.

iii. Gantry support to be designed for ultimate loading capacity including weight of gantry girder crane.

iv. Design of shoring scaffolding.

v. Protection works against flood and erosion of banks near Intake well.

vi. Design of inlet ports and/or approach conduit.

vii. Design of Approach road/ramp with suitable approach

Foundation

The foundation of Intake well should be designed and constructed considering the strata at site, maximum scour depth, weight of superstructure, water force, wind force, live load of pumps, motors and pipes, dynamic forces due to vibrations of pumps, water thrust, seismic force, and other live loads, water thrust, uplift pressure, etc. Any other forces required to be taken for safe design, which are not mentioned here, should also be taken into consideration for the design.

The work of foundation also includes dewatering of water, shoring of sides of excavation, scaffoldings, desilting of foundation, construction of coffer dam, etc. for which no extra payment shall be made to contractor. The concrete mix for RCC work in foundation should not be weaker than M30 and the stresses in concrete shall be taken as per IS: 3370 and other relevant BIS codes/ specifications. Minimum Cement content for M30 concrete shall not be less than 400kg/cum.

Blasting in foundation shall be permitted only by the competent authority if the contractor guarantees that such operation shall not pose any threat of damages.

The foundation of Intake well shall be taken down to such depth that it is safe against scouring, settlement, overturning, floating logs and sliding. The depth of foundation below the scour line shall not be less than 1.5m for adequate anchorage purpose. In case of foundation resting on rock, the minimum embedment in the rock shall be 1.5m. However, minimum depth below the scour line has to be ensured considering a sufficient factor of safety as per prevailing IS and IRC norms. The minimum bottom floor level of intake well should be such so that the water can be drawn from the lowest supply level. The laying of foundation concrete under water shall be done by skip boxes or termite pipes with prior permission of the Engineer-in-Charge. Pumping out of water shall not be permitted from the time of placing concrete up to 24 hours thereafter.

Sufficient dowel/ Anchor bars should be provided in the foundation by drilling suitable diameter holes in the foundation rock and fixing for steel bars and grouting it with suitable epoxy compounds.

Trash Rack Structure

The RCC trash rack structure shall have minimum suitable floor area and three inlet ports should be provided at suitable levels considering summer, fair-weather and monsoon seasons.

Control sluices for lower most and monsoon level ports shall be above HFL at Intake floor level whereas all upper sluices shall be provided in the steining of the trash rack and inlet lower and operating gears from the approach floor at top manually and electrically. The opening in the trash rack structure has to be provided in such a way that it is

not obstructed due to any floating logs/ big boulders or other floating bodies. And it should be able to draw designed flow from the river at each of the ports. The top level of trash rack structure has to connect to Intake well with an approach. Rose pieces shall be designed for 50% clogging.

The RCC Intake well with RCC framed structure shall be having a diameter specified in scope of work/ detailed specifications with a minimum discharge floor area for required number of suitable motor pump sets with all its appliances and appurtenances and minimum 6.00m height. These shall be placed such as to have a clear space of 1.5m between the pumps/ motors and the minimum distance of the pump/ motor from the wall should not be less than 1.5m and should be designed to carry the static, dynamic and impact load excluding dead load of floor. The motor floor should have a minimum floor area to accommodate starters and capacitors for motor at floor level; motor floor slab has to be designed accordingly. The arrangement of beams for slab should be given in such a way that these do not obstruct motor foundation and column assembly of pumps, valves and pipelines. The head room on discharge floor shall not be less than 3.0m below the bottom of the beams of motor floor. Sufficient sand filling in sunk wells are to be made as per IRC clause 1207.

Trash Rack Structure with suitable arrangement for arresting silt and other floating body shall be made at intake well as per design criteria. A suitable air and water jet arrangement with suitable compressor for scouring of Intake and trash rack arrangement is also to be made.

Staircase

An RCC staircase of minimum 1.0m width with SS railing for a suitable live load of discharge floor level to bottom of the Intake well should be provided, on the inner circumference of the steining. The stairs shall be provided in such a way that there is no obstruction to inlet ports or operation rods, etc. The intermediate landings should be provided at about 3.0m vertical intervals. The stairs should be provided with railing as per specifications.

Inlet ports

The well staining shall be provided with 3 no. inlet ports with suitable thimble plates, gates, manual/ electrically operated operating rods and strainers at various levels and directions to draw water from maximum draw down level to highest water level. Each inlet port shall be designed to allow a flow of raw water at the design rate of flow. The location and R.L. of these ports may be altered if required during construction by the Engineer-in-Charge. These inlet ports/ pipes shall be provided with sluices gates/valves with spindles, wheels and works gears so as to operate them from pipe floor level. The size of suction well and grit chamber shall be sufficient so that no vortex eddies or cavitation's effects are experienced on pumps.

Pipe floor or Discharge floor

Pipe floor should be provided to house all discharge pipes, valves and manifold for all the pumping sets as per requirement. The floor slab should be designed for a uniformly distributed expected live load and moving loads, suitably designed openings should be provided for passing column pipes through the slab. The suitable provision for concrete bedding and thrust blocks for all discharge pipes should also be provided. A door entrance opening with steel collapsible gate should be provided in the well steining above pump discharge floor level towards the approach road/ ramp. An opening towards the approach ramp should be provided at motor floor just above the entrance of discharge floor.

The arrangement should be such that easy handling of motors/ pumps is possible with the help of gantry crane. An arrangement with suitable pump for desilting of intakewell shall be provided.

Motor floor and Pump house

Pump house above intake Well will be circular in shape. It shall be constructed of RCC framed structure with M30 grade concrete column and beams. Columns will be constructed at suitable spacing as per design requirement. The wall of the pump house shall be of Brick masonry 20cm thick in 1:6 mortar with plaster minimum 20mm plaster in CM 1:4

A floor slab at motor floor with minimum 1.2m cantilever projection as balcony all around on the outer side of wall steining should be provided with Tubular MS steel railing of Dia. 50mm (Heavy class) in minimum 3 rows on RCC vertical posts of M20 Grade with height 1.20m. The RCC post will be fixed at 1.5m c/c. The colour/ specification of colour of railing shall be as per the direction of Engineer in Charge.

At floor motors of pumping sets, starters, control panels, bus bar channels etc. shall be installed. The floor slab shall be designed for expected live loads plus point loads for pumping sets etc. 50% impact and vibration factor be taken separately. All the civil work involved for installation of pumps and to lay electric cables of motor shall be provided by the contractor.

The motor floor shall be of mosaic with 1.50% ironite. An opening in floor with steel frames and cover/ Chequered plates for unloading of motorsand machineries. The inner dimension of well at motor floor level shall be such that a sufficient floor area is available at this floor level. Suitable rubber mats as per BIS specifications are to be provided in both the motor as well as discharge floors. Top slab of Pump house will be extended 1.5m around pump house all around with MS railing throughout as speciifed above with suitable opening to go to roof of pump house from MS round staircase of suitable size and specification as decided by Engineer in charge, which will be fixed on slab of approach bridge in such a way that there is no obstruction for movement as well as turning of vehicle near the face of intake well. Quality of paint with Colour for railing shall be as per direction of Engineer in charge. After water proofing Mosaic flooring will be done on the roof of the pump house with proper slope to avoid stagnation of water.

Good architectural appearance should be provided to balcony and chhajja to the satisfaction of department. A main entrance of of suitable size as required shall be provided. Sufficient doors, ventilator and windows frames fully glazed shall be provided in such a way to provide at least 30% opening in floor area for light and ventilation. All doors/ windows ventilators shall be provided with fine graded netting for fly/ insect proofing. The height of pump house shall be such as to have a roof top level, so as to have a minimum of 6 m from motor floor level to inside roof level.

Note- If the contractor is using pumps with stool/ stand, in which no separate discharge and motor floors are required, then he may submit the drawings accordingly for approval. The roof of pump house shall be of RCC and the live load over the roof should not be less than 150 Kg/Sqm.

Sufficient space for storage of spare parts and tools and plants should be provided in shape of almirah in the pump house. The total area or racks and almirah shall be wooden with sun-mica finish of good quality with locking arrangement. Suitable comfortable sitting arrangement shall be provided for operator within or outside pump house. If it is provided outside, accordingly proper roofing shall be provided.

Arrangement of Gantry Girder

Cantilever projection not less than 0.3m should be provided all around on the inner side of the wall at suitable height from motor floor level to support Gantry girder crane. Rails of suitable size should be provided all around this projection over which the gantry girder will move along the circumference. The gantry girder should be provided with electrically operated chain pulley in such a manner that it can move along the gantry girder. The length of chain pulley shall be capable of lifting the machines/pipes/other articles from the bottom of the Intake well discharge floor, motor floor level or from any height between the bottoms of Intake to gantry girder level.

The various guidelines to be considered for designing and constructing intake well cum pump house shall be as below : -

i. Floor area should be sufficient to have panels, switch gears and capacitors, etc. considering space for future demand of raw water which is to be drawn through intake well.

ii. Size of the intake well should be such that it has adequate capacity to draw raw water as per the requirement specified in the scope of raw water supply scheme.

iii. Size of the Intake well should have adequate capacity for installation & maintenance of pumps/motors with minimum internal diameter as specified in the scope of the work to house the pumping equipment, as per specifications capable to draw raw water as per the requirement given in the scope of the work.

iv. Minimum dimensions of intake well are specified in the scope of work. However, Intakewell shall be designed for the retention period of minimum 15 minute or as required from the design consideration and should have grit / silt removal arrangement with mechanical scraping belt or bucket for de-gritting/desilting the well.

v. Suitable sluice valves/gates at the entrance of Intake well have to be provided to control the flow with with electrically operated arrangement.

vi. Proper arrangement for fresh air circulation should be there so that suffocation does not occur during the maintenance of the well.

vii. An RCC staircase in Intake well with proper railing up-to bottom, with landing at intermittent points has to be provided. Railing shall have RCC post of 1.2m with minimum four rows of Stainless steel (Heavy quality) railings of suitable dia/ tubular section as per direction of Engineer in Charge.

viii. An electrically operated gantry of suitable capacity is to be provided for maintenance purposes.

ix. Proper ventilation/lighting arrangement as per the BIS requirements for the pumps and motors are to be provided.

x. Installation of mechanical operated Rolling shutter of suitable size of heavy gauge with painting as per direction of Engineer in charge, Installation of uPVC sliding window shutters with glass and mosquito net (heavy sectioons) in required no. and size, uPVC exhaust fans in the wall of pump house. Wall mounted fans 4 nos, of bigger size of reputed make shall be installed in pump house. Mobile Air cooler 1No. of bigger size to be kept in the pump house

xi. Proper lightening arrester arrangement and fire extenguishers as per BIS standards are to be made for Intake well cum Pump House.

xii. Internal Diameter of the intake well shall be as per the required inflow; however, it shall be minimum as specified in the scope of the work.

xiii. The intake well shall have inlet port arrangements at different levels with screens, racks, sluice valve etc. with mechanical/electrical operated arrangement.

Ancillary Works and Other General Considerations

The scope of work includes all the necessary works required including ancillary works if any, to be executed to make available the desired quantity of water into the Intake and/or Jack well. No extra payment will be made on this account to the contractor and rates for such work shallbe deemed to be included in the quoted rates.

The job implies all works from detailed preliminary investigations to the final commissioning of the structure including all ancillary works such as construction of coffer dam for diversion of flow, dewatering, bailing out of water, etc. including testing during trial run period and defects liability and O&M period of this contract.

The contractor is required to arrange complete construction of civil works, supply and installations of all the pumping sets. Testing will also have to be carried out at contractor's cost and it will be the responsibility of the contractor to ensure that respective standards are achieved.

The Intake well is required to house suitable pumps and other accessories, therefore

coordination with the pump suppliers and hydraulic model studies will be essential for deciding type of pumps and arrangements of pump installations with 50% standby capacity. The contractor will have to carry out any modification, addition and alteration to the design and structure to meet such requirement, if needed, without causing any extra liabilities on the GMC He will design the Intake well cum pump house in such a manner that it also meets the efficient vortex free hydraulic and structural requirements of pumps/ motors to be installed to meet such requirements as per provisions of CPHEEO Manual. The contractor will submit QAP minimum from three manufacturers with all details including maximum consumption of elecricity power, and power factor. Bidder shall ensure that Power consumption shall not be more than guaranteed power consumption i.e minimum power consumption required to run whole scheme upto the maximum output and which will be mentioned in the data sheet provided by the manufacturer.

The trash rack of the Intake well should be suitable for ultimate capacity of the scheme to draw raw water through different ports at same level. In the Intake well cum pump house, suitable pumps (Energy Efficient) should be installed for design capacity. These pumps will operate individually or in all possible combinations. The civil structure will have to be such that no vortex is formed and there is no interference or pump starvation by operating these pumps in any combination or independently. However, for structural design suitable static loading for each motor pump assembly may be considered at the points where pumps will be installed. In addition to this a live load at motor and discharge pipe at floor may also be considered. The essential vibration effect factors will also be considered.

Chapter 4

WATER TREATMENT PLANT

All the work shall be carried out as per relevant clause of CPHEEO Manual

(1) Designing (aesthetically), providing and Constructing and commissioning Conventional Water Treatment Plant comprising of rapid sand gravity filters alongwith clariflocculator**. The system should also be designed for 20 % overload. The unit flow should be calculated for 22 hours working. The Job includes the** all Civil, works including cost of providing and applying Epoxy Paint (Food Grade) to inside surface of water retaining structures in contact with chlorine and providing anti - Termite treatment to entire structure below Ground Level, Mechanical and Electrical components of Various sub-works as given below: including Necessary hydraulic testing, structural testing Equipment testing, etc. complete as directed by Engineer-in charge (Turn-key job).

1.1 Scope of Work

The scope comprises all necessary site investigations, functional planning, supply of plant, general design, detailed design, manufacture, supply, delivery to site, installation, construction, testing and commissioning of all works required for the Water Treatment Plant of capacity as specified contract including all associated mechanical and electrical plants, equipment and services, civil and building works, pipe lines and appurtenances from the raw water inlet up to clear water reservoir and pump house.

The principal requirement is a spacious and convenient layout. The structure should represent a pleasing appearance with aesthetic features forming a balance between function and form. The interiors of the structure shall be eye appealing and in keeping with the objectives of the plant viz. production of potable and wholesome water. While designing and constructing, it should be ensured that all materials, design, construction and fabrication details for different units including doors and windows conform to the IS specifications and codes of practice wherever available and in their absence to the established standards.

The scope of work includes but shall not be limited to the provision of the following:

1.2 General

Contractor shall study available data with department and if required carryout site investigations to collect additional field data, if any.

1.3 Planning and Design

The Contractor shall carryout the planning and designs for most economical type of treatment plant to generate filtered water as per requirement specified in contract.

The detailed design including hydraulic and structural design, development and preparation of detailed plant working drawings, diagrams and cable schedules and detailed structural steel fabrication drawings, preparation of design reports, manufacture and testing at places of manufacture, transport, delivery, erection, building-in, setting to work, commissioning, testing of all plant required for the Water Treatment Works.

The contract is including but not limited to, pipelines, pumping installations, blowers, compressors, machinery, apparatus, station pipe work, lifting, handling, ventilation equipment, electrical equipment, interfacing lighting systems, earthing, fire safety and lightning protection systems, materials, articles, fittings and accessories, ancillary works of all kind and nature required for installations of the highest possible operative standards and for compliance with the standards prescribed in the Specification and with the particulars and guarantees entered by the Contractor in the schedules.

The contractor has to Design & Construct following components, if required, subject to fulfillment of the water quality parameters as specified in the contract.

(i) Cascade Aerator or any other suitable system, for removal of iron in incoming Raw water if required. It shall be designed in circular shape with circular Gullet to collect water.

(ii) Dosing equipment of coagulants, including chemicals with facilities to store Alum and chemicals for 3 months.

(iii) Flash mixer.

(iv) Clarifiers and flocculators. It shall be designed with properly designed inlet and outlet arrangement with due consideration to better efficiency of settling, mechanical sludge cleaning arrangement with sludge effluent pipe etc. complete. It shall be designed for 25% overloading including inlet and outlet.

(v) Rapid gravity sand filters, filter galleries, wash water tank, filter back wash system and filter control block. Filter house shall be designed with rapid gravity sand filters in even numbers. The minimum walkway near the filter should be 1.2m wide and pipe gallery should be minimum 2.00m or more to accommodate all the pipes and valves and have a proper slope to drain out the wash water through wash water pipe.

(vi) Module Chamber Room: It shall be designed to accommodate module chambers of all filters including weir, wash water tank filling pump and air blowers. The height should be sufficient so that the wash water tank may be constructed above it.

(vii) Clear water sump to store treated water. It shall be designed for 45 minutes to 60 minutes detention subject to the minimum capacity as specified in the contract.

(viii) Clear water Pump House to accommodate the clear water pumps, electrical panels, control room etc. It should have sufficient space to accommodate required number of Pump and arrangement, to pump treated/ clear water to OHTs and IPS. Pumping arrangement will, at least, have minimum numbers of centrifugal pumps as specified including cable ducts, surge protection (If so required), NRV, suction pipe etc. complete. It should have sufficient height to accommodate the Mono Rail/ gantry crane of suitable capacity. It should have additional storeroom, Switch Board room and office room and toilet units of appropriate sizes.

(ix) Chlorination plant building, facilities for post and/ or pre-chlorination and storage of chlorine cylinders for 30 days requirement. It should have sufficient space to accommodate liquid or gaseous feed chlorinator. The cylinder storing room should have sufficient capacity to accommodate cylinder/ toner for at least 30 days storage. It should be forced ventilated, easily accessible from at least two sides. It should have an arrangement near the ramp to unload the truck by providing chain pulley arrangement at a height of minimum 5.00m.

(x) Monitoring and instrumentation with their housing.

(xi) Electrical works including incoming cable from substation to main control panel, motor control centers, local control and panels, power and, plant ventilation and lighting etc. It should be done as per details givenand as per ISS/ B.I.S./ I.E. Rules.

(xii) Pipelines, valve chambers, service water installation, sampling and water quality monitoring, instrumentation and other miscellaneous works associated with the plant.

(xiii) Storm water drain and sewers with appurtenances.

(xiv) Administrative building, laboratory, store, and for control arrangements housing building, including services such as electrical, Lighting, water supply, sanitation Laboratory block as per recommendation of CPHEEO Manual on Water Supply and Treatment, provided with one Chemist room, one office room and remaining part for storing the chemicals and conducting the test and toilet.

(xv) Internal roads with storm water drains, landscaping and area lighting etc.

(xvi) Training of PHE personnel in operation and maintenance of the plant during the last three months of O&M period.

(xvii) By-Pass Channel: Following bypass channels shall be designed and constructed. The channels shall be designed as a one unit provided with sluice gates arrangement for isolation of channels. These channels shall be designed for capacity with 20% overload.

- From raw water channel to flocculator.
- From raw water channel to filters by-passing the flocculator.
- From flocculator to sump by-passing filters.

(xviii) The overflow from various units shall be connected to the storm water drains for safe disposal to the near water bodies up to 100m distance with environmental acceptance.

(xix) Storage Tank for Neutralization of Chlorine - Solution storage tank of minimum dimension 3.00 x 3.00 x 2.00m size should be provided with fire resisting tiles cladding. It should be easily accessible from chlorine house or storage room.

(xx) Wastewater Pipe from Flocculator, Filter, Clear Water Sump - The wastewater line from all the units shall be of R.C.C. NP2 class and shall be provided up to the nearby drain/ nala.

1.4 Commissioning and Testing of Plant

1.4.1 Basic Information

1.4.1.1 The source of water will be as specified in Contract.

1.4.1.2 The contractor is required to assess the raw water quality parameters themselves before designing the plant.

1.4.1.3 Location of Treatment Plant - The location of the WTP is as specified in Contract.

1.4.1.4 Bearing Capacity of Soil - It is the responsibility of the contractor to test and check the bearing capacity of soil and submit it with the design. The bearing capacity should be tested by plate load bearing test method by any NIT/ Govt. Engg. College/ NABL Laboratory or any accrediting institution as per ISO/ IEC: 17011 or laboratory accredited as per ISO/ IEC: 17025.

1.4.1.5 Treated Water Quality - The plant as a whole and the units individually should show efficiency as per requirement given in BIS specification and manual on water supply & treatment by CPHEEO, New Delhi. In case of any confusion BIS specification shall get the precedence to all manuals. The quality of treated water after filtration and chlorination shall be as per IS:10500 with up to date amendment.

1.4.1.6 Guarantees of Clarified Water and Final Treated Water - All works for the processing and treatment of raw water shall be designed to provide designed discharge. The performance tests on the treatment works shall be carried out at the flow inclusive of over loading of treated water as per CPHEEO Manual on Water Supply and Treatment.

1.5 Detailed Specification of Components of Treatment Plant

All the units of Water Treatment Plant shall be designed on the basis of CPHEEO Manual on Water Supply and Treatment.

1. Inlet Chamber/ Cascade Aerator

It shall be provided to collect the raw water form. Source. It shall be designed for designed Capacity. It shall be in R.C.C. of minimum grade M-30. The Inlet chamber/ Cascade Aerator shall serve the purpose of dissipating the kinetic energy as well as to increase the dissolve oxygen of incoming water and also provide necessary static head for flow through treatment plant, during normal and emergency loading conditions. The residual head at the end of pumping main shall be 2 m, which may be kept, is consideration during the design.

2. Inlet Channel

It shall be designed for designed capacity with flow measuring arrangement of Wier/ parshall flume type with dial type flow indicator / ultrasonic flow measuring arrangement. The raw water channel shall be constructed in R.C.C. of minimum grade M-30, supported on columns in R.C.C. of minimum grade M-25. A puddle collar shall be provided to receive water.

3. Flow Measuring Arrangement

The Wier/ flume shall be designed for designed flow and free board of not less than 30 cm. The measuring flume shall have a side chamber for float. The flume shall be of open channel type. The entire construction shall be in R.C.C. and shall be perfectly watertight and suitably supported on R.C.C. columns. The walkway of 1.0m width with G.I. pipe railing as per specifications given shall be provided. The dial type flow indicator shall be provided near the flume to indicate the flow. The flow indicator shall have a capacity of measuring the maximum flow as per the project requirement as specified in including 50% overloading. The weir plate will have brass edges which shall be graduated to read in litres per hour. The weir shall operate with clear free fall condition. It shall be ensured that the weir plates do not get submerged due to afflux. The dial of indicator shall be properly lighted. Necessary arrangements for proper desalting of stilling chamber shall be provided.

4. Bypass Channel

In order to provide the flexibility in operation the bypass channels shall be provided. The by-pass channel shall be designed for designed capacity. It shall connect the following units:-

- From raw water channel to flocculator.
- From raw water channel to filters by-passing the flocculator.
- From flocculator to sump by-passing filters.

The necessary penstock with simple operating arrangement shall be provided as per IS: 3042 or some alternative arrangement that suits to the proposal. It shall be constructed in minimum M-30 grade concrete and columns shall be minimum M- 25 grade concrete.

5. Flash Mixer

It shall be designed for detention period of 30 to 60 seconds. It shall be circular or square tank with ratio of impeller diameter to tank diameter as 0.20 to 0.40 and the shaft speed of propeller to impart tangential velocity greater than 3m/s at the tip of the blade. The ratio of the tank height to the diameter shall be 3:1. The power-mixing device shall be capable of creating velocity gradient of 300 per second.

The power requirement of flash mixer shall be as per recommendation given in IS: 7090.

The paddles made of structural steel confirming to IS: 226. The agitator shall be mechanically driven consist of electric motor with continuous duty operating through a reduction gear. To achieve good results the chemical should be added just near the tip of blade. The design should be such that there should be no possibility of short circuiting in the tank. It shall be designed for a full flow of designed capacity.

A desludging pipe of RCC NP2 with sluice valve confirming to B.I.S. shall be provided. The length of pipe shall be as per requirement of layout plan, and it shall be connected to waste water sump. Hand railing along operating platform (1.0m diameter all around) of 0.75m height shall be provided. The M.S. ladder of 0.75m wide shall also be provided. The specification of ladder and railing are given separately. It shall be constructed in RCC minimum M-25 grade concrete. Protective cover of G.I Sheet of 8 gauge shall be provided for motor.

6. Inflow Arrangement

Coagulated water from flash mixer will be conveyed to the inlet of clarifloculator through precast R.C.C. pressure pipe or CI or DI D/F or MS pipe of min. 8mm thk of suitable diameter. The pipes shall be laid over firm bedding and at uniform grade. The velocity of flow through this conduit shall preferably be between 1.0m/s.

7. Sedimentation and Clarifloculator

It shall be designed for the detention time as per the provisions in CPHEEO Manual. It shall have properly designed inlet and outlet arrangement with due consideration to better efficiency of settling, mechanical sludge cleaning arrangement with sludge effluent pipe etc. complete. Combined units of flocculation and sedimentation shall be provided immediately after flash mixer, but it should be designed on concept of tapering velocity gradient and settling velocity with suitable motor operated mechanical cleaning device. Clarifloculator (s) shall be provided for the total designed flow with provision for 25% overloading under emergency condition as per CPHEEO manual. The inlet and outlet arrangement will also be designed for 25% overloading.

8. Flocculator

It shall be designed to provide a detention time of 30 minutes at design flow. Other design parameter viz depth of water, velocity of flow, paddle area, peripheral velocity of blade. Velocity gradient G and the factor GT shall be generally in accordance with CPHEEO manual Para 7.4.3.2. Mechanical Type Flocculator. Coagulated water will be admitted through the central hallow shaft near the water surface and then will flow radially outwards uniformly in all directions through slots. The velocity of flow through these slots should be about 0.2 to 0.3 m/s. Flocculation paddles 2 or 4 in numbers depending upon design will be mounted on a vertical shaft, which will be rotated by A.C. motor 3 phase 415 v. The partition wall and the floor slab of flocculator shall be of RCC. Area of opening at the bottom of partition wall should be large enough to maintain sufficiently low velocity of flocculated water.

9. Clarifier

The Clarifier shall be designed to provide a minimum detention period of 2.0 to 2.5 hours and surface loading of 30 to 40 m3/ m2/ day at design flow. The outlet shall comprise of MS V-notches and effluent launder or rectangular weir and outlet flume. A weir loading of 300 to 600 m3/day/m length may be provided.

The outer circular wall and the floor slab of the clarifier shall be of R.C.C. The floor slab shall have a minimum slope of 1: 12. A peripheral walkway 1.2m shall be provided with railing shall be provided.

10.1 Sedimentation Tank/ Clarifiers/ Flocculators/ Clariflocculator Scraping Mechanism

Sludge scrapers and mechanical flocculators Sludge scraper mechanism and discharge arrangements shall be designed for raw water suspended solids loadings up to 20000 mg/l for the plain sedimentation tank and 1000 mg/l solids in clariflocculator. The design shall allow for starting up under a sludge sit down situations, and to accept without distortion any normal torsional or bending loads which may occur during erection and operation, scraper mechanism shall extend to the entire diameter of the tank.

10.2 Drainage of Plain Sedimentation Tanks and Clarifier

Each tank shall be provided with facilities for complete gravity drainage of all water and sludge from both the main body of tank and from sludge concentrates. All valves, pipe work required for discharge to and connection with the main drainage system shall be provided, so that a tank may be emptied within 6 to 8 hours. These drains shall discharge into the main works drainage duct to be provided by the Contractor.

10.3 Rotating Bridge

The clarifier scraper bridge structure shall be made of welded and bolted structural M.S. It will be a lattice girder bridge pivoted at the center and trolley fitted with pneumatic solid rubber tired wheel/ MS Wheel and resting on clarifier wall.

The bridge shall be 1.2m wide made of welded and bolted R.S. section of suitable size.

The scraper arm with scraper blade of suitable size fitted with rubber squeezer shall be suspended from the top of bridge having M.S. grill walkway/ 8mm thick M.S. chequered plate walkway ground flocculator drive gear. The scraper arm shall spread across the radius of the clarifier zone and flocculator zone. The bridge shall be so designed that it should also act as a hand railing or otherwise additional double row 25mm MS pipe railing light duty shall be provided. The driving mechanism for the bridge shall rotate over the steel fabricated track located over peripheral wall or directly over wall by rubber tyre wheel and comprise of GEC/ Kirloskar/ Crompton/ Jyoti or other approved make A.C. motor of suitable rating, 4 pole horizontal foot mounted class 'B' insulation T.E.F.C. squirrel cage motor having degree of protection as IP 55 suitable for 415 + 10 V-3 phase 50Hz having flexible coupling single speed worm reduction gear, duplex chain sprocket drive, plumber block with bearing M.S. ideal and driving shaft. The central bearing shall comprise of combined radial and thrust bearing housed in high grade CI/ MSEP housing with G.M. bush for rigidity.

The bridge shall be designed to take its own dead weight with uniformly distributed loading of 400kg/sqm. The bridge should be so braced that deflection at mid span is less than 80mm. The central bearing assembly shall be adequately infraction in all respect catch trays shall be provided to prevent slippage into water.

Five rings special current Collector shall be provided for each clariflocculator for transmission of electric power through pipe laid under floor from outside to the sub distribution board located over rotating board.

200mm dia RCC NP2 pipe and specials as required shall be provided for each clarifier under the floor slab between sludge outlet channel around inlet well to sludge outlet chamber at periphery.

200mm dia D/F PN-1 manually operated D.I. Non-rising spindle sluice valve confirming to B.I.S. standards and bearing ISI mark is to be provided for each clariflocculator.

One M.S. steel operating shaft with 25mm dia extension complete with C.I. head stock, 200 mm dia hand wheel and brass indicator plate for operation of sluice valve for each clarifloculator is to be provided.

For each clarifier bridge 2 No. 450mm wide rung ladder made out of M.S. black medium class pipe to IS 1239 painted with primer and paint, fixed to clarifier bridge structure for access inside the clarifier zone/ flocculation zone be provided.

Lamella HDPE Plate Settlers:

The Lamella Clarifier is a passive system which removes particulates from water. It is designed to fit into rectangular clarification basins and uses a series of 55° inclined lamella sheets. Lamellas Plates are made from a hydrophobic geotextile material, a proven, NSF/ANSI/CAN 61 and NSF/ANSI/CAN 372 certified, high-density polyethylene (HDPE). These sheets provide a large effective settling area within a small footprint. The Lamella Clarifier removes over 80% of the solids and achieves turbidity levels as low as 1 NTU which maximizes the filterability of the water.

The contractor shall provide and install the settling equipment for the pre-treatment process settling basin(s), as designed by the equipment manufacturer. The settling equipment shall include, but not be limited to,

a. Inclined settler sheets w/ corresponding side panels and tensioning system	b. Effluent troughs
c. Trough covers with integral v-notch weir	d. Mounting brackets
e. Tensioning angles and fasteners and related appurtenances	

This equipment shall be provided as an integral package by a single manufacturer who shall take responsibility for coordination of all components to assure proper functioning of the settling equipment as a system. The manufacture of the lamella clarifier shall provide all support members made from SS304 including horizontal support beams and vertical columns as needed.

The feed to the sedimentation basin shall be admitted from the flocculation basin through openings in the sedimentation wall. The individual lamella sheets shall be constructed of polyethylene geotextile material with HDPE hooks sonically welded to the vertical edges. The lamella sheets shall be designed to slide into mating hooks which are part of HDPE side panels located on either side of the lamella sheets. A pair of side panels with corresponding, supported sheets constitutes a lamella sheet module. A tensioning system shall be provided to keep the lamella sheet modules at a uniform width. The sheets shall operate in counter-current mode with clarified water flowing upward and settled solids moving downward. Incoming water shall be fed near the bottom of a sheet entering through feed orifices in the side panels to allow the settled solids to fall to the floor of the basin while minimizeng re-entrainment by the incoming flow. Sheets shall be capable of being individually removed with no need to remove any other components or structures of the lamella clarifier. Lamella sheet shall be kept at 55° angle.

The manufacturer of the lamella clarifier shall provide a trough cover that is designed such that an operator may walk onto the troughs. The trough cover shall incorporate an integral v-notch weir that separates the operator from the waterline so the equipment can be serviced without contacting the clarified effluent water. Local safety regulations shall be followed at any time to access the clarifier. The maintenance access surface shall above the water line to separate the operator from the water. Orifices on the trough cover shall allow washing, rain or excess water to drain into the trough. Trough cover shall be designed to minimize debris and reduce algae growth. The trough cover shall be made from UV resistant, impervious high strength, completely corrosion-resistant, reinforced polymer material. The trough cover shall be resistant to erosion and corrosion.

To assure system integrity and responsibility, all items of equipment described in this section shall be manufactured by a single manufacturer regularly engaged in the production of the specified equipment. In accordance with general accepted construction practices, the equipment supplier shall be solely and completely responsible for conditions of the job site, including safety of all persons and property affected directly or indirectly by his operations during the performance of the work. This requirement will apply continuously, 24 hours per day, until acceptance of the work by the owner and shall not be limited to normal working hours.

The equipment shall be handled and stored at the jobsite in accordance with the manufacturer's recommendations. The equipment shall be suitably covered and protected at all times, except during installation. The equipment shall be designed so that individual sheets can be removed without disassembly of the side panels or removal of other plates.

The CONTRACTOR shall install the system in strict accordance with: (1) the manufacturer's written instructions and recommendations and the manufacturer's installation drawings; (2) the oral and written directions provided by the manufacturer's technical representative who is supervising and observing the WORK; and (3) any additional requirements specified herein.

11. **Filtration**

Traces of suspended matter in the clarified water shall be removed by filtration. Filter designs shall be down-flow open rapid gravity units type or Single dual media filters of the divided bed type and shall operate on declining rate principle.

Each filter shall be hydraulically designed for an overload of 20%. Filters shall be capable of giving a run time of at least 24 hours. The number of filters shall be in even numbers. All filters shall be of identical shape and size. In the case of vertical flow rapid gravity sand filters; the average rate of filtration shall be 80 to 100 lpm/sqm. The main clarified water channel shall connect to the two filter inlet channels if necessary via a chamber. All filters shall be identical in internal dimensions and shall be designed for washing using filtered water and air scour. The air scour may precede or be concurrent with the wash water. The filters shall be designed so that when receiving clarified water of the standards the minimum period between washes is not less than 24 hours.

12. Number of Filter

The filters should be in even numbers and designed for economy to provide treated water as per requirement specified in contract

12.1 **Rate of Filtration**

The rate of filtration shall be taken as 4.8 to 6m/hr (80 to 100 lpm/m2). The inlet and outlet arrangements shall be designed to permit 100% overload for emergency conditions.

12.2 **Capacity of Filter Units**

The capacity of filter should be such that the number of units can take care of the total quantity of water to be filtered and with optimum efficiency to keep the filters working without undue overloading at any time.

12.3 **Size of Filter Unit**

Where the filters are located on both sides of pipe gallery the ratio of length to width of a filter box shall be about 1.25 to 1.33. A minimum overall depth of 2.6m including 0.5m free board shall be provided. The filters shall be constructed in R.C.C. of minimum grade M-30.

12.4 **Filter Media**

Filter media will consist of sand gravel. The Specification of media shall be as per CPHEEO manual Para 7.6.3.6 to 7.6.3.10.

12.5 **Under Drainage System**

The design parameters in under drainage system shall be as given in Para 7.6.3.9 of CPHEEO manual. The under drain system with central manifold or laterals either perforated in the bottom or having umbrella type strainers on top shall be provided. The central manifold and laterals shall be of cast iron, concrete or other suitable materials.

13. Wash Water Gutter (Troughs)

It shall be of RCC with M-30 grade it shall be designed in such a way that the Horizontal travel of dirty water over surface of filter shall be kept in between 0.6 to 1.00 m before reaching the Gutter. The upper edge of wash water gutter should be placed sufficiently near to the surface of sand so that large quantity of dirty water is not left in the filter after the completion of washing. At the same time the top of the wash water gutter should be placed at sufficient height above the surface of the sand so that the sand will not be washed into the gutter. The height should be considered by considering the expansion of sand.

14. **Filter Back Wash**

The backwash shall be arranged at such a pressure that the sand should expand to about 130 to 150% of its undisturbed volume. The backwash shall be of air water type. The air shall be introduced at a rate of 36 to 45m/h and pressure of 0.35 kg/Sqcm for a duration of 5 minutes and then the wash water shall be introduced through the same under drains at a rate of 36 to “cubic meter per hour” (cum/h) for 10 minutes. For introducing air and water the piping may be same or separate. it shall be worked out by the firm with due consideration to the economy.

15. Wash Water Tank

It is advisable and economical also if wash water tank is constructed above the clear water control chamber. Its capacity will depend on the total loss of head due to expansion of sand, loss in under drainage system, loss in incoming pipe and height of wash water gutter with respect to

under drainage system. The bottom of wash water tank shall be at a height of about 9.00 m to 12.00 m, the capacity of wash water should be varied from 2 to 6% of filtered water and should be sufficient for at least 10 minutes wash of one filter or 5 to 6 minutes wash of two filters. It shall be of RCC with minimum grade M-30. The most preferable shape is Rectangular. The corners of tank shall be rounded off. The top slab of tank shall be of RCC minimum grade M- 30 with sufficient number of manholes and ventilators of 100mm diameter C.I./ HDPE/ MSEP with cowls and aluminium ladder of 0.45m wide from top to bottom of tank.

A float operated mechanical gauge shall be provided. The arrangement should be fixed in such a way that during excessive wind flow it should not be disturbed. It shall have meter scale painted with enamel paint with black and red colour on white colour background. A RCC Staircase / MS ladder from ground to top of tank shall be provided by contractor. It shall have the railing either on both sides or on one side as per site conditions.

15.1. **Appurtenances**

15.2 **Filter Gauges**

It shall be provided to measure accurately the rate or flow through each filter box and to determine the loss of head occurring at any time during the filter run. It must be simple in operation and easy in handling by the maintenance staff.

15.3 **Walkway**

Walkway all-round the filters minimum 1.20m width shall be provided. it shall be fitted with railing of 0.75m height.

15.4 **Pipe Gallery**

Effluent wash and wastewater pipes all together with the sluice valves are placed in the pipe gallery should be well designed with minimum 2.00m width provided with a ladder or steel rungs to make it for the maintenance staff easily accessible to the bottom. It should be well ventilated. It should have a sufficient slope to drain out the wash water or other leakage water. It shall be connected with drain. The system will avoid the unnecessary congestion of the pipes and avoid the hindrances in cleaning of pipe galleries.

15.5 **Effluent and Wash Water Pipe**

The effluent pipe shall be designed for a velocity of 0.9 to 1.8m/s and wash water pipe for a velocity of 1.5m/s. These shall be MS / D.I. double-flanged pipes confirming to I.S.8329 - 2000 and D.I. specials shall be confirming to I.S.1538-1976. The sluice valve provided shall be confirming to I.S. 14846-2000.

15.6 **Module Chamber of Clear Water Control Chamber**

It shall shave sufficient space to accommodate sluice valves of effluent pipes, effluent discharge weir and clear water effluent pipe feeding to clear water sumps. It shall have sufficient circulation space minimum 2.00m of ground floor where the module chambers shall be minimum ceiling height of floor shall be 4.00m. The ground floor shall be easily accessible by providing aluminium/ MS ladders. Arrangement for effluent sluice valves, air blowers piping, wash water tank etc.

15.7. **Air Blower**

The Air blowers shall be designed for a free flow of air at the rate of 36 to 45 Cum/hr at a pressure of 0.35 kg/sqcm for a duration of 5 minutes. Minimum Two air blowers including100% standby arrangements shall be provided. The material used for the pipe and specials should be MSEP, GI, DI class k-7 or C.I. Class LA double flanged and D.I. Sluice valves confirming to I.S. 14846.

15.8 **Back Wash Water Pump**

It shall be designed for 2.00 Hr. i.e.; pump shall be of capacity to fill wash water tank in 2 Hr.

There shall be two pumps where one will act as stand by unit. The pump shall be of centrifugal type with a suitable motor to operate on 415+10% volt, 50 Hz. B class of insulation and degree of protection at IP-56 with suitable Auto Transformer Starter panel, Cable, suitable rating capacitors and Stabilizer. These pumps shall be provided and fixed by the contractor. The

pump shall be provided suitably with a NRV and sluice valve of PN 1.0. The size of the pipe to fill the wash water tank shall be designed by the contractor. The pipe shall be MS/GI/ with bell mouth at the top. The delivery pipe in wash water tank connecting to different filter to sluice valve shall be designed by contractor and made of MS/ DIDF/. The scour and overflow pipe also be designed and provided by the contractor. The overflow pipe shall be connected with the clear water sump pipeline or clear water sump whichever in nearer. Scour pipe shall be connected with the wastewater line of filter going to waste water sump of chamber with necessary CI sluice valves.

15.9 **Modules**

The module chamber shall be R.C.C. constructed for each filter bed separately. It shall have two compartments. In one compartment it shall have clear water effluent pipe with sluice valve. It shall be operated from the operating platform. The second compartment will be used for collecting the clear water spill over the weir; the height of weir shall be kept in order to avoid the negative head. It shall be line with tiles of standard make, preferably in sky blue colour. The top of the module chamber shall be covered with M.S. framed cover divided in two parts, fixed with glass of minimum 4mm thick the cover shall be so fixed that it should be easily opened at the time of maintenance. The module chamber shall also have arrangement for draining out the dirty water (at the time of maintenance).

15.10. **Filter Control**

Filters shall be of the declining rate type with filtration rates varying by +20% of the average over a filter run length. At the start of the filter run, when the bed is clear the filtration rate shall be 20% above the average value whilst at the end of the filter run the filtration rate shall be of 20% below the average value.

Clarified water shall be fed to each filter from the inlet channel through a submerged penstock opening.

At the outlet of each filter the filtered water shall flow through an adjustable valve or 'setting valve', which shall limit the maximum flow to 20% above average flow. The valve setting shall be adjusted during commissioning in order to achieve the required range of filtration rates between clean bed and dirty bed conditions to within + 20% of the average flow though the filters.

A separate valve drain shall be provided to drain the underflow chamber. Emergency access to the underside of the filter floor shall be provided.

15.11. **Staircase**

The stairs made of brick masonry in cement mortar 1:4 with anti-skid tiles shall be provided. The staircase shall be provided to connect ground level to the floor of control room and wash water pump or air blower floor to the operating plate from of control room.

16. **Chemical House**

The chemicals to be added to raw water for coagulation and flocculation will be filter alum, hydraulic lime and other coagulant aid if necessary. Looking to the characteristics of the water only filter alum shall be sufficient. But in certain conditions where the raw water becomes acidic in nature then hydraulic lime shall be mixed.

The minimum storage shall be for 3 months for filter alum and 3 months for hydraulic lime and other chemicals. Store for the chemicals shall be provided below chemical dosing room. The minimum ceiling height for ground floor shall be 4m and for first floor it shall be 5m. In order to avoid moisture, the minimum plinth height shall be 0.60m. The floor should be made damp proof and it should be filled about 0.6 0m with sand and bituminous coating shall be provided over it in order to avoid the moisture. The building shall be constructed in R.C.C. framed structure in M-25 grade concrete and panels shall be of second class brick masonry in cement mortar 1:6 and plaster in cement mortar 1:4. The specification of brick shall be as given in the specifications. The roof of this storage room shall have an opening of 1.0 x 1.0m to facilitate lifting of chemicals. The arrangement for lifting and weighting of chemicals shall be provided by contractor from first floor of chemical house. Adequate ventilation and lighting will be provided. It should be easily accessible to the trucks for unloading the chemicals. The main entrance shall be 3.00m wide and 3.50 m height with rolling shutter. The height of alum stack should not be kept more than 2.0 m with suitable open space for passage.

First floor to accommodate alum and lime tanks. Chain pulley block etc. and labaratory (min. 5 m height) shall be provided. However, any other suitable arrangement/ plan for chemical house can also be adopted with min. area requirement to be fulfilled as per the following formula or as specified in the contract; which ever is more

Alum & Lime storage area (A) of required size in SQM shall be as follows-		
A= 1.2 X (D_A X MLD of plant + D_L x MLD of plant) x 90	SQM	
Alum density x 2 (ht)		
Where, D_A is rate of Alum dosing in PPM, D_L is rate of lime dosing in PPM.		

16.1. **Equipment for Handling Chemicals**

The Platform type-weighting machine of capacity 1.0 tonne for weighing the chemicals shall be provided. For transportation of chemicals from storeroom to the solution tank, an electric operated hoist of 2.00 tonne capacity of approved make shall be provided. The chemical should be loaded in the steel tray attached to wire rope.

16.2 **Solution Tank**

The solution tank will be designed for the worst condition of turbidity for alum dosing to the designed flow. The minimum number of solution tank shall be two so that one will be standby unit. The capacity of tank shall be designed for 8 Hrs. capacity with 0.30m free board. The feeding arrangement shall be automatic based on flow. It shall be designed for 10% strength of solution. The solution tank shall be constructed in R.C.C. minimum M-30 grade and inside surface shall be lined with acid resistance material like F.R.P. or epoxy resin. The lime solution tank shall be designed for a dose of 5% strength of lime solution using 87% pure hydrated lime for the design flow for the period of 8 hrs. Each tank shall have platform at least 0.75m wide to allow the worker to stand for handling the chemicals and preparation of solution the platform shall have railing up to a minimum height of 0.75m. The height of the solution tank shall not be more than 1.50m from the first floor to the platform. A M.S. ladder 0.60m wide shall be provided and it shall be provided from solution tank platform to top of solution tank.

16.3 **Dissolving Trays**

The weighted chemical shall be placed into the SS tray. These trays shall be made up of SS with perforations both at sides and at the bottom. The weight of these trays shall be such so as to handle easily by the workers.

16.4 **Chemical Feed Devices**

The solution feed device will depend upon the point of application. The Constant head box/ pump type dosing system shall be provided. Chemical feeder in which the solution from the chemical solution tank shall be flown by pump through a strainer and through the float valve into the orifice box. It shall include the necessary piping arrangement for drains, overflow, and delivery pipe as per I.S. standard. It shall also have necessary agitating arrangement coupled with motor as per standard.

16.5 **Chlorination**

Arrangement for Pre & Post chlorination shall be made. The chlorinator shall be designed for a dosing of maximum 5mg/l. It shall be designed for designed water flow. The chlorinator shall be vacuum type chlorinator with 100% standby. If any change in requirement of chlorine comes the firm may suggest and quote the rate accordingly, the liquid chlorine shall be supplied in cylinders/ toners. The contractor shall have to make arrangement for Brand new chlorine toners with a nominal capacity of holding 900kg of liquid chlorine for minimum one months requirement. The toners should be as per relevant IS standard specifications. The contractor should also submit the required test certificate and other certificate to enable these cylinders to be put into the use. These toners shall be taken over by the department only after these have worked up to the satisfaction of the department after the O&M period. The chlorinator shall be fixed up by the firm with all required accessories. Due to corrosive nature of chlorine, it should be conveyed through either heavy wrought or steel pipe on flexible annealed copper to be tested for 35kg/cm2 working pressure. The long pipeline shall be avoided. The gasket used shall be made of Antimony Lead (with 2 to 3% Antimony) or asbestos sheet. Screwed fitting shall be forged steel construction.

Pressure indicators shall have Teflon diaphragms or silver foil protectors. Pressure reducing valves shall be of bronze or metal with a Teflon diaphragm.

- The dosing rate shall be manually set, and each chlorinator shall be equipped with a 0 to 5 mg/l scale and a manual dose setter over the complete range.
- Mal-operation of the duty chlorination system shall be indicated in the chlorination room and the monitoring room in order that manual changeover to the standby system can be initiated.

17. **Chlorine House**

It shall be situated in an isolated place and near to the chlorine feeding place in order to avoid the long piping. The ventilation shall be provided at the bottom of the floor. It shall be well lighted. In the proposed chlorinator an auxiliary water system shall be provided. The structure shall be R.C.C. framed with masonry panels in cement mortar 1:6 and plastered in cement mortar 1:4. An exhaust at sufficient height from bottom of 300mm diameter shall also be provided. The ventilator shall be of aluminium Z section fixed with 4 mm thick glass in order to avoid the corrosion; it shall be opened outside, a ramp on the main door of 2.0m wide to connect the G.L. to the plinth level of chlorine house shall be provided.

- 100% Standby shall be provided.
- 100 kg chlorine cylinder for capacity upto 5 mld and chlorine tonner for capacity above 5 mld.
- Gaseous operated chlorinators for Disinfection 1 working +1 standby for Pre & post chlorination with max. dosing up to 2PPM and 3PPM of CL2 respectively 30 days storage.
- Minimum area for chlorinator room shall be 12 Sqm or as per the actual requirement looking to the numbers of chlorinator..

17.1. **Room To Store Chlorine Containers**

The capacity of the room shall be to store the minimum chlorine toners for one months looking to the requirement of 5mg/l average dose. The minimum space of room shall be provided with rails and trolley as per requirement. It shall be constructed at an isolated place near to the chlorine house for the chlorinator. It shall be constructed in R.C.C. M-25 concrete framed structure and shall have same specification as for chlorine room. Suitable number of exhaust fans of 300mm dia of the standard make painted with anticorrosive paint shall be provided.

Tonner storage area of required size shall be as follows-

B= 5m min ,

L = ((0.8 x N)+((N-1)x0.60) + 1+3.5) meter.

Where B= width of tonner room, L= Length of tonner room and N= Number of tonners of capacity 900 kg required as per MLD wise consumption of chlorine.

17.2 **Tank For Neutralisation Of Chlorine**

A solution tank of minimum size 3.0 x 3.0 x 2.0m shall be provided. It shall be of R.C.C. with epoxy painted on all sidewalls and floor. It should be very near to the chlorine and storage room and shall have easy access without obstruction.

17.3 **Emergency Kit**

It shall consist of various tools appliances like gasket, Yokes Studs. Tie rods, hoods, clamps, spanners, mild steel channel kits, screws pins etc. complete. It shall cover the total precautionary arrangement parts, which shall be required at the time of chlorine leakage. All the Gadgets shall be designed for using in controlling or stopping the leakages from valves, fusible plugs, and sidewalls of cylinder used for handling chlorine.

17.4 **Fire Extinguishing - Arrangement**

Suitable fire extinguisers at least 2 nos or as per requirement shall be provided and placed at different position in pump house and chlorine storage room besides buckets filled with sand and placed at different places in clear water control chamber, chlorine room pump house shall also be provided. Additionally

17.5 **Water Sampling**

Continuous comparison of water samples is required, and the equipment provided shall include a suitable drained bench containing two clarity bowls complete with all interconnecting pipe work, automatic sampling pumps and drainage facilities for samples drawn from:

Raw water at the inlet works;

The bench shall be sited in the reception area of the administration building. In addition, a suitable sampling arrangement shall be included in the laboratory to enable samples to be obtained as follows:

a) Raw water at the inlet works;
b) Clear water reservoir outlet;

The sampling arrangement shall include all interconnecting pipe work, and taps, adequate sink and drainage, all incorporated into a satisfactory sample bench.

In addition, convenient means shall be provided to obtain samples manually from each filter bed outlet and sample cocks shall be provided locally at the clear water reservoir outlet.

17.6 **Safety Equipment**

1) Self-contained air-breathing apparatus with gas mask 6 numbers.

2) Gas leakage detector orthotolodine impregnated paper type leak detection system eight numbers.

   Two chlorine gas leak detectors shall be supplied and installed, each with a single, detector cell, one for the drum tonner room, and one for the chlorination room to alarm in the event of a chlorine leak.

   The chlorine leak detectors in the drum room shall be mounted at either end of the drum room. The chlorine leak detectors shall have two adjustable alarm levels sensitive to chlorine concentrations above 1ml/m³, and the range of adjustment of alarms shall facilitate selection of the following alarms:

   - low level - 2ml/m³
   - high level - 4ml/m³

   The low-level alarm shall initiate local audible and visual alarms. The high-level alarm shall initiate local audible and visual alarms, audible and visual alarms outside the buildings, alarms in the monitoring room.

3) Compressed air cylinder recharging facilities comprising of 40 litres capacity cylinder with recharging kit to refill service cylinder or breathing apparatus - 2No.

4) Protective Clothing - Rubber & P.V.C. clothing 2 No.

5) First Aid Facility - Emergency Oxygen Kit 2 No. complete for artificial respiration to neutralize Inhaled chlorine effect - 4 No. First Aid printed chart. It shall be mounted on a glass framed wooden board in chlorine room.

6) Emergency safety showers & eye wash- the contractor shall provide two safety showers and eye bath units. These units shall be installed at location approved by Engineer-in-charge.

17.7 . **Unloading Arrangement**

An arrangement with chain pulley block of 2 tonne capacity at a height of minimum 5.00m supported on M.S. beam section. It shall be provided at the entrance of chlorine storage room to unload the toners.

18. **Clear Water Sump**

The clear water from the module chamber shall be conveyed to the clear water sumps either through pipe or channel desig1 ed for a velocity 0.9 to 1.8m/s. The selection of pipe or channel shall be made on the basis of economy. The pipe if openly laid according to level shall be D.I. class K-7 Tyton joints and if covered then RCC. If the channel is provided, then it should be of RCC M-30 grade and covered with the RCC cover slab and it shall be in horseshoe shape as far as possible. The levels shall be so fixed up so that the tank up to water depth shall be totally underground. The tank shall be designed for a capacity for a minimum detention period of 45-60 minutes or as mentioned in BOQ. A minimum free board of 0.5m should be provided below the roof beam. The shape of tank may be circular or rectangular as per design and economy consideration. It should be constructed in RCC with a minimum M- 30 grade. The sump should be covered at top with RCC slab. The top slab shall

have adequate number of manhole chambers of size 0.9 x 1.20m in each compartment fixed with M.S. cover and frame painted with primer & anticorrosive paint and locking arrangement. There should be 100mm diameter C.I. or M.S. ventilators painted with primer and anticorrosive paint with C.I. cowls in each quadrant of beams on the roof shall be provided. The top of roof shall be sloped outward to drain rainwater easily. An aluminium ladder 0.45m wide in one manhole of both chambers shall be provided by contractor. The inlet and outlet pipes shall be located at diagonally opposite end in order to minimize the short-circuiting and turbulence effect. An overflow pipe designed to maintain level shall be provided. The scour pipe shall be laid from bottom of clear water sump to nearest drain.

All the pipes shall be fitted with sluice valve as per I.S. standard and wherever necessary provided with inspection chambers.

A locally mounted dial type float operated level indicator to indicate the water level shall be provided along with automatic level indication in control room.

18.1 **Clear Water Pump House, office, panel room, store room.**

It shall have sufficient space to accommodate required numbers of pumps for conveying water to OHTs and IPS. There shall be one pump house of suitable size and the minimum height shall be provided in order to accommodate the function of Mono rail/ Gantry. The elevation of building should be such as to give an architectural view. Sufficient minimum space between the two pumps shall be provided for circulation and the distance of pump from the sidewall should also be minimum 2.00m. The distance of pumps from rear wall shall be decided to accommodate the suction pipe, NRV, sluice valve to each pump individually. Space for LT/ HT ACB/ OCB shall also be provided for generator set if used in emergency. The floor of pump house and all other units shall be mosaic/ ironite laid over 1:2:4 cement concrete 40mm thick. The main gate shall be of size 3.00 x 3.50m rolling shutter. Office room, L.T. switch gear room and storeroom shall be with Aluminum door as per I.S. specification including door stoppers, tower bolts etc. complete. The door of the W.C. block shall be either M.S. Fabricated or flush doors. The window in pump house shall be about 20% of total area provided in each panel (except where doors are provided). The window shall also be the Z-Section/ Aluminum fitted with 4mm thick glass, tower bolts and all other necessary arrangements as per I.S. specification. Suitable number of exhaust fan of 450 mm diameter shall be provided. A ramp shall be provided to connect the ground level to the floor of pump house in main room only and in other doors and in depressed floor of pump house wherever necessary steps shall be provided. The minimum plinth height of pump house shall be 0.60m. It shall provide easy loading and unloading of pumps / motors from trucks.

Pump house of required size over the sump or by the side shall be of following size;
B= 4m Min. up to 10MLD and
for above, B= (Q+2.0+space for accomodation of valves & header pipe) M
L= (PxN)+(Nx0.75) +2m Clearance from both sides.
Where, B- Width of Pump House, L- Length of Pump house, P- length of pump foundation, Q- Width of Pump foundation, N- nos. of pump.

Pump Electrical Panel room, Maintenance way, office, store etc. at Pump House.
Minimum area shall be as follow or as specified in contract,

Sr. Nno.	Plant Capacity	Area of office room (M2)	Area of panel room (M2)	Area of store room (M2)	Area of maintenance way (M2)	Total area (M2)
1	Up to 5 MLD	15	10	15	10	50
2	Above 5 MLD & up to 10 MLD	20	15	15	10	60
3	Above 10 MLD & up to 20 MLD	25	20	20	15	80
4	Above 20 MLD & up to 50 MLD	30	25	25	20	100
5	Above 50 MLD & up to 100 MLD	30	30	25	30	115

Office can also be placed in any other suitable location in place of Pump House.

18.2 . **Laboratory Block and Control Room**

The laboratory provided to put up the instruments and in one corners of wall wash basins as per I.S. Specification shall be fitted with water supply connection. It shall have two number exhaust fan of 300 mm diameter. Below the platform an almirah for storing the chemicals, fitted with wooden/ Aluminum frames and door shall be provided. If the laboratory block is provided isolated to the module chamber in other building than a water sampling table, for visual examination and collection of raw, settled filtered and chlorinated water shall be provided in laboratory room. The testing platform and stands shall be cladded with acid resistance tiles.

Min. area of control room – 15m2 or as per actual requirement.

18.3. **Site Drainage**

The site drainage system shall be designed to dispose of overflow from tanks and rainwater in a manner to prevent damage to any structure. The drainage may use pipelines, culverts, conduits or open channels to carry the water to a safe disposal site leading to nearby water body up to 100m. Open channels shall be lined, and pipelines shall be of RCC-NP2, grades.

18.4. **Landscaping**

The campus of WTP shall be designed keeping a green belt on the periphery of the WTP preferable with trees (of local species) planted inside the boundary/ compound wall. Other open areas within the WTP campus shall be maintained with green grass, shrubs and plantation of plants (flowers, herbs, etc.) to maintain a pleasing appearance. The green area shall be maintained in prime condition and failure to do so may invoke penalty as decided by Engineer- in-Charge.

18.5. **External Lighting**

External lighting shall be employed throughout the site to illuminate all site roads, turning areas, car parks, paths, tanks and building perimeters.

The following average levels of illumination are required :

- Car Parking area: 5 lux
- Access Roads, Pedestrian Walkways: 10 lux
- Lorry Loading and turning Areas: 20 lux
- Top of Tanks: 50 lux
- Building Perimeters: 50 lux

The access road lighting scheme shall be designed in accordance with the requirements of BS 5489: Part 3: 1989 (group B5/6). Column heights shall be 5 meters and each lantern shall incorporate a photocell for control. At each position where a section of road lighting columns is fed a selector switch shall be provided having the following functions:

- ON - Access road lighting permanently
- ON/ OFF - Access road lighting permanently
- OFF AUTO - Access road lighting under photocell control

Some flexibility in positioning of road lighting units is allowed but generally units shall be positioned at intersections and junctions with spacing not exceeding 30 metres + 10%.

5.0 Vertical Turbine Pumps for Raw / Clear water pumping

Design, manufacturer, supply, erection, testing commissioning of Vertical turbine pumping sets for pumping clear water including all electrical, mechanical equipment's accessories and civil works viz., foundation of pump, motor, cable, pipes, fittings for suction, delivery butterfly valves, reflex valve, specials etc., complete as per details given in this NIT. The Pumps shall be energy efficient.

The rising main shall be equipped with the specially designed surge protection valves to be installed at the tail end of the pump without interfering with the flow and for which no extra payment shall be made. The surge protection system should be supported with the surge analysis of the complete rising main.

Housing based tampered proof air valves to be used for which no extra payment shall be made.

1.0 DETAILED SPECIFICATIONS OF PUMPS SETS:

The pumps shall be vertical turbine wet pit type and non pull out design with single / multistage bowl assembly directly coupled with vertical Solid shaft motors. The pumps should be of standard make only.

1.1 GENERAL SPECIFICATIONS

The pumps shall be forced water/ Self water lubricated complete with bowl assembly, column pipe such floor discharge head, line shaft, foundation plate/sole plate, basket strainer, motor foot stool and all necessary accessories. The pump shall be designed so as to have a maximum flow capacity not less than 110% of the rated flow capacity. The pumps shall also be designed for continuously operation at any point of head capacity curve between 90% and 110% or higher of pump rated flow, without undue vibration or overheating and thrust bearing should be antifriction type point head.

A. BOWLS:

The bowl shall be made of close grained cast iron up to 65 mtr pump head above that it should be of Cast Steel ASTM A216 grade WCB material, smoothly finished and free any casting defects. The bowls shall be capable of withstanding hydrostatic pressure equal to twice the pressure at rates capacity or 1.5 times of the shut off head whichever is greater. The water passage in the bowls shall be smooth finished

B. IMPELLERS:

Impellers shall be closed/ semi open type made of cf8m/ss statically and dynamically balanced. Impellers shall be free from any casting defect and shall be properly machined. All the water passage shall be smooth finished. The impellers shall be secured to shaft with tapered lock collect or key & split rings.

C. IMPLLER SHAFT:

The impeller shaft shall be of stainless steel with renewable stainless-steel sleeves at bearing portion. The impeller shaft shall be guided by bearing provided in each bowl. The butting faces of the shaft shall be machined surer to the axis and the shaft shall be chamfered at the edged. The shaft shall have a surface finish of 0.75 micron as per IS: 3078/1967.

D. LINE SHAFT :

The line shaft shall be made of Stainless steel. The shafts shall be furnished with interchangeable suctions having a length of 1.25 M / 2.5 M / 3M. The butting faces of shaft shall be machined square to shaft axis and the shaft ends shall be chamfered on the edges. To ensure to correct alignment of shafts they shall be perfectly straight.

The shaft shall not have the surface roughness more than 0.75 microns as per IS: 3073/1967. The shaft shall have the adequate strength to withstand all the forces at + 10% of the critical speed of shaft.

E. COLUMN PIPE :

Column pipe shall be manufactured from the heavy class M.S. pipe confirming to relevant India Standard Specification MS IS :2062 Fabrication. The column pipe shall not exceed 3.0 meters. in length & made of 6 mm thick sheet the dia. of bolts to be used in flange joints should not be less than 20mm.

F. LINE SHAFT BEARING :

Line shaft bearings shall be designed to be lubricated by forced water. line shaft bearing shall be cut less Nitrile rubber lined bearing with bronze shell.

G. LUBRICATION :

The pumps are forced water lubricated.

H. DISCHARGE HEAD :

The discharge head shall be of fabricated MS IS:2062 Fabrication standard

I. STUFFING BOX :

A packing gland shall be provided at the top of stuffing box. Shaft sleeves shall be provided on the top shaft. The stuffing box shall be of sufficient depth to permit adequate packing. The space between the pump motor main coupling and the stuffing box shall be sufficient to permit removal of packing gland and insertion of new packing without dismantling the pump.

J. MOTOR STOOL :

The motor stool shall be of fabricated mild steel shall be designed to take care of all static and dynamic loads on it.

K. PRESSURE INDICATION DEVICES :

Each pump shall be provided with pressure gauge of best quality makes to give indications of delivery pressure. The pressure gauges shall be of Borden type, dial size 150mm.

L. LABORATORY TEST :

Laboratory pump test shall be carried out as per IS: 9137 / ISO 9906 Gr 2B for each pump to assess the pump discharge V/s head, horse power and efficiency figures. The pump shall be subjected to a test pressure of 1.5 times of the shut of pressure or twice the working (rated) pressure which ever higher.

M. FIELD TEST :

The field test shall be carried out as per IS : 1710 and 5126.

The successful BIDDER will ensure the Engineer-in-charge of work or any other Senior Engineer nominated by the COMMISSIONER / CHIEF MUNICIPAL OFFICER, MUNICIPAL CORPORATION /NAGAR PARISHAD -------- will inspect and witness tests conducted on the pumps and motors at manufacturer's place with respect to their characteristic and performance as specified by the department.

1.2 GUARANTEED PERFORMANCE & TECHNICAL PARTICULARS:

The contractor shall submit the details of guaranteed performance & technical particulars as desired in the Performa enclosed vide schedule's with the TENDER along with the preliminary out line drawing indicating principal dimensions & weight of pumping equipment's and cross section drawing indicating the assembly of pumps & manor parts thereof with materials of constructions and special features. Complete descriptive and illustrated literature on the equipment and accessories offered.

1.3 SPECIAL NOTES FOR BIDDER:

1.3.1 Pump should be capable of throttling. The throttle point discharge should be quoted. The extent to which pump can be throttle and the corresponding discharge to be mentioned.

1.3.2 Duty point discharge should be specifically mentioned along with the head at which the same will be attained. Efficiency at different operating heads and discharge should be mentioned.

1.3.3 The accessories like surface, discharge head/underground discharge head with elbow, prelub tank with fittings, motor stand with NRR and thrust bearing housing, column assembly bowl, assembly, basket strainer, as per specifications shall be quoted, individually. The drawing to the scale showing the proposed arrangements for the pumps and the positions of various parts with detail drawing must accompany the TENDERs along with detailed specification, make, guarantee period etc.

1.3.4 Head capacity curve shall be rising type and shut off head must be higher than the maximum operating head for paralleled operations of pumps. Performance covers for each individual pump sets and also for the four pump

sets working in parallel should be given for full load and for throttled conditions also. All pumps to be of identical in all respect.

2.0 SPECIFICATIONS FOR 415-V INDUCTION MOTORS :

2.1 TYPE :.

The motor shall be vertical hollow shaft squirrel cage type induction motors suitable to operate on 415 V/6.6KVA, 3 Phase, 50 Cycle A.C. Supply at 1500 RPM directly coupled with vertical turbine pump. The motor shall generally confirm to latest revision of IS : 325/1978 and other relevant ISS. Duly fitted with space heater and RTDs & BTDs. The motor shall be of reputed manufacturer.

2.2 VARIATION IN SUPPLY VOLTAGE :

The motors shall be capable of delivering rated output and rated power factor with following variations :

Voltage	:	± 10%
Frequency	:	± 5%
Combined	:	As per IS 325

2.3 RATED CAPACITY:

The minimum conditions rated capacity of motors shall be such that it meets the power requirements of pumps in the complete range of its operation. It shall also provide on additional power requirement on the motor. By 5% at the maximum power requirement or by 10% at the duty point of operation whichever is maximum. The contractor shall ascertain the K.W. requirement and provide the motors of suitable capacity.

2.4 ACCELERATION CHARACTERISTICS:

The acceleration characteristics of motor shall be matched with the driven equipment so that acceleration is obtained without over heating of motor.

2.5 METHOD OF STARTING:

The motors shall be designed for star/delta/soft starting at full voltage with starting current not exceeding four times the rated full load current. The motor shall also be designed for a minimum pull out torque of 200%.

2.6 CONDITION OF START:

Motor when started with the drive imposing its full starting torque under the specified supply voltage variation shall be capable of withstanding at least one successive starts from hot condition to start from cold condition without damage to the winding.

2.7 CLASS OF INSULATION:

The motor winding shall be provided with insulation conforming to thermal class "F". The maximum temperature rise of the winding shall not exceed the limits specified from class "B" insulation. The insulation can be given tropical and fungicidal treatment for successful operation of motor in hot humid tropical climate. It shall of thermos setting type and shall remain unaffected by heat. The coils shall be highly uniform with uniform insulation strength and uniform dielectric losses.

2.8 MOTOR CONSTRUCTION:

The motor construction shall be suitable for easy dismantling and reassemble at site with the help of simple over head crane. The motor shall be of core pack construction attached to the stator frame to facilitate easy removal and replacement of the winding for maintenance purpose. The over head for winding at both ends of the core shall be accessible for usual inspection without resorting to major dismantling.

2.9 MOTOR FRAME:

Motor frames shall be rigid fabricated steel they shall be suitably annealed to eliminate any residual stress introduced during process of fabrication and machining.

2.10 STARTOR LAMINATIONS:

Stator laminations shall be made from suitable grade sheet steel varnished on inner side and shall be adequately designed to overheating during starting and running conditions stipulated above.

2.11 ROTOR SHORT CIRCUITING RINGS:

Rotor short circuiting and rings shall be such that it is free to move with expansion of bars without distortion. The connections of the bars to the end rings shall be made by bracing.

2.12 LOCKING ROTOR WITH STAND TIME:

Locked rotor with stand time under hot conditions at 110% voltage shall be more than starting time at minimum permissible voltage by at least two seconds.

2.13 TYPE OF ENCLOSURE & DEGREE OF PROTECTION:

The degree of protection provided by the enclosures of motor shall conform to IS: 4691. the enclosure for the motors shall be screen protected Drip Proof (SPDP) IP 23.

2.14 SHAFT INSULATION:

Suitable insulation shall be provided on shaft/bearing house to prevent shaft current. The insulation provided shall be such that it shall retain its dielectrically properties even after its handled for number of times during dismantling and reassemble.

2.15 BEARING ASSEMBLY:

Bearing assembly shall be such that it prevents dust and water from getting to the bearing. Further, bearing lubricant shall not find access to the motor winding. The bearing assembly shall be provided with proper lubricating nipples.

2.16 EARTHING:

The motor body shall have two separate earthing terminals for earthing in compliance with I.E. Rules.

2.17 DIMENSIONS OF MOTORS:

Motors shall be properly dimensioned to have greater stability and low vibration limit.

2.18 TESTING

All the motors shall be routine tested at manufacturers workshop and test certificate shall be provided with motors.

(Bidder can suggest other options of pumps in the light of less Electrical expenditure and better efficiency without compromising on head and discharge. The change shall have to be approved by the Competent authority. However no separate cost shall be payable for the same.)

6.0 HORIZONTAL SPLIT CASING CENTRIFUGAL PUMPS FOR CLEAR WATER PUMPING

1.1 GENERAL DESIGN CONDITIONS

The pumps shall be high head single /double suction horizontally split casing type based on system study requirement to facilitate easy inspection & maintenance. The pumps shall be energy efficient WITH I3 standarads of motors. The pumps shall be designed to operate satisfactorily while handling a minimum suction lift from all causes nits of pump sets of combined capacity and total Head for pumping clear water from WTP as per the requirement of proposed works

The pump shall be horizontally split with the suction and delivery branches cast INLINE on the bottom half of the casing. The top half should be constructed to allow easy dismantling. There by providing the facility of inspection and repair to the equipment without any difficulty.

The rotating elements of pumps will be dynamically balanced and over stressing should not occur due to sudden failure of power, Reverse rotation should not damage the pumps.

The pumps shall be so designed as to have a maximum flow capacity of not less than 110% of the rated flow capacity.

The pumps shall be designed for continuous operation at any point of head capacity curve between 60% & 120% or 60%-120% of pump rates flow without under vibration or overheating.

The pumps shall be so designed as to have a stable non overloading characteristics, capacity head curve shall be continuously from shut-off point to operating point and shall be suitable for parallel operation of pumps without any haunting possibility. The shut-off head should not exceed 120% of duty point head.

The impeller adjustment shall be designed in such a way that impellers run free in any installed condition.

1.2 GENERAL SPECIFICATION

The pump shall be complete with suction pipe, base plate and all other necessary accessories.

The pumps should generally comply with the requirement of following standard.

1. IS 1520-1972/ANSI : Horizontal Centrifugal Pumps for clear, cold and fresh water.
2. IS 5120-1968/ANSI : Technical requirement of Rotor Dynamic special purpose pumps.

1.3 PUMP CASING

The pump in cast iron Stainless Steel CF8 (304 equivalent) fitted construction specifically designed and guaranteed for quiet operation. Suitable standard operations at 225°F (107°C) and 175 psig (12 BAR) working pressure or optional operations at up to 250°F (121°C) and 280 psig (19 BAR) or 400 psig (27 BAR) working pressures. Working pressures shall not be de-rated at temperatures up to 250°F (121°C). The pump internals shall be capable of being serviced without disturbing piping connections or electrical motor connections.

Pump volute shall be of a cast iron (rated for 175 psig [12 BAR] Max WP) or ductile iron (rated for 280 psig [19 BAR] or 400 psig [27 BAR] Max WP) axially-split design with flanges (175 psig [12 BAR] drilled for125 PSI ANSI companion flanges or optional 280 psig [19 BAR] and 400 psig [27 BAR] working pressures are drilled 250 PSI flange drilled) and mounting feet integral cast into the bottom half of the casing. Suction and discharge flanges shall be on a common centerline in both the horizontal and vertical planes, and the volute shall include Bronze Casing Wear Rings, priming port, gauge ports at nozzles, and vent and drain ports. The upper half casing shall be capable of being removed without disturbing piping connections or electrical motor connections.

Pump volute shall be of a Stainless steel CF8M (316) (rated for 175 psig [12 BAR] Max WP) or duplex/super duplex (rated for 280 psig [19 BAR] or 400 psig [27 BAR] Max WP) axially-split design with flanges (175 psig [12 BAR] drilled for125 PSI ANSI companion flanges or optional 280 psig [19 BAR] and 400 psig [27 BAR] working pressures are drilled 250 PSI flange drilled) and mounting feet integral cast into the bottom half of the casing. Suction and discharge flanges shall be on a common centerline in both the horizontal and vertical planes, and the volute shall include AISI 316 casing wear rings or Duplex casing wear rings or Super Duplex Wear Rings, priming port, gauge ports at nozzles, and vent and drain ports. The upper half casing shall be capable of being removed without disturbing piping connections or electrical motor connections.

1.4 IMPELLERS

Impeller shall be of the enclosed double suction type made of Stainless Steel CF8 (304) /conforming to relevant IS or with suitable coating both hydraulically and dynamically balanced to ANSI/HI 1.1-1.5-1994, section 1.4.6.1.3.1, figure 1.106, balance grade G6.3 keyed to the shaft and fixed in the axial position. Impeller shall be fitted with 2 wear ring made of the same material, easily replaceable

1.5 PUMP SHAFT

The impeller shaft shall be a Carbon Steel shaft protected by a Stainless steel (304) shaft sleeve/ conforming to relevant IS. The shaft sleeve shall protect completely the shaft from the fluid handled by the pump.

1.6 INTERNAL PARTS

Pump shall be equipped with a pair of externally flushed mechanical seal assemblies over the shaft sleeve Or as per relevant IS. Seal assemblies shall be a MR2 arrangement specifically design for the pump having a stainless steel housing, EPDM bellows and seat gasket, stainless steel spring, and be of a carbon-Silicon carbide design with the carbon face rotating against a stationary silicon carbide face.

1.7 PUMP BEARING

A bearing housing shall supply support for a pair of heavy-duty regreaseable ball bearings. An inboard single row bearing will absorb thermal expansive forces while an outboard single row bearing will be clamped in place to absorb both radial and thrust loads and keep the rotating element in proper axial alignment. Bearings shall be replaceable without disturbing the system piping and shall be regreaseable without removal of the bearings from the bearing housing. The bearings could be oil lubricated, and might be fitted with a constant oil level system. The bearing housing is fitted with bearing temperature and 2 dimension vibration sensors port. The temperature sensor should be able to measure the temperature at the skin of the bearing.

1.8 PUMP COUPLING

A center drop-out type coupling, capable of absorbing torsional vibration, shall be employed between the pump and motor. On variable speed applications the coupler sleeve should be constructed of an EPDM material to maximize performance life.

The coupling shall be shielded by a dual rated ANSI B15.1, Section 8 and OSHA 1910.219 compliant coupling guard, orange painted and completely protecting all rotating elements for the coupling.

Pump seal flushing lines shall be mounted on the upper half pump casing. Sealing from an external source shall be possible for lubrication and/or cooling.

1.9 SOLE/BASE PLATE

Base plate shall be of structural steel or fabricated steel channel with fully enclosed sides and ends, and securely welded cross members fabricated as per IS 226/ANSI. Grouting area shall be fully open. The combined pump and motor base plate shall be sufficiently stiff as to limit the susceptibility of vibration. The minimum base plate stiffness shall conform to ANSI/HI 1.3.4-1997 for Horizontal Baseplate Design standards. The base plate shall be fitted with 4 jacking screws, in order to ease the motor alignment in both lateral and longitudinal direction.

1.10 PRESSURE INDICATION DEVICE

Pump and motors shall be factory aligned, and shall be realigned after installation by the manufacturer's representative. Motors shall be of electrical characteristics as scheduled, denoted and/or as indicated on the electrical plans and specifications. Pump characteristics shall be such that the head of the pump under varying conditions shall not exceed the rated horsepower of the drive motor.

1.11 BOLTS, NUTS & WASHERS

All bolts, nuts and washers shall be of superior quality conforming to relevant Indian Standard Specification.

1.12 INSPECTION & TESTING

All the inspection, examination and testing shall be carried out in accordance with relevant Standard Specification**.**

1. LABORATORY TEST

Laboratory pump test shall be carried out as per IS: 5120-1968 each pump to assess the pump discharge Vs head, horse power and efficiency figure. The pump casing shall be subjected to a pressure test of 1.5 times the working pressure at duty point.

2. FIELD TEST

The field test shall be carried out as per IS: 1520-1972 & IS: 5120-1968. These test May be witnessed by the Engineer-in-charge or his authorized representative if they desire. The tolerance as specified in relevant IS code of practice shall not be application on the efficiency & KW output. The contractor shall have to demonstrate the quoted efficiency of pump during testing.

1.13 GUARANTEES PERFORMANCE & TECHNICAL PARTICULARS

The contractor shall submit the details of guaranteed performance and technical particulars as desired in the performance enclosed vide schedule 'B' along with the TENDER & the preliminary out line drawing indicating principal dimension and weight of pumping equipment and cross-section, drawing indicating the assembly of pumps & major parts thereof with materials of construction and special features, complete descriptive and illustrated literature on the equipment and accessories offered.

The pump(s) selected shall conform to relevant IS/ ANSI/HI 9.6.3.1 standards for Preferred Operating Region (POR) unless otherwise approved by the engineer. The pump NPSH shall conform to the ANSI/HI 9.6.1-1997 standards for Centrifugal and Vertical Pumps for NPSH Margin.

The pump curves should comply with the ANSI/HI 14.6, grade 2B. The supplier should be able to provide curves under ANSI/HI 14.6, grade 1B without derating the pump performance and efficiency curves.

Pump rotation shall be right hand or left hand as viewed from the pump end.

The seismic capability of the pump shall allow it to withstand a horizontal load of 0.5g, excluding piping and/or fasteners used to anchor the pump to mounting pads or to the floor, without adversely affecting pump operation.

Each pump shall be factory hydrostatically tested per Hydraulic Institute standards and name-plated before shipment. It shall then be thoroughly cleaned and painted with at least one coat of high-grade paint prior to shipment.

PART - II

SPECIFICATION FOR 415 V INDUCTION MOTORS

2.1 TYPE

The motor shall be horizontal SOLID Totally Enclosed Fan-Cooled (TEFC) for clear water pumps suitable to operate on 415 V/6.6KV, 3 phase, 50 Hz. AC supply (with allowable variation of 10%) directly coupled with pumps. The rotations of clear water pumps shall be 1500 RPM. The motor rating and efficiency (IE3) generally conform to latest revision of IS 12615 and other relevant I.S.

2.2 VARIATION IN SUPPLY VOLTAGE

The motors shall be capable of deliver integrated output and rated power factor with following variations:

VOLTAGE	:	±10%
FREQUENCY	:	±5%
COMBINED	:	±10%
PHASE IN BALANCE	:	±5%

2.3 RATES CAPACITY

The minimum continuous rated capacity of motors shall be such that it meets the power requirements of pumps in the complete range of its operations. It shall also provide additional power requirement in the motor by 5% at the maximum power requirement or by 10% at the duty point of operation whichever is maximum. The contractor shall ascertain the KW requirement and provide the motors of suitable capacity.

2.4 ACCELARATION CHARACTERISTICS

The accelerating characteristics of motor shall be matched with the driven equipment so that acceleration is obtained without over heating of motors.

2.5 METHOD OF STARTING

The motors shall be designed for star-delta/soft starting/VFD at full voltage with starting current not exceeding 2.0 to 2.5 times the rated full load current. The motor shall also be designed for a minimum pull out torque of 200%. S of motors shall be as follows:

Starter shall be aopted as follows:

1.	DOL STARTER	:	Up to 15 KW
2.	STAR DELTA STARTER	:	18.5 to 22 KW
3.	SOFT STARTER	:	22 KW to 200 KW (Electronic Soft Starter)
4.	FCMA / HFSR SOFT STARTER (With by pass arrangement)	:	200 KW and Above Rating
5.	VFD	:	As per design / process requirement.

2.6 NUMBER OF START

Motors when started with the drive imposing its full starting torque under the specified supply voltage variation shall be capable of withstanding at least two successive starts from hot condition and one start from cold condition without damage to the winding.

2.7 CLASS OF INSULATION

The motor winding shall be provided with insulation conforming to thermal class F. The maximum temperature rise of the winding shall not exceed the limits specified for class 'B' insulation. The insulation shall be given tropical and fungicidal treatment for successful operation of motor in hot, humid tropical climate. It shall be of thermo-setting type and shall remain unaffected by heat. The coils shall be highly uniform with uniform insulation strength and uniform dielectric lose. The dielectric losses shall be low and the star delta measurement should be not exceed 1% at 440V.

2.8 MOTOR CONSTRUCTION

The motor construction shall be suitable for easy dismantling and reassembly at site with the help of simple over head crane. The motor shall be of core pack construction attached to the stator frame to facilitate easy removal and replacement of the winding for maintenance purpose. The over head for winding at both ends of the core shall be accessible for usual inspection without resorting to major dismantling.

2.9 MOTOR FRAME

Motor frame shall be of rigid casted steel. They shall be suitably annealed to eliminate up any residual stresses introduced during process of fabrication and machining.

2.10 STATOR LAMINATION

Stator laminations shall be made of suitable grade sheet varnished on either side and shall be adequately designed to avoid over heating during the starting and running conditions stipulated above.

2.11 ROTOR

Rotor should be desisted dynamically balanced and having carbon steel shaft hydraulically fitted.

2.12 LOCKED ROTOR WITH STAND TIME

Locked rotor with stand time under hot conditions at 110% voltage shall be more than starting time at minimum permissible voltage by at least two seconds.

2.13 TYPE OF ENCLOSURE & DEGREE OF PROTECTIONS

The degree of protection provided by the enclosures of motor shall conform to IS : 4691. The enclosure for the motors shall be totally Enclosed Fan Cooled (TEFC) type, having of protection IP. 55.

2.14 SHAFT INSULATION

Suitable insulation shall be provided on shaft bearing housing to prevent shaft current. The insulation provided shall be such that it shall retain its dialectical properties even after its handling for number of times during dismantling and reassembly.

2.15 BEARING ASSEMBLY

Bearing assembly shall be such that it prevents dust and water from getting into the bearing. Further, bearing lubricant shall not find access to the motor winding. The bearing assembly shall be provided with proper lubricating nipples.

2.16 EARTHING

The motor body shall have two separate earthing terminals for earthing in compliance with I.E. RULES.

2.17 TERMINAL BOXES

Separate terminal boxes shall be provided for main-Terminals of the motors and for R.T.D. and for space heaters. The terminals box for main terminals of motor shall be segregated type suitable for 3 core. 440 V. Aluminum conductor PVC insulated armored cables. The terminal boxes shall be spacious, dust & house proof designed and properly insulated. Adequate clearance should be given between live motor terminals and covers.

2.18 TEMPERATURE DETECTORS

Motors shall be provided with embedded temperature detectors, two for each phase winding at the location where the high temperatures May be expected in the stator winding. The temperature detectors shall also be provided in bearing assembly for monitoring the bearing temperature. The temperature detectors shall be connected with digital temperature scanners with alarm and trip points in the control panel.

2.19 ANTI - CONDENSATION HEATERS

Motors shall be have space heaters suitable for 240 V. single phase 50 Hz. A.C. supply, space heaters shall have adequate capacity to maintain motor internal temperature above due point to prevent moisture condensation on insulation during shut down periods.

2.20 DIMENSIONS OF MOTORS

Motors shall be properly dimensioned to have greater stability and low vibration limits. Mounting dimensions should confirm to IS : 2254.

2.21 COUPLING TO PUMPS

The motors shall be coupled to the pumps by means of polished steel shaft and flexible coupling. The size of line shaft and flexible coupling shall be calculated on the basis of maximum combined shear stress as per the relevant IS and shall not exceed 30% of the elastic limit in tension or 10% of ultimate tensile stress, shaft shall be designed taking into consideration that critical speed of the shaft which shall be higher than the operating or runaway speed.

2.22 DETAILS OF MOTORS TO BE FURNISHED

The TENDERs shall furnish along with their offer, the details of efficiency, total losses and power at different loads etc. as required in the form of guaranteed performance and technical particulars of motors in schedule 'B'.

2.23 TESTING

Routene test will done on all motors and one motor of each rating shall be done Type test at the supplier's workshop in the presence of Engineer-in-charge of work or his authorized representative or a third party inspection directed by Nagar Nigam / UADD.

PART - VI DELIVERY PIPES

6.1 GENERAL

The scope of the work is providing, laying & jointing of all delivery pipes, specials valves of all the pump and their connection to the common manifold as shown in drawing inclusive of the jointing of the manifolds to the respective conveyance mains fixing with one main butterfly valve.

1. The contractor shall design and fabricate the common manifold which would be made out of MS 6mm thick plates conforming to IS: 226 in such a way that it gives minimum frictional loss of head

to the flow of water and also avoid cavitations or vortices in the manifold. The manifolds should be in line and coated with suitable material to protect it from corrosion in case of clear water ultimate carrying capacity shall be ,,,,,,,,,,MLD.

2. The contractor shall provide an expansion joint of his own design duly approved by the Engineer-in-charge.
3. Flanged joints shall be adopted for valves and butt welding joints or flanged joints in jointing of delivery pipe with dismantling joints.
4. Concrete saddles for valves and thrust blocks shall be provided by the contractor as per the design and drawings to be furnished by the contractor and subsequently approved by Engineer-in-charge.

6.2 MATERIAL OF CONSTRUCTION

All the pipes of works shall be fabricated out of steel plates conforming to IS:226-1962. The fabrication of pipes shall generally conform to IS: 3589-1966.

6.3 TESTING

The pipes and common manifold shall be hydraulically tested to a pressure of 2 times the working pressure. Test certificate to that effect shall be furnished by the contractor.

The Engineer-in-charge witness the above test if so desired, the contractor shall arrange for such test in presence of Engineer-in-charge.

PART - VII RATING AND NAME PLATES

7.1 RATING PLATE

Each main and auxiliary item of plate shall have permanently attached to it a rating plates in a conspicuous position. This shall be a non-corrodible material preferably chromium plates steel. The inscription shall be engraved in black on the plate.

7.2 NAME PLATE

1. Each item of plant shall be provided with a name plate or label designating the service of the particular equipment. The shape and size of the plate and inscription shall be approved by the Engineer-in-charge.
2. Such name plate shall be non-corrodible material preferably chromium plated steel having engraved black lettering.
3. In case of indoor equipment like circuit breakers, starters etc. the plate shall be of transparent plastic material with black lettering engraved on the back.
4. The name plate shall be screwed to the body of the equipment.

7.0 GENERAL SPECIFIFACTION FOR SCADA & INSTRUMENTATION

1. **General**

SCADA system for water supply scheme include all Raw Water Source/Intake (existing & proposed), all WTPs (existing & proposed) to all overhead tanks (existing & proposed).
The SCADA system shall be designed in accordance with 'open system' standards, have a distributed architecture, provide facilities for smooth interfacing with existing and future control and monitoring schemes. The SCADA system shall be of complete and integrated design, facilities shall be provided for control, Supervision, Operation and Maintenance of the WTPs, ESRs and CWS. A high level of system automation shall be applied to the SCADA system in order to achieve minimal supervision and manual intervention.
This part covers the general specifications required for the design, supply, installation, inspection and testing for the instrumentation and controls hereinafter specified to perform the intended function. Work shall include all necessary materials, equipment, labor and services. In order to insure the inter changeability of parts, the maintenance of quality, the ease of communications between the various subsystems and the establishment of minimum standards of quality, all equipment manufacturers' standard offerings, shall be of the same product family of a single manufacturer. In order to issue this compatibility between all equipment, it shall be the responsibility of the Contractor to enforce this requirement with the equipment suppliers. Equipment shall be fabricated, assembled, installed and placed in proper operating condition in full conformity with these specifications, engineering data, instructions and recommendations of the equipment manufacturer. The Contractor shall engage a single system supplier, who shall provide all of the services, equipment and appurtenances required to achieve a fully integrated and operational instrumentation and control system.

Component	**Parameters to be Covered Under PLC SCADA**		**Instruments to be installed, monitored and reported**
	Monitoring	**Regulation/Control Operation**	
Intake Well	Electricity parameter -Current, Voltage, Power Factor, Energy consumption Other Parameters - Level, Flow & totalized flow, Discharge Pressure for each pumps & Pressure at delivery end	On/ Off Operation of Pumps	Electromagnetic Flow Meter, Pressure Sensor, level sensor & Multi-function meter (MFM)/Energy Meter, Actuator Valve, RTU
WTP	Raw Water Parameters - Flow of totalised value at inlet, pH & Turbidity Clear Water Parameters - pH, Turbidity, Residual Chlorine Level for CWR & Back wash tank	Complete Regulation of Filtration process including Backwash Arrangement, Clarifier Drain with actuator enabled for each line of Filter Beds, Automation of dosing pumps such as lime, alum dosing	LOH-Level Sensor (Filter Bed & Backwash), ROF (Raw & Clear Water), Turbidity, Chlorine & pH analyser (Raw & Clear Water), Filter bed console panel for each filter bed with HMI, Level Sensor for CWR & Back wash tank, level switch for lime & alum dosing tank, RTU, SCADA OWS at MCC room

Clear Water Sump cum Pump House & IPS	Electricity parameter Current, Voltage, Power Factor, Energy Other Parameters -Flow, Pressure, Residual Chlorine	On/ Off operation of Pumps	Electromagnetic Flow Meter, Pressure Sensor (individual pumps & common header), Level Sensor and RTU
ESR/GSR	Flowmeter, Water Level at container, residual chlorine, Pressure at inlet, Flow control valve at inlet	_	Electromagnetic Flow Meter, Level Sensor, Chlorine analyser, Pressure Transmitter and RTU
Note: 1. Final scope of PLC SCADA & Instrumentation shall remain same as detailed in the BOQ of bid document. 2. RTU/PLC Panel shall provide at each remote location with 4 hours' battery backup. 3. Master Control Center (MCC)/ SCADA control room shall located/constructed at WTP premises. 4. Internet connectivity charges & montholy rental and SIM charges & montholy rentel during O&M period shall be paid by contractor.			

Brief Scope of SCADA & Instrumentation

The key deliverables related to SCADA and I&C works in the Project are;

- Provision of PLC/RTU system at – each Intake, each Water Treatment Plant (WTP), each Pumping Stations and each ESR/GSR.
- Provision of SCADA licenced software, OPC licenced for 5 years, Historian servers Operator works station and Engineering Work station.
- PLC/RTU system with Control panel, required IOs and 7 inch HMI.
- Programming, HMI development of Works incorporating additional equipment and Control philosophy
- Energy Meter for electrical parameters such as Current, Voltage, Power Factor, Power consumtion etc.
- Quality (pH, Turbidity & Chlorine) and Quantity (ULT/RLT, Flowmeter, PT) instruments for Intake, WTP & ESR/GSR
- Actuator valves for Intake well common header, WTPs for different locations like each line of filter beds (filter bed inlet line, filter bed outlet line, backwash water line, air pressure line & sludge drain line), WTP outlet common header/CWPS outlet, etc.
- LOH meter-ULT, ROF meter and filter bed console panel with HMI for each filter bed of WTP
- Automation of service pumps such as lime dosing pumps, alum dosing pumps etc. through Level Switch.
- The Backwash and backwash pumps should be operated through SCADA.
- All the valves and gate of rapid sand filter bed shall be electrically operated from the filter control consoles panel and tagged to the PLC with SCADA on flow rate(RoF) & LoH meter basis.
- Configuration of regular reports like daily, monthly, yearly and ad-hoc reports using standard software for providing Management Information. Following information shall be reported
  - Power consumption
  - Pumping ON/OFF control and Status
  - Valve Actuator ON/OFF operation and Status
  - Equipment failures
  - Pump Condition Monitoring
  - Electrical parameters monitoring.
  - Instrument failures
  - Control System /communication failure/ Power failure at pump station.
  - Water consumption report
  - Others as per requirement to be defined during the Project.
- Install necessary Instrumentation in areas as per BOQ, cabling, JBs and all other works required for successful completion.
- Contactor shall provide API to UADD server & integrate the local SCADA system with UADD server.

5. Sign Board

The Contractor shall provide a sign board at the site of the works of approved size and design which provides (i) the name of the project (ii) the names and addresses of the Employer, the Contractor and the Consultant; (iii) the name and short description of the project, (iv) the amount of the Contract Price; and v) the starting and completion dates.

6. Protection of Utilities

The Contractor is required to carefully examine the location of the Works and their alignments and to make special enquiry's with all authorities concerning all utility lines such as water, sewers, gas pipe, telephone (underground and/or overhead) lines, electric cable (underground and/or overhead) lines, etc.; and to determine and verify to his own satisfaction the character, sizes, position and lengths of such utilities from authentic records. The Contractor shall be wholly responsible for the protection and/or facilitating relocation of such utilities as may be required, and shall not make any claim for extra work or extra time that may be required to protect or facilitate relocating such utilities. If any major shifting or realignment of water, sewers, gas pipes, electric and telephone lines is necessary due to their interference with the proposed Works, the same may be done by the Employer. The cost of such relocation will be borne by the Employer. All the electrical work shall be done by class – I licensed electrical contractor.

In case the alignment of the pipeline crosses the high tension electrical transmission lines belonging to the MPMKVV or other authorities, the Contractor shall take all precautions necessary to see that the work is carried out with care and safety, without disturbing such transmission lines. The Contractor will be responsible to carry out all construction activities in such reaches in consultation with the owners of such facilities. However, satisfactory completion of the entire work will be the responsibility of the Contractor.

7. General Design Requirements

1) Instrumentation and control systems shall be so designed, manufactured and installed so as to ensure the highest standards of operational reliability.
2) The Contractor shall provide all instrumentation and control equipment required in accordance with the specification together with local control panels, main Instrument Control Panel (ICP), motor control center, wiring, ancillaries and things of every kind required for a complete, reliable, economic and fully satisfactory system.
3) Instruments mounted in field and on panels shall be suitable for continuous operation. All electronic components shall be adequately rated and circuits shall be designed so that change of component characteristics shall not affect plant operation.
4) Apparatus mounted in panels shall be suitable for continuous operation at the maximum internal panel temperature expected in service and due account shall be taken of heat given off by other equipment. All electronic components shall be adequately rated and circuits shall be designed so that change of component characteristics shall not affect plant performance. In the selection of solid state equipments, special consideration shall be given to the effects of heat and the need for artificial cooling. The air temperature ranges within which such apparatus is designed to operate without effect on performance shall be stated.
5) Where electronic units, such as amplifiers and oscillators, are incorporated in an instrument they shall be fully solid state.
6) The standards of reliability for moving armature relays and electromechanical timers shall not be less than as specified in IS: 2959 for medium voltage Contactors of Class III mechanical endurance. The number of no-load operations which can be made before it becomes necessary to replace mechanical parts shall be not less than 5 million and the number of on-load operations shall be not less than 20,000
7) It is the responsibility of the Contractor to provide / arrange necessary power points required for instrumentation etc. while executing the work.
8) It is considered that solid-state devices are more likely to ensure the desired standards of reliability than electromechanical devices although the latter are not precluded if they can be shown to have the order of reliability specified above.
9) All relays and timers, whether solid-state or not, shall be mounted on plug-in base to facilitate replacement.
10) Routine maintenance and fault-finding in telemetry and computer equipment shall be facilitated by the provision of built-in diagnostic or self-checking system.
11) All I&C equipment shall be new, of proven design, reputed make, and shall be suitable for continuous operation. Unless otherwise specified, all instruments shall be tropicalised. The outdoors equipment shall be designed to withstand tropical rain. Wherever necessary space heaters, dust and waterproof cabinets shall be provided. Instruments offered shall be complete with all the necessary mounting accessories. All instruments should be designed for ambient temperature of 50 Deg Centigrade.
12) For transmitting instruments, output signal shall be 4 - 20 mA & RS485.

13) Electronic instruments shall utilize solid state electronic components, integrated circuits, microprocessors, etc. and shall be of proven design.
14) All digital outputs from the instruments shall be volt free.
15) All instruments of submersible type shall be protected to IP-68.
16) analogue displays shall be of the digital type with no moving parts utilizing LCD technology.
17) All All the instrumentation to be supplied under this project for measuring flow, level and pressure, residual chlorine, water pH, turbidity, energy, rpm, vibration etc. shall be of robust design, inherently free of faults and requiring as little maintenance and adjustments as possible for effective and reliable operation.
18) Instruments, supports etc. are all to be of materials resistant to or protected from the temperature and humidity to be encountered in the atmosphere present in the tropical climate.
19) The unit of flow meter shall be in litres/sec. and the integration shall be in cubic metres. Ranges shall be selected to show normal plant operation between 50% and 75% of the full scale reading.
20) Instruments are to be of a type which will maintain reasonable accuracy without adjustment. Normally accuracy shall be within plus or minus 1% of the full scale deflection throughout the full range of measurement, unless otherwise specified.
21) All necessary sensors, transmitters / converters, RTUs, indicators, cabling, etc. for the instrumentation system shall be provided as required by the manufacturer.
22) Electronic equipment shall be of the solid-state type and of the manufacturer's latest design.
23) Instruments shall be provided with mounting hardware and floor stands, wall brackets, or instrument racks and all necessary accessories as per site requirement.
24) All work shall be in strict accordance with codes and reference standards as described above.
25) Transmitters shall be provided with either integral indicators or conduit mounted indicators in metric process units, accurate to one percent.
26) Electronic equipment utilizing printed circuitry shall be suitably coated to prevent contamination by dust, moisture and fungus. Solid state components shall be conservatively ambient atmosphere fluctuations and 0 to 100 percent relative humidity. The field mounted equipment and system components shall be designed for installation in dusty, humid and highly corrosive service conditions.
27) Equipments shall be heavy-duty type, designed for continuous industrial service. Equipment shall be of the latest equipment models which are currently in production. All equipment shall be of modular construction and shall capable of field expansion. All identical components shall be by the same manufacturer.
28) The control room equipment will be installed in air-conditioned areas; however, the equipment shall nevertheless be designed to operate satisfactorily up to 50degree C ambient temperature and 90 percent humidity assuming air conditioning may not be available.
29) Field cabinets and enclosures shall be IP55 gasketed with multi-point latching doors.
30) Unless otherwise stated, overall accuracy of all measurement system shall be ±1% of measured value, and repeatability shall be ± 0.5%.
31) The instruments shall be designed to permit maximum interchangeability of parts and ease of access during inspection and maintenance.
32) The field instruments i.e. the instruments mounted outside the control panel shall be mounted at a convenient height of approximately 1.2m above grade platform.
33) Unless otherwise stated, field mounted electrical and electronic instruments shall be weatherproof to IP-65. The degree of weather protection for panel mounted instruments shall be IP-52 of 18:13947, Part-I.
34) The instruments shall be designed to work in the ambient conditions of temperature, humidity and chlorine contamination that may prevail. The instruments shall be given enough protection against corrosion and all the wetted parts of the instruments shall be non-corrosive.
35) Lockable enclosures shall be provided for the field mounted instruments.
36) All field instruments and cabinet/panel-mounted instruments shall have tag plates/ name plates permanently attached to them.
37) All wetted parts of the sensors shall be made of non-corrosive material capable of working with chlorine content of 5 ppm.
38) For all the instruments installed in the field and outside the buildings, Lightning Protection Units (LPU) shall be provided at both the ends of connecting cable for protection against static discharges/ lighting of electromagnetic interferences. xl. The performance of all instruments shall be unaffected for the ±10% variation in supply voltage and ±5% variation in frequency simultaneously.
39) Unless otherwise specified, good compression glands shall be used for glanding the cable in field instruments and instrument control panel.

40) The transmitters shall be provided with on-line test terminals. Zero and span adjustments shall be provided for all instruments.
41) Access ladders/ platforms shall be provided for maintenance and operation of all instruments.
42) HMI shall be provided for disply status of each sensors, and other primary instruments.
43) On return of power after an electrical power failure, all instrumentation and controls shall function automatically without any reset action required by personnel. The Contractor's attention is drawn to the requirements of installations which require to be "FAIL SAFE".
44) The mode of operation of any device used for alarm signaling shall be such that on failure of the power supply, the devices shall de-energize and actuate the alarm even if no alarm condition exists.
45) It is preferred that alarm and control relays with the same function reference number shall receive their power supply from the same sub-circuit fuse so that failure of power supply to the control relay causes an immediate alarm Signal-Contacts used for initiation of alarm states or controls shall be integral with transmitters. If this facility is not available, the trip amplifiers shall be used. Contacts integral with indicating instruments such as pressure indicators, may be used in certain instances subject to the approval of the Engineer's Representative and provided such instances are brought to the attention of the Engineer's Representative.
46) Tested Calibration certificates along with calibration procedure of all instruments shall be provided by Contractor.

8. Electrical Requirements

Equipment shall be designed to operate on 230 volts AC/12-48 VDC power source except where specifically noted. All regulators and power supplies required for compliance with the above shall be provided between power supply and interconnected instrument loop. Where equipment requires voltage regulation, constant voltage transformers shall be supplied.

Analogue transmitter and controller outputs shall be 4-20 milliamps & RS485 into a load of 0-750 ohms, unless specifically noted otherwise.

Equipment shall be designed and constructed so that in the event of a power interruption, the equipment shall resume normal operation without manual resetting when power is restored.

9. Quality Assurance

The system supplies shall have in-house engineering, programming, fabrication, and testing capability to expedite the project in its entirety. The supplier shall have experience on similar projects and capable of maintaining or upgrading the telemetry / SCADA system and other instrumentation systems.

The proposed SCADA system for all WTP, ESR/GSRs and pumping stations/Intake shall be open system architecture utilizing standard operating and communication systems like radio telemetry or GPRS. The Contractor shall provide documentation of successful and field performance of the proposed system, including hardware and software, for application of similar type, size, and complexity. The Contractor shall demonstrate that such system has been successfully employed in water applications. Unproven or proprietary systems shall not be acceptable.

The Contractor shall employ full time personnel for detail engineering, co-operating, drafting, procurement and expediting, scheduling, testing, inspection, installation, start-up service for calibration and commissioning and warranty compliance for the period specified.

10. Shipping Precautions

Special instructions for proper field handling, storage and installation of instrumentation required by manufacturer for proper protection, shall be securely attached to each piece of equipment prior to packaging and shipment. None of the instrumentation shall be shipped to the site until the room(s) are environmentally suitable.

11. Test Equipment

Test equipment shall be provided, together with items such as instruction manuals, carrying / storage cases, unit battery charger where applicable, special tools, calibration fixtures, cord extenders, patch cords and test leads, which are not specified but are necessary for checking field operation of equipment supplied under this Project.

- One hand held pressure calibrator
- One portable (DMM) digital multi-meter with rechargeable battery and charger and test leads, and carrying-case.
- One toolkit consists of screw driver, testers etc.

Instrumentation Circuit Routes

Signal cables shall not be run in the same conduits or duct as power cables. Installation in rigid steel conduit or steel trunking is preferred.

Signal cables which are not completely enclosed in a steel raceway shall be separated by at least 1000 mm from LV power cables, and at least 3000mm from HV cables. Wherever signal and power cables cross they shall do so at right angles.

All cables running from the field instruments to the RTU panel shall be a single, Continuous length, without joints, except at marshalling boxes. The boxes shall have terminal blocks with 20 percent spares in addition to terminals for all wires including spare wires. Special care shall be exercised to carry earthing lines through marshalling boxes and control panels.

12. Instrumentation Earthing

The Earthing arrangements shall, unless otherwise specified, comprise:

a. Chemical Earthing shall be provided at each remote site for instruments protection.
b. All instruments shall be properly earthed to nearest earthpit using 6sq mm copper earthing cable.
c. Earthing resistance measured report shall be provided by contactor using calibrated Megger.

13. Installation

Instrumentation and necessary equipment shall be installed in accordance with the manufacturer's instructions. The locations of equipment shall be determined in the field. Exact locations shall be suggested by the ULB during construction. Obtain in the field all relevant to the placing of Instruments RTU's, PLCs and etc. Contractor shall furnish all labour and materials necessary to complete the work in the approved manner.

Brackets and hangers required for mounting of equipment shall be provided. They shall be installed in a workmanlike manner and not interfere with any other equipment.

The screen on each process instrumentation cable shall be continuous from source to destination and be earthed as directed by the manufacturer of the instrumentation equipment but in no case shall more than one earth point be employed from each screen.

14. Tests

Test plan shall be submitted for approval as part of the project plan. Furnish the services of the servicemen, all special tools, calibration equipment and labour to perform the tests. Certified copies of the tests shall be providing in duplicate to the ULB.

15. Source Quality Control

The manufacturers of the equipment and fabricators of RTU cabinet supplied under this Contrast shall allow concern authority to inspect and witness the testing of the equipment at the site of fabrication. Equipment shall include the cabinets, special control systems, pressure measuring devices, level measuring devices, transmitters and other pertinent systems and / or devices. A minimum of ten working days notification shall be provided to ULB prior to testing.

Detailed Technical Specifications for Field Instruments: -

DETAILED TECHNICAL SPECIFICATION:

The Detailed Technical Specifications provided below shall supplement the Technical Specifications specified as above. Whenever there is a conflict, the provisions of Detailed Technical Specifications shall prevail over other.

All the field instruments shall be procured at site after the third party inspection by reputed Third Party Inspection Agency with the presence of consultant & Client. The charges for the same shall be borne by the contractor.

1. PRESSURE GAUGE

Pressure gauges shall relevant BIS or equivalent. Pressure gauges and transmitters shall have over range protection up to 1.5 times the maximum line pressure and shall be capable of withstanding full line pressure on any side with the other side vented to atmosphere without damage or effect on the calibration. No plastic material shall be used in their construction. Internal parts shall be of stainless steel, bronze or approved corrosion-resistant material. Where necessary, a special diaphragm shall be used to segregate the gauge tube from corrosive fluid media. In chlorine applications, the diaphragm shall be in silver or tantalum. The minimum diameter for any pressure gauge shall be 150 mm unless specified otherwise or where the gauge forms part of a standard item of equipment.

Where compensation of more than 2% of the instrument span is needed for the difference in level between the instrument and the tapping point, the reading shall be suitably adjusted and the amount of compensation shall be marked on the dial. The zero and span of a pressure transmitter shall not change by more than 0.1% of the span per degree Celsius change in ambient temperature. After application for 10 minutes of pressure at 130% of maximum pressure, the change in zero and span shall not exceed 0.1% of the span.
Pressure transmitters shall have accuracy typically better than 0.25% of span, depending on the application and shall be protected to BS EN 60529:1992, IP 65 standard or higher. For transmitters installed in locations liable to flooding or underwater applications, they shall be to IP 68 standard and shall operate up to a maximum submergence of 20 meters of water.
Pressure transmitters shall provide a 4 to 20mA d.c./RS485 output proportional to the pressure range at a maximum load of 750 ohms.

2. Open Channel Ultrasonic Flow meter:

a. Ultrasonic flow measuring system shall consist of a flow sensor, flow transmitter cum computing unit, prefabricated cable connecting the sensor and transmitter, and any other item required for completing the flow measurement system.
b. The flow sensor shall be suitable for flange or bracket mounting as required. It shall have ambient temperature compensation and adjustable datum setting facilities.
c. The flow transmitter cum computing unit shall be provided in an enclosure conforming to IP 67. It shall be programmable with an integral programming keyboard and shall provide an isolated 4-20 mA /RS485/ HART (if required) current output signal to Control System.
d. The design and application of ultrasonic flow meters shall take into account the partial flume or channel construction, the material, size, shape, environment, process fluid or material, the presence of foam, granules, size etc.
e. The installation shall avoid any degradation of performance from spurious reflections, absorption, sound velocity variations, sensor detection area, temperature fluctuations, specific gravity changes and condensation. For applications where spurious reflections are unavoidable the control unit shall be provided with facilities for spurious reflection rejection.
f. If turbulence exists, shielding, stilling tubes or other measures shall be provided to avoid effects on the measurement.

Ultrasonic Flow transmitter	Service : Water, Non-contact type
Transmitter :	
Type	Open channel Ultrasonic flowmeter
Principle	Pulse Time of flight
Output	4-20 mA /RS485/ HART(if required)
Number of sensor inputs	1
Material	Polycarbonate for field housing / Die cast Aluminum
Ingress Protection	IP66 / IP 67 field housing
Accuracy	+/- 0.5% of the maximum span of the sensor
Power supply	12-48 VDC
Temperature	from 0 deg C to 50 deg C
Display (If required)	LCD graphic display. Menu guided operation.
Configuration	Using Keypad on display
Sensor	
Range	0 – 5m
Max Temperature	55 deg C
Max Pressure	4 bar abs
Materials	PVDF/PP
Process connection	G 1"/ 1" NPT or manufacturers standards
Degree of protection	IP67/IP68
Mounting	Top Mounted

3. Electromagnetic Flow Meter

Item includes: -

a. Bi-directional electromagnetic flow meters shall be designed, manufactured and calibrated as per ISO/IEC 17025 standard. Meters shall also have actual Flow rate & Totalized value for effective water management purposes. The accuracy shall be +/- 0.25% of reading or higher.

b. The sensors shall be rated IP68. The transmitter shall be rated IP66/IP67 in line with local operating conditions. Installations shall be made with cables and /or conduits that guarantee the integrity of the system under all operational conditions. The transmitter/converter shall be wall-mounted type with a 4 line display for the indication of Actual Flow rate & Totalized value. A glass window within the protection enclosure with optical switches which shall be provided for local reading purposes. The non-corrosive, Polycarbonate Housing material of the enclosure shall be sufficient to guarantee five years operational life. Magnetic flow meters should be supplied with built in software feature to analyze and continuously monitor the health of the sensor, display errors in text format. Transmitter should be capable of performing the verification program on demand or on request without taking meter off the line or without any additional external hardware/accessories.

c. The transmitter shall be capable of being fully programmable. It shall have a set-up menu so that all relevant parameters may be user-set from the self-prompting driven menu.

d. The manufacturer shall have a full system of local offices in India and full service capability in the ULBs throughout the Madhya Pradesh. Full contact details for key personnel, both national and local shall be furnished on request.

e. The flow meter manufacturer / supplier shall provide full data on each meter required, including optimizing and sizing programs calculation sheet. The proposed flow meter model no. by the manufacturer shall be available on their official website with complete technical catalogue/operating manual for flow meter (sensor / transmitter). The official latest meter sizing program shall be available on official website of flow meter supplier.

f. All safety precaution with lighting protection shall be provided to all the flow meters, the failure / damage to flow meter for any reason for the warranty and period of annual maintenance contract shall be with Manufacture, hence, cost of insurance against all failure / damage (if any) shall be borne by Manufacture no additional payment will be made. Separate Control Panel Unit to accommodate flow transmitter and all other accessories shall be provided by Contractor. Thickness of control panel shall not be less than 2 mm.

g. Excavation for fixing Flow meters in all types of soils & surfaces in all types of site conditions including drainage, dewatering, etc.

h. Scope includes supply, installation and commissioning of Electromagnetic Flow Meters along with all required accessories like spool, flanges, etc.. Flowmeter are to be installed in running line. Necessary excavation, dewatering, disposal of debris are included in contracvtor scope.

i. ULB has provided the size of pipe is indicative. The actual size, O.D.I.D., thickness shall be measured by Manufacture before supply of meter, any deviation and delay or damage cost due to non-fitting of meter shall be borne by Manufacture. The liquidity damage due to delay in fitting for wrong selection of fittings and accessories lies with Manufacture.

j. The sensor / transmitter cables shall be capable of withstanding the climatic condition as applicable at site and should be weatherproof. The cable shall be installed in a suitable UPVC / HDPE pipes or duct to minimize the risk of damage during excavations for other works. All the cable laid at a minimum depth of 0.3 m below the ground. Maximum length between the sensor & the transmitter shall not be more than that recommended by the meter Manufacture.

k. The Manufacture will have a full system of local offices in India and full service capability in the ULBs throughout the Madya Pradesh. Full contact details for key personnel, both national and local shall be furnished on request.

l. Detailed technical specification of Full bore electromagnetic flow meter are as below:

Technical Specifications for Full bore Electromagnetic Flow meter		
General		
1	Application (Suitable for)	Water
2	Op Temperature	0 to 60 Deg C Max
3	Op Pressure	As per requirement
4	Measurement	Instantaneous water flow Rate, Totalized Flow
Flow Sensor		

5	Type	In-Line Full Bore Electromagnetic flow meter.
6	Power supply type	Mains operated
7	Sizes (mm) /Qty	Refer Bill of Quantity for details
8	Sensor Type	Pulsed DC
9	Process Connection	Flanged type
10	Weather Protection Class	IP68
11	Full Scale Velocity	0.4 to 10 m/s, vendor to provide the detailed sizing calculations based on the process details.
12	Electrodes	Measuring electrodes – 2 Nos. Grounding electrode- 1 No. and Empty pipe detection electrode -1 No. MOC: For Raw/Drinking Water: SS316L For Waste Water/ Sewage: Alloy C22
13	Coil Housing	Powder-coated die-cast aluminium Anti corrosive grade / SS304 with IP 68 protection class with 20 meter cable for remote mount transmitter connection
14	Flanges	Carbon Steel PN 10 as per EN 1092-1
15	Flow sensor Grounding	In- built Grounding Electrode SS316L is must. Grounding rings are also acceptable
16	Flow Tube material	SS304
17	Flow Tube Liner	For Raw / Drinking Water: Polyurethane or Hard Rubber For Waste Water/ Sewage: PTFE / PFA
Flow Transmitter Unit		
18	Type	Microprocessor based with facility to configure the range
19	Type of Display	2 -Line display with 3 optical keys (touch control)
20	Transmitter Type	Remote Wall Mounted with 5 cable glands, integrated fixing points for wall mounting, with sensor interface
21	Units For Display	Flow Rate m3/Hr , Totalized Flow m3
22	Input	From Flow Tube
23	Accuracy	+/- 0.5 **%** of Measured Value
24	Operation	Via local display
25	Output	4-20mA and RS485/HART **(**if required**)**
26	Power supply	0-250 VAC , 45-65Hz – Optional - 11 to 40 V DC
27	Self-diagnostic feature	Required, Following minimum diagnostic features shall be provided Empty pipe detection, event logbook, system diagnostic features
28	Transmitter Enclosure material	Non corrosive, polycarbonate.
29	Communication	4-20 mA & RS485/ HART(if required)
30	Memory	100% secure data storage
31	Weather Protection Class	IP 66/67 (NEMA 4X) for flow transmitter
33	Flow calibration rig	Duly Accredited according to ISO/IEC 17025 / NABL

4. Radar Type Microwave Level transmitter:

Radar Level transmitter	Service :Water (Non-Contact Type)
Transmitter	
Type	Microwave Level measurement
Principle	Pulse Time of flight
Output	4-20 mA /RS485/ HART
Housing	Die Cast Aluminum
Ingress Protection	IP65 / IP 66 / IP 67

Accuracy	+/- 3 mm
Display	4 line LCD display. Menu guided operation.
Configuration	Using Keypad on display
Range	Liquids 0 to 30m
Temperature range	0°C ... +60°C
Max Pressure	3 bar abs
Materials	Sensor: PVDF Seal: VITON/EPDM
Process connection	Threaded or universal flange dependent on model selection as per site condition

5. Ultrasonic Level transmitter:

Ultrasonic Level transmitter	
Transmitter	
Type	Ultrasonic
Principle	Pulse Time of flight
Output	4-20 mA/RS485/ HART(if required)
Housing	Die Cast Aluminum
Ingress Protection	IP67
Accuracy	+/- 2 mm or 4 mm depending on selection or 0.2% of set measuring distancelam Whichever is greater.
Area Classification	Non-Hazardous
Temperature range	0°C ... +50°C
Display	4 line LCD display. Menu guided operation.
Configuration	Using Keypad on display
Sensor	
Range	Liquids 0 to 5m and 0 -10m depending on Tank size
Temperature range	0°C ... +50°C
Materials	Sensor: PVDF/PP Seal: EPDM
Process connection	Threaded or universal flange dependent on model selection
Degree of protection	IP67/IP66

6. Pressure Transmitters:

Sr.No	Specifications	Pressure Range - as per site requirement
1	Process Temperature range	0 to +60 deg.C.
2	Output Signal	4-20 mA /RS485/ HART(if required)
3	Supply voltage	11.5 to 45 V DC
4	Reference accuracy	± 0.1% of the set Span
5	Housing	Die Cast Alu. Housing
6	Diaphragm Material	SS316L/Equivalent
7	Display Operation	3 Key Push button for configuration COMMUNICATOR. A 4-line liquid crystal display (LCD) is used for display and operation
8	Long Term Stability	± 0.25% URL/year for 1 year, 5 year

7. Online Turbidity Analyzer

1. Online sensor based on Nephelometric 90° scattering light method.
2. The sensor must be easy plug and play with digital communication based on inductive energy transfer with IP68 rating suitable for measuring range 0 to 40 NTU for clear water & 0 to 1000 NTU for raw water with accuracy of <2% of measured value or 0.1 NTU.
3. The sensor shall store on-board all the calibration data and other diagnostic information
4. The sensor must have provision for to store at-least 5 user calibration points.
5. The offered transmitter shall be 4 wire digital with possibility of connecting multiple sensor inputs for additional parameters – referred as multichannel / multi parameter type.
6. The transmitter shall be suitable for outdoor installation with IP66/67 rating.

Turbidity Measurement	
Transmitter	
Type	Turbidity and suspended solids transmitter
Principle	Nephelometric measuring principle 90° scattered light according to ISO7027/ EN 27027 or equivalent
Output	4-20 mA & RS485/ HART(if required)
Supply voltage	100 / 115 / 230 V AC, 48 ... 62 Hz ; 12-48 V DC
Material	Field Housing : ABS PC / PC FR , shall be Non corrosive type
Display	LC display, two lines, with status indicators
Electromagnetic compatibility	Interference emission and interference immunity acc. to EN 61326-1:1998/latest
Protection class of field housing	IP 65/IP66
Ambient temperature	0 ... +50 °C
Self-Diagnostic feature	Required
Sensor	
Measurement range	0 – 999 NTU
Material	Sensor shaft : PVC / PPS GF40 Optical window : sapphire Cable : TPEO
Max Process temperature	50°C
Max Process pressure	10bar
Connection	Fixed cable connection
Ingres protection	IP68
Additional Certifications	Calibration certification
Resolution	0.001 NTU, 0.01 ppm, 0.1 g/l, 0.1%
Measurement error	< 2% of meas. value (min. 0.02 NTU)
Flow Through Assembly	OEM

8. pH Analyzer:

a. Online pH sensor based on potentiometric measurement using glass sensor.
b. All pH analyzer instruments shall be provided with integral LCD display and shall be designed for field mounting. The measuring / transmitting equipment shall be installed local to the respective sampling point. It should have Flow through assembly unit of OEM.
c. The instrument can compensate for changes in the ambient temperature.
d. The sensor shall have digital communication based on inductive energy transfer;
e. The sensor shall have inbuilt memory to store calibration data, and other additional diagnostic information.
f. The offered transmitter shall be 4 wire digital with possibility of connecting multiple sensor inputs for additional parameters – referred as multichannel / multi parameter type
g. The transmitter shall be suitable for outdoor installation with IP66/67 rating.

pH Analyzer	
Transmitter	
Type	Glass electrode
Principle	Glass electrode with dirt repellent PTFE diaphragm
Output	4-20 mA & RS485/ HART(if required)
Supply	100 / 115 / 230 V AC, 48 ...62 Hz ; 24 VDC
Material	Field Housing : ABS Polycarbonate non corrosive
Display	LCD/LED display, two lines, with status indicators
Protection class of field housing	IP 67/IP66
Ambient temperature	0 ... +60 °C
Diagnostic feature	Required, 3 point calibration
Sensor :	
Measurement range	0 - 14 Ph
Material	Glass
Max Process temperature	60°C

Max Process pressure	6 bar
temperature sensor	NTC / Pt100
Connection	Inductive digital connection with Transmitter
Ingres protection	IP68
Resolution	0.01pH, Temp 0.1°C
Measurement Error	± 0.5% of Measuring range
Flow Through Assembly	OEM

9. Residual Chlorine Analyzer

a. Online chlorine analyser based on Amperometric/membrane based measurement of active chlorine converted to free chlorine by means of pH compensation**.**
b. Complete measuring system including chlorine sensor, pH compensation
c. with suitable flow assembly for mounting these sensors along with transmitter**.**
d. Flow through assembly should have auto cleaning feature and flow should be maintained as 1L/M

Chlorine Analyzer	
Transmitter	
Type	Residual chlorine with PH compensation
Principle	Amperometric measurement of free chlorine.
Output	4-20 mA & RS485/ HART(if required)
Supply voltage	100 / 115 / 230 V AC ,48 ... 62 Hz ; 12-48 V AC
Material	Field Housing : ABS PC Fr
Display	LC display, two lines, with status indicators
Protection class of field housing	IP 65/IP66
Ambient temperature	0 ... +60 °C
Sensor	
Measurement range	0.01 - 5ppm free chlorine
Material	Sensor shaft : PVC
	Membrane : PTFE
	Membrane cap :PBT (GF30); PVDF
Process temperature	0°C - 50°C
Max Process pressure	1 bar
temperature sensor	NTC / Pt100
Connection	Inductive digital connection with Transmitter/Analog cable connection
Ingres protection	IP68
Resolution	0.01mg/l
Measurement Error	± 0.5% of Measuring range
Flow through Assembly	OEM

ENERGY MEASURING (POWER ANALYSER) INSTRUMENTS (Energy Meter)
Power Analyzer/ Energy Meter

Measurement Parameters	VLN , V1, V2, V3, VLL, V12, V23, V31, A,A1, A2, A3 W, W1, W2, W3 VAR, VAR 1, VAR 2, VAR 3 VA, VA 1, VA 2, VA 3 PF, PF 1, PF 2, PF 3 Frequency F
Sensing / Measurement	True RMS, 1 sec update time, 4 Quadrant Power & Energy
Accuracy (Wh)	Class 1.0 as per IEC 62052-11, 62053-21
Input Voltage	4 Voltage inputs V1, V2, V3, VN, 110 or 415 VL-L nominal (Range 80 to 600 VL-L

Aux. Supply (control power)	44 to 300 V ac / dc (3 VA Max)
Input Current	Current Inputs A1, A2, A3, 20 mA to 6 A (field Configurable)
Overload	10A Max Continuous
Frequency	45 to 65 Hz
Resolution	RMS 4 Digit
Communication	RS-485/ RTU Protocol
Isolation	2000 volts AC for one minute between communication & other circuits
Demand	integration period multiple of 5 minute form 5 to 30 minutes 15 sec. update time
Safety	Measurement category III, Pollution degree2, Protection against shock by double insulation at user accessible
Pumping Hours	Measurement Possible

10. **PLC BASED CONTROL PANELS**

Access doors or panels shall have heavy duty hinges and approved latching or fastening means to allow access. Fabrication shall be of 2mm thick, steel, suitably braced internally for structural rigidity and strength. The final finish shall be smooth, free of runs, and uniform in tone and thickness. Control panel enclosures shall be rated IP54/IP55.

All wiring shall be numbered in accordance with the numbering system used on the wiring/ connection diagrams. Wiring and connection diagrams shall conform to ISA S5.4 Instrument loop diagrams and shall be submitted as part of the shop drawings for approval of the Engineer.

All wiring shall terminate in a master terminal board, rigid type and numbered. The master terminal board shall have a minimum of 25 percent spares. Terminal blocks shall be barrier type with the appropriate voltage rating (600 Volts minimum). They shall be the raised channel mounted type rail mounting (marshalling).

Wire and tube markers shall be the sleeve type with heat impressed letters and numbers. DIN rail mounted terminal strips shall be provided for the purpose of connecting all control and signal wiring. Direct interlock wiring between equipment will not be allowed. Only one side of a terminal block row shall be used for terminal wiring. The field wiring side of the terminal shall not be within 150 mm of the side panel or adjacent terminal.

Nameplates shall be provided and mounted on each RTU enclosure. The nameplates shall be approximately 25mm x 75mm constructed of black and white laminated, phenolic material having engraved letters approximately 6 mm high, extending through the black face into the white layer. Nameplates shall be attached to panels by self-tapping Type 316 stainless steel screws.

All components shall be mounted in a manner that will permit servicing, adjustment, testing and removal without disconnecting, moving or removing any other component. Components shall be mounted on plates on the inside of the panels in such a manner that allows for removal of the components without removal of the plate. Components shall not be mounted directly to the enclosure. Mounting shall be rigid and stable unless shock mounting is required otherwise by the manufacturer to protect equipment from vibration.

Components mounting shall be oriented in accordance with industries standard practices. All internal components shall be identified with suitable plastic or metal engraved tags attached with drive pins adjustment to (not on) each component identifying the component in accordance with the drawing specifications and supplier's data.

Control Panel requirements for in-plant SCADA System :

a. Programmable Logic Controller (PLC) based Instrument Control Panel (ICP) shall be provided at control room of water pumping stations and Master Control Room locations covered under scope of project with manual override facility.

b. Technical Particulars

Service	For raw and pure water pumping station or as per reuirement
Type	Free standing type
Construction	Prefabricated and modular construction
Sheet Material	Cold rolled sheet steel/ COLD ROLLED CLOSE ANNEAL (CRCA)
Internal	Required

Lighting	
Cable Entry	From bottom
Access	Front and Rear

The following instruments shall be mounted on front fascia of the ICPs for the control of pumping station:

1. Auto/ Manual/ Off selector switch for each pump
2. Level measuring system of raw and pure water sump for automatic operations.
3. Digital indicators/HMI for the following:
   a) Level in each compartment / sump
   b) Pressure in Pure water pumping main
   c) Position of Valve in discharge of pure water pump.
   d) Position of Valve in the pumping main
4. Digital flow indicator/HMI for flow in pumping main of pure Water
5. Alarm annunciator,
6. HMI/SCADA of the PLC system
7. Temperature Scanners for motor winding and bearing temperatures and pump bearing temperatures of each raw / pure water pump.
8. Electronic microprocessor based power meter/Energy Meter shall be used for each motor, incomer & other major loads to provide complete details of different electrical parameters. Suitable CT/ PT is required to wire the power meter. These meters shall communicate and integrate with local in-plant SCADA.
9. Pump start & stop operation will be automatically done in required sequence & all safety requirements shall be monitored continuously. Only single push button command is required to be given for start or stop of the pump.
10. For automated operation of pump start or stop discharge valve shall be fitted with electric/ pneumatic actuators. It is responsibility of the Contractor to integrate these actuators with local SCADA system.
11. A common emergency stop pushbutton for emergency system stop for pure water pump.

11. INTERFACING OF EXISTING AND INSTRUMENTS

If there are existing flowmeters/instruments installed at any location than it is responsibility of the Contractor to acquire instruments data from the new and as well as existing flowmeter/instruments after repairs / refurbish or replaced (as per instruments condition) and integrate it with SCADA system and all the required accessories like sensors, transmitters, cables, power points etc. to be provided to interface with the meter and communicate the data with Central Server Station. Consent from ULB is required before taking up such work.

It is the responsibility of Contractor to provide complete interfacing required to receive the data from existing and new instruments.

12. VALVE ACTUATORS

Actuators shall be suitable for the medium, climatic, environmental and pressure conditions of the system in which they are to be fitted. Actuators shall be provided with:

a) AC Electric Motor.
b) Reduction gear unit.
c) Torque switch mechanism.
d) Limit switch mechanism complete with set of limit switches and additional two spare sets for suitable position.
e) Hand wheel, for manual operation.
f) Valve position indicator.
g) Hand-auto lever with suitable locking arrangement.
h) 10 W single phase space heater in the switch compartment.
i) Blinking light throughout the valve operation.
j) Junction box for terminating power and control cables.
k) With additional accessories for integrating with PLC system.

The actuator shall be suitable for operation on 415V, 3 phase, 50 Hz power supply. The motor winding insulation shall conform to class B as per relevant BS and motor shall be protected by suitable thermal overload relays. The actuator shall be capable of producing not less than 1 1/2 times the required operator torque at the required time cycle of valve operation. The transmission shaft connecting the actuator to the valve shall be

provided with 2 bearings one at actuator end and one at valve end with universal couplings at suitable places. The required numbers of switch/contacts meet requirements for PLC system. The electric motor shall be of the squirrel cage type as per IS 325 with insulation to IS 1271

Class B. The windings shall be impregnated to render them non-hygroscopic and oil resistant. All internal metal parts shall be painted. The motor shall be rated for 15 minutes. They shall also be suitable for operating on the specified electric supply and shall satisfactorily open and close the valve under variations of electric supply specified.

Motor shall be protected by suitable overload protection device. The reversing contactor starter and local controls shall be integral with the valve actuator. The starter shall comprise mechanically and electrically interlocked reversing contactors of appropriate rating fed from a 110 Volt control transformer. The common connection of the contactor coils at the transformer shall be grounded. HRC cartridge type primary and secondary fuses shall be provided.

Local control shall comprise pushbuttons for open, close and stop operations and a Lockable Local/Remote/off selector switch. The control schematics shall be subject to approval. Internal wiring shall be of 650/1100-volt grade PVC insulated stranded copper conductor of minimum 1.5 sq. mm for control circuits and of minimum 4 sqm copper for the power circuit. Each wire shall be number identified at each end. The terminals shall be of stud type.

Cable entries shall be suitable for PVC insulated/ sheathed, armored cables. A separate terminal box shall be provided for the heater. A separate terminal box shall be provided for cabling to control circuits.

The actuator enclosure shall be fully weatherproof and hose proof to IP 67 and shall be fitted with an anticondensation heater, which shall be switched off when the motor is running. The torque switch mechanism shall function as follows to stop the motor on closing or opening of the valve, or upon actuation by the torque when the valve disc is restricted in its attempt to open or close.

The torque switch in the closing direction shall interrupt the control circuit if mechanical overload occurs during the closing cycle or when the valve is fully closed.

The torque switch in the opening direction shall interrupt the control circuit if mechanical overload occurs during the opening cycle or when the valve is fully open.

The mechanism shall facilitate adjustment of the torque at which the switches are required to operate. Nonadjustable limit switches shall stop the motor and give indication when the disc has attained the fully open or closed position.

The adjustable limit switches shall have control rated 2A, 48 V DC for specified system interlock, at the desired value position in both the opening and closing directions.

Motor Operator shall be provided with clearly visible local valve position indicators mounted on the operator assembly to give an indication whether the valve is fully open, fully closed or in an intermediate position. Settings and emergency operation shall be possible with the use of a hand wheel. The Hand wheel shall be of stainless steel and the drive mechanically independent of the motor drive and any gearing should limit the operating torque at the hand wheel to less than 15 kg and be such as to permit emergency manual operation in a reasonable time. During electric operation the hand wheel shall not rotate.

Actuators shall be adjusted at the manufacturer's works to ensure that they provide the correct, fully, open position and fully closed position. Mechanical adjustable stops shall be provided to prevent over-travel of the valve in the open and closed positions.

The contractor shall supply the required dia. of valves at his cost. **The valves shall be accepted after the third party inspection by reputed Third Party Inspection Agency with** the presence of PDMC & Client. The charges for the same shall be **borne** by the contractor.

13. INTERFACING OF VALVE ACTUATORS

At ESR / UGRs / MBRs Inlet and Outlet Water Pumping Locations or any other locations within distribution system, there are existing valve actuators installed. It will be the responsibility of the Contractor to interface with the telemetry system and control the valve through telemetry / SCADA system. It is inclusive all the required accessories like sensors, transmitters, cables, power points etc. to be provided to interface with the valve actuators and communicate the data with Central Server Station. Consent from ULB is required before taking up such work.

It is the responsibility of Contractor to provide complete interfacing required to receive the data from such valve actuators.

14. ALL REQUIRED COMMUNICATION INTERFACE, SCADA INTERFACES, TELEMETRY INTERFACES, RTUS / PLC HARDWARES, CONTROLS

General Description of Proposed System

The Telemetry / SCADA specified herein shall be designed, manufactured, installed, tested and shall provide along with the following functions:

- To provide Master Control Centre (MCC) to collect, store, control and monitor the Water supply scheme for all ESR/ UGRs/ GSRs/Intake/WTP covering all pumping station (as per BOQ) locations and incuding BPTs.
- To monitor Real-time water distribution & management process
- To perform calculations based on sensor and manual operator data inputs.
- Store and retrieve all the information of each location.
- Compile and prepare daily, weekly and monthly water & energy report for analysis
- Provide mimic / graphic display to indicate water supply and distribution system through extensive graphics support, which also reduces the learning curve of an average operator. All the data logged at central server shall be analysed through representation of this data in various meaningful user defined graphs, trends and customized analytical requirements of the system.
- All the required field instruments to measure flow, level and pressure, residual chlorine etc. shall be provided, calibrated, tested and installed. The signals from the field instruments both existing and newly proposed instrumentation shall be wired to the PLCs / RTU's with necessary programming etc. Complete for further communication and data integration with Server & Central SCADA system.
- The system shall be open system architecture utilizing standard operating and communication systems like GPRS or radio telemetry. The system supplier shall provide documentation of successful field performance of the proposed system, including hardware and software applications. Unproven or proprietary systems shall not be acceptable.

15. The SCADA Documents Requirements;

a) QAP, Technical Datasheet & GAD for each instruments
b) Layout drawings for each piece of equipment fabricated or assembled by the Contractor, showing the position of each component with required clearance where applicable, and with overall dimensions.
c) Wiring diagrams indicating each component of the system and all wiring and cabling thereto, showing manufacturers, types, duties, ranges and nomenclature, referencing the P&I diagram where applicable, with inputs, output, cable wiring and terminal identifications clearly marked.
d) PLC/ RTU IO instrumentation drawings, cable along with sizings and cable layout trenches as per site condition.
e) Communication details / protocol details for all communicating instruments viz. PLC, GPRS telemetry, SCADA, etc. shall be provided.
f) Complete input and output list for associated field device, range (if applicable), critical/non-critical alarm status etc.
g) Description of quality control methods and approvals.
h) Calibration certificates for all the instruments along with test procedures.
i) Detailed works and acceptance test procedures.
j) Programme for manufacture, delivery, installation, interfacing and commissioning.

16. RTUs / PLCs Panel General Specifications

Based and requirements Contractor shall be responsible to supply, install, commission the RTU's / PLC's as per the General Requirements as below: -

Type	Microprocessor/Micro-controller based
Input Supply (Operating Voltage)	12 – 48 VDC
Input/Output Module	As per site condition and Expandable I/O bytes with minimum 20% spare
Temperature Range	0 C to 55 C
Permissible Humidity	0 % to 95 %
Microprocessor	32 bit RISC or better microcontroller
Clock Frequency	Min. 64 MHz
Program Memory	Min 128 KB

Data Memory	Min 128 KB
Retentive Data Memory	Min 8 KB
Flash Memory	Min 4 MB Integrated
Expandable Memory	Upto 2 GB
Enclosure	Standard Wall Mount / DIN Rail Mount
Cooling Method	Self Cooling
Communication Protocol	MODBUS/DNP3/ETHERNET
Design Type	Modular
Communication Ports	
RS – 485	Min 1 No.
Ethernet	Min 1 No.
USB	Min 1 No.
RS – 232/HART(4-20 mA)	If Required
Isolation	1000 VDC
On Board RTC	In-built
External Optional Module Connectivity	GSM/GPRS Module/GPS Module
LED's Indication	Power, Communication, Digital Channel LED's, etc
Telemetry Protocol	IEC 60870-5104/101, HTTP/HTTPS, DNP 3.0, MQTT etc. for remote data transmission
PLC Programming	Open Standard IEC 61131-3
Time Synchronization	RTU/PLC should support time Synchronization
Reset Facility	Yes
Electromagnetic Compatibility	According to EMC Directive 2014/30/EU or equivalent
Analog Input Hardware Specification	
	Type 4 – 20 mA
	Single Ended
	Resolution : 12 bit
	Accuracy – 0.25
	Input Impedance : 50 Ohms
Digital Input Hardware Specification	
	Type : Potential Free (Sink-Dry Contact)
	Optical Isolation
	Input Voltage Range : 12-48 VDC
Ingress Protection	IP 20
Other Specification	Inbuilt Web-Server & FTP Server ; SMS Triggering with Alarm, Data filling in case of network failure

1. RTUs / PLC enclosures shall be corrosion resistant welded for outdoor locations. Enclosures shall be fabricated from a minimum of 14 gauge cold rolled steel with manufacturer's standard color.
2. RTUs / PLC protected from lightning or other transient voltages by a surge protection device.
3. All power supplies required for operation shall be provided. Power supplies shall be sized to have a minimum of 40% spare capacity providing increased reliability and allowing for the addition of future equipment.
4. All wiring shall be in complete conformance with the standards as specified in general specifications. All wiring neatly tied and fastened and for ease of servicing and maintenance, all wiring shall be color coded and uniquely numbered.
5. Battery Backup system for RTUs / PLCs shall be provided and working in conjunction with the unit's power supply, shall be an intelligent battery backup system including voltage converter, battery health logic module, charger and sufficiently sized battery. Battery system shall provide a seamless switchover to battery upon detection of main power supply failure. Once main power is restored, the unit shall provide seamless switchback to normal power source and recharge the battery. Battery health logic module shall individually monitor main power supply, battery and converter voltages for low voltage conditions, and provide low voltage cutoff to protect battery from an unrecoverable depletion.

6. Battery system shall be of sufficient capacity to provide a minimum of 4 hours of backup in the event of a failure of the main power source. To avoid battery damage and erroneous data transmissions when operating on battery, should the battery voltage drop below 10.8 V, the PLC shall be inhibited from operation. Recovery shall be automatic upon restoration of normal power.
7. All RTUs / PLCs shall have powerful communication Capabilities.
8. It shall be comprising of CPU and signal conditioning ends for analog and digital signals.
9. It shall have a HMI display for indicating local data.
10. All RTUs / PLCs signals, status signals, alarm and process variable data shall be transmitted and received between the central location and the remote locations via the radio telemetry system. The system shall convert commands, alarms and variable analog data to digital blocks and transmit this information between the Central and the multiple remote locations.
11. RTUs / PLCs shall be furnished completely configured and tested for providing the specified communication, monitoring, display, input/output, annunciation, etc. for the SCADA control room/MCC requirements for operation of the in-plant SCADA system (pumping station). Any additional components required for operation, whether specifically referenced herein or not, shall be provided.
12. RTUs / PLCs system shall be based on a robust, field proven, current technology hardware platform allowing utilization of the latest advances in technology and permitting the most open programming and communication architectures. The PLC system shall be modular and scalable to be efficiently applied at each of the specified sites within the system.
13. RTUs / PLCs shall store field /system parameters including, logic configuration, set points, time delays, alarm and event data, counters and tantalizers, etc. in field programmable non-volatile memory. Sufficient non-volatile memory must be provided to protect at least 8,000 variables. The PLC shall also provide enough protected memory for time stamped data logging.

RTUs / PLCs allow telemetry operations over radio frequency communication media or GPRS media affording the most efficient and reliable operation. Telemetry System communication architecture can be based on any one or a combination of these communication media.

The Remote Terminal Units (RTUs) shall be micro-processor based units of modern field proven design. RTUs must be modular in design to allow expansion of any count communications and I/O interfaces and capable of a variety of data transfer methods using various communications media as above.

RTU/PLC Panel shall have provision of online UPS for protection and minimum 4 hours' battery backup.

B. DATA TRANSFER

The RTUs proposed MUST support a number of data transfer philosophies that include,

1. **Standard polling** – where the master station **continuously** requests some/all real-time data values.
2. **Exception Reporting** – where the RTU initiates messages.
3. **Time tagged data uploads** – where the master station requests data that has been logged based on its priority/class. All data is logged in an event log buffer with a time stamp of 1mS resolution (system wide accuracy of 10mS or better).
4. **Dynamic message routing** –RTUs must be able to 'learn' of changes to network routing and dynamically change message routes.
5. **Peer-to- peer communications** – allowing RTUs to communicate with each other rather than simply with a master station.
6. **Store and forward communications** – allowing RTUs to 'pass on' messages for other RTUs. This is used to extend the coverage of point to multipoint style networks and also allows linear communications paths (where RTUs are typically 'daisy chained').
7. **Transparent port connections** – allowing RTUs to 'connect' ports between remote locations for the transfer of non-RTU data/messages. This requires the RTU to receive and package a non-RTU message within a RTU message packet for transfer to another RTU location for output (the output message is stripped of the RTU message headers, CRC etc).

C. POWER SUPPLIES

The RTU power supply shall accept AC or DC input and provide all necessary power to support the RTU operation. The power supply outputs shall be fully isolated from the input power source and include both monitoring and control of the various outputs to allow advanced power management functionality. The power supply shall be able to run from its standby battery for a minimum of 6 hours.

D. DATA LOGGING

The RTU shall have data logging capability of any I/O or register data value. The event log entry shall be time tagged with standard resolution time stamp and each entry shall include the following parameters:

- year, month, day, hour, minute, second, millisecond,
- data value, (2 bytes)

The event log capacity shall be configurable in the RTU memory setup, with a minimum capability of 10,000 event logs before any data is overwritten. The RTU logic shall allow the packing/archiving of 'old data' using the data priority field to mark data older than a specified period as deleted.

The RTU event logs shall be retrievable using any/all arguments of;

- date/time,
- priority,
- user type,

E . LOCAL COMMUNICATIONS

The RTU shall support a number of device drivers that allow communications with local devices such as meters, PLCs, and other intelligent electronic devices (IEDs). These interfaces will typically operate on any of the RTU communications ports using data transfer protocols such as DNP3.0/ MODBUS/Ethernet TCP/IP etc. The RTU shall allow the use of multiple communications protocols.

F. NETWORK COMMUNICATIONS

The RTU shall be capable of industry standard SCADA protocols. The RTU shall be configurable to manage network communications in slave mode, master mode or as a data concentrator (where a data concentrator operates in both slave and master modes). The RTUs shall maintain 'images' of network data received from other RTUs to allow reference of such data (and its communications status) to be included in the RTU logic.

G. RTU DIAGNOSTICS

The RTU shall have in-built diagnostics that allow local and remote interrogation of diagnostic information. Basic diagnostic information shall be presented to the user via the module LEDs and advanced diagnostic information via the RTU programming and configuration interface program.

The advanced diagnostic information for I/O modules should include the health/presence of each RTU module and the current state/value of each I/O channel (including power supply I/O data). Diagnostic information specific to the processor module and RTU in general should include the scan rate of I/O and logic, the current time and date, the RTU firmware version, firmware options (i.e. protocols, application extensions etc), list error status flags. Communications statistics should be accessible for each RTU in the network that include the number of message successes; message fails and fails since last success.

The RTU shall have port monitoring capability that allows local and remote capture of messages to/from any RTU port for display in the programming and configuration interface.

H. SPARE CAPACITY

Each RTU/PLC Panel shall be sized to accommodate all of the I/O listed in the scope of works and include an additional 20% spare installed I/O capacity.

SECURITY

The RTU shall have two types of security, one for the programming and configuration interface and the other for the RTU ports.

I. Programming and Configuration Interface - User Security

The programming and configuration interface shall support at least six levels of user access that can be configured by a system administrator. These should include as a minimum,

1. Unlimited Access Allows unlimited access, can read and write all RTU parameters including configuration parameters. RTU parameters include :- Local Registers, Network Registers, System Registers, Hardware (I/O) Registers, Event Logs, and Logic Program's.
2. Read All, Write All (except config.) Allows reading and writing of all RTU parameters, except for System Registers and Ladder Logic. Thus reconfiguring an RTU is not possible from this access level.
3. Read All, Limited Write Access Allows reading all RTU parameters. Writing is limited to some Local Registers. No other parameters can be written at this access level.

Whenever the interface program is started, the user should be prompted to enter a username and password.
This is checked against the User Database, to verify the password, and to provide the authorized access level.

J. RTU Security

Each RTU Communication Port shall have a configurable access level. The default access level can be any of the 6 levels (0-5) specified above to control read and write access to that port.
Thus, if a RTU port has default access level 4, then only some of the local data registers can be read via that port (eg. in energy delivery applications this functionality allows the energy vendor to allow the customer access to 'some' of the RTU data).

K. I/O Modules:

The proposed I/O modules should be have identical Channel density and shall be from the same product family as that of processor.
While defining the channel density of the I/O modules, I/O requirement for each location shall be taken into consideration. In addition to the required I/O at each location, 20 % spare I/O's shall be provided in each panel.
Human Machine Interface (HMI): The panels requiring local display and control of the process shall be provided with the suitable color touch-screen HMI mounted on the front panel and interfaced with the PLC processor. The HMI software should be designed in such a way that it will cater to the local monitoring and control requirements.
PLC panels should be provided with required space heathers, ventilation and fan arrangements, Electrical Service socket and earthing arrangements and shall be wired as per the relevant standards and norms adopted for such panels in the industry.

L. PLC Control Panel for Pumping Stations/Filter Bed:

The Panel shall be stand alone, floor mounting type with IP 55 protection and shall house PLC, HMI-7inch colour touch screen, 24 VDC, 5 A, SMPS, Modem / Router, Relay boards, MCBs for Incoming Power (DP), PLC Power, I/C & O/G for 24 VDC supply, service socket, 2 additional DP MCBs for auxiliary supply. PLC Panel for filter bed shall be pedestal desktop type panel only for ease of operation. Necessary UPS/ Power back arrangement shall be provided with PLC's to avoid any data loss.

M. PLCs for SCADA system (pumping station/Filter Bed)

The PLC system shall have I/O resources to support a wide variety of control operation of water pumping. Each PLC shall be supplied with the required I/O to meet the specified requirements.

N. Control Modes

i. Control Logic

The pure water pumping station comprises of duty and standby pumps. Programmable Logic Controller (PLC) based control system shall be provided for automatic and semi-automatic operation of pumping station. The Contractor shall design and size the PLC system, its memory, inputs/ outputs etc. considering the requirements of pumping station i.e. considering additional pumps (may be required to add in future) and associated equipment and instruments. However, the program to be loaded in the PLC shall be suitable for the operation of specified no. of pumps. In the event of failure of automatic controls or by operator choice, it shall be possible to operate the pumping station independent of the PLC function. The control system should be designed to recover fully to a normal operational state on restoration of power, following a power failure without manual intervention.
"Remote"/"local" selector switch shall be provided on the switchgear panel for each pump.
"Auto"/off / Manual" selector switch shall be provided for selection of operation mode for each pump on the Instrument Control Panel.
"Auto / Manual" selector switch for the pumping station operations.
On the SCADA System operator station, a facility shall be provided for selecting various combinations of working and standby pumps.
More combinations shall be provided to cover all the required combinations.

ii. Manual Control Mode

In this mode, the raw / pure water pumps and valves shall be operated manually from the local PC based SCADA system. The control commands shall be routed through the PLC. The PLC shall prevent simultaneous starts or stops of the pumps and shall carry out automatically any start up or shut down routines necessary in order to start or stop a pump.

iii. Local Control Mode

In the 'Local' mode, the pumps shall be operated through pushbuttons provided on the Switchgear Panel. The pumps shall be operated independent of the PLC function. The motorized valves shall be operated through pushbuttons provided on the valve actuator.

The pumps shall stop automatically if the level in the raw water & pure water sump is at 'low low' level in order to prevent dry running of the pumps in all the control modes.

iv. I/O Schedule

The PLC shall be capable of having following minimum specifications
Digital Inputs: As per site requirement with 20% spare
Digital outputs: As per site requirement with 20% spare
Analog Inputs: As per site requirement with 20% spare
Analog Outputs: As per site requirement with 20% spare
Communication ports: 3 Nos. One for HMI, one for connecting various parameters through MODBUS /Ethernet and one for MODEM / Router.

18. INSTRUMENTATION CABLE WITH ALL REQUIRED ACCESSORIES

All cabling required for connecting all field instrument, sensors, transmitters, PLC etc. Shall be provided by Contractor. This item shall be executed as per specific requirement supported with justification with prior consent and approval of the Engineer. Specifications for cables

1. All conduit and cable entering control panel shall be gland sealed to prevent the intrusion of gas and moisture.
2. All signal cables for carrying 4-20 mA , 0-5/10 V, low level transducer outputs, etc shall be copper PVC insulated twisted pairs, individually screened with tinned copper drain wire, overall screened, steel wire armored and overall PVC sheath. The twisted pairs shall be constructed with 24-30 twists per meter.
3. The rated working voltage shall be 100 V RMS and the maximum working voltage shall be 600 V RMS. The continuous current rating shall be atleast 5. Insulation between conductors and earth shall not be less than 10 Mega Ohms at 500 Volts.
4. Screening shall provide a minimum of 95% coverage of Copper braid or Mylar backed Aluminum foil. Individual shields in multicore cables shall be insulated from each other and from the overall shield and armoring.
5. Strict segregation shall be followed with not more than one type of signal run in any multicore cable. The different types of signals shall be segregated from each other and shall be contained in separate cables.

Application	**For connecting sensors and transmitters**
Type	Copper, PVC insulated, Shielded with aluminum polyester tape with tinned copper drain wire, Twisted pair, overall shielded with aluminum polyester tape with tinned copper drain wire PVC inners heated GI wire (0.90mm) Armored, PVC sheathed cable as per IS- 5308
No of pairs	1 pair X 1 X 1 Sqmm/ 1 pair X2 X 1 Sqmm / 1 pair X3 X1 Sq. mm
Working Voltage	100-600 VRMS
Application	**For Providing Power to field equipments**
Type	PVC insulated PVC Sheathed copper conductor armored cable1.1 KV grade as per IS 1554
No. of core	2/3/4
Gauge	1.5 Sq. Mm
Working Votage	1100 Vrms
Application	**For Providing Power to in house equipments**
Type	PVC insulated PVC Sheathed copper conductor armored cable1.1 KV grade as per IS 1554
No. of core	3
Gauge	1.5 Sq. Mm
Working Votage	1100 Vrms

19. RCC CHAMBERS

RCC Chambers of required size at site in M20 grade concrete shall be constructed on site for electromagnetic flowmeters and flow control valves. Item includes:-

1. Excavation for construction of instrument chambers in all types of soils and surfaces in all types of site conditions including drainage, dewatering, etc. including back filling.
2. The chambers shall have P.C.C of 150 mm thick below foundation, R.C.C. wall 150 mm thick, R.C.C. precast roof cover in MS angle frame in strip of 30 cm width with proper hooks for lifting, necessary number of M.S. steps embedded in wall etc. complete.

20. COMMISSIONING OF WIRELESS TELEMETRY

Either Radio telemetry or GPRS as a communication media shall be used to communicate with nos. locations of OHT's / ESR / UGRs / GSR/ remote locations of and one central server station for monitoring & SCADA purpose.

21. GPRS Modem/ Router:

The device shall act as REMOTE MANAGEMENT DEVICE.

This shall have facility to connect PLC with central SCADA server PC at central location office through GPRS or PSTN.

The PSTN / Broad band with dynamic IP connection or SIM card with GPRS enabled with Dynamic IP be created. The connection will be provided based on best available connectivity at each ESR or remote location. The Contractor shall provide connectivity device which will be able to incorporate any connection. The specifications for the device shall match following. Remote Maintenance PLC point to Point RAS or Internet remote access: and any PLC / device / equipment with TCP/IP & Modbus/Ethernet.

Remote Service: Data acquisition (Tag names) in MODBUS/RTU, MODBUS/TCP 'Tag names' enable alarm management, Basic programming, custom Web pages, reporting, Data Logging: Internal data base for data logging 21.000 points. Retrieval of the data base with files transferred by FTP put or email attachment.

- Alarms: 'Tag name' database: 128Kb. Alarm Notification by email, SMS, FTP put and/or SNMP trap. Available standard limits to configure: Very Low, Low, High, Very High + Dead zone and activation delay.
- Alarm summary and historian available in HTTP and via FTP files transfer.
- Alarm cycle management: ALM, RTN, ACQ and END.
- MMI HTTP: System and user defined Web site.
- SNMP: 'Tag Name' read/write
- FTP: Whole set of parameters are available in files
- Callback: Call back on user request or on amount of rings
- Direct or Internet call back (supports dynamic DNS)
- Firewall IP filtering
- Script-dedicated application to be programmed with the Basic language.
- Router IP forwarding, NAT, port forwarding and routing tables.
- Internet RAS connection (PPP), PAP/CHAP security. Data compression, ISP connection (Internet Service Provider) primary and secondary, supports DNS.
- Synchronization Embedded real-time clock, manual setup via http or automatic NTP setup
- File Management FTP client and server for configuration and data transfer.
- Web Site Security: DAA and session control. HTML standard supports the entire PDA browsers.
- SSI technology (Server Side Include) and BASIC scripted ASP (Active Server Pages).
- Maintenance SNMP V1 with MIB2 and/or via FTP files
- Material ARM processor @75Mhz, 8Mb SDRAM, 8Mb Flash, Din Rail Mounting
- Power supply 12 - 24VDC +/-20%, SELV; consumption: 3-6 watts
- 1x SUBD9 serial port: RS232, RS422 or RS485, 1,5kV isolation
- 1x RJ45 Ethernet 10/100 base Tx; 1,5kV isolation
- Embedded modem: PSTN or GSM/GPRS
- GPRS Modem should support 4G
- Operating Temperature range: 0° to 50°C, 80% humidity (no condensation)

22. GPRS / GSM BASED DATA LOGGER

Providing supplying, fixing and commissioning GPRS/ GSM Data logger with internal battery having (5 years battery life) dual channel for flow and pressure logging with internal pressure transducer, PC software, PC connection cable, Infra-red reading head, pressure hose, instruction manual etc.

The Contractor shall supply the required GPRS enabled Data logger to be installed at remote locations. The GPRS Data LOGGER shall be accepted after the third party inspection, the charges for the same shall be borne by the Contractor

23. LOCAL SCADA SYSTEM

Local (in-plant) SCADA system with control at required location shall be provided as per contract BOQ (if required)

The Contractor shall supply and install all necessary cables and connectors including power point connections / arrangements etc. for proper functioning of the system. Unless otherwise specified following are the minimum requirements of the system:

- To automate, supervise & control all the pumping operations
- Supervisory control of pumping parameters at local as well as at Master Control Centre central location with data transfer.
- All the control functions and control parameters at ICP.
- HMI for Operator

24. SCADA PC CONSOL FOR RUNNING SCADA SOFTWARE

A. SCADA PC Consol

The SCADA System shall be located in the control room of WTP. Control panel board shall be provided to mount all instruments and house all components of the instrumentation and control systems to execute control functions for the pumping operation and PC based software for dynamic viewing of the plant process and operations. It will offer all required supervisory control functions as well as provide GUI for plant personnel. The system shall have all the facilities such as communication ports and modules for interfacing, data recording and display with the SCADA control room/MCC. Contractor shall provide all the required furniture for the PCs & system at SCADA control room/MCC with air-conditioning unit.

B. SCADA PC Configuration: -

Processor	Intel Core i7
Make	Industrial Grade Reputed Make
Chipset	Intel
RAM	16 GB DDR
Cache	8 MB Total Cache
Optical Drive	Combo drive
HDD	1TB SATA
Ports	Display Port-1 VGA
	USB 4.0-6 Nos.
	Serial Port -1 Rear
	Parallel Port -1Rear
	Ethernet network- 1 Rear
	Stereo Line-In 1 Rear
	Stereo Line-Out 1 Rear
Ethernet	Gigabit LAN
Power supply	Power Supply - 280 W
Keyboard	Wireless Keyboard
Mouse	Wireless Mouse
Monitor	24"LCD/LED
Operating Software	Windows -10

The Contractor shall supply and install all necessary cables and connectors etc.

25. SCADA SOFTWARE

A. SCADA Server & Softwares

The Contractor shall supply all programming (PLCs) and SCADA software's necessary to provide Operation and control for water supply scheme along with control operation of valves actuators from MCC. SCADA System shall communicate with field PLC through RF/ GPRS for data, information & Control. The software required shall consist of those programs necessary for the system to efficiently perform the functions specified

herein, plus enable convenient and efficient preparation of new programs. The system shall be capable of operating in a foreground/background mode. The Contractor shall assume complete responsibility for the successful operation of all software and application programs provided as part of the system.
Full SCADA Functions with Visualization, trend representation. Data Logging, trend display and recipe management.

VBA and IL Scripts shall be supported on SCADA Software for ease of developmental Activities. SCADA Software shall have pre- installed Collection of symbols. The symbol can be extended specifically and shall support various graphics format like (wmf, png, jpg etc).
Historian Server for data archiving
Report Generation
Alarm/Event
Trends
Liftime license for SCADA & PLC software
OPC license for 5 Yrs

- The diagnostic information from SCADA shall be available through using Mobile app. The SCADA software shall be support signaling System with Various alarms, degree of severity and trigger options.
- The software shall have access to multiple databases like SQL Server, MS Access etc with role-based access control. The Software Solution shall have numerous data logger with all events with time stamped with Values.
- The SCADA Software shall utilize the web services communication for viewing or controlling automation system from any client station or mobile in LAN or WLAN which offers maximum security. The number of remote or local Viewing shall be of minimum 10 Stations.
- Using the webserver functionality additional features such as trends, alarm management, historians, recipe shall be supported on remote/client station too. The alarms reports shall be imported/exported in external files in .csv format.
- The alarm management features of the Software shall be logged in Historical log which can be displayed later. The alarm management part of the Software shall set to give notifications via SMS, Voice. E-mail etc.
- SCADA softwares shall be of reputed OEM

B. Cloud based web Pages (If Required)

Important Data to be uploaded to the cloud using state of the art protocols like IOT drivers etc. and viewed using Http protocol. Any software/hardware/drivers required for the same shall be provided by Contractor
This shall be only in view mode and no changes shall be possible in this method
It should be possible to add a link to these Web Pages on the ULB website. The link to be added by the customer using the services of the service provider/agency maintaining the website of ULB.

C. Mobile App (If Required)

Apps for IOS and android devices to be developed by the Contractor

1. Apps to be downloadable by users/citizens if required
2. Screens of the apps to be developed as per poject requirement
3. No Changes to be possible using the apps.

D. Control Room Furniture

In addition to the control system equipment, the Contractor shall provide furniture to complement or match both the colour and styling of the equipment. Control room furniture shall comply with relevant IEC standards for ergonomic design. Details of the control room furniture shall be submitted to the Engineer for approval. Fabric-covered upholstered swivel-type adjustable arm chairs with casters shall be provided along with a rigid and lockable steel cupboard for the storage of operating and maintenance manuals, drawings, logger paper, charts, disks and the like.
The visual display unit consoles or VDU desk shall incorporate suitable nos. drawer unit with drawers for operators use and for standard files.

E. SCADA control room/MCC instruments earthing

EInstruments and all related equipment earths shall be electrically bonded to the instrumentation earthing system or be provided with a separate independent isolated earth terminal bonded directly to the main station earth terminal in accordance with BS 6739.

26. PC SOFTWARES, KEYBOARDS, PRINTERS ETC

A. System Softwares

System parameters such as data, time, set points, alarm limits, loop tuning constants, etc., shall be entered or modified via the operator's terminal. Any input, which modifies the system, shall be logged on the logging printer with data and previous value.

<u>Software Licensing</u>- The computer system supplier shall include all necessary licensing for operation of all software on all computer systems provided. All packaged and system software licenses shall be registered by the Contractor in the name of the ULB.

B. Operating System

The Operating System software shall be the latest version of Microsoft Windows to provide system resource allocation and management in a Real-Time, multi user environment.

The Operating System shall provide an orderly shutdown on loss of power or low voltage and automatically restart the system, including processor and peripherals, upon restoration of power without operator intervention. The power failure protection shall protect against loss of program and/or work in progress at the time of occurrence.

C. Programmer's console

For systems, which require it for modifications or additions to be made to the operating system or application programmes, a programmer's terminal shall be provided of the type recommended by the system software and hardware manufacturers.

D. Diagnostic capability

The system shall be provided with the capability to diagnose hardware malfunctions in the central station and related peripheral equipment. Diagnostic programmes shall be password-protected to prevent unauthorized use. The diagnostic procedures shall be able to be performed by operations personnel with a minimum of training. Software routines shall be provided which can isolate a single problem to the circuit board level. The routines shall be menu-driven to allow for ease of use. Complete documentation of diagnostic procedures shall be provided as part of the system documentation.

E. Printer Requirements

Printer shall be supplied for reports, alarms and events as detailed below. It shall be possible to configure any text based output from the master station to either of these printers.

It shall be possible to add additional printers if required. Additional printers shall be capable of being assigned the functions of alarm/event logging, or report printing/ scanning of documents.

a) Printer Connections

Alarm / Event printers, shall offer at a minimum of 100 characters per second, upper and lower case modbud character set with true descenders, a minimum of 132 characters per line, and a self-test facility capable of printing automatically the entire character set.

The minimum size of printer buffers shall be 4k characters.

Each printer type shall connect to the system using the standard interfaces such as RS232 / Ethernet with both RTS/CTS control signal and XON/XOFF data transfer control methods supported.

All printer types shall be of a low noise type or shall be provided with acoustic hoods to ensure the ambient noise level never exceeds 30Dba.

Printer types shall be provided with all necessary cables and connectors. Should a printer be off-line when an output is ready, the control system shall send a message to the operator.

b) Alarm/Event printers

The alarm/event printer shall be a dot matrix/ laser printer with a minimum of four colors available. Alarms shall be highlighted by colored printing (i.e. plant alarms red/system alarms orange/return to normal green) while events and alarm acceptance shall be printed in black. This printer shall be located in the Control Room in a self-contained printer stand.

C) Report colour printers

The report colour printer shall be an A4 printer capable of producing high quality text and graphical reports for plant and management purposes and full colour high quality prints of screen display mimics, trends etc. In text mode the printer shall be capable of up to 7 pages per minute for text and 1 minute per page for graphics. The resolution shall be 600 by 600 dpi for colour and 600 by 600 dpi for black as a minimum. The printer shall be capable of receiving and printing data formatted for a Hewlett Packard printer. It shall be located in the control room of each location.

26. UNINTERRUPTED POWER SUPPLY

The monitoring system processor(s), VDUs, logging printer and communications equipment shall be powered through an uninterruptible power supply (UPS) via a dedicated distribution system. The UPS shall provide for full functioning of this equipment for minimum of one hour in the event of a power failure. UPS capacity shall be over-sized in terms of rating and power duration by 50% to provide for future additional equipment. The Uninterruptible Power Supply system shall consist of a regulated rectifier to operate from commercial AC power, a battery to store emergency power, and an all – static inverter to convert DC power from the rectifier or battery to high quality AC. The power supply system shall normally operate from the existing commercial power. When commercial power failure occurs, the battery shall immediately and without switching, take over as the source of power to the inverter which shall continue to provide high quality AC power to the equipment.

The UPS shall be following Specifications: -

Type	On Line
Input	230 VAC+/-15%
Output	230 VAC+/-1%
Power factor	0.8 Leg
Power capacity	Min 5 KVA or As per requirement of site and system with approval +/-0.5 Hz.
Frequency	50Hz+/-0.5Hz.
Wave form	Pure sine wave
Total Harmonic Distortion	<4% on linear load
Inverter	IGBT based high freq PWM
Inverter efficiency	>85%
Duty	Continuous
Transfer time	Zero secs.
Battery Voltage	48-360 VDC as per requirement
Battery Type	Scaled Maintenance Free
Operating Temp	0-45ºC
Humidity	Up to 95 % RH
Protection	Output high/ Output Low/ Output Short Ckt/Batt. Low Trip
Alarm	Battery Low
Indicators	Mains ON, Inverter Low, Fault
Metering	Output voltage

Exact sizing shall be the responsibility of the system suppliers based on total load requirements of equipments proposed.

The system batteries shall be sized to sustain operation for a minimum of 360 minutes. The system batteries shall be of the gelsel or NiCad type and shall reside in the same enclosure as the UPS.

27. AIR CONDITIONING UNIT

Split type room air conditioner of 2-ton (or suitable) shall be provided at the Central Control Room for monitoring system and at SCADA rooms in pumping stations. The air conditioner shall be sized to maintain a temperature of 20 +/- 2 degree Celsius. Air conditioner and its part shall be constructed with the strength and rigidity adequate for normal conditions of handling, transport and usage. AC shall be of reputed brand.

28. FLOW CONTROL VALVE (FCV)

A. Function

The valve will be controlled by an electronic-controller, which enables control of the requested flow rate or as per demand. The controller should enable receive as an input either pulse/volume contact-input or 4-20mA analogue signal.

The start / stop operation of the controller can be made manually or by external dry-contacts (digital) signals.

The flow set-value may be modified automatically on time basis, or by a pre-defined relation to pressure or another measured parameter. The losses in flow control valve shall not be more than 0.2 kg/ cm2 for present as well as ultimate demand

B. Main Valve

The valve shall be hydraulically operated, single chamber, direct diaphragm-sealing (weir type) or globe pattern valves. Oblique/Y Pattern valves shall not be permitted over 200 mm due to dismantling difficulties.

The valve shall consist of three major components: the body, cover and the diaphragm/trim assembly.

For proper working of FCV, a kinetic air valves of adequate pressure rating & size need to be installed.

Kinetic Air walls of adequate pressure rating and size need to be installed with FCV for proper functioning of Flow Control Valbe.

In Weir Patten valves, a diaphragm and a spring are the the only moving parts, and the diaphragm itself is the component that closes or opens the liquid flow.

The diaphragm shall form a sealed chamber in the upper portion of the valve, separating operating pressure from line pressure.

Packing glands and/or stuffing boxes are not permitted and there shall be no o-ring or similar seals on the shafts that may suffer friction which can start internal leakage in the valve.

The valves should be capable of regulating at very low flow, no hourglass-shaped disc retainers shall be permitted and no V-type, U-type or other slotted type disc guides shall be used.

The valve shall contain a resilient, nylon reinforced rubber diaphragm.

The ratio between valve flow capacity (KV) and the valve position shall be linear so to allow stable regulation even under high pressure differentials and low-flow conditions.

Disassembly and reassembly of all the valve's components shall be made possible on site, without having to remove the valve from the line.

Each valve should be hydrostatically tested at least 60% above the rated pressure. A test report should be supplied with each valve.

C. Material of Construction:

Body	Ductile iron ASTM A536 (Valve should be WRAS approved or Equivalent standard)
Circuit fittings	Brass EN12164
Control tubes	Nylon/SST316
Surface Protection	Epoxy coating min. 150 microns, color RAL 5005 Blue
Polyester coating	min.100microns, color RAL 5005 Blue
Operation	Automatic, manual override enabled
Seat Ring	Stainless steel 304/316, raised (if applicable).
Stem	Cast Steel / Stainless Steel 304/316 (replaceable inline & onsite), if applicable
Spring and Bearing Bush	Stainless Steel AISI – 302/304/316 (if applicable)
Disc guide, disc retainer & Diaphragm washer	Cast Steel / Stainless Steel AISI – 304/316, Bronze & Coated Steel
Seal	Synthetic rubber-Buna-N /EPDM (FDA/WRAS or equivalent standard approved)
Nut-Bolts and Studs	Stainless Steel AISI-304 ASTM-A 193 B7
Solenoid Valve (IP 65)	Brass, SST 316 with IP68

D. Operation

The pressure losses inflicted by the above mentioned fully opened valve should not exceed 0.3 bar at nominal flow speed of 3 m/sec. The minimal required pressure differential for reliable closure of the valve should not exceed 0.1 bar.

E. Maintenance

The valve supplier should propose a recommended seven years set of spare parts per a batch of 5 valves of the same diameter.
The valves manufacturer must provide 7 years of warranty.
The valve's pilot control loop should include a low maintenance, inline "self-cleaning" control-filter.
The typical weight of any control chambers and trim assembled as a complete unit, regardless of valve diameter, shall not exceed the permitted lifting weight for a single person as defined in the regulations. Disassembly will not require usage of sophisticated, heavy lifting devices such as cranes of any type. Should such devices be required – these are to be provided and installed at the assembly site by the supplier.

29. CONSTRUCTION OF INSTRUMENTATION ROOMS

Designing the structure with relevant standard

Construction of instrumentation room as per approved drawing in RCC framed structure in M20 grade concrete with PCC in M-10, B.B masonry (II class in C.M. 1:5) 12 & 15mm cement plaster in C.M. 1:5 inside and outside & min. one brick thick wall, steel doors with frames and alumimium framed windows with glass panels provided with external protection. 4) Vitrified tiles flooring on M-10 C.C.
One coat of primer and two coats of plastic emulsion paint of approved quality and shame dot the internal surface shall be provided Waterproof cement painting to exterior surfaces.
The reinforced concrete roofs shall have waterproofing with water proof compound. Finished roof surface shall have adequate slope to drain quickly the rain water to R.W. down take inlet points.
Building plinth shall be minimum 450 mm above average finished ground level around building

30. Laboratory equipment.

For the testing of incoming Raw water & outgoing Clear water laboratory of suitable Class as per CPHEEO Manual shall be provided by the Contractor**.** Contractor shall provide proper lab instruments with chemical & accessories **(**during O&M period**)** for checking the following water quality parameters from different WTP locations, **i.**e, inlet chamber, Clear water sump and ESR/GSRs**.**

Drinking Water Quality Standards

WATER QUALITY PARAMETERS AND BIS STANDARDS FOR VARIOUS CHEMICAL AND BIOLOGICAL CONSTITUENTS

S.No.	**Parameters**	**Drinking water IS 10500: 2012**	
		Permissible Limit	**Maximum Limit**
1	Odor	Agreeable	Agreeable
2	Taste	Agreeable	Agreeable
3	PH	6.5 to 8.5	No relaxation
4	TDS (9mg/l)	500	2000
5	Hardness (as CaCO3) (mg/1)	200	600
6	Alkalinity (as CaCO3) (mg/1)	200	600
7	Nitrate (mg/1)	45	No relaxation
8	Sulfate (mg/l)	200	400
9	Fluoride (mg/l)	1	1.5
10	Chloride (mg/l)	250	1000
11	Turbidity (NTU)	5	10
12	Arsenic (mg/l)	0.01	0.05
13	Copper (mg/l)	0.05	1.5
14	Cadmium (mg/1)	0.003	No relaxation
15	Chromium (mg/l)	0.05	No relaxation
16	Lead (mg/1)	0.01	No relaxation
17	Iron (mg/1)	0.3	No relaxation
18	Zinc (mg/1)	5	15
19	Fecal Coliform (cfu)	0	0
20	E. Coli (cfu)	0	0

8.0 GENERAL SPECIFICATION FOR ELECTRICAL SYSTEM

The scope of work of this specification includes design, manufacture, quality assurance, shop assembly, shop testing, packing, Transportation, Insurance, delivery at site, site storage and preservation, installation, testing, commissioning, performance testing, acceptance testing, training of Employer's personnel, handing over to Employer with warranty as per the specifications hereunder, each complete with all auxiliaries, accessories, mandatory spare parts, tools and warranting a trouble free safe operation of the installation.

Any item or works though not specifically mentioned in this specification but required to complete the equipment & systems for its trouble free & safe operation for meeting the intent of the specification shall be deemed included in scope of Contractor.

Design of electrical systems shall be as per the norms & rules set by different regulatory bodies, IS, BIS recommendations, standard practice. Wherever felt necessary guidelines laid down in other internationally accepted manuals/text books are followed. Design criteria considered for different items of work shall involved the below.

1. Compliance of Electrical Items

All the electrical apparatus to be used should comply to:

Indian Electricity Rules
Indian Electricity Acts
Factory acts
BIS & International Standards
National Electrical Codes of Practice;
Any other regulations laid down by local authorities;
International Electro-technical Commission (IEC).

IMPORTANT CONDITIONS:-

1 The BIDDER shall submit design calculation for each and every equipment so as to assess and decide suitable offer.

2 A licensed class A electrical contractor authorized under I.E. Rules shall only carry out the work.

3 The Successful BIDDER on award of contract shall have to prepare and submit the detailed drawing of the work duly approved by the Chief Electrical Inspector and Electrical Adviser, Govt. of M.P. After completion of work the representative of the Chief Electrical Inspector and Electrical Adviser shall inspect the same. The inspection the same. The inspection fee shall be borne by the contractor and electrical substation shall be charged only after approval and permission of the competent authority as per I.E. Rules.

4 Supply and inspection of all the equipments shall be as per relevant BIS/ I.S. Specification and latest I.E. Rules.

5 Make, Materials, Technical specification, Circuit diameter grams and connection details of each and every equipment and its major parts offered should be clearly specified in the TENDER.

6 Test certificates guarantee, certificate and operation manual shall be submitted along with the supply of equipment.

7 After commissioning of all the equipment successful trial will have to be given for at least 72 Hours.

8 Maintenance and training of department staff:-
After installation, commission and official testing of electric substation and other equipment satisfactorily, the contractor shall have to run and maintain and electric substation to the complete satisfaction of the Engineer in charge for a period of at least 7 days round the clock through his experienced and competent staff under supervision of his experienced and qualified engineer.

9 Any work equipment not specified in particular but considered necessary to complete the works as per specification and I.E. Rules are also included in this TENDER.

2. Power Supply Information

The proposed power supplies are as follows:

i.	33 kV / 0.433 KV AC System	Voltage variation +10 to -10% Freq. Variation +5 to -5%	Three Phase, 50 Hz, effectively earthed AC System.
ii.	11 kV / 0.433 KV AC System	Voltage variation +10 to -10% Freq. Variation +5 to -5%	Three Phase, 50 Hz, effectively earthed AC System.
iii.	415V AC System	Voltage variation +10 to -10%	Three Phase and neutral, 50 Hz,

		Freq. Variation +5 to -5 %	effectively earthed system.
iv.	240V AC System	Voltage variation+10% to -10% Freq. Variation + 5% to -5 %	Single phase and neutral, 50 Hz, effectively earthed system
v.	24V DC Supply	Voltage variation +/- 10%	Unearthed two wire system

Permanent power supply is to be obtained from the local power supply authorities at HT (33 KV / 11 KV) or LT (415V) voltage as per the guidelines of the power supply authorities depending upon the load requirement of the pumping stations. Its Contractor responsibility to lay power line from near by Grid Sub Station to Plant as per direction of local electrical power authority. 2 Pole / 4 Pole structure shall be provided with all required equipmebnts such as LA, DOF, AB Switch, ACSR conductor etc.

3. Design Criteria

The Design Criteria described below covers the electrical equipment/system for the proposed Project as follows: The following assumptions have been made to arrive at the estimated load of the different sites.

			Load Factor
(i)	Main motor	:	0.9 or based on BKW
(ii)	Auxiliary motor	:	0.9
(iii)	Auxiliary load, valve motors, etc.	:	0.9
(iv)	Lighting load	:	1.25
	Diversity Factor		
(i)	Main and auxiliary motor	:	1.0
(ii)	Auxiliary load, valve motors, etc.	:	1.5
(iii)	Lighting load	:	1.2
	Power factor of LV Motors	:	As per Manufacturer's Catalogue
	Efficiency of LV Motors		Premium Energy Efficient Motors (Energy efficient eff 3)

(Energy efficient, high performance motors shall be provided for optimum utilization of energy.)

Transformer shall be design for operation of complete plant load with 10 % design margin. Step down voltage shall be selected as per design requirement. Transformers efficiency shall be as per IS 1180 (Part 1) : 2014, Level 3.

DG set shall be capable to cater the emergency load the plant in case of main power failure.
All power cables (XLPE Type) shall be sized based on continuous current capacity, permissible voltage drop and short circuit current rating. The voltage drop shall be limited to 3 % at rated equipment current rating.
The other rating factors for variation in ground temperature, variation in ambient air temperature, grouping of cables, depth of laying, etc. shall also be considered for cable sizing.
For all indoor and outdoor areas LED fittings to be provided. Decorative fixtures/ receptacles shall be used for office areas and industrial type fixtures/ receptacles shall be used for remaining areas. Street/ area lighting shall be controlled by time switch for automatic switching off luminaries.
The values of fault level for designing the electrical system shall be based on transformers capacities, its impedance and system fault level. Fault clearing time for sizing of earth conductor will be taken as one second.
The material of earthing conductor shall be hot dip Galvanized Steel. All connections between the earth conductors buried in earth/concrete and between earth conductor and earth leads shall be of welded type. While sizing the buried earth conductor, a corrosion allowance of at least 20% shall be taken. Plant earthing system shall be designed such that the overall earthing grid resistance is maximum one (1) ohm.

4. SWITCHGEAR PANEL

A. General Requirement

This specification covers technical specifications of 33 or 11 kV Indoor switchgear panels, as per relevant IEC62271-100, IEC62271-200 and Indian standards.
The switchgear boards shall have a single front, single tier, fully compartmentalized, metal enclosed construction complying with IEC 62271-200, comprising of a row of free standing floor mounted panels.

B. Design

The switchgear assembly shall be dust, moisture, rodent and vermin proof, with the truck in any position SERVICE, TEST or ISOLATED and all doors and covers closed. All door and removable back cover shall have gaskets all round.

The panels shall have a degree of protection of IP-4X. All louvers, if provided, shall have very fine brass or GI mesh screen. Tight fitting gourmet / gaskets are to be provided at all openings in relay compartment. Numerical Relays shall be fully flush mounted on the switchgear panels at a suitable height. Switchgear along with the enclosure shall be type tested for temperature shall test as per IEC and should be suitable for an ambient of -7 deg C to 35deg C.

The switchgear construction shall be such that the operating personnel are not endangered by breaker operation and internal explosions and the front of the panels shall be specially designed to withstand these. Pressure relief device shall be provided in high voltage compartment of a panel, so that in case of a fault in a compartment, the gases produced are safely vented out, thereby minimizing the possibility of it's spreading to other compartments and panels.

Enclosure shall be constructed with rolled steel sections. The doors and covers shall be constructed from cold rolled steel sheets of 2.0 mm or higher thickness. Gland plates shall be 2.5 mm thick made out of hot rolled or cold rolled steel sheets and for non-magnetic material it shall be 3.0 mm.

The switchgear shall be cooled by natural air flow and forced cooling is not acceptable for any rating. The Switchgear, Circuit Breaker, Vacuum Interrupter shall be of same make to ensure full life cycle optimization.

Total height of the switchgear panel shall generally be 2600 mm. The height of switches, pushbuttons and other hand operated devices shall not exceed 2000 mm and shall not be less than 700 mm.

Necessary guide channels shall be provided in the breaker compartments for proper alignment of plug and socket contacts when truck is being moved to SERVICE position. A crank or lever arrangement shall preferably be provided for smooth and positive movement of truck between Service and Isolated positions. Padlocking arrangement should be provided for the racking mechanism.

Safety shutters complying with IEC 62271-200 shall be provided to cover up the fixed high voltage contacts on busbar and cable sides when the truck is moved to TEST position. Padlocking arrangement should be provided for the shutter mechanism individually. The shutters shall move automatically, through a linkage with the movement of the truck. It shall however, be possible to open the shutters of busbar side and cable side individually against spring pressure for testing purpose after defeating the interlock with truck movement deliberately. Shutter shall be of metallic type and earthed.

Switchgear construction shall have a bushing or other sealing arrangement between the circuit breaker compartment and the busbar/ cable compartments, so that there is no air communication around the isolating contacts in the shutter area with the truck in service position.

The breaker and the auxiliary compartments provided on the front side shall have strong hinged doors. Busbar and cabling compartments access shall be provided on the rear side shall have separate bolted covers for easy maintenance and safety. Breaker compartment doors shall have padlocking facility. The panels shall have manual ON and OFF facility and same shall be operable with the CB door closed.

In the Service position, the truck shall be so secured that it is not displaced by short circuit forces. Busbars, jumpers and other components of the switchgear shall also be properly supported to withstand all possible short circuit forces corresponding to the short circuit rating specified.

Suitable foundation frames made out of steel channels shall be supplied along with necessary anchor bolts and other hardware, for mounting of the switchgear panels.

5. Circuit Breakers (VCB Type)

Circuit breaker shall be re-strike free, stored energy operated and trip free type. Motor wound closing spring charging shall only be acceptable. An electrical anti-pumping relay shall be provided for breaker, even if it has built-in mechanical anti-pumping features.

During closing, main poles shall not rebound objectionably and mechanism shall not require adjustments. Necessary dampers shall be provided to withstand the impact at the end of opening stroke. Slow closing facility shall preferably be provided for checking and adjustment of arc chutes and poles when the breaker is completely withdrawn and isolated.

Plug and socket isolating Contacts for main power circuit shall be silver plated, of self-aligning type, of robust design and capable of withstanding the specified short circuit currents. They shall preferably be shrouded with an insulating material. Plug and socket contacts for auxiliary circuits shall also be silver plated, sturdy and of self-aligning type having a high degree of reliability. Thickness of silver plating shall not be less than 5 microns.

All working part of the mechanism shall be of corrosion resisting material. Bearings which require greasing shall be equipped with pressure type grease fittings. Bearing pins, bolts, nuts and other parts shall be adequately secured

and locked to prevent loosening or change in adjustment due to repeated operation of the breaker and the mechanism.

The operating mechanism shall be such that failure of any auxiliary spring shall not prevent tripping and shall not lead to closing or tripping of circuit breaker. Failure of any auxiliary spring shall also not cause damage to the circuit breaker or endanger the operator.

Mechanical indicators shall be provided on the breaker trucks to indicate OPEN / CLOSED conditions of the circuit breaker and CHARGED / DISCHARGED conditions of the closing spring. An operation counter shall also be provided. These shall be visible without opening the breaker compartment door.

The closing coil and spring charging motor shall operate satisfactorily at all values of control supply voltage between 85-110% of the rated voltage. The shunt trip coil shall operate satisfactorily under all operating conditions of the circuit breaker up to its rated short circuit breaking current at all values of control supply voltage between 70-110% of the rated voltage. The trip coil shall be so designed that it does not get energized when its healthiness is monitored by indicating lamps *(*Red*)* and trip coil supervision relay.

6. <u>Control and Interlocks</u>

Panel shall have two separate limit switches, one for the Service position and the other for isolated position. Each of these limit switches shall have at least four *(4)* contacts which shall close in the respective positions.

Auxiliary Contacts of breaker may be mounted in the fixed portion or in the withdrawable truck as per the standard practice of the manufacturer and shall be directly operated by the breaker operating mechanism.

Auxiliary contacts mounted in the fixed portion shall not be operable by the operating mechanism, once the truck is withdrawn from the service position, but remain in the position corresponding to breaker open position. Auxiliary contacts mounted on the truck portion and dedicated for Employer's use shall be wired out in series with a contact denoting breaker service position. With truck withdrawn, the auxiliary contacts shall be operable by hand for testing. There shall be at least Two *(*2*)* NO and Two *(*2*)* NC breaker original Auxiliary contacts made available for Employer's use.

The contacts of all limit switches and all breaker auxiliary contacts located on truck portion and fixed portion shall be silver plated, rated to make, carry and break 1.0A, 240V AC. Contacts of control plug and socket shall be capable of carrying the above current continuously.

Movement of truck between SERVICE and ISOLATED positions shall be mechanically prevented when the breaker is closed. An attempt to withdraw a closed breaker shall not trip it.

Closing of the breaker shall be possible only when truck is either in ISOLATED or in-service position and shall not be possible when truck is in between. Further, closing shall be possible only when the auxiliary circuits to breaker truck have been connected up, and closing spring is fully charged.

7. <u>Busbars and Insulators</u>

All busbar and jumper connections shall be of high conductivity aluminum alloy / Copper of adequate size and supporting type test report shall be submitted for approval. They shall be adequately supported on insulators to withstand electrical and mechanical stresses due to specified short circuit currents

Busbar cross-section shall be uniform throughout the length of switchgear. Busbars and other high voltage connection shall be sufficiently corona free at maximum working voltage.

Contact surfaces at all joints shall be silver plated or properly cleaned and non-oxide grease applied to ensure an efficient and trouble free connection. All bolted joints shall have necessary plain and spring washers. All connection hardware shall have high corrosion resistance. Bimetallic connectors or any other technically proven method shall be used for aluminum to copper connections.

Panels Bus bar chamber should design in such way that it should not require any support insulators. All busbars shall be color coded for phase identification.

8. <u>Earthing and Earthing Devices</u>

A copper earthing bus shall be provided at the bottom and shall extend throughout the length of switch board. It shall be bolted/ welded to the framework of each panel and each breaker earthing contact bar.

The earth bus shall have sufficient cross section to carry the momentary short-circuit and short time fault currents to earth as indicated under switchgear parameters without exceeding the allowable temperature rise.

Suitable arrangement shall be provided at each end of the earth bus for bolting to Employer's earthing conductors. All joint splices to the earth bus shall be made through at least two bolts and taps by proper lug and bolt connection.

All non-current carrying metal work of the switchboard shall be effectively bonded to the earth bus. Electrical continuity of the whole switchgear enclosure frame work and the truck shall be maintained even after painting.

VT and CT secondary neutral point earthing shall be at one place only on the terminal block. Such earthing shall be made through links so that earthing of one secondary circuit may be removed without disturbing the earthing of other circuits.

9. Paintng

All sheet steel work shall be pretreated, in tanks, in accordance with IS: 6005. Degreasing shall be done by alkaline cleaning. Rust and scales shall be removed by pickling with acid. After pickling, the parts shall be washed in running water. Then these shall be rinsed in slightly alkaline hot water and dried. The phosphate coating shall be "Class-C" as specified in IS: 6005. The phosphate surfaces shall be rinsed and passivated. After passivation, Electrostatic Powder Coating shall be used. Powder should meet requirements of IS 13871 (Powder coating specification). Finishing paint shade for complete panels shall be RAL7032 for all boards, unless required otherwise by the Employer. The paint thickness shall not be less than 50 microns. Finished parts shall be suitably packed and wrapped with protective covering to protect the finished surfaces from scratches, grease, dirt and oil spots during testing, transportation, handling and erection.

10. **InstrumentTransformers**

All current and voltage transformers shall be completely encapsulated cast resin insulated type, suitable for continuous operation at the ambient temperature prevailing inside the switchgear enclosure, when the switchboard is operating at its rated load and the outside ambient temperature is 35 deg. C. The class of insulation shall be E or better.

All instrument transformers shall withstand the power frequency and impulse test voltage specified for the switchgear assembly. The current transformer shall further have the dynamic and short time ratings at least equal to those specified for the associated switchgear and shall safely withstand the thermal and mechanical stress produced by maximum fault currents specified when mounted inside the switchgear for circuit breaker modules. All instrument transformers shall have clear indelible polarity markings. All secondary terminals shall be wired to separate terminals on an accessible terminal block.

11. Transformer Protection Relay

The relay shall have The relay shall have instantaneous as well as time delayed three over current (50) and one earth fault (50N) protections.

The over current element should have the minimum setting adjustable between 5% to 200% of CT secondary rated current.

The relay shall have selectable directional & non-directional feature

The earth fault element should be suitable for residually connected CT or CBCT input. The relay shall be suitable for detection of earth fault currents in the range of 1% to 100% of the CT rated current.

Transformer troubles like Buchholz, Winding temperature, Oil temperature & Pressure Relief Device trips (as applicable) shall be wired to separate binary inputs of the relay and shall be configured to issue trip command to the breaker.

Trip circuit supervision shall be provided to monitor the circuit breaker trip circuit both in pre-trip and post-trip conditions.

12. Power Cable Termination

Cable termination compartment shall receive the stranded Aluminum conductor, XLPE insulated, shielded, armored, PVC jacketed, single core / three core, unearthed / earthed grade power cable(s).

A minimum clearance of about 600 mm shall be kept between the cable lug bottom ends and gland plates for stress cone formation for XLPE cables. Inter-phase clearance in the cable termination compartment shall be adequate to meet electrical and mechanical requirement besides facilitating easy connections and disconnection of cables. Dimensional drawing of cable connection compartment showing the location of lug, glands, CTs, gland plates etc. and the electrical clearances available shall be submitted for Employer's approval during detail engineering.

Cable termination compartment shall have provision for termination of power cables of sizes as indicated during detailed engineering with removable undrilled gland plates. For all single core cables gland plates shall be of non-magnetic material. Cable entry shall be from bottom. Any change will be intimated later.

13. Space Heaters

Strip type space heaters of adequate capacity shall be provided inside each panel to prevent moisture condensation on the wiring and panel mounted equipment Space heaters shall be rated for 240 V, 1 phase, 50 Hz supply. Heaters inside the panels shall not be mounted close to the wiring or any panel mounted equipment Heaters shall

be complete with either miniature circuit breakers or with isolating switches, HRC fuse on phase and link on the neutral of the heater supply

An adjustable type thermostat shall be provided in the heater control circuit with temperature range of 0-90 C The indication shall be provided for monitoring the healthiness of Space heater.

14. Rating Plate

The switchgear shall be provided with durable and legible nameplates containing all technical parameters. In addition to this, the Circuit breaker and its operating device shall be provided with a rating plate marked with but not limited to following data*:* -

Manufacturer's name, type of breaker
Type Designation or Serial number and year of manufacturing
Applicable relevant Standard *(*IS*/* IEC*)*
Rated voltage
Rated normal continuous current
Rated insulation level
Rated frequency
Rated short time breaking current with rated duration
Total weight of breaker

Tests

All routine tests shall be carried out as per relevant standard. Type test report shall be provided.

15. TRANSFORMER

A. General Requirement

This specification covers the technical details of 11or 33KV /0.433kV oil immersed, ONAN, outdoor type, cooper wound, three phase, 50 Hz Transformers with Off Circuit Tap Changer (OCTC) upto 1000 KVA transformer and with On Load Tap Changer (OLTC) for above 1000 KVA transformer.

The transformer and accessories shall be designed to facilitate operation, inspection, maintenance and repairs. The design shall incorporate every precaution and provision for the safety of equipment as well as topological environment condition of Site Location.

All outdoor apparatus, including bushing insulators with their mountings, shall be designed so as to avoid any accumulation of water and snow.

B. Design

The transformers shall be suitable for outdoor installation with three phase, 50 Hz, 11 kV or 33KV system in which the neutral is effectively earthed and they should be suitable for service with fluctuations in supply voltage up to plus 10% to minus 10%.

The transformer shall be capable to cater complete load of plant with 10 % design margin.

At WTP 1 Working and 1 Standby transformer shall be provided, both the transformers shall be capable to carry complete plant load.

The transformers shall conform to the following specific parameters:

Sl. No.	Item	Distribution Transformers
1	System voltage (Max.)	12 kV or 36 KV
2	Rated Voltage (HV)	11 kV or 33KV
3	Rated Voltage (LV)	6.6 KV or 433 V
4	Rating	As per design
5	Frequency	50 Hz +/- 3%
6	Quantity	1sets
7	No. of Phases	Three
8	Connection HV	Delta
9	Connection LV	Star (Neutral brought out)
10	Vector group	Dyn 11
11	Type of cooling	ONAN

Audible sound levels (decibels) at rated voltage and frequency for liquid immersed distribution transformers shall be as below 75 dB as per (NEMA Standards):

All de-ratings factors shall be considered like altitude, Temperature for deriving the transformer ratings

C. Technical Requirement

C.1 Core Material

The core shall be stack / wound type of high-grade Cold Rolled Grain Oriented or Amorphous Core annealed steel lamination having low loss and good grain properties, coated with hot oil proof insulation, bolted together and to the frames firmly to prevent vibration or noise. The core shall be stress relieved by annealing under inert atmosphere if required. The complete design of core must ensure permanency of the core loss with continuous working of the transformers. The value of the maximum flux density allowed in the design and grade of lamination used shall be clearly stated.

No-load current shall not exceed 2% of full load current and will be measured by energizing the transformer at rated voltage and frequency. Increase of 12.5% of rated voltage shall not increase the no-load current by 5% of full load current.

C.2 Winding

Material: HV and LV windings shall be wound from Super Enamel covered /Double Paper covered Aluminum / Electrolytic Copper conductor.LV winding shall be such that neutral formation will be at top. The winding construction of single HV coil wound over LV coil is preferable. Inter layer insulation shall be Nomex /Epoxy dotted Kraft Paper.

Proper bonding of inter layer insulation with the conductor shall be ensured. Test for bonding strength shall be conducted. The core/coil assembly shall be securely held in position to avoid any movement under short circuit conditions. Joints in the winding shall be avoided. However, if jointing is necessary the joints shall be properly brazed and the resistance of the joints shall be less than that of parent conductor. In case of foil windings, welding of leads to foil can be done within the winding.

C.3 Tapping Ranges and Methods

Tapping shall be provided on the higher voltage winding for variation of HV voltage within range of *(+)* 2.5% to *(-)* 5.0 % in steps of 2.5%. Tap changing shall be carried out by means of an externally operated self-position switch and when the transformer is in de-energized condition. Switch position No.1 shall correspond to the maximum plus tapping. Each tap change shall result in variation of 2.5% in voltage. Arrangement for pad locking shall be provided. Suitable aluminium anodized plate shall be fixed for tap changing switch to know the position number of tap.

C.4 OIL

The insulating oil shall comply with the requirements of IS 335. Use of recycled oil is not acceptable.

Oil shall be filtered and tested for break down voltage (BDV) and moisture content before filling. The oil shall be filled under vacuum.

The design and all materials and processes used in the manufacture of the transformer, shall be such as to reduce to a minimum the risk of the development of acidity in the oil.

C.5 Temperature Rise

The temperature rises over ambient shall not exceed the limits as per IS 2026 and IS 1180:2021.

The transformer shall be capable of giving continuous rated output without exceeding the specified temperature rise.

C.6 Tank

Transformer tank construction shall conform in all respect of IS. The internal clearance of tank shall be such, that it shall facilitate easy lifting of core with coils from the tank without dismantling LV bushings. All joints of tank and fittings shall be oil tight and no bulging should occur during service.

Inside of tank shall be painted with varnish/hot oil resistant paint.

The top cover of the tank shall be slightly sloping to drain rain water and Snow.

The tank plate and the lifting lugs shall be of such strength that the complete transformer filled with oil may be lifted by means of lifting shackle.

C.7 Plain Tank

The transformer tank shall be of robust construction rectangular/octagonal/round/ elliptical in shape and shall be built up of electrically tested welded mild steel plates of thickness of 6 mm for the bottom and top and not less than 4 mm for the sides for distribution transformers. Tolerances as per IS1852 shall be applicable.

In case of rectangular tanks, the corners shall be fully welded at the corners from inside and outside of the tank to withstand a pressure of 1.0 kg/cm2 for 30 minutes.

Under operating conditions, the pressure generated inside the tank should not exceed 0.4 kg/ sq. cm positive or negative. There must be sufficient space from the core to the top cover to take care of oil expansion. The space above oil level in the tank shall be filled with dry air or nitrogen conforming to commercial grade of IS 1747.The tank shall be reinforced by welded flats on all the outside walls on the edge of the tank.

C.8 Conservator

The dehydrating agent shall be silica gel. The moisture absorption shall be indicated by a change in the color of the silica gel crystals which should be easily visible from a distance. Volume of breather shall be suitable for 1 kg for transformers.

The capacity of a conservator tank shall be designed keeping in view the total quantity of oil and its contraction and expansion due to temperature variations. The total volume of conservator shall be such as to contain10% quantity of the oil. Normally 3% quantity the oil shall be contained in the conservator.

The cover of main tank shall be provided with an air release plug to enable air trapped within to be released, unless the conservator is so located as to eliminate the possibility of air being trapped within the main tank.

The inside diameter of the pipe connecting the conservator to the main tank should be within 20 to 50 mm and it should be projected into the conservator so that its end is approximately 20 mm above the bottom of the conservator so as to create a sump for collection of impurities. The minimum oil level (corresponding to -5 0C) should be above the sump level

C.9 Surface Preparation and Painting

All paints, when applied in a normal full coat, shall be free from runs, sags, wrinkles, patchiness, brush marks or other defects.

All primers shall be well marked into the surface, particularly in areas where painting is evident and the first priming coat shall be applied as soon as possible after cleaning. The paint shall be applied by airless spray according to manufacturer 's recommendations. However, where ever airless spray is not possible, conventional spray be used with prior approval of purchaser.

C.10 Paint Material

Following are the types of paint which may be suitably used for the items to be painted at shop and supply of matching paint to site: Heat resistant paint (Hot oil proof) for inside surface

For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of synthetic enamel/polyurethane base paint. These paints can be either air drying or staving.

For highly polluted areas, chemical atmosphere paint as above with one coat of high build Micaceous iron oxide (MIO) as an intermediate coat may be used.

C.11 Bushing

The bushings shall conform to the relevant standards specified and shall be of outdoor type. The bushing rods and nuts shall be made of brass material 12 mm diameter for both HT and LT bushings. The bushings shall be fixed to the transformers on side with straight pockets and in the same plane or the top cover for transformers above 100 kVA. The tests as per latest IS 2099 and IS 7421 shall be conducted on the transformer bushings.

Bushing shall be of porcelain/epoxy material. Polymer insulator bushings conforming with relevant IEC can also be used. Bushings of plain shades as per IS 3347 shall be mounted on the side of the Tank and not on top cover. Dimensions of the bushings of the voltage class shall conform to the Standards specified and dimension of clamping arrangement shall be as per IS 4257.Arcing horns shall be provided on HV bushings.

The bushings shall be of reputed make supplied by those manufacturers who are having manufacturing and testing facilities for insulators. The terminal arrangement shall not require a separate oil chamber not connected to oil in the main tank.

Minimum Current ratings for Bushing shall be equivalent to 1.5 to 2 times of the full load current of the transformer

C.12 Terminal Connectors

The LV and HV bushing stems shall be provided with suitable terminal connectors as per IS 5082 so as to connect the jumper without disturbing the bushing stem. Connectors shall be with eye bolts so as to receive conductor for HV. Terminal connectors shall be type tested as per IS 5561.

C.13 Terminal Markings

High voltage phase windings shall be marked both in the terminal boards inside the tank and on the outside with capital letter 1U, 1V, 1W and low voltage winding for the same phase marked by corresponding small

letter 2u, 2v, 2w. The neutral point terminal shall be indicated by the letter 2n. Neutral terminal is to be brought out and connected to local grounding terminal by an earthing strip.

C.14 Current Transformers

Current transformer shall be mounted inside the tank or outside with suitable marshalling box on LV side of the transformer. The current transformers shall comply with IS 2705.

All secondary leads of bushing mounted CT‘s shall be brought to a terminal box near each bushing.The CT terminals shall have shorting facility.CT should not get saturated up to 200% of rated current.

C.15 Fasteners

All bolts, studs, screw threads, pipe threads, bolt heads and nuts shall comply with the appropriate Indian Standards for metric threads, or the technical equivalent.

Bolts or studs shall not be less than 6 mm in diameter except when used for small wiring terminals. All nuts and pins shall be adequately locked.

Wherever possible bolts shall be fitted in such a manner that in the event of failure of locking resulting in the nuts working loose and falling off, the bolt will remain in position.

All ferrous bolts, nuts and washers placed in outdoor positions shall be treated to prevent corrosion, by hot dip galvanizing, except high tensile steel bolts and spring washers which shall be electro-galvanized/plated. Appropriate precautions shall be taken to prevent electrolytic action between dissimilar metals.

C.16 TESTS

All routine tests shall be done as per IS 2026 on both the transformers in presence of TPI and Client representative at manufacture works. Type Test Report shall also be furnished on same or heigher rating transformer. In cas e type test reports are not available same shall be done at rencogniez gouvernent agenceies.

16. LT PANEL BOARD

A. General

This Specification covers the technical requirements for the Main Low Voltage switchboard. The Low Voltage Switchboard shall be with aluminium bus bars, indoor type, free standing, with Plinth, floor mounting type, extensible on either side.

LT panels/ Main distribution boards including and above 800A (switchgear) shall fully comply with requirements of TTA equipment (Type Tested Assemblies) as per IEC-61439 or CRPI approved as per IS 8623 & IEC 60947 -1 2004 or equivalent.

Switchboards shall in accordance with the standards (IEC 61439-1 & 2 (Low Voltage switchgear and control gear assemblies) for Main LT switchboards & for Front operating switchboards) suitable for operation at three phase 4 wire, 415 volt, 50 Hz, neutral solidly grounded at transformer system with a short circuit level withstand as per schedule of quantities and drawings. The main distribution boards shall be suitable for operating voltage up to 690 V.

Switchboards shall comply to Form 4B for compartmentalized boards and Form 1B for non-compartmentalized boards as per IEC 61439-1. For form of separation only metallic covers shall be used, Hylem / PVC sheets shall not be allowed.

The enclosures shall be designed to take care of normal stress as well as abnormal electro-mechanical stress due to short circuit conditions. All covers and doors provided shall offer adequate safety to operating persons and provide ingress protection of IP 42 unless otherwise stated. Ventilating openings and vent outlets, if provided, shall be arranged such that same ingress protection of IP 42 is retained.

Entire switchgear used in switchboards shall be completely fuse free. All accessible bares terminals shall be provided with integral shrouds and shall be finger touch proof.

Bimetallic connectors shall be provided for termination of cable/Bus with aluminum conductors on copper bus bars.

B. Design Consideration

Main distribution boards shall be rated on the basis of voltage, current, frequency and the symmetrical breaking capacity of incorporated switching devices etc. as per IEC 61439.

All equipment and components of the main distribution boards shall be capable of continuous operation at their full current and voltage ratings and without detriment or malfunction at system continuous deviation of up to and including the following percentages of the normal values.

- Voltages ± 10 %
- Frequency ± 5 %

All components shall be capable of withstanding the dynamic, thermal and dielectric stresses resulting from prospective short circuit currents without damage or injury to personnel.

C. General Arrangement & Assembly

Unless otherwise specified the panels shall be designed to accept bottom/top entry of cables and shall be of rear access type. The rear access shall only be provided for access for termination of cables, all other equipments shall be accessible / openable from the front of panel. Where front access type is specified, the panel shall be designed completely for front access only. Enclosure manufacturing shall be as per IEC 61439 part-1

Each section shall be arranged typically as described below and as per the details shown in the drawings*:*

D. Constructional Features

The Switchboards shall be metal clad totally enclosed, floor mounted free standing type of modular extensible design suitable for indoor mounting. The Switchboards shall be designed for a temperature rise restricted to 45° C above ambient of 35° C.

All doors and covers shall also be fully gasketedwith metal based neoprene gaskets to ensure proper compression of the gaskets. All hinged doors shall have earth braid connected to the cubicle. Good quality door handles fitted with toggles to operate rods to latch with suitable slots in both top and bottom of switchboards shall be provided. Latching rods and associated brackets shall be cadmium plated.

Switchboard panels and cubicles shall be fabricated with CRCA Sheet Steel of thickness not less than 2.0 mm and shall be folded and braced as necessary to provide a rigid support for all components. The doors and covers shall be fabricated from CRCA sheet steel of thickness not less than 2.0 mm *(*or as per IEC 61439 for mechanical impact IK09*)*.

All panels and covers shall be properly fitted and square with the frame. The holes in the panel shall be correctly positioned.

Type test certificates shall be available for the proposed type of enclosure. The type test certificates shall be clearly state type, model and main characteristics of the assembly,

references of the standards applied for the tests, results obtained and the observations made during & after the tests.

Design drawings, component catalogues and type test certificates shall be submitted for Client approval prior to placement of order for the main distribution boards.

Each switchgear cubicles shall be fitted with label in front and back identifying the circuit, switchgear type, rating and duty. All operating device shall be located in front of switchgear only. Operating devices such as handles, push buttons, or similar shall be located at such a height that they can easily be operated and their center line shall be located within a zone between 0,2 m and 2 m above the base of the ASSEMBLY.

E. Draw out Features

Air Circuit Breakers shall be provided in fully draw-out cubicles, unless otherwise stated. These cubicles shall be such that drawout is possible without disconnection of the wires and cables.

The power and control circuits shall have self-aligning and self-isolating contacts. The fixed and moving contacts shall be easily accessible for operation and maintenance. Mechanical interlocks shall be provided on the drawout cubicles to ensure safety and compliance to relevant Standards. The MCCB's shall be provided in fixed type cubicles.

F. Instrument Accommodation

Instruments and indicating lamps shall not be mounted on the Circuit Breaker Compartment door for which a separate and adequate compartment shall be provided and the instrumentation shall be accessible for testing and maintenance without danger of accidental contact with live parts of the Switchboard.

For MCCB's instruments and indicating lamps can be provided on the compartment doors.

The current transformers for metering and for protection shall be mounted on the solid copper/aluminum busbars with proper supports.

G. Wiring

All wiring for relays and meters shall be with PVC insulated copper conductor FRLS wires. The wiring shall be coded and labelled with approved ferrules for identification. The minimum size of copper conductor control wires FRLS shall be 1.5 sq. mm. Wiring shall be terminated with ferrules on terminal block. CTs shall be provided with shorting facilities.

H. Cable Terminations

Knockout holes of appropriate size and number shall be provided in the Switchboard in conformity with the location of incoming and outgoing conduits/cables.

The cable terminations of the Circuit Breakers shall be brought out to terminal cable sockets suitably located in the cable chamber

The cable terminations for the MCCB's shall be brought out to the rear in the case of rear access switchboards or in the cable compartment in the case of front access Switchboards.

The Switchboards shall be complete with tinned brass cable sockets, tinned brass compression glands, gland plates, supporting clamps and brackets etc. for termination of 1100volt grade aluminum conductor XLPE cables.

Removable gland plates shall be provided for power and control cables. The gland plates shall be 3 mm thick and for single core cables shall be of non-magnetic material.

I. Space Heaters

Anti- condensation heaters shall be fitted in each cubicle together with an ON/OFF isolating switch suitable for electrical operation at 230 volts A.C 50 Hz single phase of sufficient capacity to raise the internal ambient temperature by 50° C. The electrical apparatus so protected shall be designed so that the maximum permitted rise in temperature is not exceeded if the heaters are energized while the switchboard is in operation. As a general rule, the heaters shall be placed at the bottom of the cubicle.

J. Ventilation

The fan if provided shall be interlocked with switchgear operation. The design of switch board cubical shall incorporate suitable measures like decreasing current density of conductors, increasing cubical volume for effective heat dissipation etc. in order to restrict temperature rise within the required limit.

K. Earthing

Continuous internal copper earth bus sized for prospective fault current to be provided with arrangement for connecting to station earth at two points. Hinged doors / frames to be connected to earth through adequately sized flexible braids.

L. Name Plates And Labels

Suitable engraved white on black name plates and identification labels of metal for all Switchboards and Circuits shall be provided. These shall indicate the feeder number and feeder designation etc.

M. Indicating Lamps

Indicating lamps shall be of the LED type. Color of lens shall be as per enclosed drawing/ data sheet.

N. Push Button

All push buttons shall be of push to actuate type having 2 `NO' and 2 `NC' self-reset contacts. They shall be provided with designation plates, engraved with their functions. Push button contacts shall be rated for 10 amps at 415V A.C. and 0.6 Amp.

O. Control and Selector Switches

Control and instrument switches shall be of the rotary type and shall be provided with properly designated plate. Control switches shall have momentary contacts spring return to centre with pistol grip handle. Instrument and selector switches shall have stay put contacts.

P. Tests

a. Type Test

The type tests certificates as specified in IEC 61439-1 standards for TTA Panels and IEC-60439-1 standards for PTTA Panels shall be provided.

b. Routine test

The panel assembler shall perform the routine test and provide the test certificates as defined in IEC standards. The routine test shall be done as per standard in presence of TPI and Client representative.

17. <u>DG Sets</u>

A. General Requirement

This specification covers the technical details of Silent type with Acoustic enclosure D.G. sets, 415V, control panel complete in all respects as indicated in these specifications.

The design construction, manufacture, inspection, testing and performance shall comply with all the currently applicable statues, safety codes, relevant Bureau of Indian Standards (BIS) British Standards (BS),

International Electro Technical Commission (IEC) Publication, CPWD DG set specifications, CPCB norms, NEMA, VDE and DEMA Standards.

The DG set shall be silent type with acoustic, radiator cooled, prime duty suitable for continuous operations & shall be capable to deliver the average load 80%, manually and automatically operated. The DG set shall comprise of diesel engine, coupled to four pole alternator on a single frame with integrated microprocessor based genset monitoring and control system having self-regulated, brushless/static excitation system.

All equipment shall be of the class most suitable for working under the conditions specified and shall withstand the atmospheric conditions without deterioration.

B. Design Consideration

DG set will be used as emergency source of power supply and for operation of emergency load in case of main power failure. DG Set will come into operation automatically in the event of AC power failure in the station. DG set will be capable for operation of Lighting, EOT Crane, PLC / SCADA system, dewatering Pumps etc. DG set will be selected for continuous duty. DG set will be provided along with AMF panel.

C. Diesel Engine

The engine shall conform to IS;10000/ ISO 3046/ BS;649 /BS 5514 amended up to date. Necessary certificate indicating the compliance of the above capacity requirement for the engine model so selected along with compliance of Noise and Emission norms as per latest CPCB guidelines for DG set.

The engine shall be fitted with following accessories subject to the design of the manufacturer:

(i) Dynamically balanced Fly wheel

(ii) Necessary flexible coupling and guard for alternator and engine (applicable only for double bearing alternator)

(iii) Air cleaner (dry/ oil bath type) as per manufacturer standard,

(iv) Electronic governor to maintain engine speed at all conditions of load.

(v) Daily fuel service tank of minimum capacity as per design, fabricated from M.S. sheet with inlet, outlet connections air vent tap, drain plug and level indicator (gauge) M.S. fuel piping from tank to engine with valves, unions, reducers, flexible hose connection and floor mounting pedestals, twin fuel filters and fuel injectors.

(vii) Suitable self-starter for 12 V/ 24 V DC.

Battery charging alternator unit and voltage regulator, suitable for starting batteries, battery racks with interconnecting leads and terminals.

(ix) Necessary gear driven oil pump for lubricating oil, priming of engine bearing as well as fuel systems as per manufacturer recommendations.

(x) Naturally aspirated/ turbo charger (as per manufacturer standard)

(xi) Lubrication oil cooler

(xii) Lubrication oil filters with replaceable elements.

(xiii) Crank case heater as per manufacturer recommendations.

(xvii) Engine Control Panel: fitted and having digital display for following: -

(a) Start/stop key switch.

(b) Lube oil pressure indication

(c) Water temp. indication

(d) RPM indication

(e) Engine Hours indications (f) Battery charging indication

(g) Low lub. Oil trip indication (h) High water temp. indication

(e) Over speed indication.

(xviii) All moving parts of the engine shall be mechanically guarded in such a manner that a human finger cannot touch any moving part.

D. Performance Requirement

The equipment shall be capable of delivering continuously at the generator terminals, (100% output at 80% load at unlimited time) as per design except for periodical maintenance when operating under the site and ambient conditions described in this specification as per relevant IS/BS 5514. Genset should have minimum 50% single step loading capacity and it should be able to take full load within 20 seconds from start.

The design parameters of the generator and excitation system shall be so chosen, that the set is stable while running at any load between no-load and full load. It should have isochronous speed control with load sensing governing system and should be capable of paralleling between sets at isochronous speed.

Engine shall be of standard design of the original manufacturers. Engine should be heavy duty four stroke; turbo charged Electrical start.

E. Oil System

It shall be fed through engine driven fuel pump. A replaceable element of fuel filter shall be suitably located to permit easy servicing. The daily service tank shall be complete with necessary supports, gauges, connecting pipe work etc. In case of Top Mounted tanks, non-return valves are must in fuel supply and return line of specified value. Pipe sealant should be used for sealing for all connections. No Teflon tape to be used. If piping length is more than 10 meters, detail engineering is required in consultation with OEM/ Manufacture. Fuel system shall be such used which comply to the latest environment flue gas data limitations amended upto date.

F. Lubricating Oil System

It shall be such that if the engine starts after a long shut down lubrication failure does not occur. A timer based auto priming pump (operative on AC/DC) shall be provided to keep engine lubricated, bearings primed all the time as per manufacturer's recommendation.

G. Cooling System

DG sets shall be radiator fan cooled, engine driven with silent enclosures.

H. Starting System

This shall comprise of necessary set of heavy duty batteries 12V/ 24V DC (as per manufacturer standard), and suitable starter motors, axial type gear to match with the toothed ring on the fly wheel. A timer in the control panel to protect the starter motor from excessively long cranking runs shall be integrated with the engine protection system. Battery capacity shall be suitable for meeting the needs of starting system (as 5 attempt starting), as well as the requirements of control panel, indications and auxiliaries such as priming pump as applicable etc. The system shall be capable of starting the DG set within 5-10 sec., in winter condition with an ambient temperature down to minus 7 degree C.

The battery charger shall be suitable to charge required numbers of batteries at 12V/ 24 volts complete with, transformer, rectifier, charge rate selector switch, indicating ammeter & voltmeter etc. Connections between the battery charger & batteries shall be provided with suitable copper leads with lugs etc.

I. Governing System

The governor shall be electronic type of class A1 or equivalent as per IS3046/BS 5514 & as per manufacturer.

J. Exhaust System

Engine exhaust system shall be fitted with hospital or most critical grade silencer available as per manufacturer; ducting, bends, thermally insulated aluminium clad exhaust piping etc. shall be provided along with structural support with stays for the engine.

K. Exhaust Stack Height

In order to dispose exhaust above building height, minimum exhaust stack height shall be followed as per MOEF & CPCB requirements.

L. Engine Mounted Instrument Panel

Engine will be supplied with engine mounted microprocessor based fully integrated generating set monitoring, metering and control system which should be equipped with digital electronic governor along with digital AVR to facilitate discreet control of speed and voltage. It should be equipped with starting control including integrated fuel ramping to limit the black smoke frequency overshoot with optimized cold weather starting. The indicating panel should have communication network facility to facilitate remote hooking on a common network. Engine instrument panel should be equipped with digital alarm and status mirage to monitor and display the following parameters of engine and generator.

M. Alternator

The alternator shall be industrial type, SPDP, IP23, Class 'H or F' insulation as per OEM, self-ventilated, air cooled, rotating field, salient pole, brushless, machine with exciter and shall be rated continuous duty with temperature rise class 'F' or as per OEM, limits at rated load. The alternator should comply to standard IEC, VDE, BS, ANSI.

The alternator shall be brushless synchronous and suitable for 3 phase, 415volts, 50 Hz. 0.8 p.f. and 1500 RPM.

The alternator shall be suitable for coupling directly to the engine. It shall be drip proof screen protected as per IP23.

The alternator shall be continuously rated and shall have class H or F (as per OEM)‖ insulation designed and built to withstand tropical conditions. It shall be confirming to BS: 5000 (Part-99) or IS: 4722-1992. The output of the alternator shall be above rated kW at 0.8 lagging power factor at site conditions and shall be suitable for sustaining of 10% overload for 1 hour in any 12 hours' period without damage.

Suitable Nos. Resistant Temperature Detector (RTDs) to measure the winding temperature and BTDS to measure bearing temperature shall be provided as per OEM standards. Space heaters shall also be provided where required.

The leads of embedded RTDs shall be wired upto the terminal block in a separate terminal box. Manufacturer shall indicate the setting values for each RTD / BTD for alarm and trip.

All external nuts and bolts shall be of high tensile steel only.

Alternator shall be provided with anti-condensation space heater of adequate rating suitable for 240 V+/-10%, 50 Hz, 1 Phase AC supply and shall be wired up to a separate terminal box.

The independent earth terminals on the frame complete with nuts, spring washer and plain washer shall be provided.

Alternator shall be provided with suitable flanges for termination of cables. Suitable arrangement shall be provided in the terminal box for cable.

N. Excitation System

The alternator shall be provided with a complete diode type brushless excitation system, capable of supplying the excitation current of the generator under all conditions of output from no load to full load and capable of maintaining voltage of the generator constant at one particular value/ or as per manufacturer's standards.

The excitation system shall comprise a shaft driven AC exciter with rotating rectifiers. The rectifiers shall have in-built protection for over-voltage.

The alternator should be complete with shunt and booster excitation. The exciter shall be fast response type and shall be designed to have a low time constant to minimize voltage transients under severe load changes. The excitation voltage response ratio shall be at least 0.8.

The rated current of the main exciter shall be at least 10% more than the alternator rated exciter current and it shall have a 40% overload capability for 10 seconds.

Self-excited type shall be required during starting and normal running of the alternator.

O. Automatic Voltage Regulator

An automatic high speed, dead band type voltage regulator shall be provided complete with all accessories. The regulation system shall be provided with equipment for automatic and manual control.

The regulator shall regulate the output voltage from generator current and potential signals. Series compounding transformer shall be provided to enable maintaining adequate terminal voltage in the event of terminal faults. Alternatively, excitation system shall be provided with arrangement for field forcing. Contractor shall co-ordinate suitable of protection relays for generator with the operational characteristics of automatic voltage regulator, especially under short circuit conditions.

Voltage regulation and steady stage modulation shall be within + 1% of the line voltage with manual voltage adjustment capability within + 5%.

Necessary equipment for field suppression and surge protection shall be provided.

The response time of the exciter and the generator shall be properly matched to avoid hunting.AVR system shall be provided with equipment for automatic and remote operation/ control as required/ as per manufacturer's standards.

Necessary equipment shall be furnished for the following: -

To prevent automatic rise of field voltage in case of failure of potential supply.

Circuit to assure correct division of reactive power for parallel operation. The excitation and voltage regulation shall be designed to cause necessary de-excitation in case of short circuit. Cross current compensation circuit shall be provided.

P. Automatic Mode

While the normal mains supply is healthy the diesel alternator set will be at rest and the load will be supplied by the mains.

The AMF system shall monitor the main supply voltage & when the main supply voltage drops below a certain preset value, the system shall sense these conditions & shall give automatic start command to the control systems.

After a time delay of 1.5seconds (or set as per site conditions) from the main supply failure the diesel engine shall start.

When the DG set attains its rated speed and voltage a closing signal shall be given to the ACB / MCCB of the DG set.

When the voltage in the mains gets restored, its quality is monitored for about one minute and if proven satisfactory the main supply breaker shall close automatically transfer the load from Diesel engine to the main supply at DG Sync.Panel.

The set shall stop after idle running of one minute after restoration of main supply.

The diesel alternator set reverts to its standby conditions & it's ready to start should the mains supply fail again.

Q. Manual mode:

In a manual mode, it shall be feasible to start-up the generator set by theoperator on pressing the start push button.

Alternator circuit breakers close and trip operations shall also be throughoperator only by pressing the appropriate button on the panel and closureshall be feasible only after alternator has built up full voltage. If the load isalready on 'mains', pressure on 'close' button shall be ineffective.

Engine shut down, otherwise due to faults, shall be manual by pressing a'stop' button.

R. Test mode:

When under 'test' mode pressing of 'test' button shall complete the start-up sequence simulation and start the engine. The simulation will be that of mains failure. Sequence 33.1 (a) and (b), (c) shall be completed.

Engine shall build up voltage but the set shall not take load by closing of alternator circuit breaker. When the load is on the mains, monitoring of performance for voltage/ frequency etc. shall be feasible without supply to load.

If during test mode, the power supply has failed, the load shall automatically get transferred to alternator.

S. Test on Diesel Generator Set

The routine test shall be done as per standard in presence of TPI and Client representative.

18. CABLING SYSTEM

A. General

HV Cable shall be 33 KV or 11 kV Grade, high conductivity stranded compacted circular aluminium conductor, 3 core, XLPE insulated, inner PVC sheathed, galvanized steel strip armoured with overall separate extruded FRLS, PVC outer sheath, conforming generally to IS: 7098 (Part-II) - 1985 and amendment there of suitable for 33 kv or 11 kV 3 phase 50 Hz earthed system.

Two distinct sheaths i.e. inner and outer shall be provided. Outer sheathing shall be designed to afford high degree of mechanical protection and shall also be heat, oil, chemicals and weather resistant, commonacids, alkalies and saline solution shall not have adverse effect on the material used for PVC outer sheathing.

LV Cable should be aluminium armoured earthed XLPE type with fire retardant compound Aluminium conductor FRLS overall sheathed 1100 volts grade conforming to the quality as specified in the schedule of work. All cables, accessories and other materials should conform to IS: 7098 amended up to date Specifications. No joint allowed in any of the cabling work.

B. Design Requirement

All power cables shall be sized for continuous current carrying capacity at the ambient temperature of 50 Deg C. The design current of any circuit shall exceed the full load current of the supplied device by at least 10%. Power cables shall be sized to limit the maximum voltage drop to no more than 3 %.

C. Construction of Cable

i. Conductor

The conductor shall be made from stranded aluminium to form compact circular conductor having resistance within limits as specified in Table-2 of IS: 8130 /1984 and any amendment thereof.

ii . Conductor Shield

The conductor shall have a semi-conducting screen, which will ensure perfectly smooth profile to avoid stress concentration. The conductor screen shall be extruded in the same operation as the insulation.

iii. Insulation

The XLPE insulation shall be suitable for specified system voltage. The manufacturing process shall ensure that insulation shall be free from voids. The insulation shall withstand mechanical and thermal stresses under steady state and transient operating conditions. The extrusion method shall give very smooth interface between semi-conducting screen and insulation. The insulation of the cables shall be of high standard quality and conform to Clause-11 of IS: 7098 (Part-II)/1985 or latest amendment thereof.

iv. Insulaton Shield

To confine electrical field to the insulation, insulation screening consisting of two parts, namely metallic (non-magnetic) and non-metallic (semi conducting) shall be provided. The non-metallic semi-conducting shield shall be put over the insulation of each core. The insulation shield shall be extruded in the same operation as the conductor shield and the insulation by triple extrusion process. The insulation shield shall be bonded and Strippable, on adequate heat treatment. Metallic shield shall be provided over non-metallic portion as per provision of clause 12.4 of IS: 7098 (Part-II)/1985 and amendment thereof.

v. Inner Sheath

The sheath shall be suitable to withstand the operating conditions and the desired temperature rating of the cable. It shall be of adequate thickness, consistent quality and free from all defects.

vi. Armour

Galvanized steel strip armouring shall be provided. The dimensions of steel strip shall be as per table 4 of IS: 7098 (Part-II)/1985 and its latest amendment and strip shall conform to latest provisions of IS: 3975 - 1988 and amendment thereof.

vii. Outer Sheath

Extruded FRLS, PVC outer sheath of type ST-2 as per IS: 5831/1984 and its latest amendment shall be applied over armouring with suitable additives to prevent attack by rodent and termite and its thickness shall be in accordance with Clause -17.32 of IS:7098 (Part-III)/1985 and latest amendment thereof.
be used for the manufacture of cable and to check that manufacturer is complying with quality control.

viii. Inspection

All Cables shall be inspected at site and checked for any damage during transit.

D. CABLE LAYING

i. Installation Of Cables

Cable shall be laid in ground, trenches, cable trays and on walls as specified. Installation shall include all supports and clamps as required. The work shall be in accordance to CPWD General Specifications for Electrical works - Part II (External) amended upto date. Cables shall not be fixed on walls directly but laid on cable trays. Cables shall be plaid in separate cable trays as per voltage ratings. Power and Control cables shall also be laid in separate cable trays. Cable pit chamber shall be cover with pre cast covers.

ii. Laying Cables in Ground

Cables shall be laid by skilled experienced workmen using adequate rollers to minimize stretching of the cables. The cable drums shall be placed on jacks before unwinding the cable. With great care it shall be unrolled on over wooden rollers placed in trenches at intervals not exceeding 2 meters. Cables shall be laid at depth of 1 mtr and 0.75 meters below ground level for HT &LT cables respectively. A cushion of sand total of 250mm shall be provided both above and below the cable, joint boxes and other accessories. Cable shall not be laid in the same trench or alongside a water main. The cable shall be laid in excavated trench over 80mm layer of sand cushion. The relative position of the cables, laid in the same trench shall preserved. At all changes in direction in horizontal and vertical planes, the cables shall be bent smooth with a radius of bent not less than 12 times the diameter of cables. Minimum 3-meter-long loop shall be provided at both end of cable. Distinguishing marks may be made on the cable ends for identifications of phases. Insulation, tapes of appropriate voltage and in red, yellow and blue colours shall be wrapped just below the sockets for phase identifications.

iii. Laying in pipes/ closedducts

Cables laid in locations such as road crossing, and any other places subject to traffic, entry in to buildings, paved areas etc., cables shall be protected and laid in suitable size of pipes or closed ducts. RCC NP-3/NP4 Hume Pipe or HDPE pipe shall be used as protection pipe for cables. The size of the pipe shall not be less than 20 cm in diameter for a single cable. The maximum distance between two nos adjacent brick masonry chambers is 30/40mtr of size 100cm wide x 150cm long along Cable length with circular/ rectangular manhole with heavy duty CI manhole cover & CI manhole frame of weight not less than 54 KG each, shall be provided on continuous along length of Hume Pipe/s.

The DWC HDPE pipe shall consist of two layers, the outer layer will be corrugated and the inner layer shall be plain and smooth.DWC HDPE pipe shall be so designed and constructed that in normal use their performance is reliable and without danger to the user or surroundings. The DWC HDPE pipe and fittings hall withstand and the stresses likely to occur during transport, storage, recommended installed on practice and application.

The base HDPE resin used for the outer and inner layer of the DWCHDPE Pipe shall conform to any designation of IS: 7328or to any equivalent standard meeting the requirements given inTableNo.1, when tested as per the standards given their in. However, the manufacturers shall furnish the designation for the HDPE resin as per IS: 7328 as applicable.

vi. Laying in open ducts

Open ducts with suitable removable covers (RCC slabs or cheered plates) are generally provided in sub-stations, switch rooms, between the buildings etc., for taking the cables. The cable ducts should be of suitable dimensions for the number of cables involved. Laying of cables with different voltage ratings in the same duct shall be avoided. Splices or joints of any type shall not be permitted inside the ducts. The cables shall be laid directly in the duct such that unnecessary crossing of cables is avoided. Cables shall be fixed with clamps on the walls of the duct or taken in hooks/ brackets/ troughs in ducts. The size and material of saddles/clamps shall be as given under laying on surface.

V. Laying on surface

This method may be adopted in places like switch rooms etc. This may also be necessitated in the works of additions and/or alterations to the existing installation, where other methods of laying may not be feasible.

Cables may be laid in surface by any of the following methods as specified: -

Directly clamped by saddles / clamps or tie belt.

Supported on cradles.

vi. Laid on troughs/ trays duly clamped.

The saddles and clamps used for fixing the cables on surface shall be 1 mm thick with fixing interval of 45 cm for cable of overall diameter up to 26 mm and 3mm thick 25mm wide with fixing interval of 60 cm for cable of overall diameter up to 45 mm. For cable of overall diameters above 45 mm the clamps shall be minimum of 3 mm thick 40 mm wide and fixing interval 60 cm. Additional clamping shall be provided at 30 cm from the center of bend on both sides. Saddles shall be secured with screws to suitable approved plugs. Clamps shall be secured with nuts on to the bolts, grouted in the supporting structure in an approved manner. In the case of single core cables, the clamps shall be of non-magnetic material. Suitable non-corrosive packing shall be used for clamping unarmored cables to prevent damage to the cable sheath. Cables shall be fixed neatly without undue sag or kinks. All MS components used in fixing the cables shall be either galvanized or given a coat of red oxide primer and finished with 2 coats of approved paint.

vii. Laying on cable tray

The cable trays may be either of perforated sheet type or of ladder type. The width of cable tray shall be chosen, so as to accommodate all the cables in required tiers plus 30 to 50% additional width for future expansion. This additional width shall be minimum 100 mm. The overall width of one cable tray shall be limited to 800 mm. All cable trays should be Hot dipped galvanized type. The cable tray shall be bonded to the earth terminal of the switchboards.

Factory fabricated bends, reducers, tee/cross junctions, etc., shall be provided as per good engineering practice. The radius of bends, junctions etc., shall not be less than the minimum permissible radius of bending of the largest size of cable to be carried by the cable tray.

The cable tray shall be measured on unit length basis, along the center line of the cable tray, including bends, reducers, tees, cross joints.

viii. Laying in covered duct

The contractor shall fix the cables by using suitable clamps at angle iron. The size and material of saddles/clamps shall be as given under laying on surface. Cable trays shall have fixed on angle iron supports and taking / laying cables in Cable trays.

ix. Cables Tags

Cable tags shall be made out of 2mm thick aluminum sheets/PVC, each tag 1-1/2 inch in dia with one hole of 2.5mm dia, 6mm below the periphery. Cable designations are to be punched with letter/number punches and the tags are to be tied inside the panels beyond the glanding as well as below the glands at cable entries. Trays tags are to be tied at all bends.

X. Testing Of Cables

Routine and Acceptance Test will done at Manufacturer permiosese on all Cables. Prior to installation burying of cables, following tests shall be carried out. Insulation test between phases, phase & neutral, phase & earth
for each length of cable.

a. Before laying.
b. After laying.
c. After jointing.

On completion of cable laying work, the following tests shall be conducted in the presence of the Engineer in Charge.

a. Insulation Resistance Test (Sectional and overall).
b. Continuity Resistance Test.
c. Earth Test.

All tests shall be carried out in accordance with relevant Indian Standard code of practice and Indian Electricity Rules.

19. EARTHING SYSTEM

A. General

This specification cover earthing stations (earth mats and earth electrodes), laying copper/GI earth strips and connecting the power panels, DBs, switch boards and other equipments such as pumps and PLCs etc.

Indian TNS system of earthing as shown on IS: 3043 – 1987 shall be followed for the entire installation under the scope.

All the non-current carrying metal parts of electrical installation shall be earthed properly. All metal conduits, cable trays, trunking, cable sheaths, switchgear, distribution fuse boards, light fittings and all other parts made of metal shall be bonded together and connected by means of specified earthing conductors to an efficient earthing system. All earthing shall be in conformity with Indian Electricity Rules and CEA safety regulation in vogue.

The Earthing System shall totally comprise the following:

Earth mats and Electrodes as per IEEE 80: 2000 and IS: 3043 - 1987

Earthing Leads

Earth Conductors

Residual current earth leakage circuit breakers where applicable as per IE rules 61A.

All three phase equipments shall have two separate and distinct body earths and single phase equipment shall have a single body earth. Earthing of transformer neutral, DG set etc. by copper pits.

All the earthing strips shall be buried in floor or wall as required and all the fastening (by nut bolts) of earthing strips shall be tag welded as well.

B. Standards

The following standards and rules shall be applicable:

(i)	IS:3043 - 1987	Code of Practice for earthing
(ii)	IEEE 80: 2000	Guide for Safety in AC Substation Grounding

All codes and standards mean the latest. Where not specified otherwise the installation shall generally follow the Indian Standard Code of Practice or the British Standard Code of Practice in the absence of Indian Standards.

C. Earthing Material

Materials of which the protective system is composed shall be resistant to corrosion or be adequately protected against corrosion. Adequate margin for corrosion should be taken for the selection of the size. The material shall be complying to the following requirements.

Copper - When solid or stranded copper wire is used it shall be of the grade ordinarily required for commercial electrical work generally designated as being of 97% conductivity when annealed, conforming to Indian standard specifications as per IEEE 80 A.

Galvanized Steel - Galvanized steel used shall be thoroughly protected against corrosion by hot dipped Zinc coating. The material coating shall withstand the test specified in IS 2309 – 1989.

The strips to be used shall be in maximum lengths available as manufactured normally avoiding unnecessary joints.

D. EARTH STATIONS

i. Maintenance Free Chemical Earth Station

This section covers the technical details for Maintenance Free Chemical Earth Station with following technical features:-

The system should be of DUAL PIPE TECHNOLOGY (GI)

The chemical earth electrode should be tested in CPRI (Central Power Research Institute) for Peak & RMS current.

The earth electrode manufacturer shall be ISO 9001:2008 & ISO 14001:2004 certified company.

The earth electrode shall be of GI construction Dual Pipe technology with the galvanization of 80-100 microns.

The earth electrode shall be surrounded by resistance lowering grounding minerals to increase the overall conductive surface area in order to lower the ohmic value and also to minimize the corrosion process.

The earth electrode shall be highly conductive and non-corrosive.

Earth electrode should be able to carry the peak short circuit withstand current of minimum 30KA and duly tested and certified by CPRI.

Resistance lowering grounding minerals should be of specially developed electrolytic gel which should be of highly conductive. The “chemical” shall be made of natural minerals - mixture of Sulphate, Silica, Alumina, Iron Oxide, Titanium Oxide, Calcium Oxide, Magnesium Oxide, Sodium Oxide, Zinc Oxide, Potassium Oxide, Chloride, Nickel Oxide, etc and it absorbs the moisture from the surrounding soil and is non corrosive.

The loss on ignition by mass of the chemical compound “chemical” shall be less than 20%.

The chemical compound used for back filling around the earth electrode shall be duly tested and certified by any International accredited and BIS (Bureau of Indian Standards), Govt of India accredited laboratory. The testing laboratory should be ISO 9001 & ISO 14001 certified.

Between the inner & outer pipes, it should be adequately filled with special composition of highly conductive metallic compound which will have the continuity for fast conduction & highly resistive to corrosion.

At any cost, usage of Salt & Charcoal with the earth electrode is strictly not permitted.

ii. Number of Earth Stations

Metallic covers or supports of all medium or HT apparatus or conductors shall, in all cases be connected to not less than two separate and distinct earth electrodes.

iii. Location of Earth Electrode

The following guidelines shall be followed for locating the earth electrodes. An earth electrode shall not be situated less than 3 meters from any building. The excavations for electrode shall not affect the column footings or foundations of the buildings. Entrances, pavements and road ways shall not be used for locating the earth electrode. In such cases electrode may be further away from the building.

The location of the earth electrode / mats shall be such where the soil has reasonable chance of remaining moist, as far as possible.

iv. Method of Installing Watering Arrangement

In the case of plate earth electrode, a watering pipe of 50mm diameter of B class GI Pipe shall be provided and attached to the electrode. A funnel with mesh shall be provided at the top of this pipe for watering the earth. The watering funnel attachment shall be housed in masonry enclosure of not less than 300 x 300 x 300 mm. A cast iron/MS frame with cover having identity mode “EARTH” and having locking arrangement shall be suitably embedded in the masonry enclosure.

v. Resistance To Earth

No earth electrode shall have a greater ohmic resistance than 1 Ω as measured by an approved earth testing apparatus. In rocky soil the resistance may be up to 10 Ω. The electrical resistance measured between earth connection at the main switchboard and any other point on the completed installation shall be low enough to permit the passage of current necessary to operate fuses or circuit breakers.

Note: - All joints and connection between dissimilar metal should be done by providing Bimetallic Conductor /Plate As Required Of Equivalent Area /Cross Section.

vi. Equipment Earthing

All apparatus and equipment transmitting or utilizing power shall be earthed as per the drawings.

Metallic conduit shall not be accepted as an earth continuity conductor. A separate insulated earth continuity conductor of size related to phase conductor shall be provided. Non-metallic conduit shall have an insulated earth continuity conductor of the same size as above. All metal junction and switch boxes shall have an inside earth stud to which the earth conductor shall be connected. The earth conductor shall be distinctly coloured (green) for easy identification.

Armoured cables shall be earthed by 2 bonding earth connections to the armouring at both the ends and the size of connection being as above. In multiple cables entering a panel/ DB, the cable joints shall be bonded together using a bonding wire selected on the basis of the largest size of cable in the group. In the case of unarmored cable, an earth continuity conductor shall either be run outside along the cable. Three phase power panels and distribution boards shall have 2 distinct earth connections of the size correlated to the incoming cable size. In case of single phase DB's a single earth connection is adequate. Similarly for 3-Phase and 1-Phase isolating switches there shall be 2 and 1 earth connections respectively, sizes being correlated to the incoming cable.

Three Phase motors and other 3-Phase apparatus shall have 2 distinct earth connections of size equal to incoming feeder size. For 1-Phase motor and 1-Phase apparatus, the single earth connections shall be provided of the above size.

vii. Testing

The following earth resistance values shall be measured with an approved earth megger and recorded.

Earthing system as a whole Earth Continuity

20. LIGHTING SYSTEM

A. TECHNICAL REQUIREMENTS

i. Light Fittings

Following Technical Parameters Tobe followed forall Light Fixtures:

Efficacy of the fixture must be minimum 95 lumens/Watt,

Service Life of the fixture should be minimum 50,000 burning hours,

The CRI of the fixture should be minimum 80 for indoor applications and 70 for outdoor applications,

The THD should be less than 10%,

The housing of the indoor fixtures should be extruded aluminium/standard alloy housing,

For outdoor fixtures the housing shall be of high pressure die cast aluminium housing,

The IP category should be IP20 or higher for indoor applications and IP65 or higher for outdoor applications,

The Surge Protection to be provided conforming to relevant IS standards/IEC 61643-II Class-2 & EN 61643-II Type-2,

The manufacturers name/logo should be engraved/embossed on the housing/body or Name/Logo on aluminium plate labels or Name/logo printed on housing/body,

The warranty period on complete luminaire including driver/control gear, LED, all accessories should be 5 years from the actual date of completion of work,

The Power factor should be 0.95 or higher,

The total power consumption of the fitting should not be more than 115% of rated capacity of LED light.

There shall be no dark spot while connecting & sandwiching of linear lightings at joints.

ii. Lux Level

Following lux level shall be maintained as per below table:

S. No.	Space	Type of Lighting	Illumination Level (LUX)

S. No.	Space	Type of Lighting	Illumination Level (LUX)
i.	Pump House	LED	200
ii.	Panel Room	LED	250
iii.	Office	LED	250
Iv	Stair Case, Coridor, walkway :	LED	100
V	Bathroom / Toilet	LED	30
Vi	Street Lights	LED	20
vii	Parking Area	LED	30

iii. Lighting Fixtures and Accessories

Fittings shall be designed for continuous trouble free operation under atmospheric conditions without reduction in lamp life or without deterioration of materials and internal wiring. Degree of protection of enclosure shall be IP-65 or above for outdoor fittings.

Fittings shall be so designed as to facilitate easy maintenance including cleaning, replacement of drivers.

Out door type fittings shall be provided with polycarbonate weather proof junction boxes (IP-55) and IP-54 Control gear boxes. All Medium bay, high bay & linear continuous fixtures (all fixtures mounted in double-triple height areas) shall be supplied with polycarbonate junction box suitable for loop-in, loop-out connection by 1100 V PVC insulated copper conductor wires/cables upto 4 sq.mm. the internal wiring should be completed by the manufacturer by means of standard copper wire and terminated on the terminal block.

Each fitting shall have a terminal block suitable for loop-out connection by 1100 V PVC insulated copper conductor wires upto 4 sq.mm. the internal wiring should be completed by the manufacturer by means of standard copper wire and terminated on the terminal block.

All hardwares used in the fitting shall be suitably plated or anodized and passivated.

Earthing: Each lighting fitting shall be provided with an earthing terminal. All metal or metal enclosed parts of the housing shall be bonded and connected to the earthing terminal so as to ensure satisfactory earthing continuity throughout the fixture.

Painting/Finish: All surfaces of the fittings shall be thoroughly cleaned and degreased and the fittings shall be free from scale, rust, sharp-edges, and burns.

The housing shall be powder coated as required.

Luminaire shall have dual optics with primary optics distribution of 120 degree or higher and secondary optics as per site requirement.

Luminaire shall be designed to operate continuously for minimum 12 hours without deviation in the output parameters. Luminaires an driver both shall be BIS certified individually.

Metal used in BODY of lighting fixtures shall not be less than 0.5 mm for CRCA, 1 mm for EXT. Aluminium, 1.5 mm for PDC or heavier if so required to comply with specification of standards. Sheet steel reflectors shall have a thickness of not less than 20 SWG.

Luminaire should be covered with suitable Glass or diffuser with High Transitivity. Outdoor luminaire shall be with clear toughened glass or clear polycarbonate cover.

Lighting fixtures and accessories shall be designed for continuous trouble free operation of minimum of 12 hrs under diverse atmospheric conditions without deterioration of materials.

Gasket: An extruded silicon loop gasket shall be provided in the lantern body to ensure a weather proof seal between the cover and the metal housing to exclude the entry of dust, water, insects, etc. Luminaire should conform to degree of protection of IP 65 or above. Felt gasket will not be accepted. This point is applicable only for highbays and outdoor type lumianires.

The fixture shall be suitable to work under following ambient conditions. For outdoor type luminaire it shall consist of separate optical and control gear compartment. Driver should be easily replaceable in the field conditions. Driver shall be integrated inside the luminaire and should not be kept behind light engine.

vi. Installations

Fixtures shall be installed at mounting heights as per approved drawing.

Pendent fixtures within the same room or area, shall be installed plumb and at a uniform height from the finished floor. Adjustment of height shall be made during installation. Flush mounted recessed fixtures, shall be installed so as to completely eliminate leakage of light within the fixture and between the fixture and adjacent finish.

Fixtures mounted outlet boxes shall be rigidly secured to a fixture stud in the outlet box. Hickeys or extension pieces shall be installed where required to facilitate proper installation.

Fixtures located on the exterior of the building shall be installed with non-ferrous metal screws finished to match the fixtures.

All light fittings shall be supported with appropriate fixing accessories such as clips, supporting brackets, suspension sets, nuts, washers, screws etc. for their proper installation on different types of ceiling panels. Suspension sets shall be of adjustable type suitable to carry the weight of the lighting fittings unless otherwise stated or indicated on drawings.

v. Testing at Installation

After all, lighting fixtures are installed and are connected their respective switches, test all fixtures to ensure operation on their correct switch in the presence of the Engineer. All un-operating fixtures or ones connected to the wrong or inconveniently located switch shall be correctly connected as directed by the Engineer.

22 SUPPLY OF SAFETY DEVICES:

Supply of safety devices like Rubber mat, Gloves, First aid box, Danger boards, CO2 type fire extinguisher, Sand buckets, Fire Safety Chart, List of Emergency Telephone Number, all safety equipments for Chlorination System etc complete required for the safety of plant as per specification and IE rules one set.

9.0 Trenchless Pipe laying :

Pipe laying by trenchless shall be carried out at the locations as shown on the Drawings for Railway/ National Highway crossings and/or directed by the Employer's Representative. The trenchless pipe laying will include i) conventional method or jacking method and ii) Horizontal Direction Drilling (HDD) method, applying for crossing works of railway, major road, river or other water bodies and congested area along the road. The

purpose to employ these work methods is to avoid or minimize traffic congestion and/or public nuisance along the road in congested areas or due to necessity/requirement of agencies concerned such as major road, railway and river/channel crossing by installing underground pipes without open trench. The exact locations to apply the above work method shall be instructed by the Employer's Representative, prior to the execution of the work with ample time for preparation and construction, in accordance with the actual or latent site conditions and/or the request of agencies concerned at the time of construction.

All codes and standards referred to in these Specifications shall be the latest revision thereof and codes which are not mentioned herewith but relevant to work shall be applicable.

Unless otherwise specified or shown, conventional method will be applied for larger size of MS pipes (› 355 mm) and HDD method for smaller size of HDPE (≧ 355mm). The main pipe (water pipe) shall be installed in casing pipe by jacking method. The casing pipe shall be MS pipe as specified in Section 4 in these specifications. However, lining and coating shall not be required for the casing pipe. HDD method will be applied for smaller size of HDPE pipes without casing pipe.

The works for trenchless pipe laying shall consist of:

i. Lowering & centering, installing, jointing and testing
ii. Connection with other laid pipes at the both sides of this work section composing of jointing, thrust blocks or pipe encasement, which may be required, and other works incidentals. The cost for the connection work shall be included in other relevant items of BOQ.
iii. Bill of Quantities provided with the Bid Document is indicative and the quantity, diameter, or pipe material may vary as per the site condition and as per direction of the Employer's Representative. The Contractor should satisfy himself about this aspect before quoting.

Survey and Investigations

The Contractor shall carry out topographical survey and underground investigation at the work site to determine the alignment of pipeline, locations of jacking and receiving pits, jacking capacity, and other things incidentals for completion of the work in accordance with the requirements in the technical specifications.

The surveys and investigation shall include, but not limited to the followings:

i. Topographical conditions of the work site, for horizontal and vertical control including over burden profiles (existing utilities), shall cover entire area for the work with additional area for stockpiling of materials and equipment, connection with the pipes up and down streams of jacking work, parking of vehicles, etc. to determine the exact locations of pits and pipeline alignment
ii. Collection of information for the existing underground utilities, and carrying out test diggings to confirm the location(s) of the existing utilities and groundwater with appropriate interval (not less than 20 m) and locations (not less than two locations) up to the depth of casing/main pipe.
iii. Subsurface investigation shall consist of review of available geological information and collection of site specific geo-technical information to obtain subsurface profile and characteristics of soil as required. Sampling and testing, where required, shall be carried out to determine the capacity of equipment and dimensions of casing pipe as required. For HDD method, soil characteristics to be obtained will be, but not limited to the following items:
iv. Standard classification of soils (sieve analysis).
v. Electric resistively and mineralogical constituents.
vi. The Contractor shall carry out the additional surveys and investigation in addition to the above to complete the work satisfaction to the Employer's Representative in accordance with the manufacturer's instruction/recommendations including soil investigations.

Drawings

The Contractor, upon the site survey and investigation, shall submit drawings (GFC) which include, but not limited to the followings:

Alignment of pipeline and locations of jacking pit and receiving pit

i. Location of the work site, materials and diameter of main pipe, length of pipeline
ii. Result of surveys and investigations
iii. Method and procedures applied for proposed trenchless pipelaying
iv. Manufacturer, model number with specifications and list of the equipment.

v. Depth of casing pipe or main pipe
vi. Dimensions of casing pipe and/or main (diameter, wall thickness)
vii. Dimensions of jacking and receiving pits with structural analysis of sheeting
viii. Piping work for connection with pipelines at up-and-down streams of jacking section including thrust block/encasement of pipes other incidentals as required
ix. Ancillary works as required
x. Schedule for execution of the work including mobilization, construction/installation works, testing and site cleaning after the completion of the work
xi. Names and CV of a supervisor and a chief Contractor

Pipe Jacking

Method

Jacking shall be hydraulic jacking. The jacking station will be composing of i) thrust block made of concrete. Steel plate may be used to spread the thrust force evenly to the thrust block. ii) guide frame abutting with the thrust block for fixing jacking cylinders, iii) hydraulic power unit for cylinders, iv) gantry crane where required, and other appurtenant equipment incidental. The main pipe shall be installed in the casing pipe which is provided in accordance with the above pipe jacking method. The pipe jacking shall start after completion preparation works including construction of the jacking pit.

Construction of Jacking Pit and Receiving Pit

The jacking pit shall be suitable length and width for jacking casing pipe as well as main pipe which is installed inside of the casing pipe as shown on the Drawings and directed by the Employer's Representative. The pit shall be protected by sheeting its around to prevent any collapse of wall and to support thrust force due to jacking. The sheeting shall be steel sheeting unless otherwise specified or approved by the Employer's Representative.

Thrust block or other means approved to the sheet piles shall be provided to resist the thrust force due to jacking. Supporting frame shall be installed for jacking cylinder. The bottom of the jacking pit shall be paved with lean concrete as shown. The guide rail shall be installed on the lean concrete base to fix the direction of casing pipe for jacking. The sump pit shall be provided with an appropriate size for installing enough capacity of drain pump(s).

The construction of receiving pit shall be completed before the pipe jacking reaches to the receiving pit which shall be appropriate width and length to connect with a pipeline at the downstream of the jacking work. The receiving pit shall be protected by sheeting as the same manner specified above.

The top edge of sheet pile shall be high enough from surrounding ground levels to avoid storm water entering into the pit.

The temporary cover, where required, shall be provided when works are suspended. During night time lighting shall be provided around the pit.

Casing Pipe

Unless otherwise specified or shown on the Drawings, the casing pipe shall be MS pipe specified which diameter and wall thickness shall be determined by the Contractor and approved by the Employer's Representative. The minimum size of the casing pipe shall, however be determined as the size of main pipe plus at least 40 cm or not smaller than 700mm. Wall thickness shall be not less than 12 mm.

Pipe Jacking and Installation of Main Pipe

The casing pipe shall be set to the appropriate direction on the guide railing installed on the concrete base. The lead pipe (edge) will be set at the top of casing pipe. Minor adjustment of direction could be possible to adjust angles of the lead pipe to the axis of the casing pipe. The casing pipe is jacked by mean of hydraulic power unit through jacking cylinders. Soil in the casing pipe will be excavated manually or mechanical means dependent on the Contractor's option. Attention shall be paid that excavation beyond the casing pipe shall be prohibited to prevent void forming around casing pipe or any settlement of ground surface.

In the event of encounter to obstacles, which makes difficult or impossible to continue jacking safely and correct alignment/direction, the jacking work shall be discontinued. The Contractor shall inform immediately

to the Employer's Representative upon the above accident and propose solution for his approval. No works shall be continued before the approval of the Employer's Representative.
Drain pump(s) shall be installed for dewatering in the jacking and receiving pits and shall be ready for operation at any time. The capacity of drain pump(s) shall be large enough to dewater the expected/estimated capacity for dewatering. Stand-by pump(s) shall be provided for any failure of installed pump(s) or unexpected volume of dewatering to be required.

Installation of Main Pipe
The main pipe shall be installed in the casing pipe, upon the completion of jacking work and approved, to the predetermined line and level. Void around peripheral of the main pipe to the casing pipe shall be filled with granular materials (sand) or light weight concrete. The granular materials shall be fully compacted up to the satisfaction of the Employer's Representative to prevent any void between the casing pipe and the main pipe. The pipe shall be temporary fixed during filling granular materials or light weight concrete. After completion of filling work, the both ends of casing pipe shall be closed by concrete when granular material is applied. The thickness of concrete shall be not less than 20cm.

Horizontal Directional Drilling (HDD) Method
The HDD machine shall be capable of horizontal directional drilling involving a steerable tunneling system for all pipe sizes. It shall be a multi-stage process. The first stage shall consist of drilling a small diameter pilot hole along the predetermined centre line of a proposed pipe installation. The second stage shall consist of enlarging the pilot hole to the predetermined size to accommodate the pipeline and pulling the same through the enlarged hole. The drilling head may be by mechanical cutting or by fluid cutting with high-pressure jets.
The machine may be pit launched or surface launched. The drilling machine, especially a surface launched type machine, may incorporate an automatic drill pipe loading system with fluid assisted lubricating and bore stabilizing.
The mid range guided boring machine shall have a thrust and pull back capability of between 8 to 20 tonnes and be capable to install up to 500 mm diameter pipe over a distance of 100 to 350 meters. Boring shall be fluid assisted.

The drill head shall be angled so that constant rotation of the drill string produces a straight bore.
A sonde or beacon shall be built into the head or fixed close to it and signets emitted by this are to be picked up and traced by a receiver located on the ground surface, allowing the direction, depth, and other parameters to be monitored.
Bentonite mix water and/or polymeric mixture shall be used as the drilling fluid or 'mud'. The HDPE pipe shall be drawn in behind the reamer as the original bore is enlarged.

Tracking & Guidance System
The guided boring technique shall rely on an accurate base location and guidance system. The capability of tracking devices shall be of advance electronic technology, and of a high degree of accuracy. The sonde / probe / transmitter shall be provided inside of the drilling head, which shall be used to locate and determine the depth, pitch and position of the head, and to be received by locators above the ground.

Ancillary equipment
There are numerous accessories and ancillaries, which play an important part in the success of a guided boring / directional drilling project. Those are e.g. the swivel, duct pullers, pipe size range, etc. to accompany the HDD equipment.

Construction of Jacking and Receiving Pipe
The construction of jacking and receiving pits shall be carried with similar manner to the provisions in relevant Sub-section of this Section and in accordance with the instruction/recommendations of the manufacturer.

Installation of Pipes
The Contractor shall install the pipe, upon the completion of the preparation works including construction of the jacking pit as required, in accordance with the instruction/recommendations of the Manufacturer in accordance with the designated size of pipe and distance of pipeline.

Skilled Contractor

The directional drilling method is a system, which requires skilled Contractors. The Contractor must be knowledgeable concerning down hole drilling, the impacts of drilling in various geological formations, sensing and recording instrumentation, interpreting computer data etc.

If the opinion of the Employer's Representative that the testing, pressure and leakage, is required for the complete section of the jacking work, shall provide testing equipment and materials required for testing. The testing shall be carried out under the presence of the Employer's Representative.

River/ Nalla Crossings

The pipeline at river crossings is proposed to be supported over supporting structures / steel girders as per standard drawing. The work shall be carried out as per drawing and/or as directed by the Employer's Representative. The supply and installation of structural steel work, supply and laying, jointing of double flanged pipes, RCC works and other relevant items of these works shall be payable under respective items under BOQ.

For crossings of the road side drains, wherever required, the pipeline shall be laid below the bottom of the side drain with adequate cover. The distribution pipeline at such crossings shall be laid by carrying out only required excavation below drains by caving from both sides and simultaneously inserting the pipes. No extra payment for such crossings shall be allowed and the sub items involved shall be payable under respective BOQ items.